

IP Peripheral

tda54_analytics_cluster

MODULE SPECIFICATION

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TI-Ref	tda54_analytics_cluster
Document version	0.73
Release date	11/10/2025
Department	PTD – Analytics IP
Owner	Harshil Shah

Revision History

Revision	Author	Date	Comment
0.1	Harshil Shah	5/06/2024	Initial Draft
0.2	Harshil Shah	6/23/2024	Added memory map and IO Description
0.3	Harshil Shah	8/28/2024	Added tie-off value table
0.4	Harshil Shah	10/1/2024	Updated debug and safety information, added security section, added clock and reset diagrams, added separate clock description table
0.5	Harshil Shah	10/3/2024	Added section 2.1.2.1 for Broadcast mode
0.51	Harshil Shah	10/16/2024	Added section 4.3 for logical memory map view
0.52	Harshil Shah	10/22/2024	Updated tie-off ports and values, updated IO port section to match latest RTL
0.53	Harshil Shah	11/15/2024	Updated IO port list, fixed mismatches between AC and MSMC interfaces
0.54	Harshil Shah	12/09/2024	Updated power management section 12.3
0.55	Harshil Shah	01/02/2025	Updated IO port list to match latest release, updated power management section to add separate EL2 MEMx power domains
0.56	Harshil Shah	01/26/2025	Added list of subchips inside Analytics Cluster, Added LBIST section
0.57	Harshil Shah	02/10/2025	Updated safety section, added details about LBIST, STL and MMR diagnostics as per Abhijeet Chachad's review feedback.
0.6	Harshil Shah	05/02/2025	Updated port list, added firewall information in section 7. Updated clock and reset diagram to show msmc clock connections
0.65	Harshil Shah	05/26/2025	Updates as per feedback from Apps team
0.66	Harshil Shah	06/13/2025	Updated memory map and IO list to match 0.65 release
0.67	Harshil Shah	07/03/2025	Added LBIST sequence, added CCC spec link for CCC DMSC MMR information
0.68	Harshil Shah	07/14/2025	Added firewall diagram, added el2 subbanking implementation details
0.69	Harshil Shah	08/04/2025	Fixed EL2_SNP_CFG register definition

0.70	Harshil Shah	8/14/2025	Replace master/slave terminology with initiator/target
0.71	Harshil Shah	8/25/2025	Added power domain and LPSC interface mapping table (section 12.3.1)
0.72	Harshil Shah	10/13/2025	Added description for CAPM module in Power management section
0.73	Harshil Shah	11/10/2025	Updates per Greg's review feedback

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1. Overview and Feature List

1.1 Standards and specification Links

The TDA54_ANALYTICS_CLUSTER module complies to the following IPs :

Document	Link
TDA5 PRD	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3031&filteritem=16767
TDA5 MRD	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3024&filteritem=16722
C7xv CPU spec (original ISA and delta spec for TDA54)	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=689&filtertitle=2143&filteritem=10603 https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3043&filteritem=16856
C7xv DMC spec	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3043&filteritem=16866
C7xv PMC spec	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=689&filtertitle=2143&filteritem=10609
C7xv SE spec (original IP spec and delta spec for TDA54)	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=689&filtertitle=2143&filteritem=10604 https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3043&filteritem=16915
C7xv DebugSS spec	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=689&filtertitle=2208&filteritem=11053 https://pds.design.ti.com/d/21/2143/10637/11888/15/thinman_func.pdf
C7xv UMC/EL2 spec	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3043&filteritem=17083
C7xv MMA3_1024 spec	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3043&filteritem=16865

CMMU Spec (original spec and delta spec for TDA54)	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=689&filtertitle=2145&filteritem=10614 https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=694&filtertitle=2637&filteritem=15214
TDA5 Analytics Wrap	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3043&filteritem=16864
DRU Spec	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=694&filtertitle=2637&filteritem=13792
CLEC Spec	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3043&filteritem=17418

1.2 Features Supported

This module integrates analytics engines with shared EL2 SRAM to provide Max ~192 TOPs performance.

- 4 x Analytics engine instances each providing 48 dense TOPs and 96 sparse TOPs @1.5GHz
 - Each Analytics instance has 3MB of tightly coupled L2
- 24MB of shared EL2 between all 4 Analytics instances
 - 4 mem blocks – 6MB each
 - 4 RD and 4 WR ports per mem block
 - 16 banks per mem block
 - 128B interleaving between mem blocks
- ASIL-B Safety requirement
- No Security initiator firewall on C7xV generated accesses – Firewalls are present at memory endpoints
- There is an endpoint firewall present for 24MB EL2 and 3MB local L2 (UMC) SRAM. Firewall aggregator IP is integrated inside tda54_el2_wrap to have a single programming view of this firewall while having multiple physical copy of each.
- Debug support is required for system visibility, run control and software breakpoint.

- There is a RAT (region-based address translation) module integrated for the accesses going outside of the Analytics Cluster to enable offchip accesses which requires 48-bit physical address.
 - Some un-used internal 40-bit address range needs to be mapped to desired offchip range to enable such accesses.

2. Functional Description

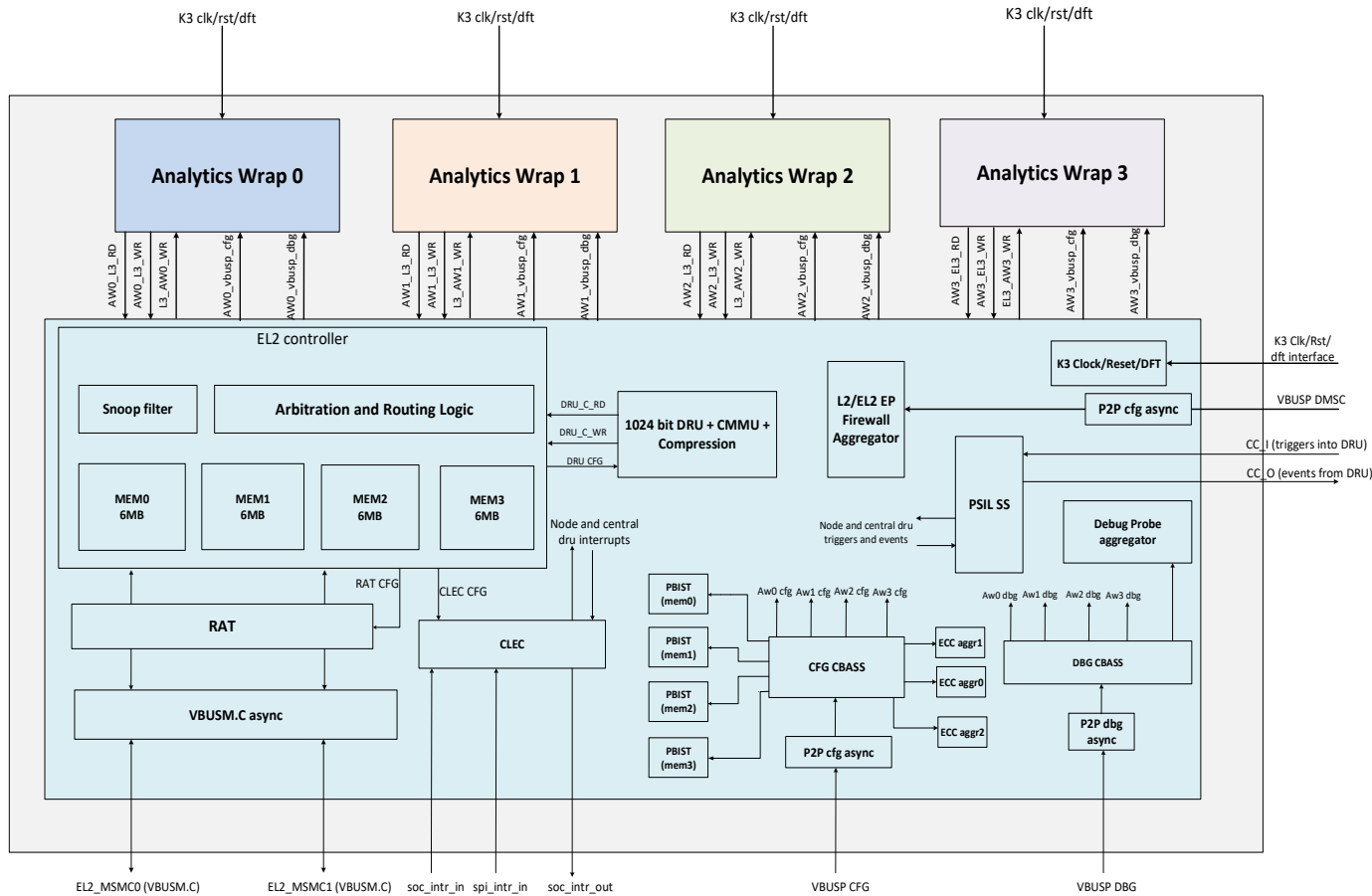


Figure 1 tda54_analytics_cluster block diagram

Analytics cluster for TDA54 includes 4 Analytics instances and one EL2 controller instance.

There is a data routing unit (DMA engine) integrated with EL2 which facilitates data transfer between DDR, EL2 and L2.

Interrupt controller (CLEC) is integrated at Analytics Cluster level and shared between all Analytics wrap instances at the cluster level.

Analytics cluster also integrates cfg, debug CBASS to route VBUSP accesses to internal end points, ECC aggregator for Safety and PBIST wrap for DFT.

There are total 17 LPSC domains within the tda54_analytics_cluster. 3 LPSC for each analytics engine (for cpu core, MMA and integration logic), and one LPSC for each EL2 mem block and one for controller and wrap.

Boundary between Analytics cluster and MSMC/SOC is asynchronous. Asynchronous bridges are placed inside Analytics cluster for VBUSM.C, VBUSP and ATB interfaces.

2.1 Summary of Features for Analytics Cluster block

2.1.1 Analytics Engine

- 1024 C7x/MMA module – provides dense 48 TOPS and sparse 96 TOPS performance @1.5GHz
- C7x ISA version: **C7604**
- 32KB L1P\$, 4-way set associative
- 128KB L1D\$, 2-way, set associative
- 2x 1024-bit Streaming engine
- 3MB of local L2 SRAM to store feature maps and feature co-efficients
- Node DRU to transfer data between EL2 and L2s.

2.1.2 EL2 Memory system

- Total 24MB of shared EL2 SRAM to store input/output feature maps and feature co-efficients
- Total 4 mem blocks, interleaved at 128B granular size
 - mem0 – 6MB
 - mem1 – 6MB
 - mem2 – 6MB
 - mem3 – 6MB

#INTERNAL#

Below information is for IP/SOC DV to help create preload scripts for these SRAMs

Number of banks per mem block = 16

Number of subbanks = 4 (for 128B wide functional access, each subbank width = 128B+ecc/4)

Subbank split = 2 (i.e. 2 macros inside memwrap to create 266 functional width)

Subbank SRAM width = 266 (each macro width = 266/subbank split = 133)

#/INTERNAL#

- Port based design to access each mem block
 - 4 Read, 4 Write ports per mem block
- Controller provides full read throughput to one SE per analytics wrap in parallel.
- Controller provides full write throughput to each DMC in analytics wrap in parallel.
- Controller includes snoop filter to track coherent DDR accesses from each L1D\$
 - Each L1D\$ is 128KB with 128B cacheline size
 - To cover all lines cached by 4 x L1D\$, snoop filter is sized to track all 4k cache lines
 - Analytics Cluster is IO coherent with external caches and doesn't support external snoops.
 - If L1D\$ has local L2 or L2 of some other core cached, that is not being tracked by this snoop filter and software needs to maintain coherency if there is data sharing in such cases.
- Controller provides two VBUSM.C interfaces to MSMC which sends 128B interleaved traffic on two interfaces
- Controller has below lister patron ports
 - umc<0-3>_el2_rd → analytics_wrap 0 to 3 read port
 - umc<0-3>_el2_wr → analytics_wrap 0 to 3 write port
 - c_dru_rd → central DRU read port
 - c_dru_wr → central DRU write port
 - mmu_el2 → central DRU's CMMU port
 - msmc_0_el2 → MSMC VBUSM.C port 0 (bi-directional)
 - msmc_1_el2 → MSMC VBUSM.C port 1 (bi-directional)

- Controller has below listed Server ports
 - el2_0_msmc → MSMC VBUSM.C port 0 (bi-directional)
 - el2_1_msmc → MSMC VBUSM.C port 1 (bi-directional)
 - el2_umc<0-3> → EL2 to Analytics_wrap 0-3 access port
 - el2_dru_mmr_cfg → central dru mmr config access
 - el2_clec_mmr_cfg → clec mmr config access

2.1.2.1 EL2 controller - UMC Broadcast space

EL2 controller has a broadcast write feature. If a write from the DRU or MSMC0 matches the rsel_tieoff_umc_broadcast value ([section 4.2](#)), then that write data will be sent as a write command to each of the attached EL2_UMC ports. For each EL2_UMC port, the upper bits of the address will be replaced with the correct value of rsel_tieoff_umc_mem0-3 so that the connected UMC will not return an addressing error. Note that broadcast writes are restricted to the UMC mem0 address base. If the attached UMC is configured to have mem0 and mem1 as separate memory regions, mem1 will not be accessible for broadcast writes unless the 2MB mem1 region tieoff is nested with the 16 MB region that is defined by the mem0 base address configuration tie-off.

Broadcast write feature allows faster transfers of feature maps and/or feature coefficients from DDR to local L2 memory using single DRU transfer instead of doing 4 sequential transfers if they are common across 4 analytics instances.

2.1.3 Central Data routing unit

- Supported Bus type VBUSM with 1024-bit data width
- 4 round robin queues and 2 priority queues
- 32 channels
- Supports 64 in-flight requests
- Constant copy feature support
- Circular buffer addressing mode support
- Static TR support
- Bit expansion/contraction support
- Forced teardown support
- Analytics Compression supported
- Virtualization supported (i.e. CMMU integrated)
 - CMMU integrated in the DRU has same features as C7xV's CMMU

#INTERNAL#

Below Interrupt table needs to be incorporated into SOC interrupt spec for it to make sense to customers since this spec doesn't cover where these interrupts are connected at SOC level.

3. Interrupt Condition

Name	Width	Type	Polarity	Category	Clock	Reset	Description
aw0_dft_pbist_cpu	1	pulse	high	event	aw0_c7xv_clk_clk	aw0_pbist_rst_mod_g_rst_n	
aw0_dft_pbist_safety_error	1	level	high	safety_event	aw0_c7xv_clk_clk	aw0_pbist_rst_mod_g_rst_n	
aw0_c7xv_core_dft_pbist_cpu	1	pulse	high	event	aw0_c7xv_clk_clk	aw0_c7xv_core_pbist_rst_mod_g_rst_n	
aw0_c7xv_core_dft_pbist_safety_error	1	level	high	safety_event	aw0_c7xv_clk_clk	aw0_c7xv_core_pbist_rst_mod_g_rst_n	
aw1_dft_pbist_cpu	1	pulse	high	event	aw1_c7xv_clk_clk	aw1_pbist_rst_mod_g_rst_n	
aw1_dft_pbist_safety_error	1	level	high	safety_event	aw1_c7xv_clk_clk	aw1_pbist_rst_mod_g_rst_n	
aw1_c7xv_core_dft_pbist_cpu	1	pulse	high	event	aw1_c7xv_clk_clk	aw1_c7xv_core_pbist_rst_mod_g_rst_n	
aw1_c7xv_core_dft_pbist_safety_error	1	level	high	safety_event	aw1_c7xv_clk_clk	aw1_c7xv_core_pbist_rst_mod_g_rst_n	
aw2_dft_pbist_cpu	1	pulse	high	event	aw2_c7xv_clk_clk	aw2_pbist_rst_mod_g_rst_n	
aw2_dft_pbist_safety_error	1	level	high	safety_event	aw2_c7xv_clk_clk	aw2_pbist_rst_mod_g_rst_n	
aw2_c7xv_core_dft_pbist_cpu	1	pulse	high	event	aw2_c7xv_clk_clk	aw2_c7xv_core_pbist_rst_mod_g_rst_n	
aw2_c7xv_core_dft_pbist_safety_error	1	level	high	safety_event	aw2_c7xv_clk_clk	aw2_c7xv_core_pbist_rst_mod_g_rst_n	
aw3_dft_pbist_cpu	1	pulse	high	event	aw3_c7xv_clk_clk	aw3_pbist_rst_mod_g_rst_n	
aw3_dft_pbist_safety_error	1	level	high	safety_event	aw3_c7xv_clk_clk	aw3_pbist_rst_mod_g_rst_n	
aw3_c7xv_core_dft_pbist_cpu	1	pulse	high	event	aw3_c7xv_clk_clk	aw3_c7xv_core_pbist_rst_mod_g_rst_n	
aw3_c7xv_core_dft_pbist_safety_error	1	level	high	safety_event	aw3_c7xv_clk_clk	aw3_c7xv_core_pbist_rst_mod_g_rst_n	
el2_soc_events_in	128	pulse	high	func	async	el2_wrap_rst_mod_g_rst_n	SOC Events in to CLEC
el2_soc_events_out_level	128	level	high	level	el2_clk_clk	el2_wrap_rst_mod_g_rst_n	Events out of CLEC - Level
el2_esm_events_out_pulse	64	pulse	high	safety_error	el2_clk_clk	el2_wrap_rst_mod_g_rst_n	Events out of CLEC - Pulse
el2_esm_events_out_level	64	level	high	safety_error	el2_clk_clk	el2_wrap_rst_mod_g_rst_n	ESM Events out of CLEC - Level
el2_arm_0_evento_in	1	pulse	high	func	async	el2_wrap_rst_mod_g_rst_n	ARM evento to CLEC
el2_arm_0_eventi_out	1	pulse	high	func	el2_clk_clk	el2_wrap_rst_mod_g_rst_n	ARM eventi from CLEC
el2_gic_spi	960	prog	high	func	async	el2_wrap_rst_mod_g_rst_n	Shared Peripheral Interrupt (SPI) inputs
el2_0_dft_pbist_cpu	1	pulse	high	event	async	el2_wrap_rst_mod_g_rst_n	
el2_0_dft_pbist_safety_error	1	level	high	safety_event	async	el2_wrap_rst_mod_g_rst_n	
el2_1_dft_pbist_cpu	1	pulse	high	event	async	el2_wrap_rst_mod_g_rst_n	
el2_1_dft_pbist_safety_error	1	level	high	safety_event	async	el2_wrap_rst_mod_g_rst_n	
el2_2_dft_pbist_cpu	1	pulse	high	event	async	el2_wrap_rst_mod_g_rst_n	
el2_2_dft_pbist_safety_error	1	level	high	safety_event	async	el2_wrap_rst_mod_g_rst_n	
el2_3_dft_pbist_cpu	1	pulse	high	event	async	el2_wrap_rst_mod_g_rst_n	
el2_3_dft_pbist_safety_error	1	level	high	safety_event	async	el2_wrap_rst_mod_g_rst_n	

#/INTERNAL#

4. Memory Map

4.1 Legend

The following sections describe the register map and register definitions for each region in the design. The state of each register after reset is shown. The key used to describe each register field is described in the following table.

Type	Description
r	A read-only field. Writes have no effect.
w	A write-only field. Reads return zeroes.
rw	A readable and writable field. Writing any value to this field results in the corresponding bits written with value written. Reads do not alter the value of the field.
rw1tc	A write ones to clear field. Writing a one to any bit will clear the corresponding bits. Reads do not alter the value of the field.
rw1ts	A write ones to set bit. Writing a one to any bit will set the corresponding bit. Reads do not alter the value of the field.
rwi	A write to increment field. Writing a value to this field increment the field value by the value written. Reads do not alter the value of the field.
rwd	A write to decrement field. Writing a value to this field decrements the field value by the value written. Reads do not alter the value of the field.
w1tc	A write-only write ones to clear field. Writing a one to any bit will clear the corresponding bits. Reads return zeroes.
w1ts	A write-only write ones to set bit. Writing a one to any bit will set the corresponding bit. Reads return zeroes.
wi	A write-only write to increment field. Writing a value to this field increment the field value by the value written. Reads return zeroes.
wd	A write-only write to decrement field. Writing a value to this field decrements the field value by the value written. Reads return zeroes.

4.2 SOC Address mapping for cluster components

Instance	Definition	Base	Size
Analytics Wrap 0	L2 SRAM + CFG + Node DRU CFG	0x6400_0000	16MB
Analytics Wrap 1	L2 SRAM + CFG + Node DRU CFG	0x6500_0000	16 MB
Analytics Wrap 2	L2 SRAM + CFG + Node DRU CFG	0x6600_0000	16MB
Analytics Wrap 3	L2 SRAM + CFG + Node DRU CFG	0x6700_0000	16MB
L2 Broadcast	For Central DRU	0x6A00_0000	8MB
Central DRU CFG + CLEC CFG	CFG	0x6A80_0000	8MB
EL2 SRAM	EL2 SRAM	0x6800_0000	24MB (total 32MB allocated)

Note: Any access not mapped to internal address range should route to SOC port

4.3 Analytics Cluster logical memory map view

Instance	Base	Implemented region size
AW0 SRAM	0x6400_0000	3MB
AW0 UMC CFG	0x6480_0000	1MB
AW0 DRU CFG	0x64C0_0000	1MB
AW1 SRAM	0x6500_0000	3MB
AW1 UMC CFG	0x6580_0000	1MB
AW1 DRU CFG	0x65C0_0000	1MB
AW2 SRAM	0x6600_0000	3MB
AW2 UMC CFG	0x6680_0000	1MB
AW2 DRU CFG	0x66C0_0000	1MB
AW3 SRAM	0x6700_0000	3MB
AW3 UMC CFG	0x6780_0000	1MB
AW3 DRU CFG	0x67C0_0000	1MB
L2 Broadcast	0x6A00_0000	3MB
EL2 CFG	0x6A80_0000	1MB
Central DRU CFG	0x6AC0_0000	1MB
RAT CFG	0x6AD0_0000	1MB
CLEC CFG	0x6AA0_0000	1MB
EL2 SRAM	0x6800_0000	24MB

4.4 Analytics Cluster memory map tie-offs

Mregion tie-offs	Tieoff value	Tie-off width	Base address	Region Size	Implemented Region
aw0_c7xv_wrap_tieoff_rsel_umc_mem0	0x64	16	0x6400_0000	16MB	3MB interleaved, 1.5MB non-interleaved*
aw0_c7xv_wrap_tieoff_rsel_umc_mem1	0x321	19	0x6420_0000	2MB	1.5 MB (non-interleaved)*
aw0_c7xv_wrap_tieoff_rsel_umc_cfg	0x648	20	0x6480_0000	1MB	1MB
aw0_c7xv_wrap_tieoff_rsel_dru_cfg	0x64C	20	0x64C0_0000	1MB	1MB
aw1_c7xv_wrap_tieoff_rsel_umc_mem0	0x65	16	0x6500_0000	16MB	3MB interleaved, 1.5MB non-interleaved*
aw1_c7xv_wrap_tieoff_rsel_umc_mem1	0x329	19	0x6520_0000	2MB	1.5 MB (non-interleaved)*
aw1_c7xv_wrap_tieoff_rsel_umc_cfg	0x658	20	0x6580_0000	1MB	1MB
aw1_c7xv_wrap_tieoff_rsel_dru_cfg	0x65C	20	0x65C0_0000	1MB	1MB
aw2_c7xv_wrap_tieoff_rsel_umc_mem0	0x66	16	0x6600_0000	16MB	3MB interleaved, 1.5MB non-interleaved*
aw2_c7xv_wrap_tieoff_rsel_umc_mem1	0x331	19	0x6620_0000	2MB	1.5 MB (non-interleaved)*

aw2_c7xv_wrap_tieoff_rsel_umc_cfg	0x668	20	0x6680_0000	1MB	1MB
aw2_c7xv_wrap_tieoff_rsel_dru_cfg	0x66C	20	0x66C0_0000	1MB	1MB
aw3_c7xv_wrap_tieoff_rsel_umc_mem0	0x67	16	0x6700_0000	16MB	3MB interleaved, 1.5MB non-interleaved*
aw3_c7xv_wrap_tieoff_rsel_umc_mem1	0x339	19	0x6720_0000	2MB	1.5 MB (non-interleaved)*
aw3_c7xv_wrap_tieoff_rsel_umc_cfg	0x678	20	0x6780_0000	1MB	1MB
aw3_c7xv_wrap_tieoff_rsel_dru_cfg	0x67C	20	0x67C0_0000	1MB	1MB
el2_addr_tieoffs_rsel_tieoff_clec_cfg	0x6AA	20	0x6AA0_0000	1MB	1MB
el2_addr_tieoffs_rsel_tieoff_dru_cfg	0x356	19	0x6AC0_0000	2MB	1MB DRU CFG, 1MB RAT CFG
el2_addr_tieoffs_rsel_tieoff_el2_cfg	0x6A8	20	0x6A80_0000	1MB	1MB
el2_addr_tieoffs_rsel_tieoff_el2_mem0	0x34	15	0x6800_0000	32MB	24MB interleaved, 6MB non-interleaved*
el2_addr_tieoffs_rsel_tieoff_el2_mem1	0xD1	17	0x6880_0000	8MB	6MB non-interleaved*
el2_addr_tieoffs_rsel_tieoff_el2_mem2	0xD2	17	0x6900_0000	8MB	6MB non-interleaved*
el2_addr_tieoffs_rsel_tieoff_el2_mem3	0xD3	17	0x6980_0000	8MB	6MB non-interleaved*
el2_addr_tieoffs_rsel_tieoff_umc0_mem	0x64	16	0x6400_0000	16MB	16MB
el2_addr_tieoffs_rsel_tieoff_umc1_mem	0x65	16	0x6500_0000	16MB	16MB
el2_addr_tieoffs_rsel_tieoff_umc2_mem	0x66	16	0x6600_0000	16MB	16MB
el2_addr_tieoffs_rsel_tieoff_umc3_mem	0x67	16	0x6700_0000	16MB	16MB
el2_addr_tieoffs_rsel_tieoff_umc_broadcast	0xD4	17	0x6A00_0000	8MB	3MB (same as UMC SRAM size)

*When interleave mode is enabled (which is default) for UMC and EL2 memory space, total implemented memory (3MB for UMC and 24MB for EL2) is mapped at the base address of awX_c7xv_wrap_tieoff_rsel_umc_mem0 for UMC and el2_addr_tieoffs_rsel_tieoff_el2_mem0 for EL2, each memory bank is then interleaved at 128B granularity. When interleave mode is off, different memory banks starts at their respective tieoff base address and may not be contiguous.

#INTERNAL#

Below information is for SOC integration and should not be added in customer facing document.

Analytics Wrap Tie-offs	Tieoff value	Tie-off width	Comments
aw0_c7xv_wrap_tieoff_clusternum	0x1	8	
aw0_c7xv_wrap_tieoff_corepacnum	0x4	8	
aw0_c7xv_wrap_tieoff_lendian	CCC_MMR	1	Driven by CCC Corepac Configuration Registers
aw0_c7xv_wrap_tieoff_el2_rd_crouteid	0x40	12	
aw0_c7xv_wrap_tieoff_el2_wr_crouteid	0x40	12	
aw0_c7xv_wrap_tieoff_relaxed_ordering	CCC_MMR	16	Driven by CCC C7x MISC Configuration Register
aw0_c7xv_wrap_tieoff_sys_mma_present	CCC_MMR	1	Driven by SOC MMR

aw0_c7xv_wrap_tieoff_sys_istp_reset_value	CCC_MMR	48	Driven by CCC Core-x Boot Vector Register
aw0_c7xv_wrap_tieoff_sys_riscv_mode	CCC_MMR	1	Driven by CCC Corepac Configuration Registers
aw1_c7xv_wrap_tieoff_clusternum	0x1	8	
aw1_c7xv_wrap_tieoff_corepacnum	0x5	8	
aw1_c7xv_wrap_tieoff_lendian	CCC_MMR	1	Driven by CCC Corepac Configuration Registers
aw1_c7xv_wrap_tieoff_el2_rd_crouteid	0x48	12	
aw1_c7xv_wrap_tieoff_el2_wr_crouteid	0x48	12	
aw1_c7xv_wrap_tieoff_relaxed_ordering	CCC_MMR	16	Driven by CCC C7x MISC Configuration Register
aw1_c7xv_wrap_tieoff_sys_mma_present	CCC_MMR	1	Driven by SOC MMR
aw1_c7xv_wrap_tieoff_sys_istp_reset_value	CCC_MMR	48	Driven by CCC Core-x Boot Vector Register
aw1_c7xv_wrap_tieoff_sys_riscv_mode	CCC_MMR	1	Driven by CCC Corepac Configuration Registers
aw2_c7xv_wrap_tieoff_clusternum	0x1	8	
aw2_c7xv_wrap_tieoff_corepacnum	0x6	8	
aw2_c7xv_wrap_tieoff_lendian	CCC_MMR	1	Driven by CCC Corepac Configuration Registers
aw2_c7xv_wrap_tieoff_el2_rd_crouteid	0x50	12	
aw2_c7xv_wrap_tieoff_el2_wr_crouteid	0x50	12	
aw2_c7xv_wrap_tieoff_relaxed_ordering	CCC_MMR	16	Driven by CCC DMSC MMRs
aw2_c7xv_wrap_tieoff_sys_mma_present	CCC_MMR	1	Driven by SOC MMR
aw2_c7xv_wrap_tieoff_sys_istp_reset_value	CCC_MMR	48	Driven by CCC Core-x Boot Vector Register
aw2_c7xv_wrap_tieoff_sys_riscv_mode	CCC_MMR	1	Driven by CCC Corepac Configuration Registers
aw3_c7xv_wrap_tieoff_clusternum	0x1	8	
aw3_c7xv_wrap_tieoff_corepacnum	0x7	8	
aw3_c7xv_wrap_tieoff_lendian	CCC_MMR	1	Driven by CCC Corepac Configuration Registers
aw3_c7xv_wrap_tieoff_el2_rd_crouteid	0x58	12	
aw3_c7xv_wrap_tieoff_el2_wr_crouteid	0x58	12	
aw3_c7xv_wrap_tieoff_relaxed_ordering	CCC_MMR	16	Driven by CCC DMSC MMRs
aw3_c7xv_wrap_tieoff_sys_mma_present	CCC_MMR	1	Driven by SOC MMR
aw3_c7xv_wrap_tieoff_sys_istp_reset_value	CCC_MMR	48	Driven by CCC Core-x Boot Vector Register
aw3_c7xv_wrap_tieoff_sys_riscv_mode	CCC_MMR	1	Driven by CCC Corepac Configuration Registers
routeid_tieoff_el2_msmc0	0x40	12	
routeid_tieoff_el2_msmc1	0x40	12	



tieoff_msmc0_el2_relaxed_ordering	CCC_MMR	16	Driven by CCC AC MISC Configuration Register
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CCC DMSC MMR information can be found in CCC spec:

[https://pds.design.ti.com/cgi-](https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3045&filteritem=17220)

[bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3045&filteritem=17220](https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=766&filtertitle=3045&filteritem=17220)

#/INTERNAL#

*For Register definition, refer to Analytics Wrap and other IP specs, Links are provided in section 1.1. Below Table provides base addresses of different memory regions inside Analytics Cluster.

4.4.1 Register Map

Interface name	Region name	Addr size	Byte offset	Request ID
ac_vbusp_cfg	aw0_c7xv_cfg_mmr	16	0x100000	0
ac_vbusp_cfg	aw0_vbusp_pbist_cfg VBUSP PBIST	10	0x100000	1
ac_vbusp_cfg	aw0_vbusp_core_pbist_cfg VBUSP PBIST	10	0x100400	1
ac_vbusp_cfg	aw0_c7xv_ecc_aggr_regs	10	0x100000	2
ac_vbusp_cfg	aw1_c7xv_cfg_mmr	16	0x110000	0
ac_vbusp_cfg	aw1_vbusp_pbist_cfg VBUSP PBIST	10	0x110000	1
ac_vbusp_cfg	aw1_vbusp_core_pbist_cfg VBUSP PBIST	10	0x110400	1
ac_vbusp_cfg	aw1_c7xv_ecc_aggr_regs	10	0x110000	2
ac_vbusp_cfg	aw2_c7xv_cfg_mmr	16	0x120000	0
ac_vbusp_cfg	aw2_vbusp_pbist_cfg VBUSP PBIST	10	0x120000	1
ac_vbusp_cfg	aw2_vbusp_core_pbist_cfg VBUSP PBIST	10	0x120400	1
ac_vbusp_cfg	aw2_c7xv_ecc_aggr_regs	10	0x120000	2
ac_vbusp_cfg	aw3_c7xv_cfg_mmr	16	0x130000	0
ac_vbusp_cfg	aw3_vbusp_pbist_cfg VBUSP PBIST	10	0x130000	1
ac_vbusp_cfg	aw3_vbusp_core_pbist_cfg VBUSP PBIST	10	0x130400	1
ac_vbusp_cfg	aw3_c7xv_ecc_aggr_regs	10	0x130000	2
ac_vbusp_cfg	el2_pbist_0_mem	10	0x140000	1
ac_vbusp_cfg	el2_pbist_1_mem	10	0x140400	1
ac_vbusp_cfg	el2_pbist_2_mem	10	0x140800	1
ac_vbusp_cfg	el2_pbist_3_mem	10	0x140c00	1
ac_vbusp_cfg	el2_ecc_aggr_regs	10	0x140000	2
ac_vbusp_cfg	el2_ecc_aggr_clk4_regs	10	0x150000	2
ac_vbusp_cfg	el2_ecc_aggr_msmc_regs	10	0x160000	2
ac_vbusp_cfg	ac_c7xv_cfg_mmr	16	0x140000	0
ac_vbusp_dbg	aw0_thinman_reg	12	0x1400000	0
ac_vbusp_dbg	aw0_columbo_reg	12	0x1401000	0
ac_vbusp_dbg	aw0_matlock_reg	12	0x1402000	0
ac_vbusp_dbg	aw0_cscti_cfg	12	0x1403000	0
ac_vbusp_dbg	aw0_ctset2_cfg	13	0x1408000	0
ac_vbusp_dbg	aw0_DBG_CT12	12	0x140a000	0
ac_vbusp_dbg	aw0_DBG_CT13	12	0x140b000	0
ac_vbusp_dbg	aw0_DBG_AGR0_mmr_slv	8	0x1440000	0
ac_vbusp_dbg	aw0_DBG_AGR0_vbus_cfg_regs	8	0x1440100	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem0	12	0x1460000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem1	12	0x1461000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem2	12	0x1462000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem3	12	0x1463000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem4	12	0x1464000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem5	12	0x1465000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem6	12	0x1466000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem7	12	0x1467000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem8	12	0x1468000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem9	12	0x1469000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem10	12	0x146a000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem11	12	0x146b000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem12	12	0x146c000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem13	12	0x146d000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem14	12	0x146e000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem15	12	0x146f000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem16	12	0x1470000	0



ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem17	12	0x1471000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem18	12	0x1472000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem19	12	0x1473000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem20	12	0x1474000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem21	12	0x1475000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem22	12	0x1476000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem23	12	0x1477000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem24	12	0x1478000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem25	12	0x1479000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem26	12	0x147a000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem27	12	0x147b000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem28	12	0x147c000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem29	12	0x147d000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem30	12	0x147e000	0
ac_vbusp_dbg	aw0_DBG_AGR0_src_vbus_mem31	12	0x147f000	0
ac_vbusp_dbg	aw1_thinman_reg	12	0x1800000	0
ac_vbusp_dbg	aw1_columbo_reg	12	0x1801000	0
ac_vbusp_dbg	aw1_matlock_reg	12	0x1802000	0
ac_vbusp_dbg	aw1_cscti_cfg	12	0x1803000	0
ac_vbusp_dbg	aw1_ctset2_cfg	13	0x1808000	0
ac_vbusp_dbg	aw1_DBG_CTI2	12	0x180a000	0
ac_vbusp_dbg	aw1_DBG_CTI3	12	0x180b000	0
ac_vbusp_dbg	aw1_DBG_AGR0_mmr_slv	8	0x1840000	0
ac_vbusp_dbg	aw1_DBG_AGR0_vbus_cfg_regs	8	0x1840100	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem0	12	0x1860000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem1	12	0x1861000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem2	12	0x1862000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem3	12	0x1863000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem4	12	0x1864000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem5	12	0x1865000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem6	12	0x1866000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem7	12	0x1867000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem8	12	0x1868000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem9	12	0x1869000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem10	12	0x186a000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem11	12	0x186b000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem12	12	0x186c000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem13	12	0x186d000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem14	12	0x186e000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem15	12	0x186f000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem16	12	0x1870000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem17	12	0x1871000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem18	12	0x1872000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem19	12	0x1873000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem20	12	0x1874000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem21	12	0x1875000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem22	12	0x1876000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem23	12	0x1877000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem24	12	0x1878000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem25	12	0x1879000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem26	12	0x187a000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem27	12	0x187b000	0

ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem28	12	0x187c000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem29	12	0x187d000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem30	12	0x187e000	0
ac_vbusp_dbg	aw1_DBG_AGR0_src_vbus_mem31	12	0x187f000	0
ac_vbusp_dbg	aw2_thinman_reg	12	0x1c00000	0
ac_vbusp_dbg	aw2_columbo_reg	12	0x1c01000	0
ac_vbusp_dbg	aw2_matlock_reg	12	0x1c02000	0
ac_vbusp_dbg	aw2_cscti_cfg	12	0x1c03000	0
ac_vbusp_dbg	aw2_ctset2_cfg	13	0x1c08000	0
ac_vbusp_dbg	aw2_DBG_CTI2	12	0x1c0a000	0
ac_vbusp_dbg	aw2_DBG_CTI3	12	0x1c0b000	0
ac_vbusp_dbg	aw2_DBG_AGR0_mmr_slv	8	0x1c40000	0
ac_vbusp_dbg	aw2_DBG_AGR0_vbus_cfg_regs	8	0x1c40100	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem0	12	0x1c60000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem1	12	0x1c61000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem2	12	0x1c62000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem3	12	0x1c63000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem4	12	0x1c64000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem5	12	0x1c65000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem6	12	0x1c66000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem7	12	0x1c67000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem8	12	0x1c68000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem9	12	0x1c69000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem10	12	0x1c6a000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem11	12	0x1c6b000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem12	12	0x1c6c000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem13	12	0x1c6d000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem14	12	0x1c6e000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem15	12	0x1c6f000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem16	12	0x1c70000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem17	12	0x1c71000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem18	12	0x1c72000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem19	12	0x1c73000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem20	12	0x1c74000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem21	12	0x1c75000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem22	12	0x1c76000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem23	12	0x1c77000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem24	12	0x1c78000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem25	12	0x1c79000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem26	12	0x1c7a000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem27	12	0x1c7b000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem28	12	0x1c7c000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem29	12	0x1c7d000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem30	12	0x1c7e000	0
ac_vbusp_dbg	aw2_DBG_AGR0_src_vbus_mem31	12	0x1c7f000	0
ac_vbusp_dbg	aw3_thinman_reg	12	0x2000000	0
ac_vbusp_dbg	aw3_columbo_reg	12	0x2001000	0
ac_vbusp_dbg	aw3_matlock_reg	12	0x2002000	0
ac_vbusp_dbg	aw3_cscti_cfg	12	0x2003000	0
ac_vbusp_dbg	aw3_ctset2_cfg	13	0x2008000	0
ac_vbusp_dbg	aw3_DBG_CTI2	12	0x200a000	0
ac_vbusp_dbg	aw3_DBG_CTI3	12	0x200b000	0



ac_vbusp_dbg	aw3_DBG_AGR0_mmr_slv	8	0x2040000	0
ac_vbusp_dbg	aw3_DBG_AGR0_vbus_cfg_regs	8	0x2040100	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem0	12	0x2060000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem1	12	0x2061000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem2	12	0x2062000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem3	12	0x2063000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem4	12	0x2064000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem5	12	0x2065000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem6	12	0x2066000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem7	12	0x2067000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem8	12	0x2068000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem9	12	0x2069000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem10	12	0x206a000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem11	12	0x206b000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem12	12	0x206c000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem13	12	0x206d000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem14	12	0x206e000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem15	12	0x206f000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem16	12	0x2070000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem17	12	0x2071000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem18	12	0x2072000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem19	12	0x2073000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem20	12	0x2074000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem21	12	0x2075000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem22	12	0x2076000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem23	12	0x2077000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem24	12	0x2078000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem25	12	0x2079000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem26	12	0x207a000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem27	12	0x207b000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem28	12	0x207c000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem29	12	0x207d000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem30	12	0x207e000	0
ac_vbusp_dbg	aw3_DBG_AGR0_src_vbus_mem31	12	0x207f000	0
ac_vbusp_dbg	el2_cscti4_cfg	12	0x3000000	0
ac_vbusp_dbg	el2_cscti5_cfg	12	0x3001000	0
ac_vbusp_dbg	el2_cscti6_cfg	12	0x3002000	0
ac_vbusp_dbg	el2_cscti7_cfg	12	0x3003000	0
ac_vbusp_dbg	el2_probe_aggr_mmr_slv	8	0x3040000	0
ac_vbusp_dbg	el2_probe_aggr_vbus_cfg_regs	8	0x3040100	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem0	12	0x3060000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem1	12	0x3061000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem2	12	0x3062000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem3	12	0x3063000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem4	12	0x3064000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem5	12	0x3065000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem6	12	0x3066000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem7	12	0x3067000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem8	12	0x3068000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem9	12	0x3069000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem10	12	0x306a000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem11	12	0x306b000	0



ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem12	12	0x306c000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem13	12	0x306d000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem14	12	0x306e000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem15	12	0x306f000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem16	12	0x3070000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem17	12	0x3071000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem18	12	0x3072000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem19	12	0x3073000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem20	12	0x3074000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem21	12	0x3075000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem22	12	0x3076000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem23	12	0x3077000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem24	12	0x3078000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem25	12	0x3079000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem26	12	0x307a000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem27	12	0x307b000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem28	12	0x307c000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem29	12	0x307d000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem30	12	0x307e000	0
ac_vbusp_dbg	el2_probe_aggr_src_vbus_mem31	12	0x307f000	0
msmc0_el2	aw0_sram	22	0x64000000	0
msmc0_el2	aw0_umc_cfg	20	0x64800000	0
msmc0_el2	aw0_dru_cfg	20	0x64c00000	0
msmc0_el2	aw1_sram	22	0x65000000	0
msmc0_el2	aw1_umc_cfg	20	0x65800000	0
msmc0_el2	aw1_dru_cfg	20	0x65c00000	0
msmc0_el2	aw2_sram	22	0x66000000	0
msmc0_el2	aw2_umc_cfg	20	0x66800000	0
msmc0_el2	aw2_dru_cfg	20	0x66c00000	0
msmc0_el2	aw3_sram	22	0x67000000	0
msmc0_el2	aw3_umc_cfg	20	0x67800000	0
msmc0_el2	aw3_dru_cfg	20	0x67c00000	0
msmc0_el2	l2_broadcast	22	0x6a000000	0
msmc0_el2	el2_cfg	17	0x6a800000	0
msmc0_el2	el2_dru_cfg	20	0x6ac00000	0
msmc0_el2	el2_rat_cfg	12	0x6ad00000	0
msmc0_el2	el2_clec_cfg	20	0x6aa00000	0
msmc0_el2	el2_sram	25	0x68000000	0

Address Offset	Register Mnemonic	Register Name
0x6a800008	EL2_CFG	EL2_CFG MMR Register
0x6a80000c	EL2_SNP_CFG	EL2_SNP_CFG MMR Register
0x6a800010	EL2_SCRB_CFG	EL2_SCRB_CFG MMR Register
0x6a801004	EL2_SBND_MSMC	EL2_SBND_MSMC MMR Register
0x6a801008	EL2_SBND_DRU	EL2_SBND_DRU MMR Register
0x6a80100c	EL2_SBND_IHOP	EL2_SBND_IHOP MMR Register
0x6a801014	EL2_SBND_MMU	EL2_SBND_MMU MMR Register
0x6a801020	EL2_SBND_UMC0	EL2_SBND_UMC0 MMR Register
0x6a801024	EL2_SBND_UMC1	EL2_SBND_UMC1 MMR Register
0x6a801028	EL2_SBND_UMC2	EL2_SBND_UMC2 MMR Register
0x6a80102c	EL2_SBND_UMC3	EL2_SBND_UMC3 MMR Register
0x6a801804	EL2_SBRST_MSMC	EL2_SBRST_MSMC MMR Register
0x6a801808	EL2_SBRST_DRU	EL2_SBRST_DRU MMR Register
0x6a80180c	EL2_SBRST_IHOP	EL2_SBRST_IHOP MMR Register
0x6a801814	EL2_SBRST_MMU	EL2_SBRST_MMU MMR Register
0x6a801820	EL2_SBRST_UMC0	EL2_SBRST_UMC0 MMR Register
0x6a801824	EL2_SBRST_UMC1	EL2_SBRST_UMC1 MMR Register
0x6a801828	EL2_SBRST_UMC2	EL2_SBRST_UMC2 MMR Register
0x6a80182c	EL2_SBRST_UMC3	EL2_SBRST_UMC3 MMR Register
0x6a802000	EL2_RAM_ECC_UNC	EL2_RAM_ECC_UNC MMR Register
0x6a802004	EL2_RAM_ECC_COR	EL2_RAM_ECC_COR MMR Register
0x6a802008	EL2_RAM_ECC_COR_WB	EL2_RAM_ECC_COR_WB MMR Register
0x6a802100	EL2_RAM_ECC_INFO_MEM0	EL2_RAM_ECC_INFO_MEM0 MMR Register
0x6a802104	EL2_RAM_ECC_INFO_MEM1	EL2_RAM_ECC_INFO_MEM1 MMR Register
0x6a802108	EL2_RAM_ECC_INFO_MEM2	EL2_RAM_ECC_INFO_MEM2 MMR Register
0x6a80210c	EL2_RAM_ECC_INFO_MEM3	EL2_RAM_ECC_INFO_MEM3 MMR Register
0x6a802200	EL2_RAM_ECC_ADDR_MEM0	EL2_RAM_ECC_ADDR_MEM0 MMR Register
0x6a802204	EL2_RAM_ECC_ADDR_MEM1	EL2_RAM_ECC_ADDR_MEM1 MMR Register
0x6a802208	EL2_RAM_ECC_ADDR_MEM2	EL2_RAM_ECC_ADDR_MEM2 MMR Register
0x6a80220c	EL2_RAM_ECC_ADDR_MEM3	EL2_RAM_ECC_ADDR_MEM3 MMR Register

4.4.2 Register Definitions

4.4.2.1 EL2_CFG (EL2_CFG MMR Register), (0x6a800008)

General Operation Configuration Register

Bit	Name	Type	Reset	Description
31:24	rsvd2	r	0x0	Reserved
23:20	asel_other	rw	0x0	Value that will be driven on the UMC_SOC_CASEL bus for accesses originating from the C7X's PMC, MMU, and SE ports
19:16	asel_dmc	rw	0x0	Value that will be driven on the UMC_SOC_CASEL bus for accesses originating from the C7X's DMC port
15:10	rsvd1	r	0x0	Reserved
9	intl	rw	0x1	When set, the memX blocks will be interleaved in the memory map, and only mem0's base address will be used to define the address of the memories. When clear, each mem block will appear as it's own separate memory space with starting address determined by the corresponding tieoff
8	bank_hash	rw	0x1	When set, the bank which will be accessed will be determined by the address hashing algorithm. When clear, the bank hashing is disabled
7:3	rsvd0	r	0x0	Reserved
2:0	def_pri	rw	0x4	Priority value that is assigned for Patrons which do not have a cpriority signal on their command bus. Mainly used by internal c7x Patrons

4.4.2.2 EL2_SNP_CFG (EL2_SNP_CFG MMR Register), (0x6a80000c)

Snoop Configuration Register

Bit	Name	Type	Reset	Description
31	snp_fltr_dis	rw	0x0	When set, the snoop filter will be disabled. When clear, the snoop filter performs it's specified operations
30:24	rsvd1	r	0x0	Reserved
23	umc3_emif_snp	rw	0x1	When set, UMC3 accesses to EMIF space will generate a CleanInvalid snoop to DMC on UMC3 accesses. When clear, UMC3 accesses will not generate a snoop to DMC
22	umc2_emif_snp	rw	0x1	When set, UMC2 accesses to EMIF space will generate a CleanInvalid snoop to DMC on UMC2 accesses. When clear, UMC2 accesses will not generate a snoop to DMC
21	umc1_emif_snp	rw	0x1	When set, UMC1 accesses to EMIF space will generate a CleanInvalid snoop to DMC on UMC1 accesses. When clear, UMC1 accesses will not generate a snoop to DMC
20	umc0_emif_snp	rw	0x1	When set, UMC0 accesses to EMIF space will generate a CleanInvalid snoop to DMC on UMC0 accesses. When clear, UMC0 accesses will not generate a snoop to DMC
19	dru_emif_snp	rw	0x1	When set, DRU accesses to EMIF space will generate a CleanInvalid snoop to DMC on DRU accesses. When clear, DRU accesses will not generate a snoop to DMC
18	mmu_emif_snp	rw	0x1	When set, MMU reads to EMIF space will generate a CleanInvalid snoop to DMC on MMU reads. When clear, MMU reads will not generate a snoop to DMC
17:2	rsvd0	r	0x0	Reserved
1	dru_snp	rw	0x1	When set, DRU writes to the internal SRAM will generate a MakeInvalid snoop to DMC. When clear, DRU writes will not generate a snoop to DMC
0	soc_snp	rw	0x1	When set, SoC writes to the internal SRAM will generate a MakeInvalid snoop to DMC. When clear, SoC writes will not generate a snoop to DMC

4.4.2.3 EL2_SCRB_CFG (EL2_SCRB_CFG MMR Register), (0x6a800010)

ECC Scrubber Configuration Register

Bit	Name	Type	Reset	Description
31	en	rw	0x1	When set, the scrubber is enabled. When clear, the scrubber is disabled and does not function
30	pdis	rw	0x0	When set, the 1024 cycle prescale counter is disabled, and the number of cycles between scrub bursts will be exactly the value in the KCYCLES field. Setting this bit is expected to only be used by the hardware verification team to stress-test interaction between the scrubber and normal functional accesses
29:24	rsvd0	r	0x0	Reserved
23:0	kcycles	rw	0x100	Number of kilocycles between scrub bursts. Specifically, there will be 1024 * KCYCLES cycles between scrub bursts delay. When set to zero, delay will be approximately 32 cycles between bursts

4.4.2.4 EL2_SBND_MSMC (EL2_SBND_MSMC MMR Register), (0x6a801004)

Starvation Bound for MSMC

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbnd	rw	0x20	Number of cycles that the Patron request may lose arbitration before declaring that it has starved

4.4.2.5 EL2_SBND_DRU (EL2_SBND_DRU MMR Register), (0x6a801008)

Starvation Bound for DRU

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbnd	rw	0x20	Number of cycles that the Patron request may lose arbitration before declaring that it has starved

4.4.2.6 EL2_SBND_IHOP (EL2_SBND_IHOP MMR Register), (0x6a80100c)

Starvation Bound for internal IHOPs

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbnd	rw	0x20	Number of cycles that the Patron request may lose arbitration before declaring that it has starved

4.4.2.7 EL2_SBND_MMU (EL2_SBND_MMU MMR Register), (0x6a801014)

Starvation Bound for EL2

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbnd	rw	0x20	Number of cycles that the Patron request may lose arbitration before declaring that it has starved

4.4.2.8 EL2_SBND_UMC0 (EL2_SBND_UMC0 MMR Register), (0x6a801020)

Starvation Bound for UMC0

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbnd	rw	0x20	Number of cycles that the Patron request may lose arbitration before declaring that it has starved

4.4.2.9 EL2_SBND_UMC1 (EL2_SBND_UMC1 MMR Register), (0x6a801024)

Starvation Bound for UMC1

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbnd	rw	0x20	Number of cycles that the Patron request may lose arbitration before declaring that it has starved

4.4.2.10 EL2_SBND_UMC2 (EL2_SBND_UMC2 MMR Register), (0x6a801028)

Starvation Bound for UMC2

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbnd	rw	0x20	Number of cycles that the Patron request may lose arbitration before declaring that it has starved

4.4.2.11 EL2_SBND_UMC3 (EL2_SBND_UMC3 MMR Register), (0x6a80102c)

Starvation Bound for UMC3

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbnd	rw	0x20	Number of cycles that the Patron request may lose arbitration before declaring that it has starved

4.4.2.12 EL2_SBRST_MSMC (EL2_SBRST_MSMC MMR Register), (0x6a801804)

Starvation Burst Count for MSMC

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbrst	rw	0x4	Maximum number of bank-to-back commands which will be marked as starvation commands from the Patron when the starvation bound counter expires

4.4.2.13 EL2_SBRST_DRU (EL2_SBRST_DRU MMR Register), (0x6a801808)

Starvation Burst Count for DRU

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbrst	rw	0x4	Maximum number of bank-to-back commands which will be marked as starvation commands from the Patron when the starvation bound counter expires

4.4.2.14 EL2_SBRST_IHOP (EL2_SBRST_IHOP MMR Register), (0x6a80180c)

Starvation Burst Count for internal IHOPs

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbrst	rw	0x4	Maximum number of bank-to-back commands which will be marked as starvation commands from the Patron when the starvation bound counter expires

4.4.2.15 EL2_SBRST_MMU (EL2_SBRST_MMU MMR Register), (0x6a801814)

Starvation Burst Count for EL2

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbrst	rw	0x4	Maximum number of bank-to-back commands which will be marked as starvation commands from the Patron when the starvation bound counter expires

4.4.2.16 EL2_SBRST_UMC0 (EL2_SBRST_UMC0 MMR Register), (0x6a801820)

Starvation Burst Count for EL2

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbrst	rw	0x4	Maximum number of bank-to-back commands which will be marked as starvation commands from the Patron when the starvation bound counter expires

4.4.2.17 EL2_SBRST_UMC1 (EL2_SBRST_UMC1 MMR Register), (0x6a801824)

Starvation Burst Count for EL2

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbrst	rw	0x4	Maximum number of bank-to-back commands which will be marked as starvation commands from the Patron when the starvation bound counter expires

4.4.2.18 EL2_SBRST_UMC2 (EL2_SBRST_UMC2 MMR Register), (0x6a801828)

Starvation Burst Count for EL2

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbrst	rw	0x4	Maximum number of bank-to-back commands which will be marked as starvation commands from the Patron when the starvation bound counter expires

4.4.2.19 EL2_SBRST_UMC3 (EL2_SBRST_UMC3 MMR Register), (0x6a80182c)

Starvation Burst Count for EL2

Bit	Name	Type	Reset	Description
31:8	rsvd0	r	0x0	Reserved
7:0	sbrst	rw	0x4	Maximum number of bank-to-back commands which will be marked as starvation commands from the Patron when the starvation bound counter expires

4.4.2.20 EL2_RAM_ECC_UNC (EL2_RAM_ECC_UNC MMR Register), (0x6a802000)

Uncorrectable Error Count

Bit	Name	Type	Reset	Description
31:16	rsvd0	r	0x0	Reserved
15:0	count	rw	0x0	Records the total number of uncorrectable errors that have been detected

4.4.2.21 EL2_RAM_ECC_COR (EL2_RAM_ECC_COR MMR Register), (0x6a802004)

Corrected Error Count

Bit	Name	Type	Reset	Description
31:0	count	rw	0x0	Records the total number of errors that have been corrected

4.4.2.22 EL2_RAM_ECC_COR_WB (EL2_RAM_ECC_COR_WB MMR Register), (0x6a802008)

Corrected Error Count Written to SRAM

Bit	Name	Type	Reset	Description
31:16	rsvd0	r	0x0	Reserved
15:0	count	rw	0x0	Records the total number of errors that were corrected and written back to SRAM by the scrubber

4.4.2.23 EL2_RAM_ECC_INFO_MEM0 (EL2_RAM_ECC_INFO_MEM0 MMR Register), (0x6a802100)

Last ECC Error Information Register from the MEM0 SRAM

Bit	Name	Type	Reset	Description
31	locked	rw	0x0	When set, the EL2_RAM_ECC_INFO_MEMX and EL2_RAM_ECC_INFO_ADDRX will be locked from logging new ECC events. This bit is set by hardware when an error is logged, and must be cleared by software before more errors will be logged in these registers. Note that EL2_RAM_ECC_UNC, EL2_RAM_ECC_COR, and EL2_RAM_ECC_WB registers will still be updated when any of the EL2_RAM_ECC_INFO_MEMX.LOCKED bits are set
30:26	rsvd2	r	0x0	Reserved
25:24	subbank	rw	0x0	Records the sub-bank which was read that produced the error
23:20	rsvd1	r	0x0	Reserved
19:16	bank	rw	0x0	Records the bank which was read that produced the error
15:14	src	rw	0x0	Records whether the error was detected on a functional access or a scrub cycle Field values: FUNC(0x0): Error detected on functional access SCRB(0x1): Error detected during a scrub cycle INJECTED(0x2): Error detected during an error injection cycle RSVD(0x3): Reserved
13:12	etype	rw	0x0	Records the type of error that was detected Field values: NONE(0x0): No errors have been detected ONEBIT(0x1): 1-bit correctable error TWOBIT(0x2): 2 or more bit uncorrectable error RSVD(0x3): Reserved
11:9	rsvd0	r	0x0	Reserved
8:0	bitloc	rw	0x0	Upon detection of an ECC error, records the location of the bit which was found to be in error for the last 1-bit correctable error that was detected within the subbank.

4.4.2.24 EL2_RAM_ECC_INFO_MEM1 (EL2_RAM_ECC_INFO_MEM1 MMR Register), (0x6a802104)

Last ECC Error Information Register from the MEM1 SRAM

Bit	Name	Type	Reset	Description
31	locked	rw	0x0	When set, the EL2_RAM_ECC_INFO_MEMX and EL2_RAM_ECC_INFO_ADDRX will be locked from logging new ECC events. This bit is set by hardware when an error is logged, and must be cleared by software before more errors will be logged in these registers. Note that EL2_RAM_ECC_UNC, EL2_RAM_ECC_COR, and EL2_RAM_ECC_WB registers will still be updated when any of the EL2_RAM_ECC_INFO_MEMX.LOCKED bits are set
30:26	rsvd2	r	0x0	Reserved
25:24	subbank	rw	0x0	Records the sub-bank which was read that produced the error
23:20	rsvd1	r	0x0	Reserved
19:16	bank	rw	0x0	Records the bank which was read that produced the error
15:14	src	rw	0x0	Records whether the error was detected on a functional access or a scrub cycle Field values: FUNC(0x0): Error detected on functional access SCRUB(0x1): Error detected during a scrub cycle INJECTED(0x2): Error detected during an error injection cycle RSVD(0x3): Reserved
13:12	etype	rw	0x0	Records the type of error that was detected Field values: NONE(0x0): No errors have been detected ONEBIT(0x1): 1-bit correctable error TWOBIT(0x2): 2 or more bit uncorrectable error RSVD(0x3): Reserved
11:9	rsvd0	r	0x0	Reserved
8:0	bitloc	rw	0x0	Upon detection of an ECC error, records the location of the bit which was found to be in error for the last 1-bit correctable error that was detected within the subbank.

4.4.2.25 EL2_RAM_ECC_INFO_MEM2 (EL2_RAM_ECC_INFO_MEM2 MMR Register), (0x6a802108)

Last ECC Error Information Register from the MEM2 SRAM

Bit	Name	Type	Reset	Description
31	locked	rw	0x0	When set, the EL2_RAM_ECC_INFO_MEMX and EL2_RAM_ECC_INFO_ADDRX will be locked from logging new ECC events. This bit is set by hardware when an error is logged, and must be cleared by software before more errors will be logged in these registers. Note that EL2_RAM_ECC_UNC, EL2_RAM_ECC_COR, and EL2_RAM_ECC_WB registers will still be updated when any of the EL2_RAM_ECC_INFO_MEMX.LOCKED bits are set
30:26	rsvd2	r	0x0	Reserved
25:24	subbank	rw	0x0	Records the sub-bank which was read that produced the error
23:20	rsvd1	r	0x0	Reserved
19:16	bank	rw	0x0	Records the bank which was read that produced the error
15:14	src	rw	0x0	Records whether the error was detected on a functional access or a scrub cycle Field values: FUNC(0x0): Error detected on functional access SCRB(0x1): Error detected during a scrub cycle INJECTED(0x2): Error detected during an error injection cycle RSVD(0x3): Reserved
13:12	etype	rw	0x0	Records the type of error that was detected Field values: NONE(0x0): No errors have been detected ONEBIT(0x1): 1-bit correctable error TWOBIT(0x2): 2 or more bit uncorrectable error RSVD(0x3): Reserved
11:9	rsvd0	r	0x0	Reserved
8:0	bitloc	rw	0x0	Upon detection of an ECC error, records the location of the bit which was found to be in error for the last 1-bit correctable error that was detected within the subbank.

4.4.2.26 EL2_RAM_ECC_INFO_MEM3 (EL2_RAM_ECC_INFO_MEM3 MMR Register), (0x6a80210c)

Last ECC Error Information Register from the MEM3 SRAM

Bit	Name	Type	Reset	Description
31	locked	rw	0x0	When set, the EL2_RAM_ECC_INFO_MEMX and EL2_RAM_ECC_INFO_ADDRX will be locked from logging new ECC events. This bit is set by hardware when an error is logged, and must be cleared by software before more errors will be logged in these registers. Note that EL2_RAM_ECC_UNC, EL2_RAM_ECC_COR, and EL2_RAM_ECC_WB registers will still be updated when any of the EL2_RAM_ECC_INFO_MEMX.LOCKED bits are set
30:26	rsvd2	r	0x0	Reserved
25:24	subbank	rw	0x0	Records the sub-bank which was read that produced the error
23:20	rsvd1	r	0x0	Reserved
19:16	bank	rw	0x0	Records the bank which was read that produced the error
15:14	src	rw	0x0	Records whether the error was detected on a functional access or a scrub cycle Field values: FUNC(0x0): Error detected on functional access SCRUB(0x1): Error detected during a scrub cycle INJECTED(0x2): Error detected during an error injection cycle RSVD(0x3): Reserved
13:12	etype	rw	0x0	Records the type of error that was detected Field values: NONE(0x0): No errors have been detected ONEBIT(0x1): 1-bit correctable error TWOBIT(0x2): 2 or more bit uncorrectable error RSVD(0x3): Reserved
11:9	rsvd0	r	0x0	Reserved
8:0	bitloc	rw	0x0	Upon detection of an ECC error, records the location of the bit which was found to be in error for the last 1-bit correctable error that was detected within the subbank.

4.4.2.27 EL2_RAM_ECC_ADDR_MEM0 (EL2_RAM_ECC_ADDR_MEM0 MMR Register), (0x6a802200)

Last ECC Error SRAM Bank Address from the MEM0 SRAM

Bit	Name	Type	Reset	Description
31:16	rsvd0	r	0x0	Reserved
15:0	addr	rw	0x0	Upon detection of an ECC error, records the row address that was accessed on the SRAM with the error

4.4.2.28 EL2_RAM_ECC_ADDR_MEM1 (EL2_RAM_ECC_ADDR_MEM1 MMR Register), (0x6a802204)

Last ECC Error SRAM Bank Address from the MEM1 SRAM

Bit	Name	Type	Reset	Description
31:16	rsvd0	r	0x0	Reserved
15:0	addr	rw	0x0	Upon detection of an ECC error, records the row address that was accessed on the SRAM with the error

4.4.2.29 EL2_RAM_ECC_ADDR_MEM2 (EL2_RAM_ECC_ADDR_MEM2 MMR Register), (0x6a802208)

Last ECC Error SRAM Bank Address from the MEM2 SRAM

Bit	Name	Type	Reset	Description
31:16	rsvd0	r	0x0	Reserved
15:0	addr	rw	0x0	Upon detection of an ECC error, records the row address that was accessed on the SRAM with the error

4.4.2.30 EL2_RAM_ECC_ADDR_MEM3 (EL2_RAM_ECC_ADDR_MEM3 MMR Register), (0x6a80220c)

Last ECC Error SRAM Bank Address from the MEM3 SRAM

Bit	Name	Type	Reset	Description
31:16	rsvd0	r	0x0	Reserved
15:0	addr	rw	0x0	Upon detection of an ECC error, records the row address that was accessed on the SRAM with the error

4.5 MMR Region: el2_dru_cfg

There are no memory-mapped registers in this region.

4.6 MMR Region: el2_rat_cfg

KSBBUS Region-Based Address Translation Region

4.6.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x6ad00000	PID	Revision Register
0x6ad00004	CONFIG	Config Register
0x6ad00020+(a*0x10), a=0..15	REGION[a]_CTRL	Region Control Register
0x6ad00020+(a*0x10)+0x4, a=0..15	REGION[a]_BASE	Region Base Register
0x6ad00020+(a*0x10)+0x8, a=0..15	REGION[a]_TRANS_L	Region Translated Lower Address
0x6ad00020+(a*0x10)+0xc, a=0..15	REGION[a]_TRANS_U	Region Translated Upper Address
0x6ad00804	DESTINATION_ID	Destination ID Register
0x6ad00820	EXCEPTION_LOGGING_CONTROL	Exception Logging Control Register
0x6ad00824	EXCEPTION_LOGGING_HEADER0	Exception Logging Header 0 Register
0x6ad00828	EXCEPTION_LOGGING_HEADER1	Exception Logging Header 1 Register
0x6ad0082c	EXCEPTION_LOGGING_DATA0	Exception Logging Data 0 Register
0x6ad00830	EXCEPTION_LOGGING_DATA1	Exception Logging Data 1 Register
0x6ad00834	EXCEPTION_LOGGING_DATA2	Exception Logging Data 2 Register
0x6ad00838	EXCEPTION_LOGGING_DATA3	Exception Logging Data 3 Register
0x6ad00840	EXCEPTION_PEND_SET	Exception Logging Interrupt Pending Set Register
0x6ad00844	EXCEPTION_PEND_CLEAR	Exception Logging Interrupt Pending Clear Register
0x6ad00848	EXCEPTION_ENABLE_SET	Exception Logging Interrupt Enable Set Register
0x6ad0084c	EXCEPTION_ENABLE_CLEAR	Exception Logging Interrupt Enable Clear Register
0x6ad00850	EOI_REG	EOI Register

4.6.2 Register Definitions

4.6.2.1 PID (Revision Register), (0x6ad00000)

The Revision Register contains the major and minor revisions for the module.

Bit	Name	Type	Reset	Description
31:30	scheme	r	0x1	PID register scheme
29:28	bu	r	0x2	Business Unit: 10 = Processors
27:16	func	r	0x680	Module ID
15:11	rtl	r	0x9	RTL revision. Will vary depending on release.
10:8	major	r	0x1	Major revision
7:6	custom	r	0x0	Custom
5:0	minor	r	0x0	Minor revision

4.6.2.2 CONFIG (Config Register), (0x6ad00004)

The Config Register contains the configuration values for the module.

Bit	Name	Type	Reset	Description
31:24	input_addr_width	r	0x30	Number of address bits for the input address bus
23:16	addr_width	r	0x30	Number of address bits - output address bus size
15:8	addrs	r	0x2	Number of addresses
7:0	regions	r	0x10	Number of regions

4.6.2.3 REGION[a]_CTRL (Region Control Register), (0x6ad00020+(a*0x10)), a=0..15

The Control for Region a

Bit	Name	Type	Reset	Description
31	en	rw	0x0	Enable for the Region
30:24	reserved	r	0x0	Reserved
23:8	base_u	rw	0x0	Upper 16 for the Base Address for Region a. This is an extension to the base address for RAT beyond 32-bit addresses.
7:6	reserved	r	0x0	Reserved
5:0	size	rw	0x0	Size of the Region in Address Bits. 0 = 1 byte, 1 = 2B, 2 = 4B, 3 = 8B, etc. up to 48 = 256TB.

4.6.2.4 REGION[a]_BASE (Region Base Register), (0x6ad00020+(a*0x10)+0x4), a=0..15

The Base Address for Region a. This is the source address for matching to a region.

Bit	Name	Type	Reset	Description
31:0	base	rw	0x0	Base Lower for the Region. It must be aligned to the programmed size.

4.6.2.5 REGION[a]_TRANS_L (Region Translated Lower Address), (0x6ad00020+(a*0x10)+0x8), a=0..15

The Translated Lower Address Bits for Region a

Bit	Name	Type	Reset	Description
31:0	lower	rw	0x0	Translated Lower Address Bits for the Region. It must be aligned to the programmed size.

4.6.2.6 REGION[a]_TRANS_U (Region Translated Upper Address), (0x6ad00020+(a*0x10)+0xc), a=0..15

The Translated Upper Address Bits for Region a

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	upper	rw	0x0	Translated Upper Address Bits for the Region

4.6.2.7 DESTINATION_ID (Destination ID Register), (0x6ad00804)

The Destination ID Register defines the destination ID value for error messages.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	dest_id	rw	0x0	The destination ID.

4.6.2.8 EXCEPTION_LOGGING_CONTROL (Exception Logging Control Register), (0x6ad00820)

The Exception Logging Control Register controls the exception logging.

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1	disable_intr	rw	0x0	Disables logging interrupt when set. This will not disable logging, so if cleared the current log should also be cleared to guarantee the next log generates the interrupt.
0	disable_f	rw	0x0	Disables logging when set. This will also disable interrupts.

4.6.2.9 EXCEPTION_LOGGING_HEADER0 (Exception Logging Header 0 Register), (0x6ad00824)

The Exception Logging Header 0 Register contains the first word of the header.

Bit	Name	Type	Reset	Description
31:24	type_f	r	0x0	Type. 4 = RAT.
23:8	src_id	r	0x0	Source ID.
7:0	dest_id	r	0x0	Destination ID.

4.6.2.10 EXCEPTION_LOGGING_HEADER1 (Exception Logging Header 1 Register), (0x6ad00828)

The Exception Logging Header 1 Register contains the second word of the header.

Bit	Name	Type	Reset	Description
31:24	group	r	0x0	Group.
23:16	code	r	0x0	Code. 1 = Boundary crossing error.
15:0	reserved	r	0x0	Reserved

4.6.2.11 EXCEPTION_LOGGING_DATA0 (Exception Logging Data 0 Register), (0x6ad0082c)

The Exception Logging Data 0 Register contains the first word of the data.

Bit	Name	Type	Reset	Description
31:0	addr_l	r	0x0	Address lower 32 bits.

4.6.2.12 EXCEPTION_LOGGING_DATA1 (Exception Logging Data 1 Register), (0x6ad00830)

The Exception Logging Data 1 Register contains the second word of the data.

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	addr_h	r	0x0	Address upper 12 bits.

4.6.2.13 EXCEPTION_LOGGING_DATA2 (Exception Logging Data 2 Register), (0x6ad00834)

The Exception Logging Data 2 Register contains the third word of the data.

Bit	Name	Type	Reset	Description
31:28	reserved	r	0x0	Reserved
27:16	routeid	r	0x0	Route ID.
15:14	reserved	r	0x0	Reserved
13	write	r	0x0	Write.
12	read	r	0x0	Read.
11	debug	r	0x0	Debug.
10	cacheable	r	0x0	Cacheable.
9	priv	r	0x0	Priv.
8	secure	r	0x0	Secure.
7:0	priv_id	r	0x0	Priv ID.

4.6.2.14 EXCEPTION_LOGGING_DATA3 (Exception Logging Data 3 Register), (0x6ad00838)

The Exception Logging Data 3 Register contains the fourth word of the data. Reading this register will clear the error pending bit.

Bit	Name	Type	Reset	Description
31:10	reserved	r	0x0	Reserved
9:0	bytecnt	r	0x0	Byte count.

4.6.2.15 EXCEPTION_PEND_SET (Exception Logging Interrupt Pending Set Register), (0x6ad00840)

The Exception Logging Interrupt Pending Set Register allows to set the pend signal.

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	pend_set	rw1ts	0x0	Write a 1 to set the exception pend signal.

4.6.2.16 EXCEPTION_PEND_CLEAR (Exception Logging Interrupt Pending Clear Register), (0x6ad00844)

The Exception Logging Interrupt Pending Clear Register allows to clear the pend signal.

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	pend_clr	rw1tc	0x0	Write a 1 to clear the exception pend signal.

4.6.2.17 EXCEPTION_ENABLE_SET (Exception Logging Interrupt Enable Set Register), (0x6ad00848)

The Exception Logging Interrupt Enable Set Register allows to set the interrupt enable signal.

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	enable_set	rw1ts	0x0	Write a 1 to set the exception interrupt enable signal.

4.6.2.18 EXCEPTION_ENABLE_CLEAR (Exception Logging Interrupt Enable Clear Register), (0x6ad0084c)

The Exception Logging Interrupt Enable Clear Register allows to clear the interrupt enable signal.

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	enable_clr	rw1tc	0x0	Write a 1 to clear the exception interrupt enable signal.

4.6.2.19 EOI_REG (EOI Register), (0x6ad00850)

EOI Register

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	eoi_wr	rw	0x0	EOI Register

4.7 MMR Region: el2_clec_cfg

4.7.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x6aa00000	GLOBAL_PID	CLEC PID register
0x6aa0a000+(a*0x4), a=0..62	GLOBAL_SET_FLAG[a]_EFR	CLEC event flag register
0x6aa0b000+(a*0x4), a=0..62	GLOBAL_DROP_FLAG[a]_EDR	CLEC dropped event register
0x6aa0c000	GLOBAL_GELRS	Global Event Lock register for secure claims
0x6aa0d000	GLOBAL_GELRNS	Global Event Lock register for non-secure claims
0x6aa80000+(a*0x100), a=0..2047	CFG[a]_MRR	Map and Routing register
0x6aa80000+(a*0x100)+0x4, a=0..2047	CFG[a]_ESR	Event send register. Ignores value written. Sends event to cpu
0x6aa80000+(a*0x100)+0x8, a=0..2047	CFG[a]_ECR	Event clear register. Ignores value written. Clears level events for that event number

4.7.2 Register Definitions

4.7.2.1 GLOBAL_PID (CLEC PID register), (0x6aa00000)

Revision parameters

Bit	Name	Type	Reset	Description
31:30	scheme	r	0x1	Scheme
29:28	bu	r	0x2	Scheme
27:16	module_id	r	0x10	Module ID
15:11	revrtl	r	0x0	RTL version
10:8	revmax	r	0x1	Major version
7:6	custom	r	0x0	Custom version
5:0	revmin	r	0x0	Minor version

4.7.2.2 GLOBAL_SET_FLAG[a]_EFR (CLEC event flag register), (0x6aa0a000+(a*0x4)), a=0..62

CLEC event flags

Bit	Name	Type	Reset	Description
31:0	evntfr	r	0x0	event flag register, read only pending status of all events regardless of they are enabled or disabled, for pulse events this flag is cleared as soon as event is sent, for level events

				this flag is cleared when input event is deasserted, writing 1 to ECR_n register also clears this bit in case of pending pulse event, description in spec section 4.2.2
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4.7.2.3 GLOBAL_DROP_FLAG[a]_EDR (CLEC dropped event register), (0x6aa0b000+(a*0x4)), a=0..62

CLEC dropped event flags

Bit	Name	Type	Reset	Description
31:0	evntdr	r	0x0	dropped event flags, for input events which are pulse if events are sent back back and second event gets dropped since first event still in the pending event queue, this bit gets set indicating that an event has been dropped, it also generates dropped event interrupt, description in spec section 4.2.3

4.7.2.4 GLOBAL_GELRS (Global Event Lock register for secure claims), (0x6aa0c000)

Event Lock reg for secure claim

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	lock	rw	0x0	GELRS.LOCK - event lock register for secure claim

4.7.2.5 GLOBAL_GELRNS (Global Event Lock register for non-secure claims), (0x6aa0d000)

Event Lock reg for non-secure claim

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	lock	rw	0x0	GELRNS.LOCK - event lock register for non-secure claim

4.7.2.6 CFG[a]_MRR (Map and Routing register), (0x6aa80000+(a*0x100)), a=0..2047

Map and routing register

Bit	Name	Type	Reset	Description
31	s	rw	0x0	MRR.S - reserved, not used for AM62A since CPU doesn't support secure mode
30	ese	rw	0x0	MRR.ESE - Event-Send enable. Controls whether CLEC sends this event when ESR is written or Event occurred on interrupt line
29:26	rsvd3	r	0x0	Reserved
25	evtpf	r	0x0	MRR.EPF - same value as EFR, read only pending status of all events regardless of they are enabled or disabled, for pulse events this flag is cleared as soon as event is sent, for level events this flag is cleared when input event is deasserted, writing 1 to ECR_n register also clears this bit in case of pending pulse event
24	is_lvl	rw	0x1	MRR.IS_LVL - Indicate that input event is level or pulse, event handling is different based on whether event is level or pulse
23:22	rsvd2	r	0x0	Reserved
21:16	rtmap	rw	0x1	MRR.RTMAP - If Bit [16] = 1 Send to None, If Bit [17] = 1 and bit [16] = 0 Send to System, If [16] = 0 and [17] = 0 send to [21:18] Corepac encoded value - value 1 and 0xF are valid since there is only one CPU
15:8	ext_evtnum	rw	0x0	MRR.EXT_EVTNUM - encoded external event number, event 0- 63 SOC events, 64-95 DRU events, event 128 ARM event others are reserved for AM62A
7:6	rsvd0	r	0x0	Reserved
5:0	c7x_evtnum	rw	0x0	MRR.C7x_EVTNUM - C7x event number to send when this event is triggered

4.7.2.7 CFG[a]_ESR (Event send register. Ignores value written. Sends event to cpu), (0x6aa80000+(a*0x100)+0x4), a=0..2047

Event send reg

Bit	Name	Type	Reset	Description
31:0	esr_n	w	0x0	ESR.ESR_n sends event number n if it is enabled, address offset 0x010000 * n + 0x03000

4.7.2.8 CFG[a]_ECR (Event clear register. Ignores value written. Clears level events for that event number), (0x6aa80000+(a*0x100)+0x8), a=0..2047

Event clear reg

Bit	Name	Type	Reset	Description
31:2	reserved	w	0x0	Reserved
1	ecr_pflag	w	0x0	ECR.EDR_Clear - clears EDR flag bit when 2 is written
0	ecr_flag	w	0x0	ECR.EFR_Clear - clears EFR flag bit when 1 is written, also clears corresponding SOC level event if sent to system on SOCinterrupt port

4.8 MMR Region: el2_pbist_0_mem

4.8.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x90000	RF0L	Register Files / Instruction Registers
0x90004	RF1L	Register Files / Instruction Registers
0x90008	RF2L	Register Files / Instruction Registers
0x9000c	RF3L	Register Files / Instruction Registers
0x90040	RF0U	Register Files / Instruction Registers
0x90044	RF1U	Register Files / Instruction Registers
0x90048	RF2U	Register Files / Instruction Registers
0x9004c	RF3U	Register Files / Instruction Registers
0x90100	A0	Variable Address Registers
0x90104	A1	Variable Address Registers
0x90108	A2	Variable Address Registers
0x9010c	A3	Variable Address Registers
0x90110	L0	Variable Loop Count Registers
0x90114	L1	Variable Loop Count Registers
0x90118	L2	Variable Loop Count Registers
0x9011c	L3	Variable Loop Count Registers
0x90120	D	Data Registers
0x90124	E	Data Registers
0x90130	CA0	Constant Address Registers
0x90134	CA1	Constant Address Registers
0x90138	CA2	Constant Address Registers
0x9013c	CA3	Constant Address Registers
0x90140	CL0	Constant Loop Count Registers
0x90144	CL1	Constant Loop Count Registers
0x90148	CL2	Constant Loop Count Registers
0x9014c	CL3	Constant Loop Count Registers
0x90150	I0	Constant Increment Registers
0x90154	I1	Constant Increment Registers
0x90158	I2	Constant Increment Registers
0x9015c	I3	Constant Increment Registers
0x90160	RAMT	RAM Configuration Register
0x90164	DLR	Datalogger Register
0x90168	CMS	Clock-Mux Select Register
0x9016c	STR	Program Control Register
0x90170	SCR	Address Scrambling Register
0x90178	CSR	Chip Select Register
0x9017c	FDLY	Fail Delay Register
0x90180	PACT	PACT Register
0x90184	PID	PBIST_ID Register
0x90188	OVER	Override Register
0x90190	FSRF	Fail Status Fail Register
0x90198	FSRC	Fail Status Count Register
0x901a0	FSRA	Fail Status Address Register
0x901a8	FSRDL0	Fail Status Data Registers
0x901b0	FSRDL1	Fail Status Data Registers
0x901b4	MARGIN_MODE	Fail Status Fail Register



0x901b8	WRENZ	Fail Status Fail Register
0x901bc	PAGE_PGS	Fail Status Fail Register
0x901c0	ROM	ROM Mask Register
0x901c4	ALGO	Algorithm Mask Register
0x901c8	RINFO	RAM Info Mask Register

4.8.2 Register Definitions

4.8.2.1 RF0L (Register Files / Instruction Registers), (0x90000)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF0L	rw	0x0	Register Files / Instruction Registers RF0 lower (RF0L)

4.8.2.2 RF1L (Register Files / Instruction Registers), (0x90004)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF1L	rw	0x0	Register Files / Instruction Registers RF1 lower (RF1L)

4.8.2.3 RF2L (Register Files / Instruction Registers), (0x90008)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF2L	rw	0x0	Register Files / Instruction Registers RF2 lower (RF2L)

4.8.2.4 RF3L (Register Files / Instruction Registers), (0x9000c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF3L	rw	0x0	Register Files / Instruction Registers RF3 lower (RF3L)

4.8.2.5 RF0U (Register Files / Instruction Registers), (0x90040)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF0U	rw	0x0	Register Files / Instruction Registers RF0 upper (RF0U)

4.8.2.6 RF1U (Register Files / Instruction Registers), (0x90044)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF1U	rw	0x0	Register Files / Instruction Registers RF1 upper (RF1U)

4.8.2.7 RF2U (Register Files / Instruction Registers), (0x90048)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF2U	rw	0x0	Register Files / Instruction Registers RF2 upper (RF2U)

4.8.2.8 RF3U (Register Files / Instruction Registers), (0x9004c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF3U	rw	0x0	Register Files / Instruction Registers RF3 upper (RF3U)

4.8.2.9 A0 (Variable Address Registers), (0x90100)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A0	rw	0x0	Variable Address Register 0 (A0)

4.8.2.10 A1 (Variable Address Registers), (0x90104)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A1	rw	0x0	Variable Address Register 1 (A1)

4.8.2.11 A2 (Variable Address Registers), (0x90108)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A2	rw	0x0	Variable Address Register 2 (A2)

4.8.2.12 A3 (Variable Address Registers), (0x9010c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A3	rw	0x0	Variable Address Register 3 (A3)

4.8.2.13 L0 (Variable Loop Count Registers), (0x90110)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L0	rw	0x0	Variable Loop Count Register 0 (L0)

4.8.2.14 L1 (Variable Loop Count Registers), (0x90114)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L1	rw	0x0	Variable Loop Count Register 1 (L1)

4.8.2.15 L2 (Variable Loop Count Registers), (0x90118)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L2	rw	0x0	Variable Loop Count Register 2 (L2)

4.8.2.16 L3 (Variable Loop Count Registers), (0x9011c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L3	rw	0x0	Variable Loop Count Register 3 (L3)

4.8.2.17 D (Data Registers), (0x90120)

Bit	Name	Type	Reset	Description
31:16	D1	rw	0x0	DD1 Data Register Upper 16 (D1)
15:0	D0	rw	0x0	DD0 Data Register Lower 16 (D0)

4.8.2.18 E (Data Registers), (0x90124)

Bit	Name	Type	Reset	Description
31:16	E1	rw	0x0	EE1 Data Register Upper 16 (E1)
15:0	E0	rw	0x0	EE0 Data Register Lower 16 (E0)

4.8.2.19 CA0 (Constant Address Registers), (0x90130)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA0	rw	0x0	Constant Address Register 0 (CA0)

4.8.2.20 CA1 (Constant Address Registers), (0x90134)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA1	rw	0x0	Constant Address Register 1 (CA1)

4.8.2.21 CA2 (Constant Address Registers), (0x90138)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA2	rw	0x0	Constant Address Register 2 (CA2)

4.8.2.22 CA3 (Constant Address Registers), (0x9013c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA3	rw	0x0	Constant Address Register 3 (CA3)

4.8.2.23 CL0 (Constant Loop Count Registers), (0x90140)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL0	rw	0x0	Constant Loop Count Register 0 (CL0)

4.8.2.24 CL1 (Constant Loop Count Registers), (0x90144)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL1	rw	0x0	Constant Loop Count Register 1 (CL1)

4.8.2.25 CL2 (Constant Loop Count Registers), (0x90148)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL2	rw	0x0	Constant Loop Count Register 2 (CL2)

4.8.2.26 CL3 (Constant Loop Count Registers), (0x9014c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL3	rw	0x0	Constant Loop Count Register 3 (CL3)

4.8.2.27 I0 (Constant Increment Registers), (0x90150)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 0 (I0)

4.8.2.28 I1 (Constant Increment Registers), (0x90154)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 1 (I1)

4.8.2.29 I2 (Constant Increment Registers), (0x90158)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 2 (I2)

4.8.2.30 I3 (Constant Increment Registers), (0x9015c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 3 (I3)

4.8.2.31 RAMT (RAM Configuration Register), (0x90160)

Bit	Name	Type	Reset	Description
31:24	RGS	rw	0x0	RAM Group Select RGS
23:16	RDS	rw	0x0	Return Data select RDS
15:8	DWR	rw	0x0	Data Width Register DWR
7:6	RESERVED_SMSR	rw	0x0	RESERVED - Sense Margin Select Register, unused
5:2	PLS	rw	0x0	Pipeline Latency Select
1:0	RLS	rw	0x0	RAM Latency Select

4.8.2.32 DLR (Datalogger Register), (0x90164)

Bit	Name	Type	Reset	Description
31:24	reserved	r	0x0	Reserved
23:16	BRP	rw	0x0	Datalogger 2 (BRP)
15:14	RESERVED_DLR1	rw	0x0	RESERVED - DLR1 upper bits, unused
13	DLR1_OMISR_Z	rw	0x0	OMISR mode disable
12	DLR1_BISOR	rw	0x0	BISOR testing mode
11	DLR1_CMISR	rw	0x0	Cumulative MISR testing mode (mainly for ROM testing)
10	DLR1_RTM	rw	0x0	Retention testing mode
9	DLR1_GNG	rw	0x1	GO / NO-GO testing mode
8	DLR1_MISR	rw	0x0	MISR testing mode (mainly for ROM testing)
7	DLR0_TSM	rw	0x0	Time stamp mode
6	DLR0_CFMM	rw	0x0	Column Fail Masking mode
5	DLR0_ECAM	rw	0x0	Emulation cache access mode
4	DLR0_CAM	rw	0x0	Config access mode
3	DLR0_TCK	rw	0x1	TCK Gated mode
2	DLR0_ROM	rw	0x0	ROM-based testing mode
1	DLR0_IDDQ	rw	0x0	IDDQ testing mode
0	DLR0_DCM	rw	0x0	Distributed Compare mode

4.8.2.33 CMS (Clock-Mux Select Register), (0x90168)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CMS	rw	0x0	Clock Mux Select (CMS)

4.8.2.34 STR (Program Control Register), (0x9016c)

Bit	Name	Type	Reset	Description
31:5	reserved	r	0x0	Reserved
4	CHK	rw	0x0	Check MISR mode
3	STEP	rw	0x0	Step / Step for emulation mode
2	STOP	rw	0x0	Stop
1	RES	rw	0x0	Resume / Emulation read
0	START	rw	0x0	Start / Time Stamp mode restart

4.8.2.35 SCR (Address Scrambling Register), (0x90170)

Bit	Name	Type	Reset	Description
63:56	SCR7	rw	0xfe	Address Scrambling Register 7
55:48	SCR6	rw	0xdc	Address Scrambling Register 6
47:40	SCR5	rw	0xba	Address Scrambling Register 5
39:32	SCR4	rw	0x98	Address Scrambling Register 4
31:24	SCR3	rw	0x76	Address Scrambling Register 3
23:16	SCR2	rw	0x54	Address Scrambling Register 2
15:8	SCR1	rw	0x32	Address Scrambling Register 1
7:0	SCR0	rw	0x10	Address Scrambling Register 0

4.8.2.36 CSR (Chip Select Register), (0x90178)

Bit	Name	Type	Reset	Description
31:24	CSR3	rw	0x0	Chip Select 3 (CSR3)
23:16	CSR2	rw	0x0	Chip Select 2 (CSR2)
15:8	CSR1	rw	0x0	Chip Select 1 (CSR1)
7:0	CSR0	rw	0x0	Chip Select 0 (CSR0)

4.8.2.37 FDLY (Fail Delay Register), (0x9017c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	FDLY	rw	0x48	Fail Delay (FDLY)

4.8.2.38 PACT (PACT Register), (0x90180)

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	PACT	rw	0x0	PBIST Activate (PACT)

4.8.2.39 PID (PBIST_ID Register), (0x90184)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7	RESERVED_PID	rw	0x0	RESERVED - PID upper bit, unused
6:0	PID	rw	0x0	PBIST ID

4.8.2.40 OVER (Override Register), (0x90188)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	ALGO	rw	0x1	PBIST Override Algorithm Override
2	MM	rw	0x0	PBIST Override Multiple Memory
1	READ	rw	0x0	PBIST Override READ Override
0	RINFO	rw	0x1	PBIST Override RINFO Override

4.8.2.41 FSRF (Fail Status Fail Register), (0x90190)

Bit	Name	Type	Reset	Description
63:33	reserved	r	0x0	Reserved
32	FRSF1	r	0x0	Fail Status Fail - Port 1 (FSRF1)
31:1	reserved	r	0x0	Reserved
0	FRSF0	r	0x0	Fail Status Fail - Port 0 (FSRF0)

4.8.2.42 FSRC (Fail Status Count Register), (0x90198)

Bit	Name	Type	Reset	Description
63:36	reserved	r	0x0	Reserved
35:32	FSRC1	r	0x0	Fail Status Count - Port 1 (FSRC1)
31:4	reserved	r	0x0	Reserved
3:0	FSRC0	r	0x0	Fail Status Count - Port 0 (FSRC0)

4.8.2.43 FSRA (Fail Status Address Register), (0x901a0)

Bit	Name	Type	Reset	Description
63:48	reserved	r	0x0	Reserved
47:32	FSRA1	r	0x0	Fail Status Address - Port 1 (FSRA1)
31:16	reserved	r	0x0	Reserved
15:0	FSRA0	r	0x0	Fail Status Address - Port 0 (FSRA0)

4.8.2.44 FSRDL0 (Fail Status Data Registers), (0x901a8)

Bit	Name	Type	Reset	Description
31:0	FSRDL0	r	0xaaaaaaaa	Fail Status Data - Port 0 (FSRDL0)

4.8.2.45 FSRDL1 (Fail Status Data Registers), (0x901b0)

Bit	Name	Type	Reset	Description
31:0	FSRDL1	r	0xaaaaaaaa	Fail Status Data - Port 1 (FSRDL1)

4.8.2.46 MARGIN_MODE (Fail Status Fail Register), (0x901b4)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:2	PBIST_DFT_READ	rw	0x0	pbist_dft_read[1:0]
1:0	PBIST_DFT_WRITE	rw	0x0	pbist_dft_write[1:0]

4.8.2.47 WRENZ (Fail Status Fail Register), (0x901b8)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	WRENZ	rw	0x0	pbist_ram_wrenz[1:0]

4.8.2.48 PAGE_PGS (Fail Status Fail Register), (0x901bc)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	PGS	rw	0x0	pbist_ram_pgs[1:0]

4.8.2.49 ROM (ROM Mask Register), (0x901c0)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	ROM	rw	0x3	ROM Mask (ROM)

4.8.2.50 ALGO (Algorithm Mask Register), (0x901c4)

Bit	Name	Type	Reset	Description
31:24	ALGO_3	rw	0xff	ROM Algorithm Mask 3 (ALGO 3)
23:16	ALGO_2	rw	0xff	ROM Algorithm Mask 2 (ALGO 2)
15:8	ALGO_1	rw	0xff	ROM Algorithm Mask 1 (ALGO 1)
7:0	ALGO_0	rw	0xff	ROM Algorithm Mask 0 (ALGO 0)

4.8.2.51 RINFO (RAM Info Mask Register), (0x901c8)

Bit	Name	Type	Reset	Description
63:56	U3	rw	0xff	RAM Info Mask Upper 3 (RINFOU3)
55:48	U2	rw	0xff	RAM Info Mask Upper 2 (RINFOU2)
47:40	U1	rw	0xff	RAM Info Mask Upper 1 (RINFOU1)
39:32	U0	rw	0xff	RAM Info Mask Upper 0 (RINFOU0)
31:24	L3	rw	0xff	RAM Info Mask Lower 3 (RINFOL3)
23:16	L2	rw	0xff	RAM Info Mask Lower 2 (RINFOL2)
15:8	L1	rw	0xff	RAM Info Mask Lower 1 (RINFOL1)
7:0	L0	rw	0xff	RAM Info Mask Lower 0 (RINFOL0)

4.9 MMR Region: el2_pbist_1_mem

4.9.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x90400	RF0L	Register Files / Instruction Registers
0x90404	RF1L	Register Files / Instruction Registers
0x90408	RF2L	Register Files / Instruction Registers
0x9040c	RF3L	Register Files / Instruction Registers
0x90440	RF0U	Register Files / Instruction Registers
0x90444	RF1U	Register Files / Instruction Registers
0x90448	RF2U	Register Files / Instruction Registers
0x9044c	RF3U	Register Files / Instruction Registers
0x90500	A0	Variable Address Registers
0x90504	A1	Variable Address Registers
0x90508	A2	Variable Address Registers
0x9050c	A3	Variable Address Registers
0x90510	L0	Variable Loop Count Registers
0x90514	L1	Variable Loop Count Registers
0x90518	L2	Variable Loop Count Registers
0x9051c	L3	Variable Loop Count Registers
0x90520	D	Data Registers
0x90524	E	Data Registers
0x90530	CA0	Constant Address Registers
0x90534	CA1	Constant Address Registers
0x90538	CA2	Constant Address Registers
0x9053c	CA3	Constant Address Registers
0x90540	CL0	Constant Loop Count Registers
0x90544	CL1	Constant Loop Count Registers
0x90548	CL2	Constant Loop Count Registers
0x9054c	CL3	Constant Loop Count Registers
0x90550	I0	Constant Increment Registers
0x90554	I1	Constant Increment Registers
0x90558	I2	Constant Increment Registers
0x9055c	I3	Constant Increment Registers
0x90560	RAMT	RAM Configuration Register
0x90564	DLR	Datalogger Register
0x90568	CMS	Clock-Mux Select Register
0x9056c	STR	Program Control Register
0x90570	SCR	Address Scrambling Register
0x90578	CSR	Chip Select Register
0x9057c	FDLY	Fail Delay Register
0x90580	PACT	PACT Register
0x90584	PID	PBIST_ID Register
0x90588	OVER	Override Register
0x90590	FSRF	Fail Status Fail Register
0x90598	FSRC	Fail Status Count Register
0x905a0	FSRA	Fail Status Address Register
0x905a8	FSRDL0	Fail Status Data Registers
0x905b0	FSRDL1	Fail Status Data Registers
0x905b4	MARGIN_MODE	Fail Status Fail Register

0x905b8	WRENZ	Fail Status Fail Register
0x905bc	PAGE_PGS	Fail Status Fail Register
0x905c0	ROM	ROM Mask Register
0x905c4	ALGO	Algorithm Mask Register
0x905c8	RINFO	RAM Info Mask Register

4.9.2 Register Definitions

4.9.2.1 RF0L (Register Files / Instruction Registers), (0x90400)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF0L	rw	0x0	Register Files / Instruction Registers RF0 lower (RF0L)

4.9.2.2 RF1L (Register Files / Instruction Registers), (0x90404)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF1L	rw	0x0	Register Files / Instruction Registers RF1 lower (RF1L)

4.9.2.3 RF2L (Register Files / Instruction Registers), (0x90408)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF2L	rw	0x0	Register Files / Instruction Registers RF2 lower (RF2L)

4.9.2.4 RF3L (Register Files / Instruction Registers), (0x9040c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF3L	rw	0x0	Register Files / Instruction Registers RF3 lower (RF3L)

4.9.2.5 RF0U (Register Files / Instruction Registers), (0x90440)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF0U	rw	0x0	Register Files / Instruction Registers RF0 upper (RF0U)

4.9.2.6 RF1U (Register Files / Instruction Registers), (0x90444)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF1U	rw	0x0	Register Files / Instruction Registers RF1 upper (RF1U)

4.9.2.7 RF2U (Register Files / Instruction Registers), (0x90448)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF2U	rw	0x0	Register Files / Instruction Registers RF2 upper (RF2U)

4.9.2.8 RF3U (Register Files / Instruction Registers), (0x9044c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF3U	rw	0x0	Register Files / Instruction Registers RF3 upper (RF3U)

4.9.2.9 A0 (Variable Address Registers), (0x90500)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A0	rw	0x0	Variable Address Register 0 (A0)

4.9.2.10 A1 (Variable Address Registers), (0x90504)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A1	rw	0x0	Variable Address Register 1 (A1)

4.9.2.11 A2 (Variable Address Registers), (0x90508)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A2	rw	0x0	Variable Address Register 2 (A2)

4.9.2.12 A3 (Variable Address Registers), (0x9050c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A3	rw	0x0	Variable Address Register 3 (A3)

4.9.2.13 L0 (Variable Loop Count Registers), (0x90510)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L0	rw	0x0	Variable Loop Count Register 0 (L0)

4.9.2.14 L1 (Variable Loop Count Registers), (0x90514)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L1	rw	0x0	Variable Loop Count Register 1 (L1)

4.9.2.15 L2 (Variable Loop Count Registers), (0x90518)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L2	rw	0x0	Variable Loop Count Register 2 (L2)

4.9.2.16 L3 (Variable Loop Count Registers), (0x9051c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L3	rw	0x0	Variable Loop Count Register 3 (L3)

4.9.2.17 D (Data Registers), (0x90520)

Bit	Name	Type	Reset	Description
31:16	D1	rw	0x0	DD1 Data Register Upper 16 (D1)
15:0	D0	rw	0x0	DD0 Data Register Lower 16 (D0)

4.9.2.18 E (Data Registers), (0x90524)

Bit	Name	Type	Reset	Description
31:16	E1	rw	0x0	EE1 Data Register Upper 16 (E1)
15:0	E0	rw	0x0	EE0 Data Register Lower 16 (E0)

4.9.2.19 CA0 (Constant Address Registers), (0x90530)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA0	rw	0x0	Constant Address Register 0 (CA0)

4.9.2.20 CA1 (Constant Address Registers), (0x90534)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA1	rw	0x0	Constant Address Register 1 (CA1)

4.9.2.21 CA2 (Constant Address Registers), (0x90538)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA2	rw	0x0	Constant Address Register 2 (CA2)

4.9.2.22 CA3 (Constant Address Registers), (0x9053c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA3	rw	0x0	Constant Address Register 3 (CA3)

4.9.2.23 CL0 (Constant Loop Count Registers), (0x90540)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL0	rw	0x0	Constant Loop Count Register 0 (CL0)

4.9.2.24 CL1 (Constant Loop Count Registers), (0x90544)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL1	rw	0x0	Constant Loop Count Register 1 (CL1)

4.9.2.25 CL2 (Constant Loop Count Registers), (0x90548)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL2	rw	0x0	Constant Loop Count Register 2 (CL2)

4.9.2.26 CL3 (Constant Loop Count Registers), (0x9054c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL3	rw	0x0	Constant Loop Count Register 3 (CL3)

4.9.2.27 I0 (Constant Increment Registers), (0x90550)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 0 (I0)

4.9.2.28 I1 (Constant Increment Registers), (0x90554)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 1 (I1)

4.9.2.29 I2 (Constant Increment Registers), (0x90558)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 2 (I2)

4.9.2.30 I3 (Constant Increment Registers), (0x9055c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 3 (I3)

4.9.2.31 RAMT (RAM Configuration Register), (0x90560)

Bit	Name	Type	Reset	Description
31:24	RGS	rw	0x0	RAM Group Select RGS
23:16	RDS	rw	0x0	Return Data select RDS
15:8	DWR	rw	0x0	Data Width Register DWR
7:6	RESERVED_SMSR	rw	0x0	RESERVED - Sense Margin Select Register, unused
5:2	PLS	rw	0x0	Pipeline Latency Select
1:0	RLS	rw	0x0	RAM Latency Select

4.9.2.32 DLR (Datalogger Register), (0x90564)

Bit	Name	Type	Reset	Description
31:24	reserved	r	0x0	Reserved
23:16	BRP	rw	0x0	Datalogger 2 (BRP)
15:14	RESERVED_DLR1	rw	0x0	RESERVED - DLR1 upper bits, unused
13	DLR1_OMISR_Z	rw	0x0	OMISR mode disable
12	DLR1_BISOR	rw	0x0	BISOR testing mode
11	DLR1_CMISR	rw	0x0	Cumulative MISR testing mode (mainly for ROM testing)
10	DLR1_RTM	rw	0x0	Retention testing mode
9	DLR1_GNG	rw	0x1	GO / NO-GO testing mode
8	DLR1_MISR	rw	0x0	MISR testing mode (mainly for ROM testing)
7	DLR0_TSM	rw	0x0	Time stamp mode
6	DLR0_CFMM	rw	0x0	Column Fail Masking mode
5	DLR0_ECAM	rw	0x0	Emulation cache access mode
4	DLR0_CAM	rw	0x0	Config access mode
3	DLR0_TCK	rw	0x1	TCK Gated mode
2	DLR0_ROM	rw	0x0	ROM-based testing mode
1	DLR0_IDDQ	rw	0x0	IDDQ testing mode
0	DLR0_DCM	rw	0x0	Distributed Compare mode

4.9.2.33 CMS (Clock-Mux Select Register), (0x90568)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CMS	rw	0x0	Clock Mux Select (CMS)

4.9.2.34 STR (Program Control Register), (0x9056c)

Bit	Name	Type	Reset	Description
31:5	reserved	r	0x0	Reserved
4	CHK	rw	0x0	Check MISR mode
3	STEP	rw	0x0	Step / Step for emulation mode
2	STOP	rw	0x0	Stop
1	RES	rw	0x0	Resume / Emulation read
0	START	rw	0x0	Start / Time Stamp mode restart

4.9.2.35 SCR (Address Scrambling Register), (0x90570)

Bit	Name	Type	Reset	Description
63:56	SCR7	rw	0xfe	Address Scrambling Register 7
55:48	SCR6	rw	0xdc	Address Scrambling Register 6
47:40	SCR5	rw	0xba	Address Scrambling Register 5
39:32	SCR4	rw	0x98	Address Scrambling Register 4
31:24	SCR3	rw	0x76	Address Scrambling Register 3
23:16	SCR2	rw	0x54	Address Scrambling Register 2
15:8	SCR1	rw	0x32	Address Scrambling Register 1
7:0	SCR0	rw	0x10	Address Scrambling Register 0

4.9.2.36 CSR (Chip Select Register), (0x90578)

Bit	Name	Type	Reset	Description
31:24	CSR3	rw	0x0	Chip Select 3 (CSR3)
23:16	CSR2	rw	0x0	Chip Select 2 (CSR2)
15:8	CSR1	rw	0x0	Chip Select 1 (CSR1)
7:0	CSR0	rw	0x0	Chip Select 0 (CSR0)

4.9.2.37 FDLY (Fail Delay Register), (0x9057c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	FDLY	rw	0x48	Fail Delay (FDLY)

4.9.2.38 PACT (PACT Register), (0x90580)

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	PACT	rw	0x0	PBIST Activate (PACT)

4.9.2.39 PID (PBIST_ID Register), (0x90584)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7	RESERVED_PID	rw	0x0	RESERVED - PID upper bit, unused
6:0	PID	rw	0x0	PBIST ID

4.9.2.40 OVER (Override Register), (0x90588)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	ALGO	rw	0x1	PBIST Override Algorithm Override
2	MM	rw	0x0	PBIST Override Multiple Memory
1	READ	rw	0x0	PBIST Override READ Override
0	RINFO	rw	0x1	PBIST Override RINFO Override

4.9.2.41 FSRF (Fail Status Fail Register), (0x90590)

Bit	Name	Type	Reset	Description
63:33	reserved	r	0x0	Reserved
32	FRSF1	r	0x0	Fail Status Fail - Port 1 (FSRF1)
31:1	reserved	r	0x0	Reserved
0	FRSF0	r	0x0	Fail Status Fail - Port 0 (FSRF0)

4.9.2.42 FSRC (Fail Status Count Register), (0x90598)

Bit	Name	Type	Reset	Description
63:36	reserved	r	0x0	Reserved
35:32	FSRC1	r	0x0	Fail Status Count - Port 1 (FSRC1)
31:4	reserved	r	0x0	Reserved
3:0	FSRC0	r	0x0	Fail Status Count - Port 0 (FSRC0)

4.9.2.43 FSRA (Fail Status Address Register), (0x905a0)

Bit	Name	Type	Reset	Description
63:48	reserved	r	0x0	Reserved
47:32	FSRA1	r	0x0	Fail Status Address - Port 1 (FSRA1)
31:16	reserved	r	0x0	Reserved
15:0	FSRA0	r	0x0	Fail Status Address - Port 0 (FSRA0)

4.9.2.44 FSRDL0 (Fail Status Data Registers), (0x905a8)

Bit	Name	Type	Reset	Description
31:0	FSRDL0	r	0xaaaaaaaa	Fail Status Data - Port 0 (FSRDL0)

4.9.2.45 FSRDL1 (Fail Status Data Registers), (0x905b0)

Bit	Name	Type	Reset	Description
31:0	FSRDL1	r	0xaaaaaaaa	Fail Status Data - Port 1 (FSRDL1)

4.9.2.46 MARGIN_MODE (Fail Status Fail Register), (0x905b4)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:2	PBIST_DFT_READ	rw	0x0	pbist_dft_read[1:0]
1:0	PBIST_DFT_WRITE	rw	0x0	pbist_dft_write[1:0]

4.9.2.47 WRENZ (Fail Status Fail Register), (0x905b8)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	WRENZ	rw	0x0	pbist_ram_wrenz[1:0]

4.9.2.48 PAGE_PGS (Fail Status Fail Register), (0x905bc)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	PGS	rw	0x0	pbist_ram_pgs[1:0]

4.9.2.49 ROM (ROM Mask Register), (0x905c0)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	ROM	rw	0x3	ROM Mask (ROM)

4.9.2.50 ALGO (Algorithm Mask Register), (0x905c4)

Bit	Name	Type	Reset	Description
31:24	ALGO_3	rw	0xff	ROM Algorithm Mask 3 (ALGO 3)
23:16	ALGO_2	rw	0xff	ROM Algorithm Mask 2 (ALGO 2)
15:8	ALGO_1	rw	0xff	ROM Algorithm Mask 1 (ALGO 1)
7:0	ALGO_0	rw	0xff	ROM Algorithm Mask 0 (ALGO 0)

4.9.2.51 RINFO (RAM Info Mask Register), (0x905c8)

Bit	Name	Type	Reset	Description
63:56	U3	rw	0xff	RAM Info Mask Upper 3 (RINFOU3)
55:48	U2	rw	0xff	RAM Info Mask Upper 2 (RINFOU2)
47:40	U1	rw	0xff	RAM Info Mask Upper 1 (RINFOU1)
39:32	U0	rw	0xff	RAM Info Mask Upper 0 (RINFOU0)
31:24	L3	rw	0xff	RAM Info Mask Lower 3 (RINFOL3)
23:16	L2	rw	0xff	RAM Info Mask Lower 2 (RINFOL2)
15:8	L1	rw	0xff	RAM Info Mask Lower 1 (RINFOL1)
7:0	L0	rw	0xff	RAM Info Mask Lower 0 (RINFOL0)

4.10 MMR Region: el2_pbist_2_mem

4.10.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x90800	RF0L	Register Files / Instruction Registers
0x90804	RF1L	Register Files / Instruction Registers
0x90808	RF2L	Register Files / Instruction Registers
0x9080c	RF3L	Register Files / Instruction Registers
0x90840	RF0U	Register Files / Instruction Registers
0x90844	RF1U	Register Files / Instruction Registers
0x90848	RF2U	Register Files / Instruction Registers
0x9084c	RF3U	Register Files / Instruction Registers
0x90900	A0	Variable Address Registers
0x90904	A1	Variable Address Registers
0x90908	A2	Variable Address Registers
0x9090c	A3	Variable Address Registers
0x90910	L0	Variable Loop Count Registers
0x90914	L1	Variable Loop Count Registers
0x90918	L2	Variable Loop Count Registers
0x9091c	L3	Variable Loop Count Registers
0x90920	D	Data Registers
0x90924	E	Data Registers
0x90930	CA0	Constant Address Registers
0x90934	CA1	Constant Address Registers
0x90938	CA2	Constant Address Registers
0x9093c	CA3	Constant Address Registers
0x90940	CL0	Constant Loop Count Registers
0x90944	CL1	Constant Loop Count Registers
0x90948	CL2	Constant Loop Count Registers
0x9094c	CL3	Constant Loop Count Registers
0x90950	I0	Constant Increment Registers
0x90954	I1	Constant Increment Registers
0x90958	I2	Constant Increment Registers
0x9095c	I3	Constant Increment Registers
0x90960	RAMT	RAM Configuration Register
0x90964	DLR	Datalogger Register
0x90968	CMS	Clock-Mux Select Register
0x9096c	STR	Program Control Register
0x90970	SCR	Address Scrambling Register
0x90978	CSR	Chip Select Register
0x9097c	FDLY	Fail Delay Register
0x90980	PACT	PACT Register
0x90984	PID	PBIST_ID Register
0x90988	OVER	Override Register
0x90990	FSRF	Fail Status Fail Register
0x90998	FSRC	Fail Status Count Register
0x909a0	FSRA	Fail Status Address Register
0x909a8	FSRDL0	Fail Status Data Registers
0x909b0	FSRDL1	Fail Status Data Registers
0x909b4	MARGIN_MODE	Fail Status Fail Register



0x909b8	WRENZ	Fail Status Fail Register
0x909bc	PAGE_PGS	Fail Status Fail Register
0x909c0	ROM	ROM Mask Register
0x909c4	ALGO	Algorithm Mask Register
0x909c8	RINFO	RAM Info Mask Register

4.10.2 Register Definitions

4.10.2.1 RF0L (Register Files / Instruction Registers), (0x90800)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF0L	rw	0x0	Register Files / Instruction Registers RF0 lower (RF0L)

4.10.2.2 RF1L (Register Files / Instruction Registers), (0x90804)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF1L	rw	0x0	Register Files / Instruction Registers RF1 lower (RF1L)

4.10.2.3 RF2L (Register Files / Instruction Registers), (0x90808)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF2L	rw	0x0	Register Files / Instruction Registers RF2 lower (RF2L)

4.10.2.4 RF3L (Register Files / Instruction Registers), (0x9080c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF3L	rw	0x0	Register Files / Instruction Registers RF3 lower (RF3L)

4.10.2.5 RF0U (Register Files / Instruction Registers), (0x90840)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF0U	rw	0x0	Register Files / Instruction Registers RF0 upper (RF0U)

4.10.2.6 RF1U (Register Files / Instruction Registers), (0x90844)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF1U	rw	0x0	Register Files / Instruction Registers RF1 upper (RF1U)

4.10.2.7 RF2U (Register Files / Instruction Registers), (0x90848)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF2U	rw	0x0	Register Files / Instruction Registers RF2 upper (RF2U)

4.10.2.8 RF3U (Register Files / Instruction Registers), (0x9084c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF3U	rw	0x0	Register Files / Instruction Registers RF3 upper (RF3U)

4.10.2.9 A0 (Variable Address Registers), (0x90900)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A0	rw	0x0	Variable Address Register 0 (A0)

4.10.2.10 A1 (Variable Address Registers), (0x90904)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A1	rw	0x0	Variable Address Register 1 (A1)

4.10.2.11 A2 (Variable Address Registers), (0x90908)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A2	rw	0x0	Variable Address Register 2 (A2)

4.10.2.12 A3 (Variable Address Registers), (0x9090c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A3	rw	0x0	Variable Address Register 3 (A3)

4.10.2.13 L0 (Variable Loop Count Registers), (0x90910)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L0	rw	0x0	Variable Loop Count Register 0 (L0)

4.10.2.14 L1 (Variable Loop Count Registers), (0x90914)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L1	rw	0x0	Variable Loop Count Register 1 (L1)

4.10.2.15 L2 (Variable Loop Count Registers), (0x90918)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L2	rw	0x0	Variable Loop Count Register 2 (L2)

4.10.2.16 L3 (Variable Loop Count Registers), (0x9091c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L3	rw	0x0	Variable Loop Count Register 3 (L3)

4.10.2.17 D (Data Registers), (0x90920)

Bit	Name	Type	Reset	Description
31:16	D1	rw	0x0	DD1 Data Register Upper 16 (D1)
15:0	D0	rw	0x0	DD0 Data Register Lower 16 (D0)

4.10.2.18 E (Data Registers), (0x90924)

Bit	Name	Type	Reset	Description
31:16	E1	rw	0x0	EE1 Data Register Upper 16 (E1)
15:0	E0	rw	0x0	EE0 Data Register Lower 16 (E0)

4.10.2.19 CA0 (Constant Address Registers), (0x90930)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA0	rw	0x0	Constant Address Register 0 (CA0)

4.10.2.20 CA1 (Constant Address Registers), (0x90934)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA1	rw	0x0	Constant Address Register 1 (CA1)

4.10.2.21 CA2 (Constant Address Registers), (0x90938)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA2	rw	0x0	Constant Address Register 2 (CA2)

4.10.2.22 CA3 (Constant Address Registers), (0x9093c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA3	rw	0x0	Constant Address Register 3 (CA3)

4.10.2.23 CL0 (Constant Loop Count Registers), (0x90940)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL0	rw	0x0	Constant Loop Count Register 0 (CL0)

4.10.2.24 CL1 (Constant Loop Count Registers), (0x90944)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL1	rw	0x0	Constant Loop Count Register 1 (CL1)

4.10.2.25 CL2 (Constant Loop Count Registers), (0x90948)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL2	rw	0x0	Constant Loop Count Register 2 (CL2)

4.10.2.26 CL3 (Constant Loop Count Registers), (0x9094c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL3	rw	0x0	Constant Loop Count Register 3 (CL3)

4.10.2.27 I0 (Constant Increment Registers), (0x90950)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 0 (I0)

4.10.2.28 I1 (Constant Increment Registers), (0x90954)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 1 (I1)

4.10.2.29 I2 (Constant Increment Registers), (0x90958)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 2 (I2)

4.10.2.30 I3 (Constant Increment Registers), (0x9095c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 3 (I3)

4.10.2.31 RAMT (RAM Configuration Register), (0x90960)

Bit	Name	Type	Reset	Description
31:24	RGS	rw	0x0	RAM Group Select RGS
23:16	RDS	rw	0x0	Return Data select RDS
15:8	DWR	rw	0x0	Data Width Register DWR
7:6	RESERVED_SMSR	rw	0x0	RESERVED - Sense Margin Select Register, unused
5:2	PLS	rw	0x0	Pipeline Latency Select
1:0	RLS	rw	0x0	RAM Latency Select

4.10.2.32 DLR (Datalogger Register), (0x90964)

Bit	Name	Type	Reset	Description
31:24	reserved	r	0x0	Reserved
23:16	BRP	rw	0x0	Datalogger 2 (BRP)
15:14	RESERVED_DLR1	rw	0x0	RESERVED - DLR1 upper bits, unused
13	DLR1_OMISR_Z	rw	0x0	OMISR mode disable
12	DLR1_BISOR	rw	0x0	BISOR testing mode
11	DLR1_CMISR	rw	0x0	Cumulative MISR testing mode (mainly for ROM testing)
10	DLR1_RTM	rw	0x0	Retention testing mode
9	DLR1_GNG	rw	0x1	GO / NO-GO testing mode
8	DLR1_MISR	rw	0x0	MISR testing mode (mainly for ROM testing)
7	DLR0_TSM	rw	0x0	Time stamp mode
6	DLR0_CFMM	rw	0x0	Column Fail Masking mode
5	DLR0_ECAM	rw	0x0	Emulation cache access mode
4	DLR0_CAM	rw	0x0	Config access mode
3	DLR0_TCK	rw	0x1	TCK Gated mode
2	DLR0_ROM	rw	0x0	ROM-based testing mode
1	DLR0_IDDQ	rw	0x0	IDDQ testing mode
0	DLR0_DCM	rw	0x0	Distributed Compare mode

4.10.2.33 CMS (Clock-Mux Select Register), (0x90968)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CMS	rw	0x0	Clock Mux Select (CMS)

4.10.2.34 STR (Program Control Register), (0x9096c)

Bit	Name	Type	Reset	Description
31:5	reserved	r	0x0	Reserved
4	CHK	rw	0x0	Check MISR mode
3	STEP	rw	0x0	Step / Step for emulation mode
2	STOP	rw	0x0	Stop
1	RES	rw	0x0	Resume / Emulation read
0	START	rw	0x0	Start / Time Stamp mode restart

4.10.2.35 SCR (Address Scrambling Register), (0x90970)

Bit	Name	Type	Reset	Description
63:56	SCR7	rw	0xfe	Address Scrambling Register 7
55:48	SCR6	rw	0xdc	Address Scrambling Register 6
47:40	SCR5	rw	0xba	Address Scrambling Register 5
39:32	SCR4	rw	0x98	Address Scrambling Register 4
31:24	SCR3	rw	0x76	Address Scrambling Register 3
23:16	SCR2	rw	0x54	Address Scrambling Register 2
15:8	SCR1	rw	0x32	Address Scrambling Register 1
7:0	SCR0	rw	0x10	Address Scrambling Register 0

4.10.2.36 CSR (Chip Select Register), (0x90978)

Bit	Name	Type	Reset	Description
31:24	CSR3	rw	0x0	Chip Select 3 (CSR3)
23:16	CSR2	rw	0x0	Chip Select 2 (CSR2)
15:8	CSR1	rw	0x0	Chip Select 1 (CSR1)
7:0	CSR0	rw	0x0	Chip Select 0 (CSR0)

4.10.2.37 FDLY (Fail Delay Register), (0x9097c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	FDLY	rw	0x48	Fail Delay (FDLY)

4.10.2.38 PACT (PACT Register), (0x90980)

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	PACT	rw	0x0	PBIST Activate (PACT)

4.10.2.39 PID (PBIST_ID Register), (0x90984)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7	RESERVED_PID	rw	0x0	RESERVED - PID upper bit, unused
6:0	PID	rw	0x0	PBIST ID

4.10.2.40 OVER (Override Register), (0x90988)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	ALGO	rw	0x1	PBIST Override Algorithm Override
2	MM	rw	0x0	PBIST Override Multiple Memory
1	READ	rw	0x0	PBIST Override READ Override
0	RINFO	rw	0x1	PBIST Override RINFO Override

4.10.2.41 FSRF (Fail Status Fail Register), (0x90990)

Bit	Name	Type	Reset	Description
63:33	reserved	r	0x0	Reserved
32	FRSF1	r	0x0	Fail Status Fail - Port 1 (FSRF1)
31:1	reserved	r	0x0	Reserved
0	FRSF0	r	0x0	Fail Status Fail - Port 0 (FSRF0)

4.10.2.42 FSRC (Fail Status Count Register), (0x90998)

Bit	Name	Type	Reset	Description
63:36	reserved	r	0x0	Reserved
35:32	FSRC1	r	0x0	Fail Status Count - Port 1 (FSRC1)
31:4	reserved	r	0x0	Reserved
3:0	FSRC0	r	0x0	Fail Status Count - Port 0 (FSRC0)

4.10.2.43 FSRA (Fail Status Address Register), (0x909a0)

Bit	Name	Type	Reset	Description
63:48	reserved	r	0x0	Reserved
47:32	FSRA1	r	0x0	Fail Status Address - Port 1 (FSRA1)
31:16	reserved	r	0x0	Reserved
15:0	FSRA0	r	0x0	Fail Status Address - Port 0 (FSRA0)

4.10.2.44 FSRDL0 (Fail Status Data Registers), (0x909a8)

Bit	Name	Type	Reset	Description
31:0	FSRDL0	r	0xaaaaaaaa	Fail Status Data - Port 0 (FSRDL0)

4.10.2.45 FSRDL1 (Fail Status Data Registers), (0x909b0)

Bit	Name	Type	Reset	Description
31:0	FSRDL1	r	0xaaaaaaaa	Fail Status Data - Port 1 (FSRDL1)

4.10.2.46 MARGIN_MODE (Fail Status Fail Register), (0x909b4)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:2	PBIST_DFT_READ	rw	0x0	pbist_dft_read[1:0]
1:0	PBIST_DFT_WRITE	rw	0x0	pbist_dft_write[1:0]

4.10.2.47 WRENZ (Fail Status Fail Register), (0x909b8)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	WRENZ	rw	0x0	pbist_ram_wrenz[1:0]

4.10.2.48 PAGE_PGS (Fail Status Fail Register), (0x909bc)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	PGS	rw	0x0	pbist_ram_pgs[1:0]

4.10.2.49 ROM (ROM Mask Register), (0x909c0)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	ROM	rw	0x3	ROM Mask (ROM)

4.10.2.50 ALGO (Algorithm Mask Register), (0x909c4)

Bit	Name	Type	Reset	Description
31:24	ALGO_3	rw	0xff	ROM Algorithm Mask 3 (ALGO 3)
23:16	ALGO_2	rw	0xff	ROM Algorithm Mask 2 (ALGO 2)
15:8	ALGO_1	rw	0xff	ROM Algorithm Mask 1 (ALGO 1)
7:0	ALGO_0	rw	0xff	ROM Algorithm Mask 0 (ALGO 0)

4.10.2.51 RINFO (RAM Info Mask Register), (0x909c8)

Bit	Name	Type	Reset	Description
63:56	U3	rw	0xff	RAM Info Mask Upper 3 (RINFOU3)
55:48	U2	rw	0xff	RAM Info Mask Upper 2 (RINFOU2)
47:40	U1	rw	0xff	RAM Info Mask Upper 1 (RINFOU1)
39:32	U0	rw	0xff	RAM Info Mask Upper 0 (RINFOU0)
31:24	L3	rw	0xff	RAM Info Mask Lower 3 (RINFOL3)
23:16	L2	rw	0xff	RAM Info Mask Lower 2 (RINFOL2)
15:8	L1	rw	0xff	RAM Info Mask Lower 1 (RINFOL1)
7:0	L0	rw	0xff	RAM Info Mask Lower 0 (RINFOL0)

4.11 MMR Region: el2_pbist_3_mem

4.11.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x90c00	RF0L	Register Files / Instruction Registers
0x90c04	RF1L	Register Files / Instruction Registers
0x90c08	RF2L	Register Files / Instruction Registers
0x90c0c	RF3L	Register Files / Instruction Registers
0x90c40	RF0U	Register Files / Instruction Registers
0x90c44	RF1U	Register Files / Instruction Registers
0x90c48	RF2U	Register Files / Instruction Registers
0x90c4c	RF3U	Register Files / Instruction Registers
0x90d00	A0	Variable Address Registers
0x90d04	A1	Variable Address Registers
0x90d08	A2	Variable Address Registers
0x90d0c	A3	Variable Address Registers
0x90d10	L0	Variable Loop Count Registers
0x90d14	L1	Variable Loop Count Registers
0x90d18	L2	Variable Loop Count Registers
0x90d1c	L3	Variable Loop Count Registers
0x90d20	D	Data Registers
0x90d24	E	Data Registers
0x90d30	CA0	Constant Address Registers
0x90d34	CA1	Constant Address Registers
0x90d38	CA2	Constant Address Registers
0x90d3c	CA3	Constant Address Registers
0x90d40	CL0	Constant Loop Count Registers
0x90d44	CL1	Constant Loop Count Registers
0x90d48	CL2	Constant Loop Count Registers
0x90d4c	CL3	Constant Loop Count Registers
0x90d50	I0	Constant Increment Registers
0x90d54	I1	Constant Increment Registers
0x90d58	I2	Constant Increment Registers
0x90d5c	I3	Constant Increment Registers
0x90d60	RAMT	RAM Configuration Register
0x90d64	DLR	Datalogger Register
0x90d68	CMS	Clock-Mux Select Register
0x90d6c	STR	Program Control Register
0x90d70	SCR	Address Scrambling Register
0x90d78	CSR	Chip Select Register
0x90d7c	FDLY	Fail Delay Register
0x90d80	PACT	PACT Register
0x90d84	PID	PBIST_ID Register
0x90d88	OVER	Override Register
0x90d90	FSRF	Fail Status Fail Register
0x90d98	FSRC	Fail Status Count Register
0x90da0	FSRA	Fail Status Address Register
0x90da8	FSRDL0	Fail Status Data Registers
0x90db0	FSRDL1	Fail Status Data Registers
0x90db4	MARGIN_MODE	Fail Status Fail Register



0x90db8	WRENZ	Fail Status Fail Register
0x90dbc	PAGE_PGS	Fail Status Fail Register
0x90dc0	ROM	ROM Mask Register
0x90dc4	ALGO	Algorithm Mask Register
0x90dc8	RINFO	RAM Info Mask Register

4.11.2 Register Definitions

4.11.2.1 RF0L (Register Files / Instruction Registers), (0x90c00)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF0L	rw	0x0	Register Files / Instruction Registers RF0 lower (RF0L)

4.11.2.2 RF1L (Register Files / Instruction Registers), (0x90c04)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF1L	rw	0x0	Register Files / Instruction Registers RF1 lower (RF1L)

4.11.2.3 RF2L (Register Files / Instruction Registers), (0x90c08)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF2L	rw	0x0	Register Files / Instruction Registers RF2 lower (RF2L)

4.11.2.4 RF3L (Register Files / Instruction Registers), (0x90c0c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF3L	rw	0x0	Register Files / Instruction Registers RF3 lower (RF3L)

4.11.2.5 RF0U (Register Files / Instruction Registers), (0x90c40)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF0U	rw	0x0	Register Files / Instruction Registers RF0 upper (RF0U)

4.11.2.6 RF1U (Register Files / Instruction Registers), (0x90c44)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF1U	rw	0x0	Register Files / Instruction Registers RF1 upper (RF1U)

4.11.2.7 RF2U (Register Files / Instruction Registers), (0x90c48)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF2U	rw	0x0	Register Files / Instruction Registers RF2 upper (RF2U)

4.11.2.8 RF3U (Register Files / Instruction Registers), (0x90c4c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	RF3U	rw	0x0	Register Files / Instruction Registers RF3 upper (RF3U)

4.11.2.9 A0 (Variable Address Registers), (0x90d00)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A0	rw	0x0	Variable Address Register 0 (A0)

4.11.2.10 A1 (Variable Address Registers), (0x90d04)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A1	rw	0x0	Variable Address Register 1 (A1)

4.11.2.11 A2 (Variable Address Registers), (0x90d08)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A2	rw	0x0	Variable Address Register 2 (A2)

4.11.2.12 A3 (Variable Address Registers), (0x90d0c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	A3	rw	0x0	Variable Address Register 3 (A3)

4.11.2.13 L0 (Variable Loop Count Registers), (0x90d10)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L0	rw	0x0	Variable Loop Count Register 0 (L0)

4.11.2.14 L1 (Variable Loop Count Registers), (0x90d14)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L1	rw	0x0	Variable Loop Count Register 1 (L1)

4.11.2.15 L2 (Variable Loop Count Registers), (0x90d18)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L2	rw	0x0	Variable Loop Count Register 2 (L2)

4.11.2.16 L3 (Variable Loop Count Registers), (0x90d1c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	L3	rw	0x0	Variable Loop Count Register 3 (L3)

4.11.2.17 D (Data Registers), (0x90d20)

Bit	Name	Type	Reset	Description
31:16	D1	rw	0x0	DD1 Data Register Upper 16 (D1)
15:0	D0	rw	0x0	DD0 Data Register Lower 16 (D0)

4.11.2.18 E (Data Registers), (0x90d24)

Bit	Name	Type	Reset	Description
31:16	E1	rw	0x0	EE1 Data Register Upper 16 (E1)
15:0	E0	rw	0x0	EE0 Data Register Lower 16 (E0)

4.11.2.19 CA0 (Constant Address Registers), (0x90d30)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA0	rw	0x0	Constant Address Register 0 (CA0)

4.11.2.20 CA1 (Constant Address Registers), (0x90d34)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA1	rw	0x0	Constant Address Register 1 (CA1)

4.11.2.21 CA2 (Constant Address Registers), (0x90d38)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA2	rw	0x0	Constant Address Register 2 (CA2)

4.11.2.22 CA3 (Constant Address Registers), (0x90d3c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CA3	rw	0x0	Constant Address Register 3 (CA3)

4.11.2.23 CL0 (Constant Loop Count Registers), (0x90d40)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL0	rw	0x0	Constant Loop Count Register 0 (CL0)

4.11.2.24 CL1 (Constant Loop Count Registers), (0x90d44)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL1	rw	0x0	Constant Loop Count Register 1 (CL1)

4.11.2.25 CL2 (Constant Loop Count Registers), (0x90d48)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL2	rw	0x0	Constant Loop Count Register 2 (CL2)

4.11.2.26 CL3 (Constant Loop Count Registers), (0x90d4c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	CL3	rw	0x0	Constant Loop Count Register 3 (CL3)

4.11.2.27 I0 (Constant Increment Registers), (0x90d50)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 0 (I0)

4.11.2.28 I1 (Constant Increment Registers), (0x90d54)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 1 (I1)

4.11.2.29 I2 (Constant Increment Registers), (0x90d58)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 2 (I2)

4.11.2.30 I3 (Constant Increment Registers), (0x90d5c)

Bit	Name	Type	Reset	Description
31:16	reserved	r	0x0	Reserved
15:0	I0	rw	0x0	Constant Increment Register 3 (I3)

4.11.2.31 RAMT (RAM Configuration Register), (0x90d60)

Bit	Name	Type	Reset	Description
31:24	RGS	rw	0x0	RAM Group Select RGS
23:16	RDS	rw	0x0	Return Data select RDS
15:8	DWR	rw	0x0	Data Width Register DWR
7:6	RESERVED_SMSR	rw	0x0	RESERVED - Sense Margin Select Register, unused
5:2	PLS	rw	0x0	Pipeline Latency Select
1:0	RLS	rw	0x0	RAM Latency Select

4.11.2.32 DLR (Datalogger Register), (0x90d64)

Bit	Name	Type	Reset	Description
31:24	reserved	r	0x0	Reserved
23:16	BRP	rw	0x0	Datalogger 2 (BRP)
15:14	RESERVED_DLR1	rw	0x0	RESERVED - DLR1 upper bits, unused
13	DLR1_OMISR_Z	rw	0x0	OMISR mode disable
12	DLR1_BISOR	rw	0x0	BISOR testing mode
11	DLR1_CMISR	rw	0x0	Cumulative MISR testing mode (mainly for ROM testing)
10	DLR1_RTM	rw	0x0	Retention testing mode
9	DLR1_GNG	rw	0x1	GO / NO-GO testing mode
8	DLR1_MISR	rw	0x0	MISR testing mode (mainly for ROM testing)
7	DLR0_TSM	rw	0x0	Time stamp mode
6	DLR0_CFMM	rw	0x0	Column Fail Masking mode
5	DLR0_ECAM	rw	0x0	Emulation cache access mode
4	DLR0_CAM	rw	0x0	Config access mode
3	DLR0_TCK	rw	0x1	TCK Gated mode
2	DLR0_ROM	rw	0x0	ROM-based testing mode
1	DLR0_IDDQ	rw	0x0	IDDQ testing mode
0	DLR0_DCM	rw	0x0	Distributed Compare mode

4.11.2.33 CMS (Clock-Mux Select Register), (0x90d68)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CMS	rw	0x0	Clock Mux Select (CMS)

4.11.2.34 STR (Program Control Register), (0x90d6c)

Bit	Name	Type	Reset	Description
31:5	reserved	r	0x0	Reserved
4	CHK	rw	0x0	Check MISR mode
3	STEP	rw	0x0	Step / Step for emulation mode
2	STOP	rw	0x0	Stop
1	RES	rw	0x0	Resume / Emulation read
0	START	rw	0x0	Start / Time Stamp mode restart

4.11.2.35 SCR (Address Scrambling Register), (0x90d70)

Bit	Name	Type	Reset	Description
63:56	SCR7	rw	0xfe	Address Scrambling Register 7
55:48	SCR6	rw	0xdc	Address Scrambling Register 6
47:40	SCR5	rw	0xba	Address Scrambling Register 5
39:32	SCR4	rw	0x98	Address Scrambling Register 4
31:24	SCR3	rw	0x76	Address Scrambling Register 3
23:16	SCR2	rw	0x54	Address Scrambling Register 2
15:8	SCR1	rw	0x32	Address Scrambling Register 1
7:0	SCR0	rw	0x10	Address Scrambling Register 0

4.11.2.36 CSR (Chip Select Register), (0x90d78)

Bit	Name	Type	Reset	Description
31:24	CSR3	rw	0x0	Chip Select 3 (CSR3)
23:16	CSR2	rw	0x0	Chip Select 2 (CSR2)
15:8	CSR1	rw	0x0	Chip Select 1 (CSR1)
7:0	CSR0	rw	0x0	Chip Select 0 (CSR0)

4.11.2.37 FDLY (Fail Delay Register), (0x90d7c)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	FDLY	rw	0x48	Fail Delay (FDLY)

4.11.2.38 PACT (PACT Register), (0x90d80)

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	PACT	rw	0x0	PBIST Activate (PACT)

4.11.2.39 PID (PBIST_ID Register), (0x90d84)

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7	RESERVED_PID	rw	0x0	RESERVED - PID upper bit, unused
6:0	PID	rw	0x0	PBIST ID

4.11.2.40 OVER (Override Register), (0x90d88)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	ALGO	rw	0x1	PBIST Override Algorithm Override
2	MM	rw	0x0	PBIST Override Multiple Memory
1	READ	rw	0x0	PBIST Override READ Override
0	RINFO	rw	0x1	PBIST Override RINFO Override

4.11.2.41 FSRF (Fail Status Fail Register), (0x90d90)

Bit	Name	Type	Reset	Description
63:33	reserved	r	0x0	Reserved
32	FRSF1	r	0x0	Fail Status Fail - Port 1 (FSRF1)
31:1	reserved	r	0x0	Reserved
0	FRSF0	r	0x0	Fail Status Fail - Port 0 (FSRF0)

4.11.2.42 FSRC (Fail Status Count Register), (0x90d98)

Bit	Name	Type	Reset	Description
63:36	reserved	r	0x0	Reserved
35:32	FSRC1	r	0x0	Fail Status Count - Port 1 (FSRC1)
31:4	reserved	r	0x0	Reserved
3:0	FSRC0	r	0x0	Fail Status Count - Port 0 (FSRC0)

4.11.2.43 FSRA (Fail Status Address Register), (0x90da0)

Bit	Name	Type	Reset	Description
63:48	reserved	r	0x0	Reserved
47:32	FSRA1	r	0x0	Fail Status Address - Port 1 (FSRA1)
31:16	reserved	r	0x0	Reserved
15:0	FSRA0	r	0x0	Fail Status Address - Port 0 (FSRA0)

4.11.2.44 FSRDL0 (Fail Status Data Registers), (0x90da8)

Bit	Name	Type	Reset	Description
31:0	FSRDL0	r	0xaaaaaaaa	Fail Status Data - Port 0 (FSRDL0)

4.11.2.45 FSRDL1 (Fail Status Data Registers), (0x90db0)

Bit	Name	Type	Reset	Description
31:0	FSRDL1	r	0xaaaaaaaa	Fail Status Data - Port 1 (FSRDL1)

4.11.2.46 MARGIN_MODE (Fail Status Fail Register), (0x90db4)

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:2	PBIST_DFT_READ	rw	0x0	pbist_dft_read[1:0]
1:0	PBIST_DFT_WRITE	rw	0x0	pbist_dft_write[1:0]

4.11.2.47 WRENZ (Fail Status Fail Register), (0x90db8)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	WRENZ	rw	0x0	pbist_ram_wrenz[1:0]

4.11.2.48 PAGE_PGS (Fail Status Fail Register), (0x90dbc)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	PGS	rw	0x0	pbist_ram_pgs[1:0]

4.11.2.49 ROM (ROM Mask Register), (0x90dc0)

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1:0	ROM	rw	0x3	ROM Mask (ROM)

4.11.2.50 ALGO (Algorithm Mask Register), (0x90dc4)

Bit	Name	Type	Reset	Description
31:24	ALGO_3	rw	0xff	ROM Algorithm Mask 3 (ALGO 3)
23:16	ALGO_2	rw	0xff	ROM Algorithm Mask 2 (ALGO 2)
15:8	ALGO_1	rw	0xff	ROM Algorithm Mask 1 (ALGO 1)
7:0	ALGO_0	rw	0xff	ROM Algorithm Mask 0 (ALGO 0)

4.11.2.51 RINFO (RAM Info Mask Register), (0x90dc8)

Bit	Name	Type	Reset	Description
63:56	U3	rw	0xff	RAM Info Mask Upper 3 (RINFOU3)
55:48	U2	rw	0xff	RAM Info Mask Upper 2 (RINFOU2)
47:40	U1	rw	0xff	RAM Info Mask Upper 1 (RINFOU1)
39:32	U0	rw	0xff	RAM Info Mask Upper 0 (RINFOU0)
31:24	L3	rw	0xff	RAM Info Mask Lower 3 (RINFOL3)
23:16	L2	rw	0xff	RAM Info Mask Lower 2 (RINFOL2)
15:8	L1	rw	0xff	RAM Info Mask Lower 1 (RINFOL1)
7:0	L0	rw	0xff	RAM Info Mask Lower 0 (RINFOL0)

4.12 MMR Region: el2_ecc_aggr_regs

4.12.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x2000	AGGR_REVISION	Aggregator Revision Register
0x2004	NOGATE	Aggregator Nogate Register
0x2008	ECC_VECTOR	ECC Vector Register
0x200c	MISC_STATUS	Misc Status
0x2010+(a*0x4), a=0..7	RESERVED_SVBUS[a]	Reserved Area for Serial VBUS Registers
0x203c	SEC_EOI_REG	EOI Register
0x2040	SEC_STATUS_REG0	Interrupt Status Register 0
0x2080	SEC_ENABLE_SET_REG0	Interrupt Enable Set Register 0
0x20c0	SEC_ENABLE_CLR_REG0	Interrupt Enable Clear Register 0
0x213c	DED_EOI_REG	EOI Register
0x2140	DED_STATUS_REG0	Interrupt Status Register 0
0x2180	DED_ENABLE_SET_REG0	Interrupt Enable Set Register 0
0x21c0	DED_ENABLE_CLR_REG0	Interrupt Enable Clear Register 0
0x2200	AGGR_ENABLE_SET	AGGR interrupt enable set Register
0x2204	AGGR_ENABLE_CLR	AGGR interrupt enable clear Register
0x2208	AGGR_STATUS_SET	AGGR interrupt status set Register
0x220c	AGGR_STATUS_CLR	AGGR interrupt status clear Register
0x2300	AUTO_CONTROL	Auto Test Mode ECC Control
0x2304	AUTO_ERROR_CTRL2	Auto Test Mode ECC Error Control2 Register
0x2308	AUTO_RESTART_CTRL	Auto Test Mode Restart Control Register
0x230c	AUTO_START_CTRL	Auto Test Mode Start Control Register
0x2310	AUTO_STATUS1	Auto Test Mode Status1 Register
0x2314	AUTO_STATUS2	Auto Test Mode Status2 Register
0x231c	AUTO_STATUS4	Auto Test Mode Status4 Register
0x2320	AUTO_ENABLE	Auto Test Mode Enable Registers
0x2360	AUTOTEST_INTR_STATUS_SET_REG	Interrupt Status Set Register
0x2364	AUTOTEST_INTR_STATUS_CLR_REG	Interrupt Status Clear Register

0x2368	AUTOTEST_INTR_ENABLE_SET_REG	Interrupt Enable Set Register
0x236c	AUTOTEST_INTR_ENABLE_CLR_REG	Interrupt Enable Clear Register
0x2370	AUTOTEST_INTR_EOI_REG	EOI Register
0x2380	AUTO_CAPABLE	Auto Test Mode Capable Status Registers

4.12.2 Register Definitions

4.12.2.1 AGGR_REVISION (Aggregator Revision Register), (0x2000)

Revision parameters

Bit	Name	Type	Reset	Description
31:30	scheme	r	0x1	Scheme
29:28	bu	r	0x2	bu
27:16	module_id	r	0x6a0	Module ID
15:11	revrtl	r	0x6	RTL version
10:8	revmaj	r	0x2	Major version
7:6	custom	r	0x0	Custom version
5:0	revmin	r	0x2	Minor version

4.12.2.2 NOGATE (Aggregator Nogate Register), (0x2004)

Nogate control

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	aggr	rw	0x0	Clock gating disable for ecc aggregator when set

4.12.2.3 ECC_VECTOR (ECC Vector Register), (0x2008)

ECC Vector Register

Bit	Name	Type	Reset	Description
31:25	reserved	r	0x0	Reserved
24	rd_svbus_done	rw1tc	0x0	Status to indicate if read on serial VBUS is complete, write of any value will clear this bit.
23:16	rd_svbus_address	rw	0x0	Read address
15	rd_svbus	rw1ts	0x0	Write 1 to trigger a read on the serial VBUS
14:11	reserved	r	0x0	Reserved
10:0	ecc_vector	rw	0x0	Value written to select the corresponding ECC RAM for control or status

4.12.2.4 MISC_STATUS (Misc Status), (0x200c)

Misc Status

Bit	Name	Type	Reset	Description
31:11	reserved	r	0x0	Reserved
10:0	num_rams	r	0x20	Indicates the number of RAMS serviced by the ECC aggregator

4.12.2.5 RESERVED_SVBUS[a] (Reserved Area for Serial VBUS Registers), (0x2010+(a*0x4)), a=0..7

Reference other documents that contain the ECC RAM wrapper and EDC controller serial vbus register sets.

Bit	Name	Type	Reset	Description
31:0	data	rw	0x0	Serial VBUS register data

4.12.2.6 SEC_EOI_REG (EOI Register), (0x203c)

EOI Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	eoi_wr	rw1ts	0x0	EOI Register

4.12.2.7 SEC_STATUS_REG0 (Interrupt Status Register 0), (0x2040)

Interrupt Status Register 0

Bit	Name	Type	Reset	Description
31	eccagg_pend	rw1ts	0x0	Interrupt Pending Status for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend



21	tda54_el2_fw_aggr_wrap_fw1_t da54_el2_fw_aggr_wrap_fw1_m mrs_tda54_el2_fw_aggr_wrap_f w1_mmrs_edc_ctrl_busecc_pen d	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw1_t a54_el2_fw_aggr_wrap_fw1_mm rs_tda54_el2_fw_aggr_wrap_fw1 mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_t da54_el2_fw_aggr_wrap_fw0_m mrs_tda54_el2_fw_aggr_wrap_f w0_mmrs_edc_ctrl_busecc_pen d	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw0_t a54_el2_fw_aggr_wrap_fw0_mm rs_tda54_el2_fw_aggr_wrap_fw0 mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_ag gr_fw_aggr_tda54_ac_wrap_fw_ aggr_tda54_ac_wrap_cbass_fw_ aggr_tda54_ac_wrap_fw_aggr_ tda54_ac_wrap_cbass_scr1_s cr_fw_aggr_tda54_ac_wrap_fw_ aggr_tda54_ac_wrap_cbass_scr 1_scr_edc_ctrl_busecc_1_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw_ag gr_fw_aggr_tda54_ac_wrap_fw_ aggr_tda54_ac_wrap_cbass_fw_ aggr_tda54_ac_wrap_fw_aggr_t da54_ac_wrap_cbass_scr1_scr_ fw_aggr_tda54_ac_wrap_fw_agg r_tda54_ac_wrap_cbass_scr1_s cr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_ag gr_fw_aggr_tda54_ac_wrap_fw_ aggr_tda54_ac_wrap_cbass_fw_ aggr_tda54_ac_wrap_fw_aggr_ tda54_ac_wrap_cbass_scr1_s cr_fw_aggr_tda54_ac_wrap_fw_ aggr_tda54_ac_wrap_cbass_scr 1_scr_edc_ctrl_busecc_0_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw_ag gr_fw_aggr_tda54_ac_wrap_fw_ aggr_tda54_ac_wrap_cbass_fw_ aggr_tda54_ac_wrap_fw_aggr_t da54_ac_wrap_cbass_scr1_scr_ fw_aggr_tda54_ac_wrap_fw_agg r_tda54_ac_wrap_cbass_scr1_s cr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ ac_vbusp_dmesc_p2p_bridge_ac_ vbusp_dmesc_p2p_bridge_dst_ busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_async_bridge_wrap_a c_vbusp_dmesc_p2p_bridge_ac_ vbusp_dmesc_p2p_bridge_dst_bu secc_pend
16	dru_1_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_1_edc_pend
15	dru_0_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_0_edc_pend
14	dru_eng_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_eng_edc_pend
13	dru_ring_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_ring_edc_pend
12	dru_rd_buf_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_rd_buf_edc_pend
11	dru_queue_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ct rl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_wrap_el2_rat_edc_ctrl busecc_pend



9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend
7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_pend	rw1ts	0x0	Interrupt Pending Status for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_busecc_pend	rw1ts	0x0	Interrupt Pending Status for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_pend	rw1ts	0x0	Interrupt Pending Status for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_pend	rw1ts	0x0	Interrupt Pending Status for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_pend	rw1ts	0x0	Interrupt Pending Status for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for el2_ram_edc_ctl_busecc_pend

4.12.2.8 SEC_ENABLE_SET_REG0 (Interrupt Enable Set Register 0), (0x2080)

Interrupt Enable Set Register 0

Bit	Name	Type	Reset	Description
31	eccagg_enable_set	rw1t s	0x0	Interrupt Enable Set Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_set	rw1t s	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_set	rw1t s	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_set	rw1t s	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_set	rw1t s	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_set	rw1t s	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_set	rw1t s	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_set	rw1t s	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_set	rw1t s	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_set	rw1t s	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend

21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_1_edc_pend
15	dru_0_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_0_edc_pend
14	dru_eng_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_eng_edc_pend
13	dru_ring_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_ring_edc_pend
12	dru_rd_buf_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend

9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend
7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_amecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for clec_sram_amecc_pend
4	el2_ram_edc_ctl_mem3_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for el2_ram_edc_ctl_busecc_pend

4.12.2.9 SEC_ENABLE_CLR_REG0 (Interrupt Enable Clear Register 0), (0x20c0)

Interrupt Enable Clear Register 0

Bit	Name	Type	Reset	Description
31	eccagg_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend

21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_1_edc_pend
15	dru_0_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_0_edc_pend
14	dru_eng_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_eng_edc_pend
13	dru_ring_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_ring_edc_pend
12	dru_rd_buf_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend

9	tda54_el2_cfg_cbass_cfg_cbass_s_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_s_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_s_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_s_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend
7	tda54_el2_cfg_cbass_cfg_cbass_s_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_s_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_s_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_s_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for el2_ram_edc_ctl_busecc_pend

4.12.2.10 DED_EOI_REG (EOI Register), (0x213c)

EOI Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	eoi_wr	rw1ts	0x0	EOI Register

4.12.2.11 DED_STATUS_REG0 (Interrupt Status Register 0), (0x2140)

Interrupt Status Register 0

Bit	Name	Type	Reset	Description
31	eccagg_pend	rw1ts	0x0	Interrupt Pending Status for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend



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9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend
7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_pend	rw1ts	0x0	Interrupt Pending Status for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_busecc_pend	rw1ts	0x0	Interrupt Pending Status for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_pend	rw1ts	0x0	Interrupt Pending Status for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_pend	rw1ts	0x0	Interrupt Pending Status for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_pend	rw1ts	0x0	Interrupt Pending Status for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for el2_ram_edc_ctl_busecc_pend

4.12.2.12 DED_ENABLE_SET_REG0 (Interrupt Enable Set Register 0), (0x2180)

Interrupt Enable Set Register 0

Bit	Name	Type	Reset	Description
31	eccagg_enable_set	rw1ts	0x0	Interrupt Enable Set Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend

21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_1_edc_pend
15	dru_0_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_0_edc_pend
14	dru_eng_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_eng_edc_pend
13	dru_ring_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_ring_edc_pend
12	dru_rd_buf_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend



9	tda54_el2_cfg_cbass_cfg_cb ass_tda54_el2_cfg_cbass_cf g_cbass_el2_rat_cfg_p2p_bri dge_el2_rat_cfg_bridge_dst_ busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_t da54_el2_cfg_cbass_cfg_cbass_e l2_rat_cfg_p2p_bridge_el2_rat_cf g_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cb ass_tda54_el2_cfg_cbass_cf g_cbass_dru_mmr_cfg_split_ p2p_bridge_dru_mmr_cfg_sp lit_bridge_dst_busecc_enabl e_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_t da54_el2_cfg_cbass_cfg_cbass_d ru_mmr_cfg_split_p2p_bridge_dru _mmr_cfg_split_bridge_dst_busec c_pend
7	tda54_el2_cfg_cbass_cfg_cb ass_tda54_el2_cfg_cbass_cf g_cbass_el2_ecc_aggr_cfg_ p2p_bridge_el2_ecc_aggr_cf g_bridge_dst_busecc_enable set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_t da54_el2_cfg_cbass_cfg_cbass_e l2_ecc_aggr_cfg_p2p_bridge_el2 _ecc_aggr_cfg_bridge_dst_busec c_pend
6	tda54_el2_cfg_cbass_cfg_cb ass_tda54_el2_cfg_cbass_cf g_cbass_dru_mmr_cfg_p2p_ bridge_dru_mmr_cfg_bridge_ src_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_t da54_el2_cfg_cbass_cfg_cbass_d ru_mmr_cfg_p2p_bridge_dru_mm r_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_s et	rw1ts	0x0	Interrupt Enable Set Register for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_bus ecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem3_busecc_ pend
3	el2_ram_edc_ctl_mem2_bus ecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem2_busecc_ pend
2	el2_ram_edc_ctl_mem1_bus ecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem1_busecc_ pend
1	el2_ram_edc_ctl_mem0_bus ecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem0_busecc_ pend
0	el2_ram_edc_ctl_busecc_en able_set	rw1ts	0x0	Interrupt Enable Set Register for el2_ram_edc_ctl_busecc_pend

4.12.2.13 DED_ENABLE_CLR_REG0 (Interrupt Enable Clear Register 0), (0x21c0)

Interrupt Enable Clear Register 0

Bit	Name	Type	Reset	Description
31	eccagg_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend

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17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_1_edc_pend
15	dru_0_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_0_edc_pend
14	dru_eng_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_eng_edc_pend
13	dru_ring_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_ring_edc_pend
12	dru_rd_buf_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend
7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for clec_sram_ramecc_pend



4	el2_ram_edc_ctl_mem3_bus_ecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_bus_ecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_bus_ecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_bus_ecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for el2_ram_edc_ctl_busecc_pend

4.12.2.14 AGGR_ENABLE_SET (AGGR interrupt enable set Register), (0x2200)

AGGR interrupt enable set Register

Bit	Name	Type	Reset	Description
31:3	reserved	r	0x0	Reserved
2	autotest	rw1ts	0x0	interrupt enable set for auto test mode
1	timeout	rw1ts	0x0	interrupt enable set for svbus timeout errors
0	parity	rw1ts	0x0	interrupt enable set for parity errors

4.12.2.15 AGGR_ENABLE_CLR (AGGR interrupt enable clear Register), (0x2204)

AGGR interrupt enable clear Register

Bit	Name	Type	Reset	Description
31:3	reserved	r	0x0	Reserved
2	autotest	rw1tc	0x0	interrupt enable clear for auto test mode
1	timeout	rw1tc	0x0	interrupt enable clear for svbus timeout errors
0	parity	rw1tc	0x0	interrupt enable clear for parity errors

4.12.2.16 AGGR_STATUS_SET (AGGR interrupt status set Register), (0x2208)

AGGR interrupt status set Register

Bit	Name	Type	Reset	Description
31:6	reserved	r	0x0	Reserved
5:4	autotest	rwi	0x0	interrupt status set for auto test mode
3:2	timeout	rwi	0x0	interrupt status set for svbus timeout errors
1:0	parity	rwi	0x0	interrupt status set for parity errors

4.12.2.17 AGGR_STATUS_CLR (AGGR interrupt status clear Register), (0x220c)

AGGR interrupt status clear Register

Bit	Name	Type	Reset	Description
31:6	reserved	r	0x0	Reserved
5:4	autotest	rwd	0x0	interrupt status clear for auto test mode
3:2	timeout	rwd	0x0	interrupt status clear for svbus timeout errors
1:0	parity	rwd	0x0	interrupt status clear for parity errors

4.12.2.18 AUTO_CONTROL (Auto Test Mode ECC Control), (0x2300)

Auto Test Mode ECC Control Register that will be copied to endpoint ECC Control Registers during auto test mode

Bit	Name	Type	Reset	Description
31:15	reserved	r	0x0	Reserved
14	auto_self_mode	r	0x1	Self Test mode for endpoints
13	auto_inject_idle	r	0x1	Inject RAM error on next idle cycle
12	auto_check_svbus_timeout	r	0x1	check for svbus timeout errors
11:8	auto_ecc_pattern	rw	0x0	data pattern for injection 0=0s 1=Fs 2=As 3=5s
7	auto_check_parity	r	0x1	check for parity errors
6	auto_error_once	r	0x1	Force Error only once
5	auto_force_n_row	r	0x0	Force Error on any RAM read
4	auto_force_ded	rw	0x0	Force Double Bit Error
3	auto_force_sec	rw	0x0	Force Single Bit Error
2	auto_enable_rmw	r	0x1	Enable rmw
1	auto_ecc_check	r	0x1	Enable ECC check
0	auto_ecc_enable	r	0x1	Enable ECC

4.12.2.19 AUTO_ERROR_CTRL2 (Auto Test Mode ECC Error Control2 Register), (0x2304)

Auto Test Mode ECC Error Control2 Register that will be copied to endpoint ECC Error Control2 Registers during auto test mode

Bit	Name	Type	Reset	Description
31:16	auto_ecc_bit2	rw	0x0	Data bit that needs to be flipped if double bit error needs to be forced
15:0	auto_ecc_bit1	rw	0x0	Data bit that needs to be flipped when force_sec is set

4.12.2.20 AUTO_RESTART_CTRL (Auto Test Mode Restart Control Register), (0x2308)

Auto Test Mode Restart Control Register

Bit	Name	Type	Reset	Description
31:16	auto_grp_restart	rw	0x0	Restart group for auto test mode or use 0 to test all groups per endpoint
15:12	reserved	r	0x0	Reserved
11:0	auto_ram_restart	rw	0x0	Restart ram id endpoint for auto test mode or use 0 to test all endpoints

4.12.2.21 AUTO_START_CTRL (Auto Test Mode Start Control Register), (0x230c)

Auto Test Mode Start Control Register

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1	auto_pause	rw	0x0	Pause auto test mode write 1 to pause test or write 0 to resume test
0	auto_start	rw	0x0	Start auto test mode write 1 to start will auto clear when complete

4.12.2.22 AUTO_STATUS1 (Auto Test Mode Status1 Register), (0x2310)

Auto Test Mode Status1 Register

Bit	Name	Type	Reset	Description
31:16	auto_max_groups	r	0x0	Max groups of all endpoints that can be auto tested
15:4	reserved	r	0x0	Reserved
3	auto_inj_timeout	r	0x0	Inject timeout status for last completed auto test mode 0=no timeout or running 1=timeout error
2	auto_inj_fail	r	0x0	Inject failure status for last completed auto test mode 0=no failure or running 1=inject failure error
1	auto_pass	r	0x0	Pass status for last completed auto test mode 0=failed or running 1=passed
0	auto_active	r	0x0	Active status for auto test mode 0=inactive and can be programmed 1=active

4.12.2.23 AUTO_STATUS2 (Auto Test Mode Status2 Register), (0x2314)

Auto Test Mode Status2 Register

Bit	Name	Type	Reset	Description
31:16	auto_fail_group	r	0x0	Endpoint group for last failed auto test mode
15:12	reserved	r	0x0	Reserved
11:0	auto_fail_ramid	r	0x0	Endpoint ram id for last failed auto test mode

4.12.2.24 AUTO_STATUS4 (Auto Test Mode Status4 Register), (0x231c)

Auto Test Mode Status4 Register Pollable

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	auto_inj_timeout	r	0x0	Inject timeout status for last completed auto test mode 0=no timeout or running 1=inject timeout error
2	auto_inj_fail	r	0x0	Inject failure status for last completed auto test mode 0=no failure or running 1=inject failure error
1	auto_pass	r	0x0	Pass status for last completed auto test mode 0=failed or running 1=passed
0	auto_active	r	0x0	Active status for auto test mode 0=inactive and can be programmed 1=active

4.12.2.25 AUTO_ENABLE (Auto Test Mode Enable Registers), (0x2320)

Auto Test Mode Enable Registers

Bit	Name	Type	Reset	Description
31:0	ramid_en	rw	0xffffffff	Auto Test Mode individual RAMID enables 0=disable testing of this ramid index 1=enable testing of this ramid index

4.12.2.26 AUTOTEST_INTR_STATUS_SET_REG (Interrupt Status Set Register), (0x2360)

Interrupt Status Set Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_pend_set	rw1ts	0x0	Interrupt Pending Status Set Register

4.12.2.27 AUTOTEST_INTR_STATUS_CLR_REG (Interrupt Status Clear Register), (0x2364)

Interrupt Status Clear Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_pend_clr	rw1tc	0x0	Interrupt Pending Status Clear

4.12.2.28 AUTOTEST_INTR_ENABLE_SET_REG (Interrupt Enable Set Register), (0x2368)

Interrupt Enable Set Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_enable_set	rw1ts	0x0	Interrupt Enable Set Register

4.12.2.29 AUTOTEST_INTR_ENABLE_CLR_REG (Interrupt Enable Clear Register), (0x236c)

Interrupt Enable Clear Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register

4.12.2.30 AUTOTEST_INTR_EOI_REG (EOI Register), (0x2370)

EOI Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	eoi_wr	rw1ts	0x0	EOI Register

4.12.2.31 AUTO_CAPABLE (Auto Test Mode Capable Status Registers), (0x2380)

Auto Test Mode Capable Status Registers

Bit	Name	Type	Reset	Description
31:0	autotest_capable	r	0xffffffff	Auto Test Mode individual RAMID capable status 0=not capable of auto testing for this ramid index 1=capable of auto testing for this ramid index

4.13 MMR Region: el2_ecc_aggr_clk4_regs

4.13.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x2400	AGGR_REVISION	Aggregator Revision Register
0x2404	NOGATE	Aggregator Nogate Register
0x2408	ECC_VECTOR	ECC Vector Register
0x240c	MISC_STATUS	Misc Status
0x2410+(a*0x4), a=0..7	RESERVED_SVBUS[a]	Reserved Area for Serial VBUS Registers
0x243c	SEC_EOI_REG	EOI Register
0x2440	SEC_STATUS_REG0	Interrupt Status Register 0
0x2480	SEC_ENABLE_SET_REG0	Interrupt Enable Set Register 0
0x24c0	SEC_ENABLE_CLR_REG0	Interrupt Enable Clear Register 0
0x253c	DED_EOI_REG	EOI Register
0x2540	DED_STATUS_REG0	Interrupt Status Register 0
0x2580	DED_ENABLE_SET_REG0	Interrupt Enable Set Register 0
0x25c0	DED_ENABLE_CLR_REG0	Interrupt Enable Clear Register 0
0x2600	AGGR_ENABLE_SET	AGGR interrupt enable set Register
0x2604	AGGR_ENABLE_CLR	AGGR interrupt enable clear Register
0x2608	AGGR_STATUS_SET	AGGR interrupt status set Register
0x260c	AGGR_STATUS_CLR	AGGR interrupt status clear Register
0x2700	AUTO_CONTROL	Auto Test Mode ECC Control
0x2704	AUTO_ERROR_CTRL2	Auto Test Mode ECC Error Control2 Register
0x2708	AUTO_RESTART_CTRL	Auto Test Mode Restart Control Register
0x270c	AUTO_START_CTRL	Auto Test Mode Start Control Register
0x2710	AUTO_STATUS1	Auto Test Mode Status1 Register
0x2714	AUTO_STATUS2	Auto Test Mode Status2 Register
0x271c	AUTO_STATUS4	Auto Test Mode Status4 Register
0x2720	AUTO_ENABLE	Auto Test Mode Enable Registers
0x2760	AUTOTEST_INTR_STATUS_SET_REG	Interrupt Status Set Register
0x2764	AUTOTEST_INTR_STATUS_CLR_REG	Interrupt Status Clear Register

0x2768	AUTOTEST_INTR_ENABLE_SET_REG	Interrupt Enable Set Register
0x276c	AUTOTEST_INTR_ENABLE_CLR_REG	Interrupt Enable Clear Register
0x2770	AUTOTEST_INTR_EOI_REG	EOI Register
0x2780	AUTO_CAPABLE	Auto Test Mode Capable Status Registers

4.13.2 Register Definitions

4.13.2.1 AGGR_REVISION (Aggregator Revision Register), (0x2400)

Revision parameters

Bit	Name	Type	Reset	Description
31:30	scheme	r	0x1	Scheme
29:28	bu	r	0x2	bu
27:16	module_id	r	0x6a0	Module ID
15:11	revrtl	r	0x6	RTL version
10:8	revmaj	r	0x2	Major version
7:6	custom	r	0x0	Custom version
5:0	revmin	r	0x2	Minor version

4.13.2.2 NOGATE (Aggregator Nogate Register), (0x2404)

Nogate control

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	aggr	rw	0x0	Clock gating disable for ecc aggregator when set

4.13.2.3 ECC_VECTOR (ECC Vector Register), (0x2408)

ECC Vector Register

Bit	Name	Type	Reset	Description
31:25	reserved	r	0x0	Reserved
24	rd_svbus_done	rw1tc	0x0	Status to indicate if read on serial VBUS is complete, write of any value will clear this bit.
23:16	rd_svbus_address	rw	0x0	Read address
15	rd_svbus	rw1ts	0x0	Write 1 to trigger a read on the serial VBUS
14:11	reserved	r	0x0	Reserved
10:0	ecc_vector	rw	0x0	Value written to select the corresponding ECC RAM for control or status

4.13.2.4 MISC_STATUS (Misc Status), (0x240c)

Misc Status

Bit	Name	Type	Reset	Description
31:11	reserved	r	0x0	Reserved
10:0	num_rams	r	0x20	Indicates the number of RAMS serviced by the ECC aggregator

4.13.2.5 RESERVED_SVBUS[a] (Reserved Area for Serial VBUS Registers), (0x2410+(a*0x4)), a=0..7

Reference other documents that contain the ECC RAM wrapper and EDC controller serial vbus register sets.

Bit	Name	Type	Reset	Description
31:0	data	rw	0x0	Serial VBUS register data

4.13.2.6 SEC_EOI_REG (EOI Register), (0x243c)

EOI Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	eoi_wr	rw1ts	0x0	EOI Register

4.13.2.7 SEC_STATUS_REG0 (Interrupt Status Register 0), (0x2440)

Interrupt Status Register 0

Bit	Name	Type	Reset	Description
31	eccagg_pend	rw1ts	0x0	Interrupt Pending Status for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_1_edc_pend
15	dru_0_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_0_edc_pend
14	dru_eng_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_eng_edc_pend
13	dru_ring_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_ring_edc_pend
12	dru_rd_buf_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_rd_buf_edc_pend
11	dru_queue_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend

4.13.2.8 SEC_ENABLE_SET_REG0 (Interrupt Enable Set Register 0), (0x2480)



**TEXAS
INSTRUMENTS**

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B i t	Name	T y p e	R e s e t	Description
31	eccagg_enable_set	rw1ts	0x0	Interrupt Enable Set Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_1_edc_pend
15	dru_0_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_0_edc_pend
14	dru_eng_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_eng_edc_pend
13	dru_ring_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_ring_edc_pend
12	dru_rd_buf_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend

7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_busecc_pend

4.13.2.9 SEC_ENABLE_CLR_REG0 (Interrupt Enable Clear Register 0), (0x24c0)

Interrupt Enable Clear Register 0

B i t	Name	T y p e	R e s e t	Description
31	eccagg_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_1_edc_pend
15	dru_0_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_0_edc_pend
14	dru_eng_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_eng_edc_pend
13	dru_rng_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_rng_edc_pend
12	dru_rd_buf_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_pend

7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for el2_ram_edc_ctl_busecc_pend

4.13.2.10 DED_EOI_REG (EOI Register), (0x253c)

EOI Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	eoi_wr	rw1ts	0x0	EOI Register

4.13.2.11 DED_STATUS_REG0 (Interrupt Status Register 0), (0x2540)

Interrupt Status Register 0

B i t	Name	T y p e	R e s e t	Description
31	eccagg_pend	rw11s	0	Interrupt Pending Status for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend	rw11s	0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_pend	rw11s	0	Interrupt Pending Status for dru_1_edc_pend
15	dru_0_edc_pend	rw11s	0	Interrupt Pending Status for dru_0_edc_pend
14	dru_eng_edc_pend	rw11s	0	Interrupt Pending Status for dru_eng_edc_pend
13	dru_ring_edc_pend	rw11s	0	Interrupt Pending Status for dru_ring_edc_pend
12	dru_rd_buf_edc_pend	rw11s	0	Interrupt Pending Status for dru_rd_buf_edc_pend
11	dru_queue_edc_pend	rw11s	0	Interrupt Pending Status for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend	rw11s	0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend

4.13.2.12 DED_ENABLE_SET_REG0 (Interrupt Enable Set Register 0), (0x2580)



**TEXAS
INSTRUMENTS**

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Bit	Name	Type	Reset	Description
31	eccagg_enable_set	rw1ts	0x0	Interrupt Enable Set Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_1_edc_pend
15	dru_0_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_0_edc_pend
14	dru_eng_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_eng_edc_pend
13	dru_rng_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_rng_edc_pend
12	dru_rd_buf_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_pend

7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_busecc_pend

4.13.2.13 DED_ENABLE_CLR_REG0 (Interrupt Enable Clear Register 0), (0x25c0)

Interrupt Enable Clear Register 0

Bit	Name	Type	Reset	Description
31	eccagg_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_1_edc_pend
15	dru_0_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_0_edc_pend
14	dru_eng_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_eng_edc_pend
13	dru_nng_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_nng_edc_pend
12	dru_rd_buf_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_pend

7	tda54_ei2_cfg_cbass_cfg_cbass_tda54_ei2_cfg_cbass_cfg_cbass_ei2_ecc_aggr_cfg_p2p_bridge_ei2_ecc_aggr_cfg_bridge_dst_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for tda54_ei2_cfg_cbass_cfg_cbass_tda54_ei2_cfg_cbass_cfg_cbass_ei2_ecc_aggr_cfg_p2p_bridge_ei2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_ei2_cfg_cbass_cfg_cbass_tda54_ei2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for tda54_ei2_cfg_cbass_cfg_cbass_tda54_ei2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for clec_sram_ramecc_pend
4	ei2_ram_edc_cti_mem3_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for ei2_ram_edc_cti_mem3_busecc_pend
3	ei2_ram_edc_cti_mem2_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for ei2_ram_edc_cti_mem2_busecc_pend
2	ei2_ram_edc_cti_mem1_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for ei2_ram_edc_cti_mem1_busecc_pend
1	ei2_ram_edc_cti_mem0_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for ei2_ram_edc_cti_mem0_busecc_pend
0	ei2_ram_edc_cti_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for ei2_ram_edc_cti_busecc_pend

4.13.2.14 AGGR_ENABLE_SET (AGGR interrupt enable set Register), (0x2600)

AGGR interrupt enable set Register

Bit	Name	Type	Reset	Description
31:3	reserved	r	0x0	Reserved
2	autotest	rw1ts	0x0	interrupt enable set for auto test mode
1	timeout	rw1ts	0x0	interrupt enable set for svbus timeout errors
0	parity	rw1ts	0x0	interrupt enable set for parity errors

4.13.2.15 AGGR_ENABLE_CLR (AGGR interrupt enable clear Register), (0x2604)

AGGR interrupt enable clear Register

Bit	Name	Type	Reset	Description
31:3	reserved	r	0x0	Reserved
2	autotest	rw1tc	0x0	interrupt enable clear for auto test mode
1	timeout	rw1tc	0x0	interrupt enable clear for svbus timeout errors
0	parity	rw1tc	0x0	interrupt enable clear for parity errors

4.13.2.16 AGGR_STATUS_SET (AGGR interrupt status set Register), (0x2608)

AGGR interrupt status set Register

Bit	Name	Type	Reset	Description
31:6	reserved	r	0x0	Reserved
5:4	autotest	rwi	0x0	interrupt status set for auto test mode
3:2	timeout	rwi	0x0	interrupt status set for svbus timeout errors
1:0	parity	rwi	0x0	interrupt status set for parity errors

4.13.2.17 AGGR_STATUS_CLR (AGGR interrupt status clear Register), (0x260c)

AGGR interrupt status clear Register

Bit	Name	Type	Reset	Description
31:6	reserved	r	0x0	Reserved
5:4	autotest	rwd	0x0	interrupt status clear for auto test mode
3:2	timeout	rwd	0x0	interrupt status clear for svbus timeout errors
1:0	parity	rwd	0x0	interrupt status clear for parity errors

4.13.2.18 AUTO_CONTROL (Auto Test Mode ECC Control), (0x2700)

Auto Test Mode ECC Control Register that will be copied to endpoint ECC Control Registers during auto test mode

Bit	Name	Type	Reset	Description
31:15	reserved	r	0x0	Reserved
14	auto_self_mode	r	0x1	Self Test mode for endpoints
13	auto_inject_idle	r	0x1	Inject RAM error on next idle cycle
12	auto_check_svbus_timeout	r	0x1	check for svbus timeout errors
11:8	auto_ecc_pattern	rw	0x0	data pattern for injection 0=0s 1=Fs 2=As 3=5s
7	auto_check_parity	r	0x1	check for parity errors
6	auto_error_once	r	0x1	Force Error only once
5	auto_force_n_row	r	0x0	Force Error on any RAM read
4	auto_force_ded	rw	0x0	Force Double Bit Error
3	auto_force_sec	rw	0x0	Force Single Bit Error
2	auto_enable_rmw	r	0x1	Enable rmw
1	auto_ecc_check	r	0x1	Enable ECC check
0	auto_ecc_enable	r	0x1	Enable ECC

4.13.2.19 AUTO_ERROR_CTRL2 (Auto Test Mode ECC Error Control2 Register), (0x2704)

Auto Test Mode ECC Error Control2 Register that will be copied to endpoint ECC Error Control2 Registers during auto test mode

Bit	Name	Type	Reset	Description
31:16	auto_ecc_bit2	rw	0x0	Data bit that needs to be flipped if double bit error needs to be forced
15:0	auto_ecc_bit1	rw	0x0	Data bit that needs to be flipped when force_sec is set

4.13.2.20 AUTO_RESTART_CTRL (Auto Test Mode Restart Control Register), (0x2708)

Auto Test Mode Restart Control Register

Bit	Name	Type	Reset	Description
31:16	auto_grp_restart	rw	0x0	Restart group for auto test mode or use 0 to test all groups per endpoint
15:12	reserved	r	0x0	Reserved
11:0	auto_ram_restart	rw	0x0	Restart ram id endpoint for auto test mode or use 0 to test all endpoints

4.13.2.21 AUTO_START_CTRL (Auto Test Mode Start Control Register), (0x270c)

Auto Test Mode Start Control Register

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1	auto_pause	rw	0x0	Pause auto test mode write 1 to pause test or write 0 to resume test
0	auto_start	rw	0x0	Start auto test mode write 1 to start will auto clear when complete

4.13.2.22 AUTO_STATUS1 (Auto Test Mode Status1 Register), (0x2710)

Auto Test Mode Status1 Register

Bit	Name	Type	Reset	Description
31:16	auto_max_groups	r	0x0	Max groups of all endpoints that can be auto tested
15:4	reserved	r	0x0	Reserved
3	auto_inj_timeout	r	0x0	Inject timeout status for last completed auto test mode 0=no timeout or running 1=inject timeout error
2	auto_inj_fail	r	0x0	Inject failure status for last completed auto test mode 0=no failure or running 1=inject failure error
1	auto_pass	r	0x0	Pass status for last completed auto test mode 0=failed or running 1=passed
0	auto_active	r	0x0	Active status for auto test mode 0=inactive and can be programmed 1=active

4.13.2.23 AUTO_STATUS2 (Auto Test Mode Status2 Register), (0x2714)

Auto Test Mode Status2 Register

Bit	Name	Type	Reset	Description
31:16	auto_fail_group	r	0x0	Endpoint group for last failed auto test mode
15:12	reserved	r	0x0	Reserved
11:0	auto_fail_ramid	r	0x0	Endpoint ram id for last failed auto test mode

4.13.2.24 AUTO_STATUS4 (Auto Test Mode Status4 Register), (0x271c)

Auto Test Mode Status4 Register Pollable

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	auto_inj_timeout	r	0x0	Inject timeout status for last completed auto test mode 0=no timeout or running 1=inject timeout error
2	auto_inj_fail	r	0x0	Inject failure status for last completed auto test mode 0=no failure or running 1=inject failure error
1	auto_pass	r	0x0	Pass status for last completed auto test mode 0=failed or running 1=passed
0	auto_active	r	0x0	Active status for auto test mode 0=inactive and can be programmed 1=active

4.13.2.25 AUTO_ENABLE (Auto Test Mode Enable Registers), (0x2720)

Auto Test Mode Enable Registers

Bit	Name	Type	Reset	Description
31:0	ramid_en	rw	0xffffffff	Auto Test Mode individual RAMID enables 0=disable testing of this ramid index 1=enable testing of this ramid index

4.13.2.26 AUTOTEST_INTR_STATUS_SET_REG (Interrupt Status Set Register), (0x2760)

Interrupt Status Set Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_pend_set	rw1ts	0x0	Interrupt Pending Status Set Register

4.13.2.27 AUTOTEST_INTR_STATUS_CLR_REG (Interrupt Status Clear Register), (0x2764)

Interrupt Status Clear Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_pend_clr	rw1tc	0x0	Interrupt Pending Status Clear

4.13.2.28 AUTOTEST_INTR_ENABLE_SET_REG (Interrupt Enable Set Register), (0x2768)

Interrupt Enable Set Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_enable_set	rw1ts	0x0	Interrupt Enable Set Register

4.13.2.29 AUTOTEST_INTR_ENABLE_CLR_REG (Interrupt Enable Clear Register), (0x276c)

Interrupt Enable Clear Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register

4.13.2.30 AUTOTEST_INTR_EOI_REG (EOI Register), (0x2770)

EOI Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	eoi_wr	rw1ts	0x0	EOI Register

4.13.2.31 AUTO_CAPABLE (Auto Test Mode Capable Status Registers), (0x2780)

Auto Test Mode Capable Status Registers

Bit	Name	Type	Reset	Description
31:0	autotest_capable	r	0xffffffff	Auto Test Mode individual RAMID capable status 0=not capable of auto testing for this ramid index 1=capable of auto testing for this ramid index

4.14 MMR Region: el2_ecc_aggr_msmc_regs

4.14.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x2800	AGGR_REVISION	Aggregator Revision Register
0x2804	NOGATE	Aggregator Nogate Register
0x2808	ECC_VECTOR	ECC Vector Register
0x280c	MISC_STATUS	Misc Status
0x2810+(a*0x4), a=0..7	RESERVED_SVBUS[a]	Reserved Area for Serial VBUS Registers
0x283c	SEC_EOI_REG	EOI Register
0x2840	SEC_STATUS_REG0	Interrupt Status Register 0
0x2880	SEC_ENABLE_SET_REG0	Interrupt Enable Set Register 0
0x28c0	SEC_ENABLE_CLR_REG0	Interrupt Enable Clear Register 0
0x293c	DED_EOI_REG	EOI Register
0x2940	DED_STATUS_REG0	Interrupt Status Register 0
0x2980	DED_ENABLE_SET_REG0	Interrupt Enable Set Register 0
0x29c0	DED_ENABLE_CLR_REG0	Interrupt Enable Clear Register 0
0x2a00	AGGR_ENABLE_SET	AGGR interrupt enable set Register
0x2a04	AGGR_ENABLE_CLR	AGGR interrupt enable clear Register
0x2a08	AGGR_STATUS_SET	AGGR interrupt status set Register
0x2a0c	AGGR_STATUS_CLR	AGGR interrupt status clear Register
0x2b00	AUTO_CONTROL	Auto Test Mode ECC Control
0x2b04	AUTO_ERROR_CTRL2	Auto Test Mode ECC Error Control2 Register
0x2b08	AUTO_RESTART_CTRL	Auto Test Mode Restart Control Register
0x2b0c	AUTO_START_CTRL	Auto Test Mode Start Control Register
0x2b10	AUTO_STATUS1	Auto Test Mode Status1 Register
0x2b14	AUTO_STATUS2	Auto Test Mode Status2 Register
0x2b1c	AUTO_STATUS4	Auto Test Mode Status4 Register
0x2b20	AUTO_ENABLE	Auto Test Mode Enable Registers
0x2b60	AUTOTEST_INTR_STATUS_SET_REG	Interrupt Status Set Register
0x2b64	AUTOTEST_INTR_STATUS_CLR_REG	Interrupt Status Clear Register

0x2b68	AUTOTEST_INTR_ENABLE_SET_REG	Interrupt Enable Set Register
0x2b6c	AUTOTEST_INTR_ENABLE_CLR_REG	Interrupt Enable Clear Register
0x2b70	AUTOTEST_INTR_EOI_REG	EOI Register
0x2b80	AUTO_CAPABLE	Auto Test Mode Capable Status Registers

4.14.2 Register Definitions

4.14.2.1 AGGR_REVISION (Aggregator Revision Register), (0x2800)

Revision parameters

Bit	Name	Type	Reset	Description
31:30	scheme	r	0x1	Scheme
29:28	bu	r	0x2	bu
27:16	module_id	r	0x6a0	Module ID
15:11	revrtl	r	0x6	RTL version
10:8	revmaj	r	0x2	Major version
7:6	custom	r	0x0	Custom version
5:0	revmin	r	0x2	Minor version

4.14.2.2 NOGATE (Aggregator Nogate Register), (0x2804)

Nogate control

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	aggr	rw	0x0	Clock gating disable for ecc aggregator when set

4.14.2.3 ECC_VECTOR (ECC Vector Register), (0x2808)

ECC Vector Register

Bit	Name	Type	Reset	Description
31:25	reserved	r	0x0	Reserved
24	rd_svbus_done	rw1tc	0x0	Status to indicate if read on serial VBUS is complete, write of any value will clear this bit.
23:16	rd_svbus_address	rw	0x0	Read address
15	rd_svbus	rw1ts	0x0	Write 1 to trigger a read on the serial VBUS
14:11	reserved	r	0x0	Reserved
10:0	ecc_vector	rw	0x0	Value written to select the corresponding ECC RAM for control or status

4.14.2.4 MISC_STATUS (Misc Status), (0x280c)

Misc Status

Bit	Name	Type	Reset	Description
31:11	reserved	r	0x0	Reserved
10:0	num_rams	r	0x20	Indicates the number of RAMS serviced by the ECC aggregator

4.14.2.5 RESERVED_SVBUS[a] (Reserved Area for Serial VBUS Registers), (0x2810+(a*0x4)), a=0..7

Reference other documents that contain the ECC RAM wrapper and EDC controller serial vbus register sets.

Bit	Name	Type	Reset	Description
31:0	data	rw	0x0	Serial VBUS register data

4.14.2.6 SEC_EOI_REG (EOI Register), (0x283c)

EOI Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	eoi_wr	rw1ts	0x0	EOI Register

4.14.2.7 SEC_STATUS_REG0 (Interrupt Status Register 0), (0x2840)

Interrupt Status Register 0

Bit	Name	Type	Reset	Description
31	eccagg_pend	rw1ts	0x0	Interrupt Pending Status for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbase_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbase_scri_scri_edc_ctrl_busecc_1_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbase_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbase_scri_scri_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbase_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbase_scri_scri_edc_ctrl_busecc_0_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbase_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbase_scri_scri_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_1_edc_pend
15	dru_0_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_0_edc_pend
14	dru_eng_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_eng_edc_pend
13	dru_ring_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_ring_edc_pend
12	dru_rd_buf_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_rd_buf_edc_pend
11	dru_queue_edc_pend	rw1ts	0x0	Interrupt Pending Status for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbase_cfg_cbase_tda54_el2_cfg_cbase_cfg_cbase_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbase_cfg_cbase_tda54_el2_cfg_cbase_cfg_cbase_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbase_cfg_cbase_tda54_el2_cfg_cbase_cfg_cbase_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend	rw1ts	0x0	Interrupt Pending Status for tda54_el2_cfg_cbase_cfg_cbase_tda54_el2_cfg_cbase_cfg_cbase_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend

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B i t	Name	T y p e	R e s e t	Description
31	eccagg_enable_set	rw1ts	0x0	Interrupt Enable Set Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_1_edc_pend
15	dru_0_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_0_edc_pend
14	dru_eng_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_eng_edc_pend
13	dru_ring_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_ring_edc_pend
12	dru_rd_buf_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_pend

7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_busecc_pend

4.14.2.9 SEC_ENABLE_CLR_REG0 (Interrupt Enable Clear Register 0), (0x28c0)

Interrupt Enable Clear Register 0

Bit	Name	Type	Reset	Description
31	eccagg_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_1_edc_pend
15	dru_0_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_0_edc_pend
14	dru_eng_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_eng_edc_pend
13	dru_nng_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_nng_edc_pend
12	dru_rd_buf_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_pend

7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for el2_ram_edc_ctl_busecc_pend

4.14.2.10 DED_EOI_REG (EOI Register), (0x293c)

EOI Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	eoi_wr	rw1ts	0x0	EOI Register

4.14.2.11 DED_STATUS_REG0 (Interrupt Status Register 0), (0x2940)

Interrupt Status Register 0

Bit	Name	Type	Reset	Description
31	eccagg_pend	rw11s	0x0	Interrupt Pending Status for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_pend	rw11s	0x0	Interrupt Pending Status for dru_1_edc_pend
15	dru_0_edc_pend	rw11s	0x0	Interrupt Pending Status for dru_0_edc_pend
14	dru_eng_edc_pend	rw11s	0x0	Interrupt Pending Status for dru_eng_edc_pend
13	dru_ring_edc_pend	rw11s	0x0	Interrupt Pending Status for dru_ring_edc_pend
12	dru_rd_buf_edc_pend	rw11s	0x0	Interrupt Pending Status for dru_rd_buf_edc_pend
11	dru_queue_edc_pend	rw11s	0x0	Interrupt Pending Status for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend	rw11s	0x0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_split_p2p_bridge_dru_mmr_cfg_split_bridge_dst_busecc_pend

7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend	r w 1 1 t s	0 x 0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend	r w 1 1 t s	0 x 0	Interrupt Pending Status for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_pend	r w 1 1 t s	0 x 0	Interrupt Pending Status for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_busecc_pend	r w 1 1 t s	0 x 0	Interrupt Pending Status for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_pend	r w 1 1 t s	0 x 0	Interrupt Pending Status for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_pend	r w 1 1 t s	0 x 0	Interrupt Pending Status for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_pend	r w 1 1 t s	0 x 0	Interrupt Pending Status for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_pend	r w 1 1 t s	0 x 0	Interrupt Pending Status for el2_ram_edc_ctl_busecc_pend

4.14.2.12 DED_ENABLE_SET_REG0 (Interrupt Enable Set Register 0), (0x2980)

Interrupt Enable Set Register 0

B i t	Name	T y p e	R e s e t	Description
31	eccagg_enable_set	rw1ts	0x0	Interrupt Enable Set Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_1_edc_pend
15	dru_0_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_0_edc_pend
14	dru_eng_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_eng_edc_pend
13	dru_rng_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_rng_edc_pend
12	dru_rd_buf_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_enable_set	rw1ts	0x0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_pend

7	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_p2p_bridge_el2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for clec_sram_ramecc_pend
4	el2_ram_edc_ctl_mem3_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem3_busecc_pend
3	el2_ram_edc_ctl_mem2_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem2_busecc_pend
2	el2_ram_edc_ctl_mem1_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem1_busecc_pend
1	el2_ram_edc_ctl_mem0_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_mem0_busecc_pend
0	el2_ram_edc_ctl_busecc_enable_set	r w 1 1 t s	0 x 0	Interrupt Enable Set Register for el2_ram_edc_ctl_busecc_pend

4.14.2.13 DED_ENABLE_CLR_REG0 (Interrupt Enable Clear Register 0), (0x29c0)

Interrupt Enable Clear Register 0

Bit	Name	Type	Reset	Description
31	eccagg_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for eccagg_pend
30	tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw10_tda54_el2_fw_aggr_wrap_fw10_mmrs_edc_ctrl_busecc_pend
29	tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw9_tda54_el2_fw_aggr_wrap_fw9_mmrs_edc_ctrl_busecc_pend
28	tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw8_tda54_el2_fw_aggr_wrap_fw8_mmrs_tda54_el2_fw_aggr_wrap_fw8_mmrs_edc_ctrl_busecc_pend
27	tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw7_tda54_el2_fw_aggr_wrap_fw7_mmrs_tda54_el2_fw_aggr_wrap_fw7_mmrs_edc_ctrl_busecc_pend
26	tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw6_tda54_el2_fw_aggr_wrap_fw6_mmrs_tda54_el2_fw_aggr_wrap_fw6_mmrs_edc_ctrl_busecc_pend
25	tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw5_tda54_el2_fw_aggr_wrap_fw5_mmrs_tda54_el2_fw_aggr_wrap_fw5_mmrs_edc_ctrl_busecc_pend
24	tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw4_tda54_el2_fw_aggr_wrap_fw4_mmrs_tda54_el2_fw_aggr_wrap_fw4_mmrs_edc_ctrl_busecc_pend
23	tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw3_tda54_el2_fw_aggr_wrap_fw3_mmrs_tda54_el2_fw_aggr_wrap_fw3_mmrs_edc_ctrl_busecc_pend
22	tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw2_tda54_el2_fw_aggr_wrap_fw2_mmrs_tda54_el2_fw_aggr_wrap_fw2_mmrs_edc_ctrl_busecc_pend
21	tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw1_tda54_el2_fw_aggr_wrap_fw1_mmrs_tda54_el2_fw_aggr_wrap_fw1_mmrs_edc_ctrl_busecc_pend
20	tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw0_tda54_el2_fw_aggr_wrap_fw0_mmrs_tda54_el2_fw_aggr_wrap_fw0_mmrs_edc_ctrl_busecc_pend
19	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_1_pend
18	tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_fw_aggr_wrap_fw_aggr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_fw_aggr_tda54_ac_wrap_cbass_scr1_scr_edc_ctrl_busecc_0_pend
17	tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_async_bridge_wrap_ac_vbusp_dmssc_p2p_bridge_ac_vbusp_dmssc_p2p_bridge_dst_busecc_pend
16	dru_1_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_1_edc_pend
15	dru_0_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_0_edc_pend
14	dru_eng_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_eng_edc_pend
13	dru_rng_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_rng_edc_pend
12	dru_rd_buf_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_rd_buf_edc_pend
11	dru_queue_edc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for dru_queue_edc_pend
10	tda54_el2_wrap_el2_rat_edc_ctrl_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_wrap_el2_rat_edc_ctrl_busecc_pend
9	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_el2_rat_cfg_bridge_dst_busecc_pend
8	tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register for tda54_el2_cfg_cbass_cfg_cbass_tda54_el2_cfg_cbass_cfg_cbass_dru_mmrc_cfg_split_p2p_bridge_dru_mmrc_cfg_split_bridge_dst_busecc_pend

7	tda54_ei2_cfg_cbass_cfg_cbass_tda54_ei2_cfg_cbass_cfg_cbass_ei2_ecc_aggr_cfg_p2p_bridge_ei2_ecc_aggr_cfg_bridge_dst_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for tda54_ei2_cfg_cbass_cfg_cbass_tda54_ei2_cfg_cbass_cfg_cbass_ei2_ecc_aggr_cfg_p2p_bridge_ei2_ecc_aggr_cfg_bridge_dst_busecc_pend
6	tda54_ei2_cfg_cbass_cfg_cbass_tda54_ei2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for tda54_ei2_cfg_cbass_cfg_cbass_tda54_ei2_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dru_mmr_cfg_bridge_src_busecc_pend
5	clec_sram_ramecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for clec_sram_ramecc_pend
4	ei2_ram_edc_cti_mem3_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for ei2_ram_edc_cti_mem3_busecc_pend
3	ei2_ram_edc_cti_mem2_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for ei2_ram_edc_cti_mem2_busecc_pend
2	ei2_ram_edc_cti_mem1_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for ei2_ram_edc_cti_mem1_busecc_pend
1	ei2_ram_edc_cti_mem0_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for ei2_ram_edc_cti_mem0_busecc_pend
0	ei2_ram_edc_cti_busecc_enable_clr	r w 1 t c	0 x 0	Interrupt Enable Clear Register for ei2_ram_edc_cti_busecc_pend

4.14.2.14 AGGR_ENABLE_SET (AGGR interrupt enable set Register), (0x2a00)

AGGR interrupt enable set Register

Bit	Name	Type	Reset	Description
31:3	reserved	r	0x0	Reserved
2	autotest	rw1ts	0x0	interrupt enable set for auto test mode
1	timeout	rw1ts	0x0	interrupt enable set for svbus timeout errors
0	parity	rw1ts	0x0	interrupt enable set for parity errors

4.14.2.15 AGGR_ENABLE_CLR (AGGR interrupt enable clear Register), (0x2a04)

AGGR interrupt enable clear Register

Bit	Name	Type	Reset	Description
31:3	reserved	r	0x0	Reserved
2	autotest	rw1tc	0x0	interrupt enable clear for auto test mode
1	timeout	rw1tc	0x0	interrupt enable clear for svbus timeout errors
0	parity	rw1tc	0x0	interrupt enable clear for parity errors

4.14.2.16 AGGR_STATUS_SET (AGGR interrupt status set Register), (0x2a08)

AGGR interrupt status set Register

Bit	Name	Type	Reset	Description
31:6	reserved	r	0x0	Reserved
5:4	autotest	rwi	0x0	interrupt status set for auto test mode
3:2	timeout	rwi	0x0	interrupt status set for svbus timeout errors
1:0	parity	rwi	0x0	interrupt status set for parity errors

4.14.2.17 AGGR_STATUS_CLR (AGGR interrupt status clear Register), (0x2a0c)

AGGR interrupt status clear Register

Bit	Name	Type	Reset	Description
31:6	reserved	r	0x0	Reserved
5:4	autotest	rwd	0x0	interrupt status clear for auto test mode
3:2	timeout	rwd	0x0	interrupt status clear for svbus timeout errors
1:0	parity	rwd	0x0	interrupt status clear for parity errors

4.14.2.18 AUTO_CONTROL (Auto Test Mode ECC Control), (0x2b00)

Auto Test Mode ECC Control Register that will be copied to endpoint ECC Control Registers during auto test mode

Bit	Name	Type	Reset	Description
31:15	reserved	r	0x0	Reserved
14	auto_self_mode	r	0x1	Self Test mode for endpoints
13	auto_inject_idle	r	0x1	Inject RAM error on next idle cycle
12	auto_check_svbus_timeout	r	0x1	check for svbus timeout errors
11:8	auto_ecc_pattern	rw	0x0	data pattern for injection 0=0s 1=Fs 2=As 3=5s
7	auto_check_parity	r	0x1	check for parity errors
6	auto_error_once	r	0x1	Force Error only once
5	auto_force_n_row	r	0x0	Force Error on any RAM read
4	auto_force_ded	rw	0x0	Force Double Bit Error
3	auto_force_sec	rw	0x0	Force Single Bit Error
2	auto_enable_rmw	r	0x1	Enable rmw
1	auto_ecc_check	r	0x1	Enable ECC check
0	auto_ecc_enable	r	0x1	Enable ECC

4.14.2.19 AUTO_ERROR_CTRL2 (Auto Test Mode ECC Error Control2 Register), (0x2b04)

Auto Test Mode ECC Error Control2 Register that will be copied to endpoint ECC Error Control2 Registers during auto test mode

Bit	Name	Type	Reset	Description
31:16	auto_ecc_bit2	rw	0x0	Data bit that needs to be flipped if double bit error needs to be forced
15:0	auto_ecc_bit1	rw	0x0	Data bit that needs to be flipped when force_sec is set

4.14.2.20 AUTO_RESTART_CTRL (Auto Test Mode Restart Control Register), (0x2b08)

Auto Test Mode Restart Control Register

Bit	Name	Type	Reset	Description
31:16	auto_grp_restart	rw	0x0	Restart group for auto test mode or use 0 to test all groups per endpoint
15:12	reserved	r	0x0	Reserved
11:0	auto_ram_restart	rw	0x0	Restart ram id endpoint for auto test mode or use 0 to test all endpoints

4.14.2.21 AUTO_START_CTRL (Auto Test Mode Start Control Register), (0x2b0c)

Auto Test Mode Start Control Register

Bit	Name	Type	Reset	Description
31:2	reserved	r	0x0	Reserved
1	auto_pause	rw	0x0	Pause auto test mode write 1 to pause test or write 0 to resume test
0	auto_start	rw	0x0	Start auto test mode write 1 to start will auto clear when complete

4.14.2.22 AUTO_STATUS1 (Auto Test Mode Status1 Register), (0x2b10)

Auto Test Mode Status1 Register

Bit	Name	Type	Reset	Description
31:16	auto_max_groups	r	0x0	Max groups of all endpoints that can be auto tested
15:4	reserved	r	0x0	Reserved
3	auto_inj_timeout	r	0x0	Inject timeout status for last completed auto test mode 0=no timeout or running 1=inject timeout error
2	auto_inj_fail	r	0x0	Inject failure status for last completed auto test mode 0=no failure or running 1=inject failure error
1	auto_pass	r	0x0	Pass status for last completed auto test mode 0=failed or running 1=passed
0	auto_active	r	0x0	Active status for auto test mode 0=inactive and can be programmed 1=active

4.14.2.23 AUTO_STATUS2 (Auto Test Mode Status2 Register), (0x2b14)

Auto Test Mode Status2 Register

Bit	Name	Type	Reset	Description
31:16	auto_fail_group	r	0x0	Endpoint group for last failed auto test mode
15:12	reserved	r	0x0	Reserved
11:0	auto_fail_ramid	r	0x0	Endpoint ram id for last failed auto test mode

4.14.2.24 AUTO_STATUS4 (Auto Test Mode Status4 Register), (0x2b1c)

Auto Test Mode Status4 Register Pollable

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	auto_inj_timeout	r	0x0	Inject timeout status for last completed auto test mode 0=no timeout or running 1=inject timeout error
2	auto_inj_fail	r	0x0	Inject failure status for last completed auto test mode 0=no failure or running 1=inject failure error
1	auto_pass	r	0x0	Pass status for last completed auto test mode 0=failed or running 1=passed
0	auto_active	r	0x0	Active status for auto test mode 0=inactive and can be programmed 1=active

4.14.2.25 AUTO_ENABLE (Auto Test Mode Enable Registers), (0x2b20)

Auto Test Mode Enable Registers

Bit	Name	Type	Reset	Description
31:0	ramid_en	rw	0xffffffff	Auto Test Mode individual RAMID enables 0=disable testing of this ramid index 1=enable testing of this ramid index

4.14.2.26 AUTOTEST_INTR_STATUS_SET_REG (Interrupt Status Set Register), (0x2b60)

Interrupt Status Set Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_pend_set	rw1ts	0x0	Interrupt Pending Status Set Register

4.14.2.27 AUTOTEST_INTR_STATUS_CLR_REG (Interrupt Status Clear Register), (0x2b64)

Interrupt Status Clear Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_pend_clr	rw1tc	0x0	Interrupt Pending Status Clear

4.14.2.28 AUTOTEST_INTR_ENABLE_SET_REG (Interrupt Enable Set Register), (0x2b68)

Interrupt Enable Set Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_enable_set	rw1ts	0x0	Interrupt Enable Set Register

4.14.2.29 AUTOTEST_INTR_ENABLE_CLR_REG (Interrupt Enable Clear Register), (0x2b6c)

Interrupt Enable Clear Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	autotest_intr_enable_clr	rw1tc	0x0	Interrupt Enable Clear Register

4.14.2.30 AUTOTEST_INTR_EOI_REG (EOI Register), (0x2b70)

EOI Register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	eoι_wr	rw1ts	0x0	EOI Register

4.14.2.31 AUTO_CAPABLE (Auto Test Mode Capable Status Registers), (0x2b80)

Auto Test Mode Capable Status Registers

Bit	Name	Type	Reset	Description
31:0	autotest_capable	r	0xffffffff	Auto Test Mode individual RAMID capable status 0=not capable of auto testing for this ramid index 1=capable of auto testing for this ramid index

4.15 MMR Region: ac_c7xv_cfg_mmr

4.15.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x140000	CAPM_CTRL	CAPM control register
0x140004	CAPM_STATUS	CAPM status register
0x140008	AC_PM_CTRL	AC power management control

4.15.2 Register Definitions

4.15.2.1 CAPM_CTRL (CAPM control register), (0x140000)

cluster level analytics power manager configuration register, provides control to step up, step down MMA power to maintain required di/dt levels

Bit	Name	Type	Reset	Description
31:13	RESERVED	r	0x0	Reserved
12:9	disable_mma_N	rw	0x0	Capability to disable APM per MMA. Hidden controls. only valid when enable bit bit0 is set. 1'b0: CAPM is enabled for MMA_N 1'b1: CAPM is disabled for MMA_N
8	perf_opt_enable	rw	0x1	1'b0 : Perf optimization disabled, Always takes num_steps regardless of other activity. 1'b1: Perf optimization is on, Default.
7	num_step	rw	0x0	1'b0 : 4 steps, Default 1'b1 : 2 steps. Fast-switch
6	step_width_unit	rw	0x0	1'b0 : 16 clock cycles 10.677ns at 1.5GHz 1'b1: 128 clock cycles 85.33ns at 1.5GHz clock
5:2	step_width	rw	0x0	width of step in # of step_width_units
1	slice_mma_enable	rw	0x0	1'b0 : Slice MMA feature is disabled. Template bits are ignored. 1'b1: Slice MMA feature is enabled
0	enable	rw	0x0	1'b0 : CAPM is disabled 1'b1: CAPM is enabled

4.15.2.2 CAPM_STATUS (CAPM status register), (0x140004)

cluster level analytics power manager status register

Bit	Name	Type	Reset	Description
31:0	capm_status	r	0x0	capm status bits, yet to be defined

4.15.2.3 AC_PM_CTRL (AC power management control), (0x140008)

Analytics Cluster power management control register

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	no_gate	rw	0x0	no gate control for all analytics cluster CBASS and bridges

4.16 MMR Region: el2_cscti4_cfg

4.16.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x3000000	CTICONTROL	
0x3000010	CTIINTACK	
0x3000014	CTIAPPSET	
0x3000018	CTIAPPCLEAR	
0x300001c	CTIAPPPULSE	
0x3000020	CTIINEN0	
0x3000024	CTIINEN1	
0x3000028	CTIINEN2	
0x300002c	CTIINEN3	
0x3000030	CTIINEN4	
0x3000034	CTIINEN5	
0x3000038	CTIINEN6	
0x300003c	CTIINEN7	
0x30000a0	CTIOUTEN0	
0x30000a4	CTIOUTEN1	
0x30000a8	CTIOUTEN2	
0x30000ac	CTIOUTEN3	
0x30000b0	CTIOUTEN4	
0x30000b4	CTIOUTEN5	
0x30000b8	CTIOUTEN6	
0x30000bc	CTIOUTEN7	
0x3000130	CTITRIGINSTATUS	
0x3000134	CTITRIGOUTSTATUS	
0x3000138	CTICHINSTATUS	
0x300013c	CTICHOUTSTATUS	
0x3000140	CTIGATE	
0x3000144	ASICCTL	
0x3000edc	ITCHINACK	
0x3000ee0	ITTRIGINACK	
0x3000ee4	ITCHOUT	
0x3000ee8	ITTRIGOUT	
0x3000eec	ITCHOUTACK	
0x3000ef0	ITTRIGOUTACK	
0x3000ef4	ITCHIN	
0x3000ef8	ITTRIGIN	
0x3000f00	ITCTRL	
0x3000fa0	CLAIMSET	
0x3000fa4	CLAIMCLR	
0x3000fb0	LAR	
0x3000fb4	LSR	
0x3000fb8	AUTHSTATUS	
0x3000fc8	DEVID	
0x3000fcc	DEVTYPE	
0x3000fd0	PERIPHID4	
0x3000fe0	PERIPHID0	
0x3000fe4	PERIPHID1	

0x3000fe8	PERIPHID2	
0x3000fec	PERIPHID3	
0x3000ff0	COMPID0	
0x3000ff4	COMPID1	
0x3000ff8	COMPID2	
0x3000ffc	COMPID3	

4.16.2 Register Definitions

4.16.2.1 CTICONTROL (0x3000000)

The CTI Control Register enables the CTI. >

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	GLBEN	rw	0x0	Enables or disables the ECT.

4.16.2.2 CTIINTACK (0x3000010)

The CTI Interrupt Acknowledge Register is write-only. Any bits written as a 1 cause the ctitrigout output signal to be acknowledged. The acknowledgement is cleared when MAPTRIGOUT is deactivated. This register is used when the ctitrigout is used as a sticky output, that is, no hardware acknowledge is supplied, and a software acknowledge is required.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	INTACK	rw	0x0	Acknowledges the corresponding ctitrigout output. There is one bit of the register for each ctitrigout output. When a 1 is written to a bit in this register, the corresponding ctitrigout is acknowledged and is cleared when MAPTRIGOUT is LOW. Writing a 0 to any of the bits in this register has no effect.

4.16.2.3 CTIAPPSET (0x3000014)

The CTI Application Trigger Set Register is read/write. A write to this register causes a channel event to be raised, corresponding to the bit written to.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPSET	rw	0x0	Setting a bit HIGH generates a channel event for the selected channel. There is one bit of the register for each channel. Read : 0 = application trigger inactive (reset). 1 = application trigger active. Write : 0 = no effect. 1 = generate channel event.

4.16.2.4 CTIAPPCLEAR (0x3000018)

The CTI Interrupt Acknowledge Register is write-only. A write to this register causes a channel event to be cleared, corresponding to the bit written to.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPCLEAR	rw	0x0	Clears corresponding bits in the CTIAPPSET register. There is one bit of the register for each channel. When a 1 is written to a bit in this register, the corresponding application trigger is disabled in the CTIAPPSET register. Writing a 0 to any of the bits in this register has no effect.

4.16.2.5 CTIAPPULSE (0x300001c)

The CTI Application Pulse Register is write-only. A write to this register causes a channel event pulse, one clock period, to be generated, corresponding to the bit written to. The pulse external to the ECT can be extended to multi-cycle by the handshaking interface circuits. This register clears itself immediately, so it can be repeatedly written to without software having to clear it.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPULSE	rw	0x0	Setting a bit HIGH generates a channel event pulse for the selected channel. There is one bit of the register for each channel. When a 1 is written to a bit in this register, a corresponding channel event pulse is generated for one clock period. Writing a 0 to any of the bits in this register has no effect.

4.16.2.6 CTIINEN0 (0x3000020)

The CTI Trigger 0 to Channel Enable Register enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.16.2.7 CTIINEN1 (0x3000024)

The CTI Trigger 1 to Channel Enable Register enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.16.2.8 CTIINEN2 (0x3000028)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.16.2.9 CTIINEN3 (0x300002c)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.16.2.10 CTIINEN4 (0x3000030)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.16.2.11 CTIINEN5 (0x3000034)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.16.2.12 CTIINEN6 (0x3000038)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in



				this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.
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4.16.2.13 CTIINEN7 (0x300003c)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.16.2.14 CTIOUTEN0 (0x30000a0)

The CTI Channel to Trigger 0 Enable Registers define which channels can generate a ctitrigin[0] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigin[1] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigin output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigin[0] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigin output. Reading this register returns the programmed value.

4.16.2.15 CTIOUTEN1 (0x30000a4)

The CTI Channel to Trigger 1 Enable Registers define which channels can generate a ctitrigin[1] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[1] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[1] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.16.2.16 CTIOUTEN2 (0x30000a8)

The CTI Channel to Trigger 2 Enable Registers define which channels can generate a cttrigout[2] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[2] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[2] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.16.2.17 CTIOUTEN3 (0x30000ac)

The CTI Channel to Trigger 3 Enable Registers define which channels can generate a cttrigout[3] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[3] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0]

				to cause a trigger event on the ctitrigger[3] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigger output. Reading this register returns the programmed value.
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4.16.2.18 CTIOUTEN4 (0x30000b0)

The CTI Channel to Trigger 4 Enable Registers define which channels can generate a ctitrigger[4] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigger[4] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigger output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigger[4] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigger output. Reading this register returns the programmed value.

4.16.2.19 CTIOUTEN5 (0x30000b4)

The CTI Channel to Trigger 5 Enable Registers define which channels can generate a ctitrigger[5] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigger[5] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigger output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigger[5] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigger output. Reading this register returns the programmed value.

4.16.2.20 CTIOUTEN6 (0x30000b8)

The CTI Channel to Trigger 6 Enable Registers define which channels can generate a ctitrigger[6] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[6] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[6] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.16.2.21 CTIOUTEN7 (0x30000bc)

The CTI Channel to Trigger 7 Enable Registers define which channels can generate a cttrigout[7] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[7] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[7] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.16.2.22 CTTRIGINSTATUS (0x3000130)

The CTI Trigger In Status Register provides the status of the cttrigin inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	TRIGINSTATUS	r	0x0	Shows the status of the cttrigin inputs. 1 = cttrigin is active. 0 = cttrigin is inactive. Because the register provides a view of the raw cttrigin inputs, the reset value is unknown. There is one bit of the field for each trigger input.

4.16.2.23 CTTRIGOUTSTATUS (0x3000134)

The CTI Trigger Out Status Register provides the status of the cttrigout outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	TRIGOUTSTATUS	r	0x0	Shows the status of the cttrigout outputs. 1 = cttrigout is active. 0 = cttrigout is inactive. There is one bit of the field for each trigger output.

4.16.2.24 CTICHINSTATUS (0x3000138)

The CTI Channel In Status Register provides the status of the ctichin inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTICHINSTATUS	r	0x0	Shows the status of the ctichin inputs. 1 = ctichin is active. 0 = ctichin is inactive. Because the register provides a view of the raw ctichin inputs, the reset value is unknown. There is one bit of the field for each channel input.

4.16.2.25 CTICHOUTSTATUS (0x300013c)

The CTI Channel Out Status Register provides the status of the CTI ctichout outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTICHOUTSTATUS	r	0x0	Shows the status of the ctichout outputs. 1 = ctichout is active. 0 = ctichout is inactive. There is one bit of the field for each channel output.

4.16.2.26 CTIGATE (0x3000140)

The Gate Enable Register prevents the channels from propagating through the CTM to other CTIs. This enables local cross-triggering, for example for causing an interrupt when the ETM trigger occurs. It can be used effectively with CTIAPPSET, CTIAPPCLEAR, and CTIAPPPULSE for asserting trigger outputs by asserting channels, without affecting the rest of the system. On reset, this register is 0xF, and channel propagation is enabled.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	CTIGATEEN3	rw	0x1	Enable CTICHOUT3. Set to 0 to disable channel propagation.
2	CTIGATEEN2	rw	0x1	Enable CTICHOUT2. Set to 0 to disable channel propagation.
1	CTIGATEEN1	rw	0x1	Enable CTICHOUT1. Set to 0 to disable channel propagation.
0	CTIGATEEN0	rw	0x1	Enable CTICHOUT0. Set to 0 to disable channel propagation.

4.16.2.27 ASICCTL (0x3000144)

Implementation-defined ASIC control, value written to the register is output on asicctl[7 : 0].

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	ASICCTL	rw	0x0	Implementation-defined ASIC control, value written to the register is output on asicctl[7 : 0]. If external multiplexing of trigger signals is implemented then the number of multiplexed signals on each trigger must be reflected within the Device ID Register. This is done within a Verilog define EXTMUXNUM.

4.16.2.28 ITCHINACK (0x3000edc)

This register is a write-only register. It can be used to set the value of the CTCHINACK outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	ITCHINACK	rw	0x0	Set the value of the CTCHINACK outputs.

4.16.2.29 ITTRIGINACK (0x3000ee0)

This register is a write-only register. It can be used to set the value of the CTTRIGINACK outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGINACK	rw	0x0	Set the value of the CTTRIGINACK outputs.

4.16.2.30 ITCHOUT (0x3000ee4)

This register is a write-only register. It can be used to set the value of the CTCHOUT outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	ITCHOUT	rw	0x0	Set the value of the CTCHOUT outputs.

4.16.2.31 ITTRIGOUT (0x3000ee8)

This register is a write-only register. It can be used to set the value of the CTTRIGOUT outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGOUT	rw	0x0	Set the value of the CTTRIGOUT outputs.

4.16.2.32 ITCHOUTACK (0x3000eec)

This register is a read-only register. It can be used to read the values of the CTCHOUTACK inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTCHOUTACK	r	0x0	Read the values of the CTCHOUTACK inputs.

4.16.2.33 ITTRIGOUTACK (0x3000ef0)

This register is a read-only register. It can be used to read the values of the CTTRIGOUTACK inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGOUTACK	r	0x0	Read the value of the CTTRIGOUTACK inputs.

4.16.2.34 ITCHIN (0x3000ef4)

This register is a read-only register. It can be used to read the values of the CTCHIN inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTCHIN	r	0x0	Read the value of the CTCHIN inputs.

4.16.2.35 ITTRIGIN (0x3000ef8)

This register is a read-only register. It can be used to read the values of the CTTRIGIN inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGIN	r	0x0	Read the values of the CTTRIGIN inputs.

4.16.2.36 ITCTRL (0x3000f00)

This register is used to enable topology detection. For more information see the CoreSight Architecture Specification. This register enables the component to switch from a functional mode, the default behavior, to integration mode where the inputs and outputs of the component can be directly controlled for the purpose of integration testing and topology solving. Note : When a device has been in integration mode, it might not function with the original behavior. After performing integration or topology detection, you must reset the system to ensure correct behavior of CoreSight and other connected system components that are affected by the integration or topology detection.

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	Integration_mode	rw	0x0	Allows the component to switch from functional mode to integration mode or back.

4.16.2.37 CLAIMSET (0x3000fa0)

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CLAIMSET	rw	0x0	This claim tag bit is implemented

4.16.2.38 CLAIMCLR (0x3000fa4)

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CLAIMCLR	rw	0x0	The value present reflects the current setting of the Claim Tag.

4.16.2.39 LAR (0x3000fb0)

This is used to enable write access to device registers. External accesses from a debugger (paddrdbg31 = 1) are not subject to the Lock Registers. A debugger does not have to unlock the component in order to write and modify the registers in the component.

Bit	Name	Type	Reset	Description
31:0	ACCESS_W	rw	0x0	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access.

4.16.2.40 LSR (0x3000fb4)

This indicates the status of the Lock control mechanism. This lock prevents accidental writes by code under debug. When locked, write access is blocked to all registers, except the Lock Access Register. External accesses from a debugger (paddrdbg31 = 1) are not subject to the Lock Registers. This register reads as 0 when read from an external debugger (paddrdbg31 = 1).

Bit	Name	Type	Reset	Description
31:3	reserved	r	0x0	Reserved
2	LOCKTYPE	r	0x0	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit
1	LOCKGRANT	r	0x1	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (paddrdbg31 = 1) since external debugger accesses are not subject to Lock Registers.
0	LOCKEXIST	r	0x1	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (paddrdbg31 = 1) since external debugger accesses are not subject to Lock Registers.

4.16.2.41 AUTHSTATUS (0x3000fb8)

Reports what functionality is currently permitted by the authentication interface.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:6	SNID	r	0x0	Indicates the security level for secure non-invasive debug
5:4	SID	r	0x0	Indicates the security level for secure invasive debug
3:2	NSNID	r	0x2	Indicates the security level for non-secure non-invasive debug
1:0	NSID	r	0x2	Indicates the security level for non-secure invasive debug

4.16.2.42 DEVID (0x3000fc8)

This register indicates the capabilities of the CTI.

Bit	Name	Type	Reset	Description
31:20	reserved	r	0x0	Reserved
19:16	NUMCH	r	0x4	Number of ECT channels available.
15:8	NUMTRIG	r	0x8	Number of ECT triggers available.
7:5	reserved	r	0x0	Reserved
4:0	EXTMUXNUM	r	0x0	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using asicctl. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog define EXTMUXNUM that the user must alter accordingly.

4.16.2.43 DEVTYPE (0x3000fcc)

It provides a debugger with information about the component when the Part Number field is not recognized. The debugger can then report this information.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	Sub_Type	r	0x1	Sub-classification within the major category
3:0	Major_Type	r	0x4	Major classification grouping for this debug/trace component

4.16.2.44 PERIPHID4 (0x3000fd0)

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	SIZE	r	0x0	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on.
3:0	DES_2	r	0x4	JEDEC continuation code indicating the designer of the component (along with the identity code)

4.16.2.45 PERIPHD0 (0x3000fe0)

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PART_0	r	0x6	Bits [7 : 0] of the component's part number. This is selected by the designer of the component.

4.16.2.46 PERIPHD1 (0x3000fe4)

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	DES_0	r	0x0	Bits 3 : 0 of the JEDEC identity code indicating the designer of the component (along with the continuation code)
3:0	PART_1	r	0x9	Bits [11 : 8] of the component's part number. This is selected by the designer of the component.

4.16.2.47 PERIPHD2 (0x3000fe8)

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	REVISION	r	0x4	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision.
3	JEDEC	r	0x1	Always set. Indicates that a JEDEC assigned value is used
2:0	DES_1	r	0x0	Bits 6 : 4 of the JEDEC identity code indicating the designer of the component (along with the continuation code)

4.16.2.48 PERIPHD3 (0x3000fec)

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	RELAND	r	0x0	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero.
3:0	CMOD	r	0x0	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero.

4.16.2.49 COMPID0 (0x3000ff0)

Reserved Reserved Reserved A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_0	r	0xd	Contains bits [7 : 0] of the component identification

4.16.2.50 COMPID1 (0x3000ff4)

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	CLASS	r	0x9	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15 : 12] of the component identification.
3:0	PRMBL_1	r	0x0	Contains bits [11 : 8] of the component identification

4.16.2.51 COMPID2 (0x3000ff8)

A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_2	r	0x5	Contains bits [23 : 16] of the component identification

4.16.2.52 COMPID3 (0x3000ffc)

A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_3	r	0x0	Contains bits [31 : 24] of the component identification

4.17 MMR Region: el2_cscti5_cfg

4.17.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x3001000	CTICONTROL	
0x3001010	CTIINTACK	
0x3001014	CTIAPPSET	
0x3001018	CTIAPPCLEAR	
0x300101c	CTIAPPPULSE	
0x3001020	CTIINEN0	
0x3001024	CTIINEN1	
0x3001028	CTIINEN2	
0x300102c	CTIINEN3	
0x3001030	CTIINEN4	
0x3001034	CTIINEN5	
0x3001038	CTIINEN6	
0x300103c	CTIINEN7	
0x30010a0	CTIOUTEN0	
0x30010a4	CTIOUTEN1	
0x30010a8	CTIOUTEN2	
0x30010ac	CTIOUTEN3	
0x30010b0	CTIOUTEN4	
0x30010b4	CTIOUTEN5	
0x30010b8	CTIOUTEN6	
0x30010bc	CTIOUTEN7	
0x3001130	CTITRIGINSTATUS	
0x3001134	CTITRIGOUTSTATUS	
0x3001138	CTICHINSTATUS	
0x300113c	CTICHOUTSTATUS	
0x3001140	CTIGATE	
0x3001144	ASICCTL	
0x3001edc	ITCHINACK	
0x3001ee0	ITTRIGINACK	
0x3001ee4	ITCHOUT	
0x3001ee8	ITTRIGOUT	
0x3001eec	ITCHOUTACK	
0x3001ef0	ITTRIGOUTACK	
0x3001ef4	ITCHIN	
0x3001ef8	ITTRIGIN	
0x3001f00	ITCTRL	
0x3001fa0	CLAIMSET	
0x3001fa4	CLAIMCLR	
0x3001fb0	LAR	
0x3001fb4	LSR	
0x3001fb8	AUTHSTATUS	
0x3001fc8	DEVID	
0x3001fcc	DEVTYPE	
0x3001fd0	PERIPHID4	
0x3001fe0	PERIPHID0	
0x3001fe4	PERIPHID1	

0x3001fe8	PERIPHID2	
0x3001fec	PERIPHID3	
0x3001ff0	COMPID0	
0x3001ff4	COMPID1	
0x3001ff8	COMPID2	
0x3001ffc	COMPID3	

4.17.2 Register Definitions

4.17.2.1 CTICONTROL (0x3001000)

The CTI Control Register enables the CTI. >

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	GLBEN	rw	0x0	Enables or disables the ECT.

4.17.2.2 CTIINTACK (0x3001010)

The CTI Interrupt Acknowledge Register is write-only. Any bits written as a 1 cause the ctitrigout output signal to be acknowledged. The acknowledgement is cleared when MAPTRIGOUT is deactivated. This register is used when the ctitrigout is used as a sticky output, that is, no hardware acknowledge is supplied, and a software acknowledge is required.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	INTACK	rw	0x0	Acknowledges the corresponding ctitrigout output. There is one bit of the register for each ctitrigout output. When a 1 is written to a bit in this register, the corresponding ctitrigout is acknowledged and is cleared when MAPTRIGOUT is LOW. Writing a 0 to any of the bits in this register has no effect.

4.17.2.3 CTIAPPSET (0x3001014)

The CTI Application Trigger Set Register is read/write. A write to this register causes a channel event to be raised, corresponding to the bit written to.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPSET	rw	0x0	Setting a bit HIGH generates a channel event for the selected channel. There is one bit of the register for each channel. Read : 0 = application trigger inactive (reset). 1 = application trigger active. Write : 0 = no effect. 1 = generate channel event.

4.17.2.4 CTIAPPCLEAR (0x3001018)

The CTI Interrupt Acknowledge Register is write-only. A write to this register causes a channel event to be cleared, corresponding to the bit written to.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPCLEAR	rw	0x0	Clears corresponding bits in the CTIAPPSET register. There is one bit of the register for each channel. When a 1 is written to a bit in this register, the corresponding application trigger is disabled in the CTIAPPSET register. Writing a 0 to any of the bits in this register has no effect.

4.17.2.5 CTIAPPULSE (0x300101c)

The CTI Application Pulse Register is write-only. A write to this register causes a channel event pulse, one clock period, to be generated, corresponding to the bit written to. The pulse external to the ECT can be extended to multi-cycle by the handshaking interface circuits. This register clears itself immediately, so it can be repeatedly written to without software having to clear it.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPULSE	rw	0x0	Setting a bit HIGH generates a channel event pulse for the selected channel. There is one bit of the register for each channel. When a 1 is written to a bit in this register, a corresponding channel event pulse is generated for one clock period. Writing a 0 to any of the bits in this register has no effect.

4.17.2.6 CTIINEN0 (0x3001020)

The CTI Trigger 0 to Channel Enable Register enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.17.2.7 CTIINEN1 (0x3001024)

The CTI Trigger 1 to Channel Enable Register enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.17.2.8 CTIINEN2 (0x3001028)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.17.2.9 CTIINEN3 (0x300102c)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.17.2.10 CTIINEN4 (0x3001030)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.17.2.11 CTIINEN5 (0x3001034)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.17.2.12 CTIINEN6 (0x3001038)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in



				this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.
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4.17.2.13 CTIINEN7 (0x300103c)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.17.2.14 CTIOUTEN0 (0x30010a0)

The CTI Channel to Trigger 0 Enable Registers define which channels can generate a ctitrigin[0] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigin[1] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigin output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigin[0] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigin output. Reading this register returns the programmed value.

4.17.2.15 CTIOUTEN1 (0x30010a4)

The CTI Channel to Trigger 1 Enable Registers define which channels can generate a ctitrigin[1] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigout[1] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigout[1] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigout output. Reading this register returns the programmed value.

4.17.2.16 CTIOUTEN2 (0x30010a8)

The CTI Channel to Trigger 2 Enable Registers define which channels can generate a ctitrigout[2] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigout[2] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigout[2] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigout output. Reading this register returns the programmed value.

4.17.2.17 CTIOUTEN3 (0x30010ac)

The CTI Channel to Trigger 3 Enable Registers define which channels can generate a ctitrigout[3] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigout[3] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigout output. For example, enabling bit 0 enables ctichin[0]

				to cause a trigger event on the ctitriggerout[3] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitriggerout output. Reading this register returns the programmed value.
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4.17.2.18 CTIOUTEN4 (0x30010b0)

The CTI Channel to Trigger 4 Enable Registers define which channels can generate a ctitriggerout[4] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitriggerout[4] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitriggerout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitriggerout[4] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitriggerout output. Reading this register returns the programmed value.

4.17.2.19 CTIOUTEN5 (0x30010b4)

The CTI Channel to Trigger 5 Enable Registers define which channels can generate a ctitriggerout[5] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitriggerout[5] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitriggerout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitriggerout[5] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitriggerout output. Reading this register returns the programmed value.

4.17.2.20 CTIOUTEN6 (0x30010b8)

The CTI Channel to Trigger 6 Enable Registers define which channels can generate a ctitriggerout[6] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[6] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[6] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.17.2.21 CTIOUTEN7 (0x30010bc)

The CTI Channel to Trigger 7 Enable Registers define which channels can generate a cttrigout[7] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[7] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[7] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.17.2.22 CTTRIGINSTATUS (0x3001130)

The CTI Trigger In Status Register provides the status of the cttrigin inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	TRIGINSTATUS	r	0x0	Shows the status of the cttrigin inputs. 1 = cttrigin is active. 0 = cttrigin is inactive. Because the register provides a view of the raw cttrigin inputs, the reset value is unknown. There is one bit of the field for each trigger input.

4.17.2.23 CTTRIGOUTSTATUS (0x3001134)

The CTI Trigger Out Status Register provides the status of the cttrigout outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	TRIGOUTSTATUS	r	0x0	Shows the status of the cttrigout outputs. 1 = cttrigout is active. 0 = cttrigout is inactive. There is one bit of the field for each trigger output.

4.17.2.24 CTICHINSTATUS (0x3001138)

The CTI Channel In Status Register provides the status of the ctichin inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTICHINSTATUS	r	0x0	Shows the status of the ctichin inputs. 1 = ctichin is active. 0 = ctichin is inactive. Because the register provides a view of the raw ctichin inputs, the reset value is unknown. There is one bit of the field for each channel input.

4.17.2.25 CTICHOUTSTATUS (0x300113c)

The CTI Channel Out Status Register provides the status of the CTI ctichout outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTICHOUTSTATUS	r	0x0	Shows the status of the ctichout outputs. 1 = ctichout is active. 0 = ctichout is inactive. There is one bit of the field for each channel output.

4.17.2.26 CTIGATE (0x3001140)

The Gate Enable Register prevents the channels from propagating through the CTM to other CTIs. This enables local cross-triggering, for example for causing an interrupt when the ETM trigger occurs. It can be used effectively with CTIAPPSET, CTIAPPCLEAR, and CTIAPPPULSE for asserting trigger outputs by asserting channels, without affecting the rest of the system. On reset, this register is 0xF, and channel propagation is enabled.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	CTIGATEEN3	rw	0x1	Enable CTICHOUT3. Set to 0 to disable channel propagation.
2	CTIGATEEN2	rw	0x1	Enable CTICHOUT2. Set to 0 to disable channel propagation.
1	CTIGATEEN1	rw	0x1	Enable CTICHOUT1. Set to 0 to disable channel propagation.
0	CTIGATEEN0	rw	0x1	Enable CTICHOUT0. Set to 0 to disable channel propagation.

4.17.2.27 ASICCTL (0x3001144)

Implementation-defined ASIC control, value written to the register is output on asicctl[7 : 0].

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	ASICCTL	rw	0x0	Implementation-defined ASIC control, value written to the register is output on asicctl[7 : 0]. If external multiplexing of trigger signals is implemented then the number of multiplexed signals on each trigger must be reflected within the Device ID Register. This is done within a Verilog define EXTMUXNUM.

4.17.2.28 ITCHINACK (0x3001edc)

This register is a write-only register. It can be used to set the value of the CTCHINACK outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	ITCHINACK	rw	0x0	Set the value of the CTCHINACK outputs.

4.17.2.29 ITTRIGINACK (0x3001ee0)

This register is a write-only register. It can be used to set the value of the CTTRIGINACK outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGINACK	rw	0x0	Set the value of the CTTRIGINACK outputs.

4.17.2.30 ITCHOUT (0x3001ee4)

This register is a write-only register. It can be used to set the value of the CTCHOUT outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	ITCHOUT	rw	0x0	Set the value of the CTCHOUT outputs.

4.17.2.31 ITTRIGOUT (0x3001ee8)

This register is a write-only register. It can be used to set the value of the CTTRIGOUT outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGOUT	rw	0x0	Set the value of the CTTRIGOUT outputs.

4.17.2.32 ITCHOUTACK (0x3001eec)

This register is a read-only register. It can be used to read the values of the CTCHOUTACK inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTCHOUTACK	r	0x0	Read the values of the CTCHOUTACK inputs.

4.17.2.33 ITTRIGOUTACK (0x3001ef0)

This register is a read-only register. It can be used to read the values of the CTTRIGOUTACK inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGOUTACK	r	0x0	Read the value of the CTTRIGOUTACK inputs.

4.17.2.34 ITCHIN (0x3001ef4)

This register is a read-only register. It can be used to read the values of the CTCHIN inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTCHIN	r	0x0	Read the value of the CTCHIN inputs.

4.17.2.35 ITTRIGIN (0x3001ef8)

This register is a read-only register. It can be used to read the values of the CTTRIGIN inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGIN	r	0x0	Read the values of the CTTRIGIN inputs.

4.17.2.36 ITCTRL (0x3001f00)

This register is used to enable topology detection. For more information see the CoreSight Architecture Specification. This register enables the component to switch from a functional mode, the default behavior, to integration mode where the inputs and outputs of the component can be directly controlled for the purpose of integration testing and topology solving. Note : When a device has been in integration mode, it might not function with the original behavior. After performing integration or topology detection, you must reset the system to ensure correct behavior of CoreSight and other connected system components that are affected by the integration or topology detection.

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	Integration_mode	rw	0x0	Allows the component to switch from functional mode to integration mode or back.

4.17.2.37 CLAIMSET (0x3001fa0)

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CLAIMSET	rw	0x0	This claim tag bit is implemented

4.17.2.38 CLAIMCLR (0x3001fa4)

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CLAIMCLR	rw	0x0	The value present reflects the current setting of the Claim Tag.

4.17.2.39 LAR (0x3001fb0)

This is used to enable write access to device registers. External accesses from a debugger (paddrdbg31 = 1) are not subject to the Lock Registers. A debugger does not have to unlock the component in order to write and modify the registers in the component.

Bit	Name	Type	Reset	Description
31:0	ACCESS_W	rw	0x0	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access.

4.17.2.40 LSR (0x3001fb4)

This indicates the status of the Lock control mechanism. This lock prevents accidental writes by code under debug. When locked, write access is blocked to all registers, except the Lock Access Register. External accesses from a debugger (paddrdbg31 = 1) are not subject to the Lock Registers. This register reads as 0 when read from an external debugger (paddrdbg31 = 1).

Bit	Name	Type	Reset	Description
31:3	reserved	r	0x0	Reserved
2	LOCKTYPE	r	0x0	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit
1	LOCKGRANT	r	0x1	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (paddrdbg31 = 1) since external debugger accesses are not subject to Lock Registers.
0	LOCKEXIST	r	0x1	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (paddrdbg31 = 1) since external debugger accesses are not subject to Lock Registers.

4.17.2.41 AUTHSTATUS (0x3001fb8)

Reports what functionality is currently permitted by the authentication interface.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:6	SNID	r	0x0	Indicates the security level for secure non-invasive debug
5:4	SID	r	0x0	Indicates the security level for secure invasive debug
3:2	NSNID	r	0x2	Indicates the security level for non-secure non-invasive debug
1:0	NSID	r	0x2	Indicates the security level for non-secure invasive debug

4.17.2.42 DEVID (0x3001fc8)

This register indicates the capabilities of the CTI.

Bit	Name	Type	Reset	Description
31:20	reserved	r	0x0	Reserved
19:16	NUMCH	r	0x4	Number of ECT channels available.
15:8	NUMTRIG	r	0x8	Number of ECT triggers available.
7:5	reserved	r	0x0	Reserved
4:0	EXTMUXNUM	r	0x0	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using asicctl. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog define EXTMUXNUM that the user must alter accordingly.

4.17.2.43 DEVTYPE (0x3001fcc)

It provides a debugger with information about the component when the Part Number field is not recognized. The debugger can then report this information.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	Sub_Type	r	0x1	Sub-classification within the major category
3:0	Major_Type	r	0x4	Major classification grouping for this debug/trace component

4.17.2.44 PERIPHID4 (0x3001fd0)

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	SIZE	r	0x0	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on.
3:0	DES_2	r	0x4	JEDEC continuation code indicating the designer of the component (along with the identity code)

4.17.2.45 PERIPHD0 (0x3001fe0)

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PART_0	r	0x6	Bits [7 : 0] of the component's part number. This is selected by the designer of the component.

4.17.2.46 PERIPHD1 (0x3001fe4)

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	DES_0	r	0x0	Bits 3 : 0 of the JEDEC identity code indicating the designer of the component (along with the continuation code)
3:0	PART_1	r	0x9	Bits [11 : 8] of the component's part number. This is selected by the designer of the component.

4.17.2.47 PERIPHD2 (0x3001fe8)

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	REVISION	r	0x4	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision.
3	JEDEC	r	0x1	Always set. Indicates that a JEDEC assigned value is used
2:0	DES_1	r	0x0	Bits 6 : 4 of the JEDEC identity code indicating the designer of the component (along with the continuation code)

4.17.2.48 PERIPHD3 (0x3001fec)

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	REVAND	r	0x0	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero.
3:0	CMOD	r	0x0	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero.

4.17.2.49 COMPID0 (0x3001ff0)

Reserved Reserved Reserved A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_0	r	0xd	Contains bits [7 : 0] of the component identification

4.17.2.50 COMPID1 (0x3001ff4)

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	CLASS	r	0x9	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15 : 12] of the component identification.
3:0	PRMBL_1	r	0x0	Contains bits [11 : 8] of the component identification

4.17.2.51 COMPID2 (0x3001ff8)

A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_2	r	0x5	Contains bits [23 : 16] of the component identification

4.17.2.52 COMPID3 (0x3001ffc)

A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_3	r	0x0	Contains bits [31 : 24] of the component identification

4.18 MMR Region: el2_cscti6_cfg

4.18.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x3002000	CTICONTROL	
0x3002010	CTIINTACK	
0x3002014	CTIAPPSET	
0x3002018	CTIAPPCLEAR	
0x300201c	CTIAPPPULSE	
0x3002020	CTIINEN0	
0x3002024	CTIINEN1	
0x3002028	CTIINEN2	
0x300202c	CTIINEN3	
0x3002030	CTIINEN4	
0x3002034	CTIINEN5	
0x3002038	CTIINEN6	
0x300203c	CTIINEN7	
0x30020a0	CTIOUTEN0	
0x30020a4	CTIOUTEN1	
0x30020a8	CTIOUTEN2	
0x30020ac	CTIOUTEN3	
0x30020b0	CTIOUTEN4	
0x30020b4	CTIOUTEN5	
0x30020b8	CTIOUTEN6	
0x30020bc	CTIOUTEN7	
0x3002130	CTITRIGINSTATUS	
0x3002134	CTITRIGOUTSTATUS	
0x3002138	CTICHINSTATUS	
0x300213c	CTICHOUTSTATUS	
0x3002140	CTIGATE	
0x3002144	ASICCTL	
0x3002edc	ITCHINACK	
0x3002ee0	ITTRIGINACK	
0x3002ee4	ITCHOUT	
0x3002ee8	ITTRIGOUT	
0x3002eec	ITCHOUTACK	
0x3002ef0	ITTRIGOUTACK	
0x3002ef4	ITCHIN	
0x3002ef8	ITTRIGIN	
0x3002f00	ITCTRL	
0x3002fa0	CLAIMSET	
0x3002fa4	CLAIMCLR	
0x3002fb0	LAR	
0x3002fb4	LSR	
0x3002fb8	AUTHSTATUS	
0x3002fc8	DEVID	
0x3002fcc	DEVTYPE	
0x3002fd0	PERIPHID4	
0x3002fe0	PERIPHID0	
0x3002fe4	PERIPHID1	

0x3002fe8	PERIPHID2	
0x3002fec	PERIPHID3	
0x3002ff0	COMPID0	
0x3002ff4	COMPID1	
0x3002ff8	COMPID2	
0x3002ffc	COMPID3	

4.18.2 Register Definitions

4.18.2.1 CTICONTROL (0x3002000)

The CTI Control Register enables the CTI. >

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	GLBEN	rw	0x0	Enables or disables the ECT.

4.18.2.2 CTIINTACK (0x3002010)

The CTI Interrupt Acknowledge Register is write-only. Any bits written as a 1 cause the ctitrigout output signal to be acknowledged. The acknowledgement is cleared when MAPTRIGOUT is deactivated. This register is used when the ctitrigout is used as a sticky output, that is, no hardware acknowledge is supplied, and a software acknowledge is required.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	INTACK	rw	0x0	Acknowledges the corresponding ctitrigout output. There is one bit of the register for each ctitrigout output. When a 1 is written to a bit in this register, the corresponding ctitrigout is acknowledged and is cleared when MAPTRIGOUT is LOW. Writing a 0 to any of the bits in this register has no effect.

4.18.2.3 CTIAPPSET (0x3002014)

The CTI Application Trigger Set Register is read/write. A write to this register causes a channel event to be raised, corresponding to the bit written to.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPSET	rw	0x0	Setting a bit HIGH generates a channel event for the selected channel. There is one bit of the register for each channel. Read : 0 = application trigger inactive (reset). 1 = application trigger active. Write : 0 = no effect. 1 = generate channel event.

4.18.2.4 CTIAPPCLEAR (0x3002018)

The CTI Interrupt Acknowledge Register is write-only. A write to this register causes a channel event to be cleared, corresponding to the bit written to.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPCLEAR	rw	0x0	Clears corresponding bits in the CTIAPPSET register. There is one bit of the register for each channel. When a 1 is written to a bit in this register, the corresponding application trigger is disabled in the CTIAPPSET register. Writing a 0 to any of the bits in this register has no effect.

4.18.2.5 CTIAPPULSE (0x300201c)

The CTI Application Pulse Register is write-only. A write to this register causes a channel event pulse, one clock period, to be generated, corresponding to the bit written to. The pulse external to the ECT can be extended to multi-cycle by the handshaking interface circuits. This register clears itself immediately, so it can be repeatedly written to without software having to clear it.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPULSE	rw	0x0	Setting a bit HIGH generates a channel event pulse for the selected channel. There is one bit of the register for each channel. When a 1 is written to a bit in this register, a corresponding channel event pulse is generated for one clock period. Writing a 0 to any of the bits in this register has no effect.

4.18.2.6 CTIINEN0 (0x3002020)

The CTI Trigger 0 to Channel Enable Register enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.18.2.7 CTIINEN1 (0x3002024)

The CTI Trigger 1 to Channel Enable Register enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.18.2.8 CTIINEN2 (0x3002028)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.18.2.9 CTIINEN3 (0x300202c)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.18.2.10 CTIINEN4 (0x3002030)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.18.2.11 CTIINEN5 (0x3002034)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.18.2.12 CTIINEN6 (0x3002038)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in



				this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.
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4.18.2.13 CTIINEN7 (0x300203c)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.18.2.14 CTIOUTEN0 (0x30020a0)

The CTI Channel to Trigger 0 Enable Registers define which channels can generate a ctitrigin[0] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigin[1] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigin output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigin[0] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigin output. Reading this register returns the programmed value.

4.18.2.15 CTIOUTEN1 (0x30020a4)

The CTI Channel to Trigger 1 Enable Registers define which channels can generate a ctitrigin[1] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[1] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[1] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.18.2.16 CTIOUTEN2 (0x30020a8)

The CTI Channel to Trigger 2 Enable Registers define which channels can generate a cttrigout[2] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[2] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[2] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.18.2.17 CTIOUTEN3 (0x30020ac)

The CTI Channel to Trigger 3 Enable Registers define which channels can generate a cttrigout[3] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[3] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0]

				to cause a trigger event on the ctitrigger[3] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigger output. Reading this register returns the programmed value.
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4.18.2.18 CTIOUTEN4 (0x30020b0)

The CTI Channel to Trigger 4 Enable Registers define which channels can generate a ctitrigger[4] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigger[4] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigger output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigger[4] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigger output. Reading this register returns the programmed value.

4.18.2.19 CTIOUTEN5 (0x30020b4)

The CTI Channel to Trigger 5 Enable Registers define which channels can generate a ctitrigger[5] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigger[5] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigger output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigger[5] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigger output. Reading this register returns the programmed value.

4.18.2.20 CTIOUTEN6 (0x30020b8)

The CTI Channel to Trigger 6 Enable Registers define which channels can generate a ctitrigger[6] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[6] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[6] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.18.2.21 CTIOUTEN7 (0x30020bc)

The CTI Channel to Trigger 7 Enable Registers define which channels can generate a cttrigout[7] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[7] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[7] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.18.2.22 CTTRIGINSTATUS (0x3002130)

The CTI Trigger In Status Register provides the status of the cttrigin inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	TRIGINSTATUS	r	0x0	Shows the status of the cttrigin inputs. 1 = cttrigin is active. 0 = cttrigin is inactive. Because the register provides a view of the raw cttrigin inputs, the reset value is unknown. There is one bit of the field for each trigger input.

4.18.2.23 CTTRIGOUTSTATUS (0x3002134)

The CTI Trigger Out Status Register provides the status of the cttrigout outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	TRIGOUTSTATUS	r	0x0	Shows the status of the cttrigout outputs. 1 = cttrigout is active. 0 = cttrigout is inactive. There is one bit of the field for each trigger output.

4.18.2.24 CTICHINSTATUS (0x3002138)

The CTI Channel In Status Register provides the status of the ctichin inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTICHINSTATUS	r	0x0	Shows the status of the ctichin inputs. 1 = ctichin is active. 0 = ctichin is inactive. Because the register provides a view of the raw ctichin inputs, the reset value is unknown. There is one bit of the field for each channel input.

4.18.2.25 CTICHOUTSTATUS (0x300213c)

The CTI Channel Out Status Register provides the status of the CTI ctichout outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTICHOUTSTATUS	r	0x0	Shows the status of the ctichout outputs. 1 = ctichout is active. 0 = ctichout is inactive. There is one bit of the field for each channel output.

4.18.2.26 CTIGATE (0x3002140)

The Gate Enable Register prevents the channels from propagating through the CTM to other CTIs. This enables local cross-triggering, for example for causing an interrupt when the ETM trigger occurs. It can be used effectively with CTIAPPSET, CTIAPPCLEAR, and CTIAPPPULSE for asserting trigger outputs by asserting channels, without affecting the rest of the system. On reset, this register is 0xF, and channel propagation is enabled.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	CTIGATEEN3	rw	0x1	Enable CTICHOUT3. Set to 0 to disable channel propagation.
2	CTIGATEEN2	rw	0x1	Enable CTICHOUT2. Set to 0 to disable channel propagation.
1	CTIGATEEN1	rw	0x1	Enable CTICHOUT1. Set to 0 to disable channel propagation.
0	CTIGATEEN0	rw	0x1	Enable CTICHOUT0. Set to 0 to disable channel propagation.

4.18.2.27 ASICCTL (0x3002144)

Implementation-defined ASIC control, value written to the register is output on `asicctl[7 : 0]`.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	ASICCTL	rw	0x0	Implementation-defined ASIC control, value written to the register is output on <code>asicctl[7 : 0]</code> . If external multiplexing of trigger signals is implemented then the number of multiplexed signals on each trigger must be reflected within the Device ID Register. This is done within a Verilog define <code>EXTMUXNUM</code> .

4.18.2.28 ITCHINACK (0x3002edc)

This register is a write-only register. It can be used to set the value of the CTCHINACK outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	ITCHINACK	rw	0x0	Set the value of the CTCHINACK outputs.

4.18.2.29 ITTRIGINACK (0x3002ee0)

This register is a write-only register. It can be used to set the value of the CTTRIGINACK outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGINACK	rw	0x0	Set the value of the CTTRIGINACK outputs.

4.18.2.30 ITCHOUT (0x3002ee4)

This register is a write-only register. It can be used to set the value of the CTCHOUT outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	ITCHOUT	rw	0x0	Set the value of the CTCHOUT outputs.

4.18.2.31 ITTRIGOUT (0x3002ee8)

This register is a write-only register. It can be used to set the value of the CTTRIGOUT outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGOUT	rw	0x0	Set the value of the CTTRIGOUT outputs.

4.18.2.32 ITCHOUTACK (0x3002eec)

This register is a read-only register. It can be used to read the values of the CTCHOUTACK inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTCHOUTACK	r	0x0	Read the values of the CTCHOUTACK inputs.

4.18.2.33 ITTRIGOUTACK (0x3002ef0)

This register is a read-only register. It can be used to read the values of the CTTRIGOUTACK inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGOUTACK	r	0x0	Read the value of the CTTRIGOUTACK inputs.

4.18.2.34 ITCHIN (0x3002ef4)

This register is a read-only register. It can be used to read the values of the CTCHIN inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTCHIN	r	0x0	Read the value of the CTCHIN inputs.

4.18.2.35 ITTRIGIN (0x3002ef8)

This register is a read-only register. It can be used to read the values of the CTTRIGIN inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGIN	r	0x0	Read the values of the CTTRIGIN inputs.

4.18.2.36 ITCTRL (0x3002f00)

This register is used to enable topology detection. For more information see the CoreSight Architecture Specification. This register enables the component to switch from a functional mode, the default behavior, to integration mode where the inputs and outputs of the component can be directly controlled for the purpose of integration testing and topology solving. Note : When a device has been in integration mode, it might not function with the original behavior. After performing integration or topology detection, you must reset the system to ensure correct behavior of CoreSight and other connected system components that are affected by the integration or topology detection.

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	Integration_mode	rw	0x0	Allows the component to switch from functional mode to integration mode or back.

4.18.2.37 CLAIMSET (0x3002fa0)

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CLAIMSET	rw	0x0	This claim tag bit is implemented

4.18.2.38 CLAIMCLR (0x3002fa4)

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CLAIMCLR	rw	0x0	The value present reflects the current setting of the Claim Tag.

4.18.2.39 LAR (0x3002fb0)

This is used to enable write access to device registers. External accesses from a debugger (paddrdbg31 = 1) are not subject to the Lock Registers. A debugger does not have to unlock the component in order to write and modify the registers in the component.

Bit	Name	Type	Reset	Description
31:0	ACCESS_W	rw	0x0	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access.

4.18.2.40 LSR (0x3002fb4)

This indicates the status of the Lock control mechanism. This lock prevents accidental writes by code under debug. When locked, write access is blocked to all registers, except the Lock Access Register. External accesses from a debugger (paddrdbg31 = 1) are not subject to the Lock Registers. This register reads as 0 when read from an external debugger (paddrdbg31 = 1).

Bit	Name	Type	Reset	Description
31:3	reserved	r	0x0	Reserved
2	LOCKTYPE	r	0x0	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit
1	LOCKGRANT	r	0x1	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (paddrdbg31 = 1) since external debugger accesses are not subject to Lock Registers.
0	LOCKEXIST	r	0x1	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (paddrdbg31 = 1) since external debugger accesses are not subject to Lock Registers.

4.18.2.41 AUTHSTATUS (0x3002fb8)

Reports what functionality is currently permitted by the authentication interface.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:6	SNID	r	0x0	Indicates the security level for secure non-invasive debug
5:4	SID	r	0x0	Indicates the security level for secure invasive debug
3:2	NSNID	r	0x2	Indicates the security level for non-secure non-invasive debug
1:0	NSID	r	0x2	Indicates the security level for non-secure invasive debug

4.18.2.42 DEVID (0x3002fc8)

This register indicates the capabilities of the CTI.

Bit	Name	Type	Reset	Description
31:20	reserved	r	0x0	Reserved
19:16	NUMCH	r	0x4	Number of ECT channels available.
15:8	NUMTRIG	r	0x8	Number of ECT triggers available.
7:5	reserved	r	0x0	Reserved
4:0	EXTMUXNUM	r	0x0	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using asicctl. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog define EXTMUXNUM that the user must alter accordingly.

4.18.2.43 DEVTYPE (0x3002fcc)

It provides a debugger with information about the component when the Part Number field is not recognized. The debugger can then report this information.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	Sub_Type	r	0x1	Sub-classification within the major category
3:0	Major_Type	r	0x4	Major classification grouping for this debug/trace component

4.18.2.44 PERIPHID4 (0x3002fd0)

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	SIZE	r	0x0	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on.
3:0	DES_2	r	0x4	JEDEC continuation code indicating the designer of the component (along with the identity code)

4.18.2.45 PERIPHD0 (0x3002fe0)

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PART_0	r	0x6	Bits [7 : 0] of the component's part number. This is selected by the designer of the component.

4.18.2.46 PERIPHD1 (0x3002fe4)

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	DES_0	r	0x0	Bits 3 : 0 of the JEDEC identity code indicating the designer of the component (along with the continuation code)
3:0	PART_1	r	0x9	Bits [11 : 8] of the component's part number. This is selected by the designer of the component.

4.18.2.47 PERIPHD2 (0x3002fe8)

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	REVISION	r	0x4	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision.
3	JEDEC	r	0x1	Always set. Indicates that a JEDEC assigned value is used
2:0	DES_1	r	0x0	Bits 6 : 4 of the JEDEC identity code indicating the designer of the component (along with the continuation code)

4.18.2.48 PERIPHD3 (0x3002fec)

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	REVAND	r	0x0	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero.
3:0	CMOD	r	0x0	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero.

4.18.2.49 COMPID0 (0x3002ff0)

Reserved Reserved Reserved A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_0	r	0xd	Contains bits [7 : 0] of the component identification

4.18.2.50 COMPID1 (0x3002ff4)

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	CLASS	r	0x9	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15 : 12] of the component identification.
3:0	PRMBL_1	r	0x0	Contains bits [11 : 8] of the component identification

4.18.2.51 COMPID2 (0x3002ff8)

A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_2	r	0x5	Contains bits [23 : 16] of the component identification

4.18.2.52 COMPID3 (0x3002ffc)

A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_3	r	0x0	Contains bits [31 : 24] of the component identification

4.19 MMR Region: el2_cscti7_cfg

4.19.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x3003000	CTICONTROL	
0x3003010	CTIINTACK	
0x3003014	CTIAPPSET	
0x3003018	CTIAPPCLEAR	
0x300301c	CTIAPPPULSE	
0x3003020	CTIINEN0	
0x3003024	CTIINEN1	
0x3003028	CTIINEN2	
0x300302c	CTIINEN3	
0x3003030	CTIINEN4	
0x3003034	CTIINEN5	
0x3003038	CTIINEN6	
0x300303c	CTIINEN7	
0x30030a0	CTIOUTEN0	
0x30030a4	CTIOUTEN1	
0x30030a8	CTIOUTEN2	
0x30030ac	CTIOUTEN3	
0x30030b0	CTIOUTEN4	
0x30030b4	CTIOUTEN5	
0x30030b8	CTIOUTEN6	
0x30030bc	CTIOUTEN7	
0x3003130	CTITRIGINSTATUS	
0x3003134	CTITRIGOUTSTATUS	
0x3003138	CTICHINSTATUS	
0x300313c	CTICHOUTSTATUS	
0x3003140	CTIGATE	
0x3003144	ASICCTL	
0x3003edc	ITCHINACK	
0x3003ee0	ITTRIGINACK	
0x3003ee4	ITCHOUT	
0x3003ee8	ITTRIGOUT	
0x3003eec	ITCHOUTACK	
0x3003ef0	ITTRIGOUTACK	
0x3003ef4	ITCHIN	
0x3003ef8	ITTRIGIN	
0x3003f00	ITCTRL	
0x3003fa0	CLAIMSET	
0x3003fa4	CLAIMCLR	
0x3003fb0	LAR	
0x3003fb4	LSR	
0x3003fb8	AUTHSTATUS	
0x3003fc8	DEVID	
0x3003fcc	DEVTYPE	
0x3003fd0	PERIPHID4	
0x3003fe0	PERIPHID0	
0x3003fe4	PERIPHID1	

0x3003fe8	PERIPHID2	
0x3003fec	PERIPHID3	
0x3003ff0	COMPID0	
0x3003ff4	COMPID1	
0x3003ff8	COMPID2	
0x3003ffc	COMPID3	

4.19.2 Register Definitions

4.19.2.1 CTICONTROL (0x3003000)

The CTI Control Register enables the CTI. >

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	GLBEN	rw	0x0	Enables or disables the ECT.

4.19.2.2 CTIINTACK (0x3003010)

The CTI Interrupt Acknowledge Register is write-only. Any bits written as a 1 cause the ctitrigout output signal to be acknowledged. The acknowledgement is cleared when MAPTRIGOUT is deactivated. This register is used when the ctitrigout is used as a sticky output, that is, no hardware acknowledge is supplied, and a software acknowledge is required.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	INTACK	rw	0x0	Acknowledges the corresponding ctitrigout output. There is one bit of the register for each ctitrigout output. When a 1 is written to a bit in this register, the corresponding ctitrigout is acknowledged and is cleared when MAPTRIGOUT is LOW. Writing a 0 to any of the bits in this register has no effect.

4.19.2.3 CTIAPPSET (0x3003014)

The CTI Application Trigger Set Register is read/write. A write to this register causes a channel event to be raised, corresponding to the bit written to.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPSET	rw	0x0	Setting a bit HIGH generates a channel event for the selected channel. There is one bit of the register for each channel. Read : 0 = application trigger inactive (reset). 1 = application trigger active. Write : 0 = no effect. 1 = generate channel event.

4.19.2.4 CTIAPPCLEAR (0x3003018)

The CTI Interrupt Acknowledge Register is write-only. A write to this register causes a channel event to be cleared, corresponding to the bit written to.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPCLEAR	rw	0x0	Clears corresponding bits in the CTIAPPSET register. There is one bit of the register for each channel. When a 1 is written to a bit in this register, the corresponding application trigger is disabled in the CTIAPPSET register. Writing a 0 to any of the bits in this register has no effect.

4.19.2.5 CTIAPPULSE (0x300301c)

The CTI Application Pulse Register is write-only. A write to this register causes a channel event pulse, one clock period, to be generated, corresponding to the bit written to. The pulse external to the ECT can be extended to multi-cycle by the handshaking interface circuits. This register clears itself immediately, so it can be repeatedly written to without software having to clear it.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	APPULSE	rw	0x0	Setting a bit HIGH generates a channel event pulse for the selected channel. There is one bit of the register for each channel. When a 1 is written to a bit in this register, a corresponding channel event pulse is generated for one clock period. Writing a 0 to any of the bits in this register has no effect.

4.19.2.6 CTIINEN0 (0x3003020)

The CTI Trigger 0 to Channel Enable Register enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.19.2.7 CTIINEN1 (0x3003024)

The CTI Trigger 1 to Channel Enable Register enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.19.2.8 CTIINEN2 (0x3003028)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.19.2.9 CTIINEN3 (0x300302c)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.19.2.10 CTIINEN4 (0x3003030)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.19.2.11 CTIINEN5 (0x3003034)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.19.2.12 CTIINEN6 (0x3003038)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in



				this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.
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4.19.2.13 CTIINEN7 (0x300303c)

The CTI Trigger to Channel Enable Register 0 enables the signalling of an event on CTM channels when the core issues a trigger, ctitrigin, to the CTI. Within this register there is one bit for each of the four channels implemented. This register does not affect the application trigger operations.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGINEN	rw	0x0	Enables a cross trigger event to the corresponding channel when an ctitrigin is activated. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, it enables the ctitrigin signal to generate an event on the respective channel of the CTM. For example, TRIGINEN[0] set to 1 enables ctitrigin onto channel 0. Writing a 0 to any of the bits in this register disables the ctitrigin signal from generating an event on the respective channel of the CTM. Reading this register returns the programmed value.

4.19.2.14 CTIOUTEN0 (0x30030a0)

The CTI Channel to Trigger 0 Enable Registers define which channels can generate a ctitriginout[0] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitriginout[1] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitriginout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitriginout[0] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitriginout output. Reading this register returns the programmed value.

4.19.2.15 CTIOUTEN1 (0x30030a4)

The CTI Channel to Trigger 1 Enable Registers define which channels can generate a ctitriginout[1] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[1] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[1] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.19.2.16 CTIOUTEN2 (0x30030a8)

The CTI Channel to Trigger 2 Enable Registers define which channels can generate a cttrigout[2] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[2] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[2] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.19.2.17 CTIOUTEN3 (0x30030ac)

The CTI Channel to Trigger 3 Enable Registers define which channels can generate a cttrigout[3] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[3] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0]

				to cause a trigger event on the ctitrigger[3] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigger output. Reading this register returns the programmed value.
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4.19.2.18 CTIOUTEN4 (0x30030b0)

The CTI Channel to Trigger 4 Enable Registers define which channels can generate a ctitrigger[4] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigger[4] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigger output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigger[4] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigger output. Reading this register returns the programmed value.

4.19.2.19 CTIOUTEN5 (0x30030b4)

The CTI Channel to Trigger 5 Enable Registers define which channels can generate a ctitrigger[5] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a ctitrigger[5] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the ctitrigger output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the ctitrigger[5] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the ctitrigger output. Reading this register returns the programmed value.

4.19.2.20 CTIOUTEN6 (0x30030b8)

The CTI Channel to Trigger 6 Enable Registers define which channels can generate a ctitrigger[6] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[6] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[6] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.19.2.21 CTIOUTEN7 (0x30030bc)

The CTI Channel to Trigger 7 Enable Registers define which channels can generate a cttrigout[7] output. Within this register there is one bit for each of the four channels implemented. This register affects the mapping from application trigger to trigger outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	TRIGOUTEN	rw	0x0	Changing the value of this bit from a 0 to a 1 enables a channel event for the corresponding channel to generate a cttrigout[7] output. There is one bit of the field for each of the four channels. When a 1 is written to a bit in this register, the channel input (ctichin) from the CTM is routed to the cttrigout output. For example, enabling bit 0 enables ctichin[0] to cause a trigger event on the cttrigout[7] output. When a 0 is written to any of the bits in this register, the channel input (ctichin) from the CTM is not routed to the cttrigout output. Reading this register returns the programmed value.

4.19.2.22 CTTRIGINSTATUS (0x3003130)

The CTI Trigger In Status Register provides the status of the cttrigin inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	TRIGINSTATUS	r	0x0	Shows the status of the cttrigin inputs. 1 = cttrigin is active. 0 = cttrigin is inactive. Because the register provides a view of the raw cttrigin inputs, the reset value is unknown. There is one bit of the field for each trigger input.

4.19.2.23 CTTRIGOUTSTATUS (0x3003134)

The CTI Trigger Out Status Register provides the status of the cttrigout outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	TRIGOUTSTATUS	r	0x0	Shows the status of the cttrigout outputs. 1 = cttrigout is active. 0 = cttrigout is inactive. There is one bit of the field for each trigger output.

4.19.2.24 CTICHINSTATUS (0x3003138)

The CTI Channel In Status Register provides the status of the ctichin inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTICHINSTATUS	r	0x0	Shows the status of the ctichin inputs. 1 = ctichin is active. 0 = ctichin is inactive. Because the register provides a view of the raw ctichin inputs, the reset value is unknown. There is one bit of the field for each channel input.

4.19.2.25 CTICHOUTSTATUS (0x300313c)

The CTI Channel Out Status Register provides the status of the CTI ctichout outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTICHOUTSTATUS	r	0x0	Shows the status of the ctichout outputs. 1 = ctichout is active. 0 = ctichout is inactive. There is one bit of the field for each channel output.

4.19.2.26 CTIGATE (0x3003140)

The Gate Enable Register prevents the channels from propagating through the CTM to other CTIs. This enables local cross-triggering, for example for causing an interrupt when the ETM trigger occurs. It can be used effectively with CTIAPPSET, CTIAPPCLEAR, and CTIAPPPULSE for asserting trigger outputs by asserting channels, without affecting the rest of the system. On reset, this register is 0xF, and channel propagation is enabled.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3	CTIGATEEN3	rw	0x1	Enable CTICHOUT3. Set to 0 to disable channel propagation.
2	CTIGATEEN2	rw	0x1	Enable CTICHOUT2. Set to 0 to disable channel propagation.
1	CTIGATEEN1	rw	0x1	Enable CTICHOUT1. Set to 0 to disable channel propagation.
0	CTIGATEEN0	rw	0x1	Enable CTICHOUT0. Set to 0 to disable channel propagation.

4.19.2.27 ASICCTL (0x3003144)

Implementation-defined ASIC control, value written to the register is output on asicctl[7 : 0].

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	ASICCTL	rw	0x0	Implementation-defined ASIC control, value written to the register is output on asicctl[7 : 0]. If external multiplexing of trigger signals is implemented then the number of multiplexed signals on each trigger must be reflected within the Device ID Register. This is done within a Verilog define EXTMUXNUM.

4.19.2.28 ITCHINACK (0x3003edc)

This register is a write-only register. It can be used to set the value of the CTCHINACK outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTCHINACK	rw	0x0	Set the value of the CTCHINACK outputs.

4.19.2.29 ITTRIGINACK (0x3003ee0)

This register is a write-only register. It can be used to set the value of the CTTRIGINACK outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGINACK	rw	0x0	Set the value of the CTTRIGINACK outputs.

4.19.2.30 ITCHOUT (0x3003ee4)

This register is a write-only register. It can be used to set the value of the CTCHOUT outputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTCHOUT	rw	0x0	Set the value of the CTCHOUT outputs.

4.19.2.31 ITTRIGOUT (0x3003ee8)

This register is a write-only register. It can be used to set the value of the CTTRIGOUT outputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGOUT	rw	0x0	Set the value of the CTTRIGOUT outputs.

4.19.2.32 ITCHOUTACK (0x3003eec)

This register is a read-only register. It can be used to read the values of the CTCHOUTACK inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTCHOUTACK	r	0x0	Read the values of the CTCHOUTACK inputs.

4.19.2.33 ITTRIGOUTACK (0x3003ef0)

This register is a read-only register. It can be used to read the values of the CTTRIGOUTACK inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGOUTACK	r	0x0	Read the value of the CTTRIGOUTACK inputs.

4.19.2.34 ITCHIN (0x3003ef4)

This register is a read-only register. It can be used to read the values of the CTCHIN inputs.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CTCHIN	r	0x0	Read the value of the CTCHIN inputs.

4.19.2.35 ITTRIGIN (0x3003ef8)

This register is a read-only register. It can be used to read the values of the CTTRIGIN inputs.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	CTTRIGIN	r	0x0	Read the values of the CTTRIGIN inputs.

4.19.2.36 ITCTRL (0x3003f00)

This register is used to enable topology detection. For more information see the CoreSight Architecture Specification. This register enables the component to switch from a functional mode, the default behavior, to integration mode where the inputs and outputs of the component can be directly controlled for the purpose of integration testing and topology solving. Note : When a device has been in integration mode, it might not function with the original behavior. After performing integration or topology detection, you must reset the system to ensure correct behavior of CoreSight and other connected system components that are affected by the integration or topology detection.

Bit	Name	Type	Reset	Description
31:1	reserved	r	0x0	Reserved
0	Integration_mode	rw	0x0	Allows the component to switch from functional mode to integration mode or back.

4.19.2.37 CLAIMSET (0x3003fa0)

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CLAIMSET	rw	0x0	This claim tag bit is implemented

4.19.2.38 CLAIMCLR (0x3003fa4)

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

Bit	Name	Type	Reset	Description
31:4	reserved	r	0x0	Reserved
3:0	CLAIMCLR	rw	0x0	The value present reflects the current setting of the Claim Tag.

4.19.2.39 LAR (0x3003fb0)

This is used to enable write access to device registers. External accesses from a debugger (paddrdbg31 = 1) are not subject to the Lock Registers. A debugger does not have to unlock the component in order to write and modify the registers in the component.

Bit	Name	Type	Reset	Description
31:0	ACCESS_W	rw	0x0	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access.

4.19.2.40 LSR (0x3003fb4)

This indicates the status of the Lock control mechanism. This lock prevents accidental writes by code under debug. When locked, write access is blocked to all registers, except the Lock Access Register. External accesses from a debugger (paddrdbg31 = 1) are not subject to the Lock Registers. This register reads as 0 when read from an external debugger (paddrdbg31 = 1).

Bit	Name	Type	Reset	Description
31:3	reserved	r	0x0	Reserved
2	LOCKTYPE	r	0x0	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit
1	LOCKGRANT	r	0x1	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (paddrdbg31 = 1) since external debugger accesses are not subject to Lock Registers.
0	LOCKEXIST	r	0x1	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (paddrdbg31 = 1) since external debugger accesses are not subject to Lock Registers.

4.19.2.41 AUTHSTATUS (0x3003fb8)

Reports what functionality is currently permitted by the authentication interface.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:6	SNID	r	0x0	Indicates the security level for secure non-invasive debug
5:4	SID	r	0x0	Indicates the security level for secure invasive debug
3:2	NSNID	r	0x2	Indicates the security level for non-secure non-invasive debug
1:0	NSID	r	0x2	Indicates the security level for non-secure invasive debug

4.19.2.42 DEVID (0x3003fc8)

This register indicates the capabilities of the CTI.

Bit	Name	Type	Reset	Description
31:20	reserved	r	0x0	Reserved
19:16	NUMCH	r	0x4	Number of ECT channels available.
15:8	NUMTRIG	r	0x8	Number of ECT triggers available.
7:5	reserved	r	0x0	Reserved
4:0	EXTMUXNUM	r	0x0	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using asicctl. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog define EXTMUXNUM that the user must alter accordingly.

4.19.2.43 DEVTYPE (0x3003fcc)

It provides a debugger with information about the component when the Part Number field is not recognized. The debugger can then report this information.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	Sub_Type	r	0x1	Sub-classification within the major category
3:0	Major_Type	r	0x4	Major classification grouping for this debug/trace component

4.19.2.44 PERIPHID4 (0x3003fd0)

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	SIZE	r	0x0	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on.
3:0	DES_2	r	0x4	JEDEC continuation code indicating the designer of the component (along with the identity code)

4.19.2.45 PERIPHD0 (0x3003fe0)

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PART_0	r	0x6	Bits [7 : 0] of the component's part number. This is selected by the designer of the component.

4.19.2.46 PERIPHD1 (0x3003fe4)

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	DES_0	r	0x0	Bits 3 : 0 of the JEDEC identity code indicating the designer of the component (along with the continuation code)
3:0	PART_1	r	0x9	Bits [11 : 8] of the component's part number. This is selected by the designer of the component.

4.19.2.47 PERIPHD2 (0x3003fe8)

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	REVISION	r	0x4	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision.
3	JEDEC	r	0x1	Always set. Indicates that a JEDEC assigned value is used
2:0	DES_1	r	0x0	Bits 6 : 4 of the JEDEC identity code indicating the designer of the component (along with the continuation code)

4.19.2.48 PERIPHD3 (0x3003fec)

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	RELAND	r	0x0	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero.
3:0	CMOD	r	0x0	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero.

4.19.2.49 COMPID0 (0x3003ff0)

Reserved Reserved Reserved A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_0	r	0xd	Contains bits [7 : 0] of the component identification

4.19.2.50 COMPID1 (0x3003ff4)

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:4	CLASS	r	0x9	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15 : 12] of the component identification.
3:0	PRMBL_1	r	0x0	Contains bits [11 : 8] of the component identification

4.19.2.51 COMPID2 (0x3003ff8)

A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_2	r	0x5	Contains bits [23 : 16] of the component identification

4.19.2.52 COMPID3 (0x3003ffc)

A component identification register, that indicates that the identification registers are present.

Bit	Name	Type	Reset	Description
31:8	reserved	r	0x0	Reserved
7:0	PRMBL_3	r	0x0	Contains bits [31 : 24] of the component identification

4.20 MMR Region: el2_probe_aggr_mmr_slv

MMR definitions

4.20.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x3040000	AGGREGATOR_ID	
0x3040004	AGGREGATOR_CNTL	
0x3040008	AGGREGATOR_OWN	
0x3040020	CONT_READ_PORT0	
0x3040024	CONT_READ_ADDR0	
0x3040028	CONT_READ_DATA0	
0x3040030	CONT_READ_PORT1	
0x3040034	CONT_READ_ADDR1	
0x3040038	CONT_READ_DATA1	
0x3040040	CONT_READ_PORT2	
0x3040044	CONT_READ_ADDR2	
0x3040048	CONT_READ_DATA2	
0x3040050	CONT_READ_PORT3	
0x3040054	CONT_READ_ADDR3	
0x3040058	CONT_READ_DATA3	

4.20.2 Register Definitions

4.20.2.1 AGGREGATOR_ID (0x3040000)

This is the standard TI peripheral ID register that exists at address 0 in the peripheral space

Bit	Name	Type	Reset	Description
31:30	scheme	r	0x1	
29:28	bu	r	0x2	
27:16	function	r	0x281	
15:11	rtl_ver	r	0x0	
10:8	major_rev	r	0x1	
7:6	custom	r	0x0	
5:0	minor_rev	r	0x0	

4.20.2.2 AGGREGATOR_CNTL (0x3040004)

This register contains the controls for serial indirect access and trace processing

Bit	Name	Type	Reset	Description
31:11	rsvd1	r	0x0	
10:9	continuous_read_num	rw	0x0	
8	continuous_read_mode	rw	0x0	
7:1	rsvd0	r	0x0	
0	trace_en	rw	0x0	

4.20.2.3 AGGREGATOR_OWN (0x3040008)

Module Ownership Control and Status Register

Bit	Name	Type	Reset	Description
31:3	rsvd	r	0x0	
2:1	own	r	0x0	
0	claim	rw	0x0	

4.20.2.4 CONT_READ_PORT0 (0x3040020)

Designates which port (peripheral) is the target of this port, address, data register group

Bit	Name	Type	Reset	Description
31:5	rsvd	r	0x0	
4:0	port	rw	0x0	

4.20.2.5 CONT_READ_ADDR0 (0x3040024)

Designates the offset (in peripheral space) that is the target of this port, address, data register group

Bit	Name	Type	Reset	Description
31:16	rsvd	r	0x0	
15:0	addr	rw	0x0	

4.20.2.6 CONT_READ_DATA0 (0x3040028)

Provides the continuous read data for this port, address, data register group

Bit	Name	Type	Reset	Description
31:0	data	r	0x0	

4.20.2.7 CONT_READ_PORT1 (0x3040030)

Designates which port (peripheral) is the target of this port, address, data register group

Bit	Name	Type	Reset	Description
31:5	rsvd	r	0x0	
4:0	port	rw	0x0	

4.20.2.8 CONT_READ_ADDR1 (0x3040034)

Designates the offset (in peripheral space) that is the target of this port, address, data register group

Bit	Name	Type	Reset	Description
31:16	rsvd	r	0x0	
15:0	addr	rw	0x0	

4.20.2.9 CONT_READ_DATA1 (0x3040038)

Provides the continuous read data for this port, address, data register group

Bit	Name	Type	Reset	Description
31:0	data	r	0x0	

4.20.2.10 CONT_READ_PORT2 (0x3040040)

Designates which port (peripheral) is the target of this port, address, data register group

Bit	Name	Type	Reset	Description
31:5	rsvd	r	0x0	
4:0	port	rw	0x0	

4.20.2.11 CONT_READ_ADDR2 (0x3040044)

Designates the offset (in peripheral space) that is the target of this port, address, data register group

Bit	Name	Type	Reset	Description
31:16	rsvd	r	0x0	
15:0	addr	rw	0x0	

4.20.2.12 CONT_READ_DATA2 (0x3040048)

Provides the continuous read data for this port, address, data register group

Bit	Name	Type	Reset	Description
31:0	data	r	0x0	

4.20.2.13 CONT_READ_PORT3 (0x3040050)

Designates which port (peripheral) is the target of this port, address, data register group

Bit	Name	Type	Reset	Description
31:5	rsvd	r	0x0	
4:0	port	rw	0x0	

4.20.2.14 CONT_READ_ADDR3 (0x3040054)

Designates the offset (in peripheral space) that is the target of this port, address, data register group

Bit	Name	Type	Reset	Description
31:16	rsvd	r	0x0	
15:0	addr	rw	0x0	

4.20.2.15 CONT_READ_DATA3 (0x3040058)

Provides the continuous read data for this port, address, data register group

Bit	Name	Type	Reset	Description
31:0	data	r	0x0	

4.21 MMR Region: el2_probe_aggr_vbus_cfg_regs

4.21.1 Register Map

Address Offset	Register Mnemonic	Register Name
0x3040100	STP_TRACE_CONTROL	
0x3040104	STP_TRACE_ID	
0x3040110	STP_SYNC_CONTROL	
0x3040114	STP_FLUSH_CONTROL	
0x3040118	STP_FEATURES	

4.21.2 Register Definitions

4.21.2.1 STP_TRACE_CONTROL (0x3040100)

This register contains the control and status settings for STP Trace control register. MID_Fifofull indicates status of internal Initiator/Channel Fifo as full. Data_Fifofull indicates the status of internal Data Fifo as full. It is possible that if MID_Fifofull is full and when new Initiator_id is seen on STPMI i/f, the data will not be accepted as MID is full. It will wait for MID to be read out corresponding to Data.

Bit	Name	Type	Reset	Description
31:25	RSVD3	r	0x0	
24	MID_FIFO_FUL	r	0x0	
23	DATA_FIFO_FULL	r	0x0	
22:6	RSVD2	r	0x0	
5	COMPEN	rw	0x0	
4:3	RSVD1	r	0x0	
2	SNCEN	r	0x1	
1	TSEN	rw	0x0	
0	RSVD0	r	0x0	

4.21.2.2 STP_TRACE_ID (0x3040104)

This register contains the trace id register settings. This value is sampled from input and is exported on ATB interface as ATID field. This is usually programmed only when top-level configuration errors results in collision between ID's.

Bit	Name	Type	Reset	Description
31:7	RSVD	r	0x0	
6:0	TRACEID	rw	0x0	

4.21.2.3 STP_SYNC_CONTROL (0x3040110)

This register contains the periodic interval after which an ASYNC packet is exported over ATB interface. This counter register controls the interval between synchronization packets. The number of bytes between which a sync packet is sent based on this counter value. Writing a value of 0x0 disables the synchronization counter, however if the client has any other synchronization scheme, that will continue to operate.

When this register is written, STPMI2ATB will immediately perform synchronization and reset the counter value to the newly programmed value ensuring subsequent synchronization occurs in the period programmed.

This async requests will result in a Initiator switch. A async request will result in async packet followed by VERSION packet which is followed by a new M8/C8 packet before any Dx packet is sent.

Bit	Name	Type	Reset	Description
31:13	RSVD	r	0x0	
12	MODE	rw	0x1	
11:0	COUNT	rw	0x780	

4.21.2.4 STP_FLUSH_CONTROL (0x3040114)

This register contains the bits to indicate flush in STPMI2ATB. It also controls priority control for other conditions in STPMI2ATB.

Bit	Name	Type	Reset	Description
31:6	RSVD1	r	0x0	
5	FORCE_FLUSH	rw	0x0	
4:2	RSVD0	r	0x7	
1	ASYNC_PE	rw	0x0	
0	AUTO_FLUSH	rw	0x0	

4.21.2.5 STP_FEATURES (0x3040118)

This register contains the bits to indicate the features implemented in STPMI2ATB. It shows the VERSION packet implemented based on STP2.0 type of encoding timestamp packet. It also indicates what protocol is followed in the IP.

Bit	Name	Type	Reset	Description
31:7	RSVD	r	0x0	
6:4	STP_TS_VERSION	r	0x4	
3:0	PROT	r	0x1	

5. Integration Overview

5.1 Memories Embedded in the Peripheral

- Within EL2 Sub-system there are
 - 4 x EL2 memory banks, each 6MB
 - DRU rams
 - Queue rams
 - Response rams
 - State rams
 - Ring rams
 - CLEC routing table ram
 - Memory size and Macros are listed in section [5.3.1](#)
- Within Analytics Wrap there are
 - 2x UMC memory banks, each 1.5 MB
 - DMC (L1D\$) – 128KB
 - PMC (L1P\$) – 32KB
 - CPU branch predictor rams
 - CPU Look up table rams
 - MMU rams
 - DRU rams
 - Memory sizes and Macros are listed in section [5.4.1 of Analytics wrap spec](#)

5.2 PLL Requirements

- Design support PLL clock at 1.5GHz frequency
 - Target Frequency = 1.5 GHz @0.75V
 - Operation at Opp high 0.85v is also planned with same target frequency

5.3 Overall Design for Test

- PBIST is supported
 - All memories are repairable
 - DC path access is supported to help reduce test time
 - BISTOR is supported
- DFT requirement is only standard ATPG support
- LBIST is inserted at Analytics wrap level

#INTERNAL#

5.3.1 Memory PBIST table

IP	Memory	Size	Macro	PBIST uses functional/test path
EL2	el2_data_mem0	6384K Bytes	s2duls0g4l1p3072x133m4b4w0c1p1d1q1s1rm4rw11zh1h0ms1mg0 s2duls0g4l1p3072x133m4b4w0c1p1d1q1s1rm4rw11zh1h0ms1mg0	func
	el2_data_mem1	6384K Bytes	s2duls0g4l1p3072x133m4b4w0c1p1d1q1s1rm4rw11zh1h0ms1mg0 s2duls0g4l1p3072x133m4b4w0c1p1d1q1s1rm4rw11zh1h0ms1mg0	func

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	el2_data_mem2	6384K Bytes	s2duls0g4l1p3072x133m4b4w0c1p1d1q1s1rm4rw11zh1h0ms1mg0 s2duls0g4l1p3072x133m4b4w0c1p1d1q1s1rm4rw11zh1h0ms1mg0	
	el2_data_mem3	6384K Bytes	s2duls0g4l1p3072x133m4b4w0c1p1d1q1s1rm4rw11zh1h0ms1mg0 s2duls0g4l1p3072x133m4b4w0c1p1d1q1s1rm4rw11zh1h0ms1mg0	
DRU	dru_queue	6768 Bytes	s2crl2s0g4u2p96x113m4b1w1c1p1d1q1s1rm4rw10zh1h0ms1mg0 s2crl2s0g4u2p96x112m4b1w1c1p1d1q1s1rm4rw10zh1h0ms1mg0	func
	dru_resp	8512 Bytes	s2crl2s0g4u2p64x119m4b1w1c1p1d1q1s1rm4rw10zh1h0ms1mg0 s2crl2s0g4u2p64x112m4b1w1c1p1d1q1s1rm4rw10zh1h0ms1mg0	func
	dru_state	4256 Bytes	s2crl2s0g4u2p32x89m4b1w1c1p1d1q1s0rm4rw10zh1h0ms1mg0 s2crl2s0g4u2p32x88m4b1w1c1p1d1q1s0rm4rw10zh1h0ms1mg0	func
	dru_ring	1064 Bytes	s2crl2s0g4u2p32x89m4b1w1c1p1d1q1s0rm4rw10zh1h0ms1mg0 s2crl2s0g4u2p32x88m4b1w1c1p1d1q1s0rm4rw10zh1h0ms1mg0	func
CLEC	clec_ram	1792 Bytes	s2crls0g4l1p512x28m4b1w0c0p1d1q1r0rm4rw10zh1h0ms1mg0	test

#/INTERNAL#

5.4 Target Processes

- Target process is 5nm

#INTERNAL#

Area requirements are not needed in customer facing documents, Power numbers should come from PET tool and should be reported at SOC level, this is IP specific power requirements.

5.5 Area Requirements

Area Target is **38.66 mm²**

5.6 Power Requirements

- Target from PET tool

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- Leakage Power: 7 W
- Max Dynamic Power: 22 W
- Dynamic Idle Power: 3 W

#/INTERNAL#

5.7 Performance Requirements

- Performance requirement is 192 TOPS Dense.
- C7604 ISA supports 4:2 Sparse mode which effectively provides 2x TOPS, i.e. 384 Sparse TOPS

6. Safety

tda54_analytics_cluster supports ASIL-B safety. There is K3 ECC aggregator integrated in each Analytics wrap and in el2_wrap for error injection support and safety error reporting. Below is the mapping for tda54_el2_wrap ecc aggregator. Mapping for tda54_analytics_wrap is in analytics_wrap spec.

Figure 2 Safety view

6.1 ECC aggregator0 mapping (aggregator for clk1 end point)

ECC aggregator End point	Memo ry ID	Controller	Controller Type
EL2	0	c7xv_1024_el2_wrap_c7xv_el2_c7xv_el2_core_c7xv_el2_edc_ctl	CBA
EL2	1	c7xv_1024_el2_wrap_c7xv_el2_c7xv_el2_core_c7xv_el2_mem0_c7xv_el2_edc_ctl	CBA
EL2	2	c7xv_1024_el2_wrap_c7xv_el2_c7xv_el2_core_c7xv_el2_mem1_c7xv_el2_edc_ctl	CBA
EL2	3	c7xv_1024_el2_wrap_c7xv_el2_c7xv_el2_core_c7xv_el2_mem2_c7xv_el2_edc_ctl	CBA
EL2	4	c7xv_1024_el2_wrap_c7xv_el2_c7xv_el2_core_c7xv_el2_mem3_c7xv_el2_edc_ctl	CBA
CLEC	5	tda54_c7xv_clec_clec_sram_ecc_ctl	RAM WR AP
CFG_CBASS	6	tda54_el2_wrap_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_src_edc_ctl	CBA
CFG_CBASS	7	tda54_el2_wrap_cfg_cbass_cfg_cbass_el2_ecc_aggr_cfg_msmc_p2p_bridge_dst_edc_ctl	CBA
CFG_CBASS	8	tda54_el2_wrap_cfg_cbass_cfg_cbass_dru_mmr_cfg_p2p_bridge_dst_edc_ctl	CBA
CFG_CBASS	9	tda54_el2_wrap_cfg_cbass_cfg_cbass_el2_rat_cfg_p2p_bridge_dst_edc_ctl	CBA
EL2_RAT	10	tda54_el2_wrap_el2_rat_edc_ctl	CBA
DRU_C	11	tda54_el2_wrap_dru_c_wrap_dru_queue_cmd_edc_ctl	CBA
DRU_C	12	tda54_el2_wrap_dru_c_wrap_dru_rd_buffer_cmd_edc_ctl	CBA
DRU_C	13	tda54_el2_wrap_dru_c_wrap_dru_ch_logic_ring_mem_ring_edc_ctl	CBA
DRU_C	14	tda54_el2_wrap_dru_c_wrap_dru_cmd_edc_ctl	CBA
DRU_C	15	tda54_el2_wrap_dru_c_wrap_dru_rd_edc_ctl	CBA
DRU_C	16	tda54_el2_wrap_dru_c_wrap_dru_wi_cmd_edc_ctl	CBA
VBUSP_DMISC_A_SYNC	17	tda54_el2_wrap_el2_async_bridge_wrap_ac_vbusp_dmisc_p2p_bridge_dst_edc_ctl	CBA
FW_AGGR	18	tda54_el2_wrap_fwaggr_wrap_fw_aggr_cbass_scr1_scr_edc_ctl	CBA
FW_AGGR	19	tda54_el2_wrap_fwaggr_wrap_fw_aggr_cbass_scr1_scr_edc_ctl	CBA
FW_AGGR	20	tda54_el2_wrap_fwaggr_wrap_fw0_mmrs_edc_ctl	CBA
FW_AGGR	21	tda54_el2_wrap_fwaggr_wrap_fw1_mmrs_edc_ctl	CBA
FW_AGGR	22	tda54_el2_wrap_fwaggr_wrap_fw2_mmrs_edc_ctl	CBA
FW_AGGR	23	tda54_el2_wrap_fwaggr_wrap_fw3_mmrs_edc_ctl	CBA
FW_AGGR	24	tda54_el2_wrap_fwaggr_wrap_fw4_mmrs_edc_ctl	CBA
FW_AGGR	25	tda54_el2_wrap_fwaggr_wrap_fw5_mmrs_edc_ctl	CBA
FW_AGGR	26	tda54_el2_wrap_fwaggr_wrap_fw6_mmrs_edc_ctl	CBA
FW_AGGR	27	tda54_el2_wrap_fwaggr_wrap_fw7_mmrs_edc_ctl	CBA
FW_AGGR	28	tda54_el2_wrap_fwaggr_wrap_fw8_mmrs_edc_ctl	CBA
FW_AGGR	29	tda54_el2_wrap_fwaggr_wrap_fw9_mmrs_edc_ctl	CBA
FW_AGGR	30	tda54_el2_wrap_fwaggr_wrap_fw10_mmrs_edc_ctl	CBA

6.2 LBIST for safety

LBIST is implemented for C7x Core wrap and Analytics wrap hierarchies. Analytics wrap LBIST will be run on boot time while C7x core wrap LBIST can be run boot time or runtime requested by application host.

There is no LBIST implemented for MMA. For MMA, STL can provide required safety coverage.

On demand LBIST is supported for C7x core wrap, sequence to run on-demand LBIST is described below.

- Safety manager sends request for LBIST run to C7x through an interrupt
- C7x enters LBIST ISR where it saves identified context data to the memory, write back L1 caches and then goes into IDLE mode
- Safety manager monitors C7x status after LBIST request and initiate LBIST after C7x is IDLE
- LBIST runs and completes on C7x core
- Safety manager resets C7x core and removes LBIST isolation
- C7x core boots - boot code should either have a check to know this boot is from LBIST run OR can have separate boot address for post LBIST boot (this will require updating reset boot vector register value before resetting C7x core)
- in LBIST boot code, all saved context is restored and RETE instruction is executed to return to original execution state

#INTERNAL#

Above is recommendation from IP team, final run sequence will be part of Safety manual based on software and safety team's inputs.

#/INTERNAL#

6.3 STL for Safety

Software Test Libraries can provide additional safety coverage for CPU and MMA. System integrator can utilize combination of LBIST and STL support to achieve required ASIL level for a given application.

STL Userguide is provided with STL package which contains details about STL coverage and recommended usage.

6.4 Safety for MMR

Below is the summary of safety diagnostics applied to MMRs within Analytics Cluster.

Diagnostic Type	Covered MMRs
Active Parity	UMC, EL2 MMRs, firewall registers
ECC	CLEC MRR registers
CPU STL	CPU MMRs
MMA STL	MMA MMRs

Software readback of static configuration	L1D, L1P, MMU, SE ECR registers, CLEC static configuration registers (other than MRR), RAT MMRs
---	---

Other non-safety related registers are ECC aggregator registers, Debug registers.

6.5 Safety on Interfaces

Diagnostic Type	Covered MMRs
Parity on command attributes	All UMC to EL2 interfaces, DRU to EL2 and MSMC to EL2 interfaces. CLEC CFG and DRU CFG interface
Redundancy on command attributes	
ECC on read/write data	

7. Security

- In ML applications, ML data (primarily weights) needs to have confidentiality aspect end-to-end
- Analytics Cluster (AC) can run software with different security level, i.e. each C7x can be running software coming from different supplier and even open source.
- Need isolation within Analytics cluster to allow protection of code/data from untrusted-C7x core running within C7x cluster
- Need isolation outside Analytics Cluster to allow protection of code/data from untrusted-ARM core outside
- To achieve this shared 24MB EL2 memory and all memory regions inside analytics wrap are protected by end point firewall.

7.1 Firewall aggregator

To implement end point firewall for EL2 and L2/CFG accesses coming from other cores, firewall aggregator IP has been integrated inside tda54_el2_wrap.

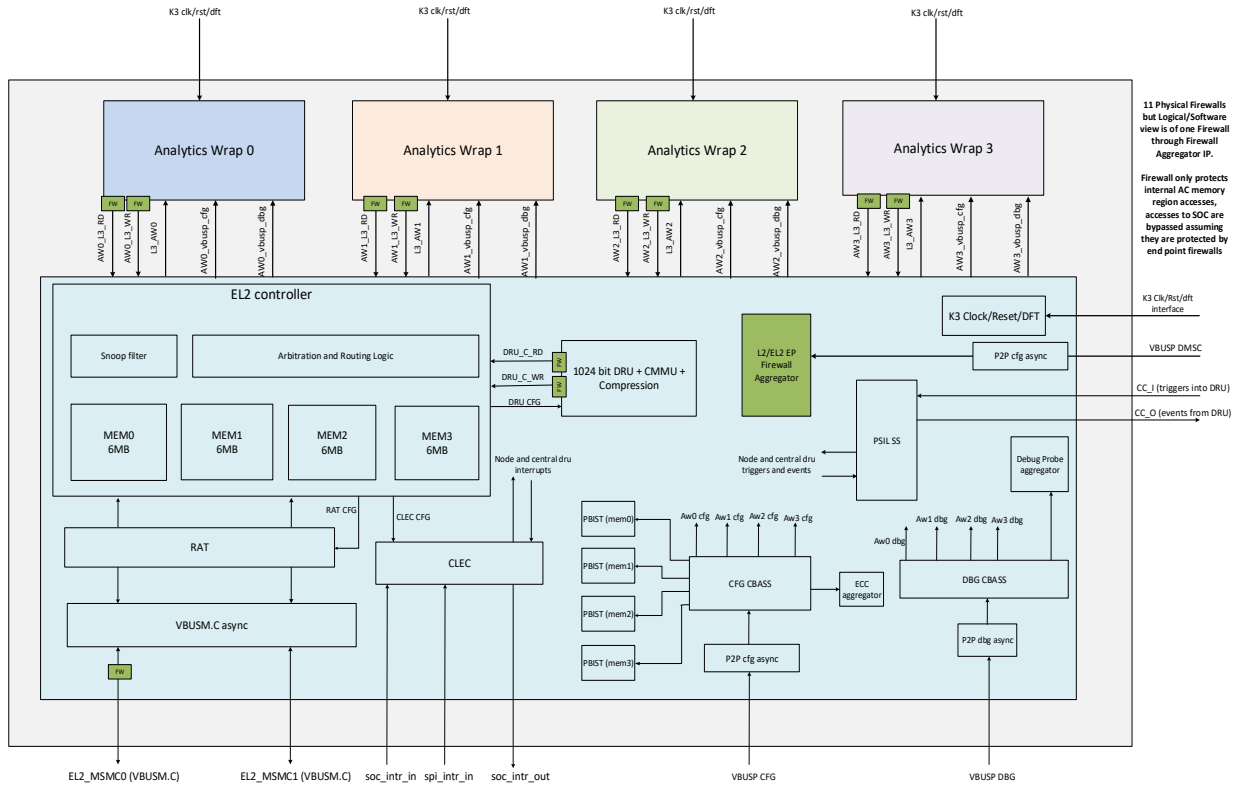
This IP provides physical design friendly implementation for end point firewall, where even if firewalls are placed locally at the interface level, they all are logically same so software view is still one end point firewall.

This IP broadcasts firewall configuration to all firewall instances and arbitrates status reads based on pending exception status or based on tieoff when none of the exception is present.

Below is the table mapping of EL2 controller interface and firewall instance inside firewall aggregator wrapper.

All Firewalls are in ELSS Power domain, so they don't need to be re-programmed when we power cycle analytics wrap Power domains. When we power cycle EL2 wrap, these firewalls need to be re-programmed

Interface	Firewall instance
umc0_el2_rd	I_fw0
umc0_el2_wr	I_fw1
umc1_el2_rd	I_fw2
umc1_el2_wr	I_fw3
umc2_el2_rd	I_fw4
umc2_el2_wr	I_fw5
umc3_el2_rd	I_fw6
umc3_el2_wr	I_fw7
dru_el2_rd	I_fw8
dru_el2_wr	I_fw9
msmc0_el2	I_fw10



8. C7xv_wrap Debug

- Refer to [compute cluster debug spec](#) and [Columbo](#), [Matlock](#), [Thinman](#) spec for debug features supported in C7x CPU.
- Below is the Debug view diagram for Analytics Cluster
- Analytics wrap spec has more details about debug architecture inside tda54_analytics_wrap subchip

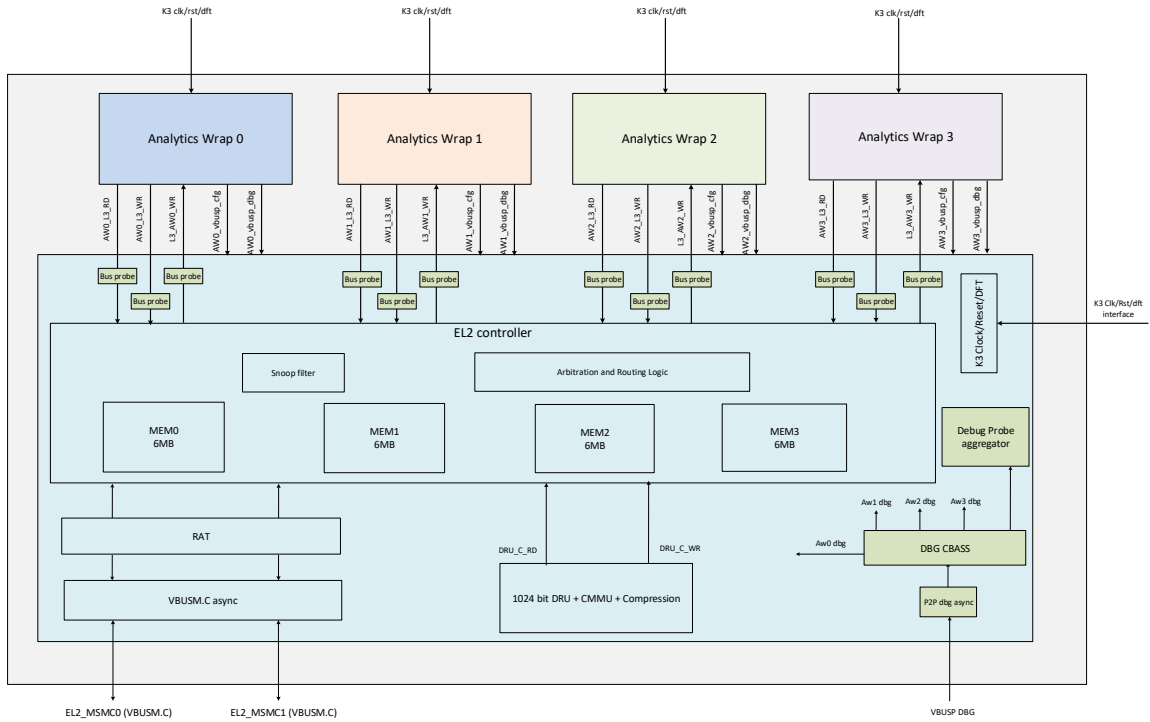


Figure 4 Debug view

8.1 CPTracer2 instance monitoring point

PROBE#	SHORT	DESCRIPTION	Interface Type	Probe Type
0	AW0_EL2_RD	AW0 to EL2 read Initiator probe	VBUSUL	cpt2_probe_vbusm_8
1	AW0_EL2_WR	AW0 to EL2 write Initiator probe	VBUSUL	cpt2_probe_vbusm_8
2	EL2_AW0	EL2 to AW0 initiator probe	VBUSUL	cpt2_probe_vbusm_8
3	AW1_EL2_RD	AW1 to EL2 read Initiator probe	VBUSUL	cpt2_probe_vbusm_8
4	AW1_EL2_WR	AW1 to EL2 write Initiator probe	VBUSUL	cpt2_probe_vbusm_8
5	EL2_AW1	EL2 to AW1 initiator probe	VBUSUL	cpt2_probe_vbusm_8
6	AW2_EL2_RD	AW2 to EL2 read Initiator probe	VBUSUL	cpt2_probe_vbusm_8
7	AW2_EL2_WR	AW2 to EL2 write Initiator probe	VBUSUL	cpt2_probe_vbusm_8
8	EL2_AW2	EL2 to AW2 initiator probe	VBUSUL	cpt2_probe_vbusm_8
9	AW3_EL2_RD	AW3 to EL2 read Initiator probe	VBUSUL	cpt2_probe_vbusm_8
10	AW3_EL2_WR	AW3 to EL2 write Initiator probe	VBUSUL	cpt2_probe_vbusm_8

11	EL2_AW3	EL2 to AW3 initiator probe	VBUSUL	cpt2_probe_vbusm_8
12 - 31	-	Reserved (not used)	NA	NA

#INTERNAL#

I/O information is needed for SOC integration and not needed for customer facing documents.

9. I/O Description

9.1 Clock ports

Clock name	Frequency	Is this a dft clock	Does this go to hard macro	Comment
aw0_c7xv_clk_clk	1.5 GHz	No	No	Functional clock for analytics wrap 0 synchronous with EL2 wrap and other analytics wraps
aw0_pll_ctrl_main_clk_clk	500 MHz	Yes	No	DFT clock for analytics wrap 0
aw0_pbist_tclk_clk		Yes	No	pbist clock for analytics wrap 0 (UMC, DRU rams)
aw0_c7xv_core_pbist_tclk_clk		Yes	No	pbist clock for analytics core wrap 0 (CPU/L1 rams)
aw1_c7xv_clk_clk	1.5 GHz	No	No	Functional clock for analytics wrap 1 synchronous with EL2 wrap and other analytics wraps
aw1_pll_ctrl_main_clk_clk	500 MHz	Yes	No	DFT clock for analytics wrap 1
aw1_pbist_tclk_clk		Yes	No	pbist clock for analytics wrap 1 (UMC, DRU rams)
aw1_c7xv_core_pbist_tclk_clk		Yes	No	pbist clock for analytics core wrap 1 (CPU/L1 rams)
aw2_c7xv_clk_clk	1.5 GHz	No	No	Functional clock for analytics wrap 2 synchronous with EL2 wrap and other analytics wraps
aw2_pll_ctrl_main_clk_clk	500 MHz	Yes	No	DFT clock for analytics wrap 2
aw2_pbist_tclk_clk		Yes	No	pbist clock for analytics wrap 2 (UMC, DRU rams)
aw2_c7xv_core_pbist_tclk_clk		Yes	No	pbist clock for analytics core wrap 2 (CPU/L1 rams)
aw3_c7xv_clk_clk	1.5 GHz	No	No	Functional clock for analytics wrap 3 synchronous with EL2 wrap and other analytics wraps
aw3_pll_ctrl_main_clk_clk	500 MHz	Yes	No	DFT clock for analytics wrap 3
aw3_pbist_tclk_clk		Yes	No	pbist clock for analytics wrap 3 (UMC, DRU rams)
aw3_c7xv_core_pbist_tclk_clk		Yes	No	pbist clock for analytics core wrap 3 (CPU/L1 rams)
el2_clk_clk	1.5 GHz	No	No	Functional clock for EL2 wrap (also called as EL2 wrap) synchronous all

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				analytics wraps. – async with MSMC (1.2GHz)
el2_pll_ctrl_main_clk_clk	500 MHz	Yes	No	DFT clock for EL2 wrap
el2_0_pbist_tclk_clk		Yes	No	EL2 PBIST controller 0 clock
el2_1_pbist_tclk_clk		Yes	No	EL2 PBIST controller 1 clock
el2_2_pbist_tclk_clk		Yes	No	EL2 PBIST controller 2 clock
el2_3_pbist_tclk_clk		Yes	No	EL2 PBIST controller 3 clock
msmc_clk	1.2 GHz	No	No	EL2 to MSMC VBUSM.C interface clock (Async to all other clocks)

9.2 Clock matrix table

Skipping DFT clocks from this matrix as they are assumed asynchronous to functional PLL clocks

Clock matrix	aw0_c7xv_clk_clk	aw1_c7xv_clk_clk	aw2_c7xv_clk_clk	aw3_c7xv_clk_clk	el2_clk_clk	msmc_clk
aw0_c7xv_clk_clk		Sync	Sync	Sync	Sync	Async
aw1_c7xv_clk_clk	Sync		Sync	Sync	Sync	Async
aw2_c7xv_clk_clk	Sync	Sync		Sync	Sync	Async
aw3_c7xv_clk_clk	Sync	Sync	Sync		Sync	Async
el2_clk_clk	Sync	Sync	Sync	Sync		Async
msmc_clk	Async	Async	Async	Async	Async	

9.3 Module I/O

#/INTERNAL#

9.4 Module I/O

The following table provides a summary of the interfaces on this module:

Interface Name	Description	Clock	Reset
aw0_ccc_aw			
aw1_ccc_aw			
aw2_ccc_aw			
aw3_ccc_aw			
ac_vbusp_cfg		el2_clk_clk	el2_wrap_rst_mod_g_rst_n
ac_vbusp_dbg		el2_clk_clk	el2_wrap_rst_mod_g_rst_n
aw0_aw_ccc			
aw0_c7xv_wrap_tieoff_clusternum			
aw0_c7xv_wrap_tieoff_corepacnum			
aw0_c7xv_wrap_tieoff_sys_mma_present			
aw0_spare	spare ports for PD		
aw0_pbist_clkstop		async	aw0_pbist_rst_mod_g_rst_n
aw0_c7xv_pwr	Power interface	async	
aw0_dft_compression_core	dft_compression	async	
aw0_c7xv_core_pwr	Power interface	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_core_testrwm	MEM_TESTRWM busdef	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_core_test1	MEM_TEST1 busdef	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_glbdbg_timestamp	Debug Timestamp Interface	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dbg_rst		async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dbg_c7xv_state		async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dbg_c7xv_susp		async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dbg_sec	ARM Trust Zone Interface	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_debug		async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dbg_chm		async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dbg_chs		async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_mem_atpg	DFT mem interface	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dftmem	MEM_MARGIN busdef	async	
aw0_dftmem_testrwm	MEM_TESTRWM busdef	async	
aw0_dftmem_dftclken	MEM_DFTCLKEN busdef	async	
aw0_dftmem_test1	MEM_TEST1 busdef	async	
aw0_dftmem_testnrm	MEM_TEST_RNM busdef	async	
aw0_pwr_bias	PWR BIAS busdef	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_efuse_ctl	efuse_ctl	async	
aw0_efuse_data	efuse_data	async	
aw0_c7xv_core_efuse_ctl	efuse_ctl	async	
aw0_c7xv_core_efuse_data	efuse_data	async	
aw0_divh_clk1_dft	divh_clk1_tft_clk_en control	async	
aw0_divh_clk4_dft	divh_clk4_tft_clk_en control	async	
aw0_divh_clk1_core_dft	divh_clk1_core_tft_clk_en control	async	
aw0_c7xv_core_clkstop	PMU Clock stp req/ack interface	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_wrap_clkstop	PMU Clock stp req/ack interface	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_clk	c7xv_clk1 clock	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_divh_clk4_obscik_out	c7xv_clk1 clock	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_pll_ctrl_main_clk	pll_ctrl clock	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dft	dft_ip	async	
aw0_dft_core	dft_core ip interface for unique clk force en control	async	
aw0_dft_mma	dft_core ip interface for unique clk force en control	async	
aw0_dft_compression_mma	dft_compression	async	
aw0_pll_ctrl_align	PLL control align signals	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dft_compression	dft_compression	async	

aw0_dft_pll_ctrl	Controls for the TFT Clock control module	async	
aw0_dft_pll_clk_bypass	DFT PLL CLK BYPASS control for the TFT Clock control module	async	
aw0_c7xv_clk_ctl	c7xv clk1 clock	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_core_clken	Clken interface	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_wrap_clken	Clken interface	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dft_local_div_clk_en	DFT clock clken control for the Dividers	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_core_rst		aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_wrap_rst		aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_mma_pwr	Power interface	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
dft_aw0_clk_bypass	EL2 Clk grid disable control	async	
dft_aw0_dft_clk_en	EL2 Clk grid disable control	async	
dft_aw0_divfactor_sel	EL2 Clk grid disable control	async	
dft_aw0_core_clk4_divfactor_sel	EL2 Clk grid disable control	async	
dft_aw0_core_clk4_disable	Clk4 clock gate disable control for dft mode	async	
dft_aw0_scrub_off	EL2 Clk grid disable control	el2_clk_clk	
aw0_c7xv_pbist_disable_scr	Target .vbusp_pbist_cfg power disable	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dft_embed_pbist		aw0_pbist_tclk_clk	aw0_pbist_rst_mod_g_rst_n
aw0_rom_ase	rom_ase	async	
aw0_dft_pbist_cpu		aw0_c7xv_clk_clk	aw0_pbist_rst_mod_g_rst_n
aw0_dft_pbist_safety_error		aw0_c7xv_clk_clk	aw0_pbist_rst_mod_g_rst_n
aw0_dft_pbist_pwr		async	aw0_pbist_rst_mod_g_rst_n
aw0_pbist_tclk		async	
aw0_pbist_rst		async	
aw0_dft_rom	EFUSE interface	async	
aw0_core_pbist_clkstop		async	aw0_pbist_rst_mod_g_rst_n
aw0_dft_lbist	dft_lbist	async	
aw0_dft_lbist_ctrl_id	dft_lbist_ctrl_id	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_dft_iso_en	dft_iso_en	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_core_dft_lbist	dft_lbist	async	
aw0_core_dft_lbist_ctrl_id	dft_lbist_ctrl_id	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_core_dft_rom	EFUSE interface	async	
aw0_core_dft_iso_en	dft_iso_en	async	
aw0_c7xv_core_bisor	bisor	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_core_pbist_disable_scr	Target .vbusp_core_pbist_cfg power disable	async	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_bisor	bisor	async	
aw0_c7xv_core_dft_embed_pbist		aw0_c7xv_core_pbist_tclk_clk	aw0_c7xv_core_pbist_rst_mod_g_rst_n
aw0_c7xv_core_dft_pbist_cpu		aw0_c7xv_clk_clk	aw0_c7xv_core_pbist_rst_mod_g_rst_n
aw0_c7xv_core_dft_pbist_safety_error		aw0_c7xv_clk_clk	aw0_c7xv_core_pbist_rst_mod_g_rst_n
aw0_c7xv_core_dft_pbist_pwr		aw0_c7xv_clk_clk	aw0_c7xv_core_pbist_rst_mod_g_rst_n
aw0_c7xv_core_pbist_tclk		aw0_c7xv_clk_clk	aw0_c7xv_core_rst_mod_g_rst_n
aw0_c7xv_core_pbist_rst		aw0_c7xv_clk_clk	aw0_c7xv_core_rst_mod_g_rst_n
aw0_c7xv_core_pwr_dis	Power SCR Disable Interface	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_mma_clkstop	PMU Clock stp req/ack interface	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_mma_clken	Clken interface	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_c7xv_mma_rst		aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_mma_r00	dft_compression for mma_array_r00 block	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_mma_r01	dft_compression for mma_array_r01 block	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_mma_r10	dft_compression for mma_array_r10 block	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n

aw0_mma_r11	dft_compression for mma_array_r11 block	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_r00_pwr	Power interface for mma_array_r00 block	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_r01_pwr	Power interface for mma_array_r01 block	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_r10_pwr	Power interface for mma_array_r10 block	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw0_r11_pwr	Power interface for mma_array_r11 block	aw0_c7xv_clk_clk	aw0_c7xv_wrap_rst_mod_g_rst_n
aw1_aw_ccc			
aw1_c7xv_wrap_tieoff_clusternum			
aw1_c7xv_wrap_tieoff_corepacnum			
aw1_c7xv_wrap_tieoff_sys_mma_present			
aw1_spare	spare ports for PD		
aw1_pbist_clkstop		async	aw1_pbist_rst_mod_g_rst_n
aw1_c7xv_pwr	Power interface	async	
aw1_dft_compression_core	dft_compression	async	
aw1_c7xv_core_pwr	Power interface	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_core_testrwm	MEM_TESTRWM busdef	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_core_test1	MEM_TEST1 busdef	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_glbdbg_timestamp	Debug Timestamp Interface	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dbg_rst		async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dbg_c7xv_state		async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dbg_c7xv_susp		async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dbg_sec	ARM Trust Zone Interface	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_debug		async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dbg_chm		async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dbg_chs		async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_mem_atpg	DFT mem interface	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dftmem	MEM_MARGIN busdef	async	
aw1_dftmem_testrwm	MEM_TESTRWM busdef	async	
aw1_dftmem_dftclken	MEM_DFTCLKEN busdef	async	
aw1_dftmem_test1	MEM_TEST1 busdef	async	
aw1_dftmem_testnrm	MEM_TEST_RNM busdef	async	
aw1_pwr_bias	PWR BIAS busdef	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_efuse_ctl	efuse_ctl	async	
aw1_efuse_data	efuse_data	async	
aw1_c7xv_core_efuse_ctl	efuse_ctl	async	
aw1_c7xv_core_efuse_data	efuse_data	async	
aw1_divh_clk1_dft	divh_clk1_tft_clk_en control	async	
aw1_divh_clk4_dft	divh_clk4_tft_clk_en control	async	
aw1_divh_clk1_core_dft	divh_clk1_core_tft_clk_en control	async	
aw1_c7xv_core_clkstop	PMU Clock stp req/ack interface	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_wrap_clkstop	PMU Clock stp req/ack interface	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_clk	c7xv_clk1 clock	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_divh_clk4_obsclk_out	c7xv_clk1 clock	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_pll_ctrl_main_clk	pll_ctrl clock	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dft	dft_ip	async	
aw1_dft_core	dft_core ip interface for unique clk force en control	async	
aw1_dft_mma	dft_core ip interface for unique clk force en control	async	
aw1_dft_compression_mma	dft_compression	async	
aw1_pll_ctrl_align	PLL control align signals	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dft_compression	dft_compression	async	
aw1_dft_pll_ctrl	Controls for the TFT Clock control module	async	

aw1_dft_pll_clk_bypass	DFT PLL CLK BYPASS control for the TFT Clock control module	async	
aw1_c7xv_clk_ctl	c7xv clk1 clock	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_core_clken	Clken interface	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_wrap_clken	Clken interface	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dft_local_div_clk_en	DFT clock clken control for the Dividers	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_core_rst		aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_wrap_rst		aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_mma_pwr	Power interface	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
dft_aw1_clk_bypass	EL2 Clk grid disable control	async	
dft_aw1_dft_clk_en	EL2 Clk grid disable control	async	
dft_aw1_divfactor_sel	EL2 Clk grid disable control	async	
dft_aw1_core_clk4_divfactor_sel	EL2 Clk grid disable control	async	
dft_aw1_core_clk4_disable	Clk4 clock gate disable control for dft mode	async	
dft_aw1_scrub_off	EL2 Clk grid disable control	el2_clk_clk	
aw1_c7xv_pbist_disable_scr	Target .vbusp_pbist_cfg power disable	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dft_embed_pbist		aw1_pbist_tclk_clk	aw1_pbist_rst_mod_g_rst_n
aw1_rom_ase	rom_ase	async	
aw1_dft_pbist_cpu		aw1_c7xv_clk_clk	aw1_pbist_rst_mod_g_rst_n
aw1_dft_pbist_safety_error		aw1_c7xv_clk_clk	aw1_pbist_rst_mod_g_rst_n
aw1_dft_pbist_pwr		async	aw1_pbist_rst_mod_g_rst_n
aw1_pbist_tclk		async	
aw1_pbist_rst		async	
aw1_dft_rom	EFUSE interface	async	
aw1_core_pbist_clkstop		async	aw1_pbist_rst_mod_g_rst_n
aw1_dft_lbist	dft_lbist	async	
aw1_dft_lbist_ctrl_id	dft_lbist_ctrl_id	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_dft_iso_en	dft_iso_en	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_core_dft_lbist	dft_lbist	async	
aw1_core_dft_lbist_ctrl_id	dft_lbist_ctrl_id	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_core_dft_rom	EFUSE interface	async	
aw1_core_dft_iso_en	dft_iso_en	async	
aw1_c7xv_core_bisor	bisor	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_core_pbist_disable_scr	Target .vbusp_core_pbist_cfg power disable	async	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_bisor	bisor	async	
aw1_c7xv_core_dft_embed_pbist		aw1_c7xv_core_pbist_tclk_clk	aw1_c7xv_core_pbist_rst_mod_g_rst_n
aw1_c7xv_core_dft_pbist_cpu		aw1_c7xv_clk_clk	aw1_c7xv_core_pbist_rst_mod_g_rst_n
aw1_c7xv_core_dft_pbist_safety_error		aw1_c7xv_clk_clk	aw1_c7xv_core_pbist_rst_mod_g_rst_n
aw1_c7xv_core_dft_pbist_pwr		aw1_c7xv_clk_clk	aw1_c7xv_core_pbist_rst_mod_g_rst_n
aw1_c7xv_core_pbist_tclk		aw1_c7xv_clk_clk	aw1_c7xv_core_rst_mod_g_rst_n
aw1_c7xv_core_pbist_rst		aw1_c7xv_clk_clk	aw1_c7xv_core_rst_mod_g_rst_n
aw1_c7xv_core_pwr_dis	Power SCR Disable Interface	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_mma_clkstop	PMU Clock stp req/ack interface	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_mma_clken	Clken interface	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_c7xv_mma_rst		aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_mma_r00	dft_compression for mma_array_r00 block	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_mma_r01	dft_compression for mma_array_r01 block	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_mma_r10	dft_compression for mma_array_r10 block	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_mma_r11	dft_compression for mma_array_r11 block	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n

aw1_r00_pwr	Power interface for mma_array_r00 block	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_r01_pwr	Power interface for mma_array_r01 block	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_r10_pwr	Power interface for mma_array_r10 block	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw1_r11_pwr	Power interface for mma_array_r11 block	aw1_c7xv_clk_clk	aw1_c7xv_wrap_rst_mod_g_rst_n
aw2_aw_ccc			
aw2_c7xv_wrap_tieoff_clusternum			
aw2_c7xv_wrap_tieoff_corepacnum			
aw2_c7xv_wrap_tieoff_sys_mma_present			
aw2_spare	spare ports for PD		
aw2_pbist_clkstop		async	aw2_pbist_rst_mod_g_rst_n
aw2_c7xv_pwr	Power interface	async	
aw2_dft_compression_core	dft_compression	async	
aw2_c7xv_core_pwr	Power interface	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_core_testrwm	MEM_TESTRWM busdef	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_core_test1	MEM_TEST1 busdef	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_glbdbg_timestamp	Debug Timestamp Interface	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dbg_rst		async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dbg_c7xv_state		async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dbg_c7xv_susp		async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dbg_sec	ARM Trust Zone Interface	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_debug		async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dbg_chm		async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dbg_chs		async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_mem_atpg	DFT mem interface	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dftmem	MEM_MARGIN busdef	async	
aw2_dftmem_testrwm	MEM_TESTRWM busdef	async	
aw2_dftmem_dftclken	MEM_DFTCLKEN busdef	async	
aw2_dftmem_test1	MEM_TEST1 busdef	async	
aw2_dftmem_testrnm	MEM_TEST_RNM busdef	async	
aw2_pwr_bias	PWR_BIAS busdef	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_efuse_ctl	efuse_ctl	async	
aw2_efuse_data	efuse_data	async	
aw2_c7xv_core_efuse_ctl	efuse_ctl	async	
aw2_c7xv_core_efuse_data	efuse_data	async	
aw2_divh_clk1_dft	divh_clk1_tft_clk_en control	async	
aw2_divh_clk4_dft	divh_clk4_tft_clk_en control	async	
aw2_divh_clk1_core_dft	divh_clk1_core_tft_clk_en control	async	
aw2_c7xv_core_clkstop	PMU Clock stp req/ack interface	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_wrap_clkstop	PMU Clock stp req/ack interface	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_clk	c7xv_clk1 clock	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_divh_clk4_obsclk_out	c7xv_clk1 clock	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_pll_ctrl_main_clk	pll_ctrl clock	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dft	dft_ip	async	
aw2_dft_core	dft_core ip interface for unique clk force en control	async	
aw2_dft_mma	dft_core ip interface for unique clk force en control	async	
aw2_dft_compression_mma	dft_compression	async	
aw2_pll_ctrl_align	PLL control align signals	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dft_compression	dft_compression	async	
aw2_dft_pll_ctrl	Controls for the TFT Clock control module	async	

aw2_dft_pll_clk_bypass	DFT PLL CLK BYPASS control for the TFT Clock control module	async	
aw2_c7xv_clk_ctl	c7xv clk1 clock	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_core_clken	Clken interface	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_wrap_clken	Clken interface	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dft_local_div_clk_en	DFT clock clken control for the Dividers	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_core_rst		aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_wrap_rst		aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_mma_pwr	Power interface	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
dft_aw2_clk_bypass	EL2 Clk grid disable control	async	
dft_aw2_dft_clk_en	EL2 Clk grid disable control	async	
dft_aw2_divfactor_sel	EL2 Clk grid disable control	async	
dft_aw2_core_clk4_divfactor_sel	EL2 Clk grid disable control	async	
dft_aw2_core_clk4_disable	Clk4 clock gate disable control for dft mode	async	
dft_aw2_scrub_off	EL2 Clk grid disable control	el2_clk_clk	
aw2_c7xv_pbist_disable_scr	Target .vbusp_pbist_cfg power disable	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dft_embed_pbist		aw2_pbist_tclk_clk	aw2_pbist_rst_mod_g_rst_n
aw2_rom_ase	rom_ase	async	
aw2_dft_pbist_cpu		aw2_c7xv_clk_clk	aw2_pbist_rst_mod_g_rst_n
aw2_dft_pbist_safety_error		aw2_c7xv_clk_clk	aw2_pbist_rst_mod_g_rst_n
aw2_dft_pbist_pwr		async	aw2_pbist_rst_mod_g_rst_n
aw2_pbist_tclk		async	
aw2_pbist_rst		async	
aw2_dft_rom	EFUSE interface	async	
aw2_core_pbist_clkstop		async	aw2_pbist_rst_mod_g_rst_n
aw2_dft_lbist	dft_lbist	async	
aw2_dft_lbist_ctrl_id	dft_lbist_ctrl_id	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_dft_iso_en	dft_iso_en	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_core_dft_lbist	dft_lbist	async	
aw2_core_dft_lbist_ctrl_id	dft_lbist_ctrl_id	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_core_dft_rom	EFUSE interface	async	
aw2_core_dft_iso_en	dft_iso_en	async	
aw2_c7xv_core_bisor	bisor	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_core_pbist_disable_scr	Target .vbusp_core_pbist_cfg power disable	async	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_bisor	bisor	async	
aw2_c7xv_core_dft_embed_pbist		aw2_c7xv_core_pbist_tclk_clk	aw2_c7xv_core_pbist_rst_mod_g_rst_n
aw2_c7xv_core_dft_pbist_cpu		aw2_c7xv_clk_clk	aw2_c7xv_core_pbist_rst_mod_g_rst_n
aw2_c7xv_core_dft_pbist_safety_error		aw2_c7xv_clk_clk	aw2_c7xv_core_pbist_rst_mod_g_rst_n
aw2_c7xv_core_dft_pbist_pwr		aw2_c7xv_clk_clk	aw2_c7xv_core_pbist_rst_mod_g_rst_n
aw2_c7xv_core_pbist_tclk		aw2_c7xv_clk_clk	aw2_c7xv_core_rst_mod_g_rst_n
aw2_c7xv_core_pbist_rst		aw2_c7xv_clk_clk	aw2_c7xv_core_rst_mod_g_rst_n
aw2_c7xv_core_pwr_dis	Power SCR Disable Interface	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_mma_clkstop	PMU Clock stp req/ack interface	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_mma_clken	Clken interface	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_c7xv_mma_rst		aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_mma_r00	dft_compression for mma_array_r00 block	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_mma_r01	dft_compression for mma_array_r01 block	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_mma_r10	dft_compression for mma_array_r10 block	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_mma_r11	dft_compression for mma_array_r11 block	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n

aw2_r00_pwr	Power interface for mma_array_r00 block	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_r01_pwr	Power interface for mma_array_r01 block	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_r10_pwr	Power interface for mma_array_r10 block	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw2_r11_pwr	Power interface for mma_array_r11 block	aw2_c7xv_clk_clk	aw2_c7xv_wrap_rst_mod_g_rst_n
aw3_aw_ccc			
aw3_c7xv_wrap_tieoff_clusternum			
aw3_c7xv_wrap_tieoff_corepacnum			
aw3_c7xv_wrap_tieoff_sys_mma_present			
aw3_spare	spare ports for PD		
aw3_pbist_clkstop		async	aw3_pbist_rst_mod_g_rst_n
aw3_c7xv_pwr	Power interface	async	
aw3_dft_compression_core	dft_compression	async	
aw3_c7xv_core_pwr	Power interface	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_core_testrwm	MEM_TESTRWM busdef	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_core_test1	MEM_TEST1 busdef	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_glbdbg_timestamp	Debug Timestamp Interface	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dbg_rst		async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dbg_c7xv_state		async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dbg_c7xv_susp		async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dbg_sec	ARM Trust Zone Interface	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_debug		async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dbg_chm		async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dbg_chs		async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_mem_atpg	DFT mem interface	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dftmem	MEM_MARGIN busdef	async	
aw3_dftmem_testrwm	MEM_TESTRWM busdef	async	
aw3_dftmem_dftclken	MEM_DFTCLKEN busdef	async	
aw3_dftmem_test1	MEM_TEST1 busdef	async	
aw3_dftmem_testrnm	MEM_TEST_RNM busdef	async	
aw3_pwr_bias	PWR BIAS busdef	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_efuse_ctl	efuse_ctl	async	
aw3_efuse_data	efuse_data	async	
aw3_c7xv_core_efuse_ctl	efuse_ctl	async	
aw3_c7xv_core_efuse_data	efuse_data	async	
aw3_divh_clk1_dft	divh_clk1_tft_clk_en control	async	
aw3_divh_clk4_dft	divh_clk4_tft_clk_en control	async	
aw3_divh_clk1_core_dft	divh_clk1_core_tft_clk_en control	async	
aw3_c7xv_core_clkstop	PMU Clock stp req/ack interface	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_wrap_clkstop	PMU Clock stp req/ack interface	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_clk	c7xv_clk1 clock	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_divh_clk4_obsclk_out	c7xv_clk1 clock	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_pll_ctrl_main_clk	pll_ctrl clock	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dft	dft_ip	async	
aw3_dft_core	dft_core ip interface for unique clk force en control	async	
aw3_dft_mma	dft_core ip interface for unique clk force en control	async	
aw3_dft_compression_mma	dft_compression	async	
aw3_pll_ctrl_align	PLL control align signals	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dft_compression	dft_compression	async	
aw3_dft_pll_ctrl	Controls for the TFT Clock control module	async	

aw3_dft_pll_clk_bypass	DFT PLL CLK BYPASS control for the TFT Clock control module	async	
aw3_c7xv_clk_ctl	c7xv clk1 clock	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_core_clken	Clken interface	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_wrap_clken	Clken interface	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dft_local_div_clk_en	DFT clock clken control for the Dividers	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_core_rst		aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_wrap_rst		aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_mma_pwr	Power interface	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
dft_aw3_clk_bypass	EL2 Clk grid disable control	async	
dft_aw3_dft_clk_en	EL2 Clk grid disable control	async	
dft_aw3_divfactor_sel	EL2 Clk grid disable control	async	
dft_aw3_core_clk4_divfactor_sel	EL2 Clk grid disable control	async	
dft_aw3_core_clk4_disable	Clk4 clock gate disable control for dft mode	async	
dft_aw3_scrub_off	EL2 Clk grid disable control	el2_clk_clk	
aw3_c7xv_pbist_disable_scr	Target .vbusp_pbist_cfg power disable	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dft_embed_pbist		aw3_pbist_tclk_clk	aw3_pbist_rst_mod_g_rst_n
aw3_rom_ase	rom_ase	async	
aw3_dft_pbist_cpu		aw3_c7xv_clk_clk	aw3_pbist_rst_mod_g_rst_n
aw3_dft_pbist_safety_error		aw3_c7xv_clk_clk	aw3_pbist_rst_mod_g_rst_n
aw3_dft_pbist_pwr		async	aw3_pbist_rst_mod_g_rst_n
aw3_pbist_tclk		async	
aw3_pbist_rst		async	
aw3_dft_rom	EFUSE interface	async	
aw3_core_pbist_clkstop		async	aw3_pbist_rst_mod_g_rst_n
aw3_dft_lbist	dft_lbist	async	
aw3_dft_lbist_ctrl_id	dft_lbist_ctrl_id	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_dft_iso_en	dft_iso_en	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_core_dft_lbist	dft_lbist	async	
aw3_core_dft_lbist_ctrl_id	dft_lbist_ctrl_id	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_core_dft_rom	EFUSE interface	async	
aw3_core_dft_iso_en	dft_iso_en	async	
aw3_c7xv_core_bisor	bisor	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_core_pbist_disable_scr	Target .vbusp_core_pbist_cfg power disable	async	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_bisor	bisor	async	
aw3_c7xv_core_dft_embed_pbist		aw3_c7xv_core_pbist_tclk_clk	aw3_c7xv_core_pbist_rst_mod_g_rst_n
aw3_c7xv_core_dft_pbist_cpu		aw3_c7xv_clk_clk	aw3_c7xv_core_pbist_rst_mod_g_rst_n
aw3_c7xv_core_dft_pbist_safety_error		aw3_c7xv_clk_clk	aw3_c7xv_core_pbist_rst_mod_g_rst_n
aw3_c7xv_core_dft_pbist_pwr		aw3_c7xv_clk_clk	aw3_c7xv_core_pbist_rst_mod_g_rst_n
aw3_c7xv_core_pbist_tclk		aw3_c7xv_clk_clk	aw3_c7xv_core_rst_mod_g_rst_n
aw3_c7xv_core_pbist_rst		aw3_c7xv_clk_clk	aw3_c7xv_core_rst_mod_g_rst_n
aw3_c7xv_core_pwr_dis	Power SCR Disable Interface	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_mma_clkstop	PMU Clock stp req/ack interface	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_mma_clken	Clken interface	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_c7xv_mma_rst		aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_mma_r00	dft_compression for mma_array_r00 block	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_mma_r01	dft_compression for mma_array_r01 block	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_mma_r10	dft_compression for mma_array_r10 block	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_mma_r11	dft_compression for mma_array_r11 block	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n

aw3_r00_pwr	Power interface for mma_array_r00 block	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_r01_pwr	Power interface for mma_array_r01 block	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_r10_pwr	Power interface for mma_array_r10 block	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
aw3_r11_pwr	Power interface for mma_array_r11 block	aw3_c7xv_clk_clk	aw3_c7xv_wrap_rst_mod_g_rst_n
ccc_ac			
el2_rat_source_id_tieoff			
msmc0_el2	msmc0_el2 vbusmc patron interface	el2_clk_clk	el2_wrap_rst_mod_g_rst_n
el2_msmc0	1 MBA Initiator Interface	msmc_clk_clk	el2_wrap_rst_mod_g_rst_n
msmc1_el2	msmc0_el2 vbusmc patron interface	el2_clk_clk	el2_wrap_rst_mod_g_rst_n
el2_msmc1	1 MBA Initiator Interface	msmc_clk_clk	el2_wrap_rst_mod_g_rst_n
vbusp_dmssc_dsp	VBUSP_FW_CFG_EXCEPTIO N subordinate VBUSP Interface	el2_clk_clk	el2_wrap_rst_mod_g_rst_n
dft_el2_scrub_off	EL2 Memory scrubber disable control	async	el2_wrap_rst_mod_g_rst_n
dft_el2_dft_clk_en	EL2 Clk grid disable control	async	el2_wrap_rst_mod_g_rst_n
dft_el2_clk_bypass	EL2 Clk grid disable control	async	el2_wrap_rst_mod_g_rst_n
dft_el2_divfactor_sel	EL2 Clk grid disable control	async	el2_wrap_rst_mod_g_rst_n
el2_divh_clk1_dft	divh_clk1_tft_clk_en control	async	el2_wrap_rst_mod_g_rst_n
el2_divh_clk4_dft	divh_clk4_tft_clk_en control	async	el2_wrap_rst_mod_g_rst_n
el2_wrap_clkstop	PMU Clock stp req/ack interface	async	el2_wrap_rst_mod_g_rst_n
el2_clk	el2_clk1 clock	async	el2_wrap_rst_mod_g_rst_n
el2_divh_clk4_obsclk_out	el2_clk1 clock	async	el2_wrap_rst_mod_g_rst_n
el2_pll_ctrl_main_clk	pll_ctrl clock	async	el2_wrap_rst_mod_g_rst_n
el2_dft	dft_ip	async	el2_wrap_rst_mod_g_rst_n
el2_pll_ctrl_align	PLL control align signals	async	el2_wrap_rst_mod_g_rst_n
el2_dft_compression	dft_compression	async	el2_wrap_rst_mod_g_rst_n
el2_dft_pll_ctrl	Controls for the TFT Clock control module	async	el2_wrap_rst_mod_g_rst_n
el2_dft_pll_clk_bypass	DFT PLL CLK BYPASS control for the TFT Clock control module	async	el2_wrap_rst_mod_g_rst_n
el2_clk_ctl	el2_clk1 clock	async	el2_wrap_rst_mod_g_rst_n
el2_wrap_clken	Clken interface	async	el2_wrap_rst_mod_g_rst_n
el2_dft_local_div_clk_en	DFT clock clken control for the Dividers	async	el2_wrap_rst_mod_g_rst_n
el2_wrap_rst		async	el2_wrap_rst_mod_g_rst_n
el2_pwr	Power interface	async	el2_wrap_rst_mod_g_rst_n
el2_mem_atpg	DFT mem interface	async	el2_wrap_rst_mod_g_rst_n
el2_dftmem	MEM_MARGIN busdef	async	el2_wrap_rst_mod_g_rst_n
el2_dftmem_testrwm	MEM_TESTRWM busdef	async	el2_wrap_rst_mod_g_rst_n
el2_dftmem_dftclken	MEM_DFTCLKEN busdef	async	el2_wrap_rst_mod_g_rst_n
el2_dftmem_test1	MEM_TEST1 busdef	async	el2_wrap_rst_mod_g_rst_n
el2_dftmem_testnm	MEM_TEST_RNM busdef	async	el2_wrap_rst_mod_g_rst_n
el2_pwr_bias	PWR_BIAS busdef	async	el2_wrap_rst_mod_g_rst_n
el2_soc_events_in	SOC Events in to CLEC	async	el2_wrap_rst_mod_g_rst_n
el2_soc_events_out_level	Events out of CLEC - Level	el2_clk_clk	el2_wrap_rst_mod_g_rst_n
el2_esm_events_out_pulse	Events out of CLEC - Pulse	el2_clk_clk	el2_wrap_rst_mod_g_rst_n
el2_esm_events_out_level	ESM Events out of CLEC - Level	el2_clk_clk	el2_wrap_rst_mod_g_rst_n
el2_arm_0_evento_in	ARM evento to CLEC	async	el2_wrap_rst_mod_g_rst_n
el2_arm_0_eventi_out	ARM eventi from CLEC	el2_clk_clk	el2_wrap_rst_mod_g_rst_n
el2_gic_spi	Shared Peripheral Interrupt (SPI) inputs	async	el2_wrap_rst_mod_g_rst_n
ac_cc_i	Triggers into the DRU	el2_pll_ctrl_main_clk_clk	el2_wrap_rst_mod_g_rst_n
ac_cc_o	Events from the DRU	el2_pll_ctrl_main_clk_clk	el2_wrap_rst_mod_g_rst_n
atb64_ac_mst_async		el2_pll_ctrl_main_clk_clk	el2_wrap_rst_mod_g_rst_n
el2_dbg_sec	ARM Trust Zone Interface	async	el2_wrap_rst_mod_por_rst_n
el2_glbdbg_timestamp	Debug Timestamp Interface	async	el2_wrap_rst_mod_por_rst_n
el2_dbg_chm		async	el2_wrap_rst_mod_por_rst_n
el2_dbg_chs		async	el2_wrap_rst_mod_por_rst_n
el2_ss_pwr	Power interface	async	
dru_c_pwr	Power interface	async	
dru_c_compression	dft_compression	async	el2_wrap_rst_mod_g_rst_n
el2_pbist_0	PMU Clock stp req/ack interface	async	el2_wrap_rst_mod_g_rst_n

pwr_el2_pbist_cfg_0	Target .el2_pbist_cfg_0 power disable	async	el2_wrap_rst_mod_g_rst_n
el2_0_dft_embed_pbist		el2_0_pbist_tclk_clk	el2_0_pbist_rst_mod_g_rst_n
el2_0_dft_pbist_cpu		async	el2_wrap_rst_mod_g_rst_n
el2_0_dft_pbist_safety_error		async	el2_wrap_rst_mod_g_rst_n
el2_0_dft_pbist_pwr		async	el2_wrap_rst_mod_g_rst_n
el2_0_pbist_tclk		async	
el2_0_pbist_rst		async	
el2_0_bisor	bisor	async	
aw0_scr_dis	aw0 power disable interface	async	el2_wrap_rst_mod_g_rst_n
pwr_el2_mem0	el2_mem0 power disable interface	async	el2_wrap_rst_mod_g_rst_n
el2_dft_mem0	dft_mem0 ip interface for unique clk force en control		
el2_pbist_1	PMU Clock stp req/ack interface	async	el2_wrap_rst_mod_g_rst_n
pwr_el2_pbist_cfg_1	Target .el2_pbist_cfg_1 power disable	async	el2_wrap_rst_mod_g_rst_n
el2_1_dft_embed_pbist		el2_1_pbist_tclk_clk	el2_1_pbist_rst_mod_g_rst_n
el2_1_dft_pbist_cpu		async	el2_wrap_rst_mod_g_rst_n
el2_1_dft_pbist_safety_error		async	el2_wrap_rst_mod_g_rst_n
el2_1_dft_pbist_pwr		async	el2_wrap_rst_mod_g_rst_n
el2_1_pbist_tclk		async	
el2_1_pbist_rst		async	
el2_1_bisor	bisor	async	
aw1_scr_dis	aw1 power disable interface	async	el2_wrap_rst_mod_g_rst_n
pwr_el2_mem1	el2_mem1 power disable interface	async	el2_wrap_rst_mod_g_rst_n
el2_dft_mem1	dft_mem1 ip interface for unique clk force en control		
el2_pbist_2	PMU Clock stp req/ack interface	async	el2_wrap_rst_mod_g_rst_n
pwr_el2_pbist_cfg_2	Target .el2_pbist_cfg_2 power disable	async	el2_wrap_rst_mod_g_rst_n
el2_2_dft_embed_pbist		el2_2_pbist_tclk_clk	el2_2_pbist_rst_mod_g_rst_n
el2_2_dft_pbist_cpu		async	el2_wrap_rst_mod_g_rst_n
el2_2_dft_pbist_safety_error		async	el2_wrap_rst_mod_g_rst_n
el2_2_dft_pbist_pwr		async	el2_wrap_rst_mod_g_rst_n
el2_2_pbist_tclk		async	
el2_2_pbist_rst		async	
el2_2_bisor	bisor	async	
aw2_scr_dis	aw2 power disable interface	async	el2_wrap_rst_mod_g_rst_n
pwr_el2_mem2	el2_mem2 power disable interface	async	el2_wrap_rst_mod_g_rst_n
el2_dft_mem2	dft_mem2 ip interface for unique clk force en control		
el2_pbist_3	PMU Clock stp req/ack interface	async	el2_wrap_rst_mod_g_rst_n
pwr_el2_pbist_cfg_3	Target .el2_pbist_cfg_3 power disable	async	el2_wrap_rst_mod_g_rst_n
el2_3_dft_embed_pbist		el2_3_pbist_tclk_clk	el2_3_pbist_rst_mod_g_rst_n
el2_3_dft_pbist_cpu		async	el2_wrap_rst_mod_g_rst_n
el2_3_dft_pbist_safety_error		async	el2_wrap_rst_mod_g_rst_n
el2_3_dft_pbist_pwr		async	el2_wrap_rst_mod_g_rst_n
el2_3_pbist_tclk		async	
el2_3_pbist_rst		async	
el2_3_bisor	bisor	async	
aw3_scr_dis	aw3 power disable interface	async	el2_wrap_rst_mod_g_rst_n
pwr_el2_mem3	el2_mem3 power disable interface	async	el2_wrap_rst_mod_g_rst_n
el2_dft_mem3	dft_mem3 ip interface for unique clk force en control		
el2_divh_clk2_dft	divh_clk2_tft_clk_en control	async	el2_wrap_rst_mod_g_rst_n

el2_divh_clk4_soc_dft	divh_clk4_soc_tft_clk_en control	async	el2_wrap_rst_mod_g_rst_n
el2_divp_clk1_msmc_dft	divp_clk1_msmc_tft_clk_en control	async	el2_wrap_rst_mod_g_rst_n
el2_divh_clk4_msmc_dft	divh_clk4_msmc_tft_clk_en control	async	el2_wrap_rst_mod_g_rst_n
msmc_clk	el2 clk1 clock	async	el2_wrap_rst_mod_g_rst_n
msmc_clk_ctl	msmc clk1 clock	async	el2_wrap_rst_mod_g_rst_n
rom_ase	rom_ase	async	
dft_rom	EFUSE interface	async	
ac_fw_bypass		async	el2_wrap_rst_mod_g_rst_n
ac_fw_id_0		el2_clk_clk	el2_wrap_rst_mod_g_rst_n
ac_fw_exp	firewall security exception bit	el2_clk_clk	el2_wrap_rst_mod_g_rst_n
ac_fw_exp_sf	firewall safety exception bit	el2_clk_clk	el2_wrap_rst_mod_g_rst_n
el2_mem0_dft_compression	dft_compression busdef	async	
el2_mem0_rst			
el2_mem0_clken	Clken interface	async	el2_wrap_rst_mod_g_rst_n
el2_mem0_clkstop	PMU Clock stp req/ack interface	async	el2_wrap_rst_mod_g_rst_n
el2_mem0_efuse_ctl	efuse_ctl busdef		
el2_mem0_efuse_data	efuse_data busdef	async	
el2_mem0_pwr	el2_pwr busdef	async	
el2_mem1_dft_compression	dft_compression busdef	async	
el2_mem1_rst			
el2_mem1_clken	Clken interface	async	el2_wrap_rst_mod_g_rst_n
el2_mem1_clkstop	PMU Clock stp req/ack interface	async	el2_wrap_rst_mod_g_rst_n
el2_mem1_efuse_ctl	efuse_ctl busdef		
el2_mem1_efuse_data	efuse_data busdef	async	
el2_mem1_pwr	el2_pwr busdef	async	
el2_mem2_dft_compression	dft_compression busdef	async	
el2_mem2_rst			
el2_mem2_clken	Clken interface	async	el2_wrap_rst_mod_g_rst_n
el2_mem2_clkstop	PMU Clock stp req/ack interface	async	el2_wrap_rst_mod_g_rst_n
el2_mem2_efuse_ctl	efuse_ctl busdef		
el2_mem2_efuse_data	efuse_data busdef	async	
el2_mem2_pwr	el2_pwr busdef	async	
el2_mem3_dft_compression	dft_compression busdef	async	
el2_mem3_rst			
el2_mem3_clken	Clken interface	async	el2_wrap_rst_mod_g_rst_n
el2_mem3_clkstop	PMU Clock stp req/ack interface	async	el2_wrap_rst_mod_g_rst_n
el2_mem3_efuse_ctl	efuse_ctl busdef		
el2_mem3_efuse_data	efuse_data busdef	async	
el2_mem3_pwr	el2_pwr busdef	async	

The following sections provide pin level detail for each of the interfaces on this module:

9.4.1 aw0_ccc_aw

Signal Name	Dir	Default Val	Description
aw0_ccc_aw_ipcfg[89:0]	In	90'd0	ip config pin

9.4.2 aw1_ccc_aw

Signal Name	Dir	Default Val	Description
aw1_ccc_aw_ipcfg[89:0]	In	90'd0	ip config pin

9.4.3 aw2_ccc_aw

Signal Name	Dir	Default Val	Description
aw2_ccc_aw_ipcfg[89:0]	In	90'd0	ip config pin

9.4.4 aw3_ccc_aw

Signal Name	Dir	Default Val	Description
aw3_ccc_aw_ipcfg[89:0]	In	90'd0	ip config pin

9.4.5 ac_vbusp_cfg

Signal Name	Dir	Default Val	Description
ac_vbusp_cfg_orderid[3:0]	In	4'd0	Order ID
ac_vbusp_cfg_routeid[11:0]	In	12'd0	Route ID
ac_vbusp_cfg_xid[11:0]	In	12'd0	Transaction ID
ac_vbusp_cfg_req	In	1'b0	Request
ac_vbusp_cfg_rsel[3:0]	In	4'd0	Region Select
ac_vbusp_cfg_dir	In	1'b0	Direction
ac_vbusp_cfg_address[21:0]	In	22'd0	Address
ac_vbusp_cfg_xcnt[2:0]	In	3'd4	Good Byte Count
ac_vbusp_cfg_byten[3:0]	In	{4{1'b1}}	Byte Enables
ac_vbusp_cfg_wdata[31:0]	In	32'd0	Write Data
ac_vbusp_cfg_wedc[6:0]	In	7'd0	ECC on write data
ac_vbusp_cfg_wready	Out	1'b1	Write Ready
ac_vbusp_cfg_rdatap[31:0]	Out	32'd0	Read Data (Pipelined)
ac_vbusp_cfg_redc[6:0]	Out	7'd0	ECC on read data
ac_vbusp_cfg_rstatus[2:0]	Out	3'd0	Read Status
ac_vbusp_cfg_rready	Out	1'b1	Read Ready
ac_vbusp_cfg_dtype[1:0]	In	2'd2	Data Type
ac_vbusp_cfg_priv[1:0]	In	2'd0	Privilege Attribute
ac_vbusp_cfg_secure	In	1'b0	Secure Attribute
ac_vbusp_cfg_emudbg	In	1'b0	Emulation Debug Attribute
ac_vbusp_cfg_interest	In	1'b0	Debug marker
ac_vbusp_cfg_flush	In	1'b0	Flush Transaction Attribute
ac_vbusp_cfg_ew	In	1'd0	Command Power Mgt EW
ac_vbusp_cfg_sorderid[3:0]	Out	4'd0	Status Order ID
ac_vbusp_cfg_srouteid[11:0]	Out	12'd0	Status Route ID
ac_vbusp_cfg_sid[11:0]	Out	12'd0	Status Transaction ID
ac_vbusp_cfg_sreq	Out	1'b0	Status Request
ac_vbusp_cfg_sbytecnt[9:0]	Out	10'd0	Status Done Indicator
ac_vbusp_cfg_sstatus[2:0]	Out	3'd0	Status Information
ac_vbusp_cfg_sinterest	Out	1'b0	Debug Marker
ac_vbusp_cfg_sready	In	1'b1	Status Ready
ac_vbusp_cfg_req_safer	In	1'b1	Req Safety Redundancy
ac_vbusp_cfg_rready_safer	Out	1'b0	Ready Safety Redundancy
ac_vbusp_cfg_wready_safer	Out	1'b0	Wready Safety Redundancy
ac_vbusp_cfg_sreq_safer	Out	1'b1	Sreq Safety Redundancy
ac_vbusp_cfg_sready_safer	In	1'b0	Sready Safety Redundancy
ac_vbusp_cfg_dir_safep	In	1'b1	Direction Safety Parity
ac_vbusp_cfg_orderid_safep	In	1'b1	Order ID Safety Parity
ac_vbusp_cfg_routeid_safep	In	1'b1	Route ID Safety Parity
ac_vbusp_cfg_xid_safep	In	1'b1	Transaction ID Safety Parity

ac_vbusp_cfg_rsel_safep	In	1'b1	Region Select Safety Parity
ac_vbusp_cfg_address_safep[1:0]	In	{2 {1'b1}}	Address Safety Parity
ac_vbusp_cfg_xcnt_safep	In	~^ac_vbusp_cfg_xcnt	Good Byte Count Safety Parity
ac_vbusp_cfg_byten_safep	In	{1 {1'b1}}	Byte Enables Safety Parity
ac_vbusp_cfg_rstatus_safep	Out	1'b1	Read Status Safety Parity
ac_vbusp_cfg_dtype_safep	In	1'b0	Data Type Safety Parity
ac_vbusp_cfg_priv_safep	In	1'b1	Privilege Attribute Safety Parity
ac_vbusp_cfg_secure_safep	In	1'b1	Secure Attribute Safety Parity
ac_vbusp_cfg_emudbg_safep	In	1'b1	Emulation Debug Attribute Safety Parity
ac_vbusp_cfg_interest_safep	In	1'b1	Debug marker Safety Parity
ac_vbusp_cfg_flush_safep	In	1'b1	Flush Transaction Attribute Safety Parity
ac_vbusp_cfg_sorderid_safep	Out	1'b1	Status Order ID Safety Parity
ac_vbusp_cfg_srouteid_safep	Out	1'b1	Status Route ID Safety Parity
ac_vbusp_cfg_sid_safep	Out	1'b1	Status Transaction ID Safety Parity
ac_vbusp_cfg_sbytecnt_safep	Out	1'b1	Status Done Indicator Safety Parity
ac_vbusp_cfg_sstatus_safep	Out	1'b1	Status Information Safety Parity
ac_vbusp_cfg_sinterest_safep	Out	1'b1	Debug Marker Safety Parity

9.4.6 ac_vbusp_dbg

Signal Name	Dir	Default Val	Description
ac_vbusp_dbg_orderid[3:0]	In	4'd0	Order ID
ac_vbusp_dbg_routeid[11:0]	In	12'd0	Route ID
ac_vbusp_dbg_xid[11:0]	In	12'd0	Transaction ID
ac_vbusp_dbg_req	In	1'b0	Request
ac_vbusp_dbg_dir	In	1'b0	Direction
ac_vbusp_dbg_address[25:0]	In	26'd0	Address
ac_vbusp_dbg_xcnt[2:0]	In	3'd4	Good Byte Count
ac_vbusp_dbg_byten[3:0]	In	{4{1'b1}}	Byte Enables
ac_vbusp_dbg_wdata[31:0]	In	32'd0	Write Data
ac_vbusp_dbg_wready	Out	1'b1	Write Ready
ac_vbusp_dbg_rdatap[31:0]	Out	32'd0	Read Data (Pipelined)
ac_vbusp_dbg_rstatus[2:0]	Out	3'd0	Read Status
ac_vbusp_dbg_rready	Out	1'b1	Read Ready
ac_vbusp_dbg_dtype[1:0]	In	2'd2	Data Type
ac_vbusp_dbg_priv[1:0]	In	2'd0	Privilege Attribute
ac_vbusp_dbg_secure	In	1'b0	Secure Attribute
ac_vbusp_dbg_emudbg	In	1'b0	Emulation Debug Attribute
ac_vbusp_dbg_interest	In	1'b0	Debug marker
ac_vbusp_dbg_flush	In	1'b0	Flush Transaction Attribute
ac_vbusp_dbg_ew	In	1'd0	Command Power Mgt EW
ac_vbusp_dbg_sorderid[3:0]	Out	4'd0	Status Order ID
ac_vbusp_dbg_srouteid[11:0]	Out	12'd0	Status Route ID
ac_vbusp_dbg_sid[11:0]	Out	12'd0	Status Transaction ID
ac_vbusp_dbg_sreq	Out	1'b0	Status Request
ac_vbusp_dbg_sbytecnt[9:0]	Out	10'd0	Status Done Indicator
ac_vbusp_dbg_sstatus[2:0]	Out	3'd0	Status Information
ac_vbusp_dbg_sinterest	Out	1'b0	Debug Marker
ac_vbusp_dbg_sready	In	1'b1	Status Ready

9.4.7 aw0_aw_ccc

Signal Name	Dir	Default Val	Description
aw0_aw_ccc_ipcfg[24:0]	Out	25'd0	ip config pin

9.4.8 aw0_c7xv_wrap_tieoff_clusternum

Signal Name	Dir	Default Val	Description
aw0_c7xv_wrap_tieoff_clusternum_ipcfg[7:0]	In	8'd0	ip config pin

9.4.9 aw0_c7xv_wrap_tieoff_corepacnum

Signal Name	Dir	Default Val	Description
aw0_c7xv_wrap_tieoff_corepacnum_ipcfg[7:0]	In	8'd0	ip config pin



9.4.10 aw0_c7xv_wrap_tieoff_sys_mma_present

Signal Name	Dir	Default Val	Description
aw0_c7xv_wrap_tieoff_sys_mma_present_ipcfg	In	1'd0	ip config pin

9.4.11 spare ports for PD

Signal Name	Dir	Default Val	Description
aw0_spare_spare_out[15:0]	Out	16'd0	spare output ports
aw0_spare_spare_in[15:0]	In	16'd0	spare input ports

9.4.12 aw0_pbist_clkstop

Signal Name	Dir	Default Val	Description
aw0_pbist_clkstop_clkstop_req	In	1'd0	Clock stop request
aw0_pbist_clkstop_clkstop_ack	Out	pbist_clkstop_clkstop_req	Clock stop acknowledge

9.4.13 Power interface

Signal Name	Dir	Default Val	Description
aw0_c7xv_pwr_pwr_log_iso_en	In	1'd0	pwr
aw0_c7xv_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw0_c7xv_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw0_c7xv_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw0_c7xv_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr
aw0_c7xv_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
aw0_c7xv_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
aw0_c7xv_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.14 dft_compression

Signal Name	Dir	Default Val	Description
aw0_dft_compression_core_dft_block_en	In	1'd1	dft_block_en
aw0_dft_compression_core_dft_partition_en	In	1'd1	dft_partition_en
aw0_dft_compression_core_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw0_dft_compression_core_dft_compression_mode	In	1'd0	dft_compression_mode
aw0_dft_compression_core_dft_compression_spread	In	1'd0	dft_compression_spread
aw0_dft_compression_core_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw0_dft_compression_core_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw0_dft_compression_core_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw0_dft_compression_core_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw0_dft_compression_core_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw0_dft_compression_core_dft_struct_intest	In	1'd0	dft_struct_intest
aw0_dft_compression_core_dft_struct_extest	In	1'd0	dft_struct_extest
aw0_dft_compression_core_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw0_dft_compression_core_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw0_dft_compression_core_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.15 Power interface

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_pwr_pwr_log_iso_en	In	1'd0	pwr
aw0_c7xv_core_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw0_c7xv_core_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw0_c7xv_core_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw0_c7xv_core_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr
aw0_c7xv_core_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
aw0_c7xv_core_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
aw0_c7xv_core_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.16 MEM_TESTRWM busdef

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_testrwm_dft_mem_testrwm	In	1'd0	memory testrwm

9.4.17 MEM_TEST1 busdef

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_test1_dft_mem_test1	In	1'd0	memory test1

9.4.18 Debug Timestamp Interface

Signal Name	Dir	Default Val	Description
aw0_glbdbg_timestamp_shft[1:0]	In	2'd0	binary interface
aw0_glbdbg_timestamp_gray[63:0]	In	64'd0	gray encoded interface

9.4.19 aw0_dbg_rst

Signal Name	Dir	Default Val	Description
aw0_dbg_rst_mod_dbg_l_rst_req_n	Out	1'd1	Module level debug local reset req

9.4.20 aw0_dbg_c7xv_state

Signal Name	Dir	Default Val	Description
aw0_dbg_c7xv_state_attention	Out	1'd0	debug event status

9.4.21 aw0_dbg_c7xv_susp

Signal Name	Dir	Default Val	Description
aw0_dbg_c7xv_susp_dbgsusp	Out	1'd0	Debug suspend mode request non real time

9.4.22 ARM Trust Zone Interface

Signal Name	Dir	Default Val	Description
aw0_dbg_sec_dbgen	In	1'd0	Invasive debug enable
aw0_dbg_sec_niden	In	1'd0	Non-invasive debug enable
aw0_dbg_sec_spiden	In	1'd0	Secure privileged invasive debug enable
aw0_dbg_sec_spniden	In	1'd0	Secure privileged non-invasive debug enable

9.4.23 aw0_debug

Signal Name	Dir	Default Val	Description
aw0_debug_debugenable	In	1'd0	enable debug

9.4.24 aw0_dbg_chm

Signal Name	Dir	Default Val	Description
aw0_dbg_chm_cti_chnl[3:0]	Out	4'd0	arm channel interface for cross triggering
aw0_dbg_chm_cti_chnl_ack[3:0]	In	4'd0	arm channel async handshake interface for cross triggering

9.4.25 aw0_dbg_chs

Signal Name	Dir	Default Val	Description
aw0_dbg_chs_cti_chnl[3:0]	In	4'd0	arm channel interface for cross triggering
aw0_dbg_chs_cti_chnl_ack[3:0]	Out	4'd0	arm channel async handshake interface for cross triggering

9.4.26 DFT mem interface

Signal Name	Dir	Default Val	Description
aw0_mem_atpg_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.27 MEM_MARGIN busdef

Signal Name	Dir	Default Val	Description
aw0_dftmem_s2cr1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cr1p compiler
aw0_dftmem_s2dc1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2dc1p compiler
aw0_dftmem_s2sr1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2sr1p compiler
aw0_dftmem_s2ss1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2ss1p compiler
aw0_dftmem_s2du1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2du1p compiler
aw0_dftmem_s2cr2p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cr2p compiler
aw0_dftmem_s2cu2p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cu2p compiler
aw0_dftmem_s2cr1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cr1p compiler
aw0_dftmem_s2dc1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2dc1p compiler
aw0_dftmem_s2sr1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2sr1p compiler
aw0_dftmem_s2ss1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2ss1p compiler
aw0_dftmem_s2du1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2du1p compiler
aw0_dftmem_s2cr2p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cr2p compiler
aw0_dftmem_s2cu2p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cu2p compiler
aw0_dftmem_s2du1p_dft_mem_wa[2:0]	In	3'd0	dft_mem_write assist for s2cr2p compiler
aw0_dftmem_s2du1p_dft_mem_wpulse[2:0]	In	3'd0	dft_mem_write assist for s2cr2p compiler
aw0_dftmem_rom_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for rom compiler

9.4.28 MEM_TESTRWM busdef

Signal Name	Dir	Default Val	Description
aw0_dftmem_testrwm_dft_mem_testrwm	In	1'd0	memory testrwm

9.4.29 MEM_DFTCLKEN busdef

Signal Name	Dir	Default Val	Description
aw0_dftmem_dftclken_dft_mem_dftclken	In	1'd0	memory dftclken

9.4.30 MEM_TEST1 busdef

Signal Name	Dir	Default Val	Description
aw0_dftmem_test1_dft_mem_test1	In	1'd0	memory test1

9.4.31 MEM_TEST_RNM busdef

Signal Name	Dir	Default Val	Description
aw0_dftmem_testnm_dft_mem_test_rnm	In	1'd0	memory test_rnm

9.4.32 PWR BIAS busdef

Signal Name	Dir	Default Val	Description
aw0_pwr_bias_pwr_mem_bc0	In	1'h0	pwr
aw0_pwr_bias_pwr_mem_bc1	In	1'h0	pwr
aw0_pwr_bias_pwr_mem_bc2	In	1'h0	pwr

9.4.33 efuse_ctl

Signal Name	Dir	Default Val	Description
aw0_efuse_ctl_fclrz	In	1'd0	efuse clear
aw0_efuse_ctl_fclk	In	1'd0	efuse clock

9.4.34 efuse_data

Signal Name	Dir	Default Val	Description
aw0_efuse_data_fdi	In	1'd0	efuse data in
aw0_efuse_data_fdo	Out	1'd0	efuse data out

9.4.35 efuse_ctl

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_efuse_ctl_fclrz	In	1'd0	efuse clear
aw0_c7xv_core_efuse_ctl_fclk	In	1'd0	efuse clock

9.4.36 efuse_data

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_efuse_data_fdi	In	1'd0	efuse data in
aw0_c7xv_core_efuse_data_fdo	Out	1'd0	efuse data out

9.4.37 divh_clk1_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw0_divh_clk1_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.38 divh_clk4_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw0_divh_clk4_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.39 divh_clk1_core_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw0_divh_clk1_core_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.40 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_clkstop_clkstop_req	In	1'd0	Clock stop request
aw0_c7xv_core_clkstop_clkstop_ack	Out	c7xv_core_clkstop_clkstop_req	Clock stop acknowledge

9.4.41 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw0_c7xv_wrap_clkstop_clkstop_req	In	1'd0	Clock stop request
aw0_c7xv_wrap_clkstop_clkstop_ack	Out	c7xv_wrap_clkstop_clkstop_req	Clock stop acknowledge

9.4.42 c7xv_clk1 clock

Signal Name	Dir	Default Val	Description
aw0_c7xv_clk_clk	In	1'd0	Single ended clock
aw0_c7xv_clk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_jpg_clk_l1

9.4.43 c7xv_clk1 clock

Signal Name	Dir	Default Val	Description
aw0_c7xv_divh_clk4_obsclk_out_clk	Out	1'd0	Single ended clock

9.4.44 pll ctrl clock

Signal Name	Dir	Default Val	Description
aw0_pll_ctrl_main_clk_clk	In	1'd0	Single ended clock

9.4.45 dft_ip

Signal Name	Dir	Default Val	Description
aw0_dft_clk_force_en	In	1'd0	dft_clk_force_en
aw0_dft_scan_en	In	1'd0	dft_scan_en
aw0_dft_async_rst_sel	In	1'd0	dft_async_rst_sel
aw0_dft_test_async_rst_n	In	1'd1	dft_test_async_rst_n
aw0_dft_tft_mcp_dis	In	1'd0	Assert High to block MCP path
aw0_dft_tft_conf_vector[67:0]	In	68'd0	clkdiv dft_tft_conf_vector
aw0_dft_tft_start_en	In	1'd0	clkdiv dft_tft_start_en
aw0_dft_delay_fault_en	In	1'd0	clkdiv dft_delay_fault_en
aw0_dft_tft_done	Out	1'd1	clkdiv dft_tft_done
aw0_dft_div_scan_en	In	1'd0	clkdiv dft_div_scan_en
aw0_dft_bypass_div_en	In	1'd0	clkdiv dft_bypass_div_en
aw0_dft_local_clk_en	In	1'd0	dft_local_clk_en
aw0_dft_pbist_clk_en	In	1'd0	dft_pbist_clk_en
aw0_dft_mode_atpg	In	1'd0	ATPG
aw0_dft_mode_iddq	In	1'd0	Iddq mode
aw0_dft_mode_burnin	In	1'd0	Burnin
aw0_dft_embed_mode_pbist	In	1'd0	Embedded pBIST Mode

9.4.46 dft_core ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
aw0_dft_core_clk_force_en	In	1'd0	dft_clk_force_en

9.4.47 dft_core ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
aw0_dft_mma_clk_force_en	In	1'd0	dft_clk_force_en

9.4.48 dft_compression

Signal Name	Dir	Default Val	Description
aw0_dft_compression_mma_dft_block_en	In	1'd1	dft_block_en
aw0_dft_compression_mma_dft_partition_en	In	1'd1	dft_partition_en
aw0_dft_compression_mma_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw0_dft_compression_mma_dft_compression_mode	In	1'd0	dft_compression_mode
aw0_dft_compression_mma_dft_compression_spread	In	1'd0	dft_compression_spread
aw0_dft_compression_mma_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw0_dft_compression_mma_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw0_dft_compression_mma_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw0_dft_compression_mma_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw0_dft_compression_mma_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw0_dft_compression_mma_dft_struct_intest	In	1'd0	dft_struct_intest
aw0_dft_compression_mma_dft_struct_extest	In	1'd0	dft_struct_extest
aw0_dft_compression_mma_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw0_dft_compression_mma_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw0_dft_compression_mma_dft_struct_so[35:0]	Out	36'd0	dft_struct_so



9.4.49 PLL control align signals

Signal Name	Dir	Default Val	Description
aw0_pll_ctrl_align_align_clk_div_rst_n	In	1'd0	Export align_clk_div
aw0_pll_ctrl_align_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.50 dft_compression

Signal Name	Dir	Default Val	Description
aw0_dft_compression_dft_block_en	In	1'd1	dft_block_en
aw0_dft_compression_dft_partition_en	In	1'd1	dft_partition_en
aw0_dft_compression_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw0_dft_compression_dft_compression_mode	In	1'd0	dft_compression_mode
aw0_dft_compression_dft_compression_spread	In	1'd0	dft_compression_spread
aw0_dft_compression_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw0_dft_compression_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw0_dft_compression_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw0_dft_compression_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw0_dft_compression_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw0_dft_compression_dft_struct_intest	In	1'd0	dft_struct_intest
aw0_dft_compression_dft_struct_extest	In	1'd0	dft_struct_extest
aw0_dft_compression_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw0_dft_compression_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw0_dft_compression_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.51 Controls for the TFT Clock control module

Signal Name	Dir	Default Val	Description
aw0_dft_pll_ctrl_dft_mode_pll_atpg	In	1'd0	dft_mode_pll_atpg
aw0_dft_pll_ctrl_dft_pll_capture_en	In	1'd0	dft_pll_capture_en
aw0_dft_pll_ctrl_dft_shift_clk_sel	In	1'd0	dft_shift_clk_sel
aw0_dft_pll_ctrl_cs_func_clk_en	In	1'd0	enable clocks to selected scan paths

9.4.52 DFT PLL CLK BYPASS control for the TFT Clock control module

Signal Name	Dir	Default Val	Description
aw0_dft_pll_clk_bypass_pll_clk_bypass	In	1'd1	DFT WIR input to kp_ti_ipg_loc_tft_clk_ctrl

9.4.53 c7xv clk1 clock

Signal Name	Dir	Default Val	Description
aw0_c7xv_clk_ctl_align_clk_div_rst_n	In	1'd0	Export align_clk_div
aw0_c7xv_clk_ctl_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.54 Clken interface

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw0_c7xv_core_clken_clk_en_ack	Out	c7xv_core_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.55 Clken interface

Signal Name	Dir	Default Val	Description
aw0_c7xv_wrap_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw0_c7xv_wrap_clken_clk_en_ack	Out	c7xv_wrap_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.56 DFT clock clken control for the Dividers

Signal Name	Dir	Default Val	Description
aw0_dft_local_div_clk_en_div_clk_en	In	1'd1	DFT local clk enable contribution to kp_ti_ipg_clk_divhdist5_tft_mux

9.4.57 aw0_c7xv_core_rst

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_rst_mod_g_rst_n	In	1'd1	Module level main reset
aw0_c7xv_core_rst_mod_por_rst_n	In	1'd1	Module level por reset
aw0_c7xv_core_rst_mod_l_rst_n	In	1'd1	Module level local reset

9.4.58 aw0_c7xv_wrap_rst

Signal Name	Dir	Default Val	Description
aw0_c7xv_wrap_rst_mod_g_rst_n	In	1'd1	Module level main reset
aw0_c7xv_wrap_rst_mod_por_rst_n	In	1'd1	Module level por reset
aw0_c7xv_wrap_rst_mod_l_rst_n	In	1'd1	Module level local reset

9.4.59 Power interface

Signal Name	Dir	Default Val	Description
aw0_mma_pwr_pwr_log_iso_en	In	1'd0	pwr
aw0_mma_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw0_mma_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw0_mma_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw0_mma_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.60 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw0_clk_bypass_wir	In	1'd0	dft wir signal

9.4.61 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw0_dft_clk_en_wir	In	1'd0	dft wir signal

9.4.62 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw0_divfactor_sel_wir	In	1'd0	dft wir signal

9.4.63 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw0_core_clk4_divfactor_sel_wir	In	1'd0	dft wir signal

9.4.64 Clk4 clock gate disable control for dft mode

Signal Name	Dir	Default Val	Description
dft_aw0_core_clk4_disable_wir	In	1'd0	dft wir signal

9.4.65 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw0_scrub_off_wir	In	1'd0	dft wir signal

9.4.66 Target .vbusp_pbist_cfg power disable

Signal Name	Dir	Default Val	Description
aw0_c7xv_pbist_disable_scr_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw0_c7xv_pbist_disable_scr_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw0_c7xv_pbist_disable_scr_disable_done	In	1'd0	Ack to SCR to stop clocks
aw0_c7xv_pbist_disable_scr_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw0_c7xv_pbist_disable_scr_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw0_c7xv_pbist_disable_scr_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.67 aw0_dft_embed_pbist

Signal Name	Dir	Default Val	Description
aw0_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
aw0_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
aw0_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
aw0_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
aw0_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
aw0_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
aw0_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
aw0_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
aw0_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
aw0_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
aw0_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
aw0_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
aw0_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
aw0_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
aw0_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
aw0_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
aw0_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
aw0_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
aw0_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.68 rom_ase

Signal Name	Dir	Default Val	Description
aw0_rom_ase_dft_mem_rom_ase	In	1'd0	dft_mem_rom_ase

9.4.69 aw0_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
aw0_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.70 aw0_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
aw0_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.71 aw0_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
aw0_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
aw0_dft_pbist_pwr_clk_en_ack	Out	dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.72 aw0_pbist_tclk

Signal Name	Dir	Default Val	Description
aw0_pbist_tclk_clk	In	1'd0	Single ended clock
aw0_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.73 aw0_pbist_rst

Signal Name	Dir	Default Val	Description
aw0_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.74 EFUSE interface

Signal Name	Dir	Default Val	Description
aw0_dft_rom_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.75 aw0_core_pbist_clkstop

Signal Name	Dir	Default Val	Description
aw0_core_pbist_clkstop_clkstop_req	In	1'd0	Clock stop request
aw0_core_pbist_clkstop_clkstop_ack	Out	core_pbist_clkstop_clkstop_req	Clock stop acknowledge

9.4.76 dft_lbist

Signal Name	Dir	Default Val	Description
aw0_dft_lbist_dft_lbist_atpg_mode	In	1'd0	ATPG control from SoC WIR
aw0_dft_lbist_dft_lbist_runbist_mode[3:0]	In	4'd0	Runbist mode enable
aw0_dft_lbist_dft_lbist_bist_reset	In	1'd0	Resets LBIST macro
aw0_dft_lbist_dft_lbist_bist_run[3:0]	In	4'd0	Starts LBIST if all bits are 1. Additional bits added to ensure that a single bit flip will not cause unintended LBIST execution.
aw0_dft_lbist_dft_lbist_scan_pc_def[3:0]	In	4'd0	Number of chain test patterns to run
aw0_dft_lbist_dft_lbist_reset_pc_def[3:0]	In	4'd0	Number of reset patterns to run.
aw0_dft_lbist_dft_lbist_set_pc_def[3:0]	In	4'd0	Number of set patterns to run
aw0_dft_lbist_dft_lbist_static_pc_def[13:0]	In	14'd0	Number of stuck-at patterns up to 16k to run
aw0_dft_lbist_dft_lbist_dc_def[1:0]	In	2'd0	Clock delay after scan_enable switching
aw0_dft_lbist_dft_lbist_prpg_def[52:0]	In	53'd0	Initial seed for PRPG
aw0_dft_lbist_dft_lbist_misr_mux_ctl[9:0]	In	10'd0	Selects block of 32 MISR bits to return to the MMR block.
aw0_dft_lbist_dft_lbist_out_mux_sel[1:0]	In	2'd0	Select to return PID, CTRL ID, or MISR result.
aw0_dft_lbist_dft_lbist_bist_done	Out	1'd0	LBIST done signal
aw0_dft_lbist_dft_lbist_bist_running	Out	1'd0	LBIST is running
aw0_dft_lbist_dft_lbist_misr_results[31:0]	Out	32'd0	32-bit section of MISR value. Selected using misr_mux_ctl input

9.4.77 dft_lbist_ctrl_id

Signal Name	Dir	Default Val	Description
aw0_dft_lbist_ctrl_id_dft_lbist_ctrl_id[5:0]	In	6'd0	Tieoff ID for this instance of LBIST

9.4.78 dft_iso_en

Signal Name	Dir	Default Val	Description
aw0_dft_iso_en_dft_iso_en_lbist	In	1'd0	DFT iso_en for pins to LBISTCTL
aw0_dft_iso_en_dft_iso_en_pbist	In	1'd0	DFT iso_en for pins to PBIST
aw0_dft_iso_en_dft_iso_en_other	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST
aw0_dft_iso_en_dft_iso_en_lbist_delayed	In	1'd0	DFT iso_en for pins to LBISTCTL
aw0_dft_iso_en_dft_iso_en_pbist_delayed	In	1'd0	DFT iso_en for pins to PBIST
aw0_dft_iso_en_dft_iso_en_other_delayed	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST

9.4.79 dft_lbist

Signal Name	Dir	Default Val	Description
aw0_core_dft_lbist_dft_lbist_atpg_mode	In	1'd0	ATPG control from SoC WIR
aw0_core_dft_lbist_dft_lbist_runbist_mode[3:0]	In	4'd0	Runbist mode enable
aw0_core_dft_lbist_dft_lbist_bist_reset	In	1'd0	Resets LBIST macro
aw0_core_dft_lbist_dft_lbist_bist_run[3:0]	In	4'd0	Starts LBIST if all bits are 1. Additional bits added to ensure that a single bit flip will not cause unintended LBIST execution.
aw0_core_dft_lbist_dft_lbist_scan_pc_def[3:0]	In	4'd0	Number of chain test patterns to run
aw0_core_dft_lbist_dft_lbist_reset_pc_def[3:0]	In	4'd0	Number of reset patterns to run.
aw0_core_dft_lbist_dft_lbist_set_pc_def[3:0]	In	4'd0	Number of set patterns to run
aw0_core_dft_lbist_dft_lbist_static_pc_def[13:0]	In	14'd0	Number of stuck-at patterns up to 16k to run
aw0_core_dft_lbist_dft_lbist_dc_def[1:0]	In	2'd0	Clock delay after scan_enable switching
aw0_core_dft_lbist_dft_lbist_prpg_def[52:0]	In	53'd0	Initial seed for PRPG
aw0_core_dft_lbist_dft_lbist_misr_mux_ctl[9:0]	In	10'd0	Selects block of 32 MISR bits to return to the MMR block.
aw0_core_dft_lbist_dft_lbist_out_mux_sel[1:0]	In	2'd0	Select to return PID, CTRL ID, or MISR result.
aw0_core_dft_lbist_dft_lbist_bist_done	Out	1'd0	LBIST done signal
aw0_core_dft_lbist_dft_lbist_bist_running	Out	1'd0	LBIST is running
aw0_core_dft_lbist_dft_lbist_misr_results[31:0]	Out	32'd0	32-bit section of MISR value. Selected using misr_mux_ctl input
aw0_core_dft_lbist_dft_lbist_divide_ratio[4:0]	In	5'd0	clkdiv divide value during lbist
aw0_core_dft_lbist_dft_lbist_load_div	In	1'd0	clkdiv divide load enable

9.4.80 dft_lbist_ctrl_id

Signal Name	Dir	Default Val	Description
aw0_core_dft_lbist_ctrl_id_dft_lbist_ctrl_id[5:0]	In	6'd0	Tieoff ID for this instance of LBIST

9.4.81 EFUSE interface

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_dft_rom_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.82 dft_iso_en

Signal Name	Dir	Default Val	Description
aw0_core_dft_iso_en_dft_iso_en_lbist	In	1'd0	DFT iso_en for pins to LBISTCTL
aw0_core_dft_iso_en_dft_iso_en_pbist	In	1'd0	DFT iso_en for pins to PBIST
aw0_core_dft_iso_en_dft_iso_en_other	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST
aw0_core_dft_iso_en_dft_iso_en_lbist_delayed	In	1'd0	DFT iso_en for pins to LBISTCTL
aw0_core_dft_iso_en_dft_iso_en_pbist_delayed	In	1'd0	DFT iso_en for pins to PBIST
aw0_core_dft_iso_en_dft_iso_en_other_delayed	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST

9.4.83 bisor

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
aw0_c7xv_core_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
aw0_c7xv_core_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
aw0_c7xv_core_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
aw0_c7xv_core_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
aw0_c7xv_core_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
aw0_c7xv_core_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
aw0_c7xv_core_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.84 Target .vbusp_core_pbist_cfg power disable

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_pbist_disable_scr_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw0_c7xv_core_pbist_disable_scr_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw0_c7xv_core_pbist_disable_scr_disable_done	In	1'd0	Ack to SCR to stop clocks
aw0_c7xv_core_pbist_disable_scr_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw0_c7xv_core_pbist_disable_scr_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw0_c7xv_core_pbist_disable_scr_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.85 bisor

Signal Name	Dir	Default Val	Description
aw0_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
aw0_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
aw0_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
aw0_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
aw0_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
aw0_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
aw0_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
aw0_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.86 aw0_c7xv_core_dft_embed_pbist

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
aw0_c7xv_core_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
aw0_c7xv_core_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
aw0_c7xv_core_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
aw0_c7xv_core_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
aw0_c7xv_core_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
aw0_c7xv_core_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
aw0_c7xv_core_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
aw0_c7xv_core_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
aw0_c7xv_core_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
aw0_c7xv_core_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
aw0_c7xv_core_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
aw0_c7xv_core_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
aw0_c7xv_core_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
aw0_c7xv_core_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
aw0_c7xv_core_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
aw0_c7xv_core_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
aw0_c7xv_core_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
aw0_c7xv_core_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.87 aw0_c7xv_core_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.88 aw0_c7xv_core_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.89 aw0_c7xv_core_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
aw0_c7xv_core_dft_pbist_pwr_clk_en_ack	Out	c7xv_core_dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.90 aw0_c7xv_core_pbist_tclk

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_pbist_tclk_clk	In	1'd0	Single ended clock
aw0_c7xv_core_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.91 aw0_c7xv_core_pbist_rst

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.92 Power SCR Disable Interface

Signal Name	Dir	Default Val	Description
aw0_c7xv_core_pwr_dis_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw0_c7xv_core_pwr_dis_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw0_c7xv_core_pwr_dis_disable_done	In	1'd0	Ack to SCR to stop clocks
aw0_c7xv_core_pwr_dis_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw0_c7xv_core_pwr_dis_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw0_c7xv_core_pwr_dis_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.93 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw0_c7xv_mma_clkstop_clkstop_req	In	1'd0	Clock stop request
aw0_c7xv_mma_clkstop_clkstop_ack	Out	c7xv_mma_clkstop_clkstop_req	Clock stop acknowledge

9.4.94 Clken interface

Signal Name	Dir	Default Val	Description
aw0_c7xv_mma_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw0_c7xv_mma_clken_clk_en_ack	Out	c7xv_mma_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.95 aw0_c7xv_mma_rst

Signal Name	Dir	Default Val	Description
aw0_c7xv_mma_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.96 dft_compression for mma_array_r00 block

Signal Name	Dir	Default Val	Description
aw0_mma_r00_dft_block_en	In	1'd1	dft_block_en
aw0_mma_r00_dft_partition_en	In	1'd1	dft_partition_en
aw0_mma_r00_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw0_mma_r00_dft_compression_mode	In	1'd0	dft_compression_mode
aw0_mma_r00_dft_compression_spread	In	1'd0	dft_compression_spread
aw0_mma_r00_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw0_mma_r00_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw0_mma_r00_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw0_mma_r00_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw0_mma_r00_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw0_mma_r00_dft_struct_intest	In	1'd0	dft_struct_intest
aw0_mma_r00_dft_struct_extest	In	1'd0	dft_struct_extest
aw0_mma_r00_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw0_mma_r00_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw0_mma_r00_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.97 dft_compression for mma_array_r01 block

Signal Name	Dir	Default Val	Description
aw0_mma_r01_dft_block_en	In	1'd1	dft_block_en
aw0_mma_r01_dft_partition_en	In	1'd1	dft_partition_en
aw0_mma_r01_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw0_mma_r01_dft_compression_mode	In	1'd0	dft_compression_mode
aw0_mma_r01_dft_compression_spread	In	1'd0	dft_compression_spread
aw0_mma_r01_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw0_mma_r01_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw0_mma_r01_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw0_mma_r01_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw0_mma_r01_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw0_mma_r01_dft_struct_intest	In	1'd0	dft_struct_intest
aw0_mma_r01_dft_struct_extest	In	1'd0	dft_struct_extest
aw0_mma_r01_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw0_mma_r01_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw0_mma_r01_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.98 dft_compression for mma_array_r10 block

Signal Name	Dir	Default Val	Description
aw0_mma_r10_dft_block_en	In	1'd1	dft_block_en
aw0_mma_r10_dft_partition_en	In	1'd1	dft_partition_en
aw0_mma_r10_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw0_mma_r10_dft_compression_mode	In	1'd0	dft_compression_mode
aw0_mma_r10_dft_compression_spread	In	1'd0	dft_compression_spread
aw0_mma_r10_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw0_mma_r10_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw0_mma_r10_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw0_mma_r10_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw0_mma_r10_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw0_mma_r10_dft_struct_intest	In	1'd0	dft_struct_intest
aw0_mma_r10_dft_struct_extest	In	1'd0	dft_struct_extest
aw0_mma_r10_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw0_mma_r10_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw0_mma_r10_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.99 dft_compression for mma_array_r11 block

Signal Name	Dir	Default Val	Description
aw0_mma_r11_dft_block_en	In	1'd1	dft_block_en
aw0_mma_r11_dft_partition_en	In	1'd1	dft_partition_en
aw0_mma_r11_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw0_mma_r11_dft_compression_mode	In	1'd0	dft_compression_mode
aw0_mma_r11_dft_compression_spread	In	1'd0	dft_compression_spread
aw0_mma_r11_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw0_mma_r11_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw0_mma_r11_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw0_mma_r11_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw0_mma_r11_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw0_mma_r11_dft_struct_intest	In	1'd0	dft_struct_intest
aw0_mma_r11_dft_struct_extest	In	1'd0	dft_struct_extest
aw0_mma_r11_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw0_mma_r11_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw0_mma_r11_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.100 Power interface for mma_array_r00 block

Signal Name	Dir	Default Val	Description
aw0_r00_pwr_pwr_log_iso_en	In	1'd0	pwr
aw0_r00_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw0_r00_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw0_r00_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw0_r00_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.101 Power interface for mma_array_r01 block

Signal Name	Dir	Default Val	Description
aw0_r01_pwr_pwr_log_iso_en	In	1'd0	pwr
aw0_r01_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw0_r01_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw0_r01_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw0_r01_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.102 Power interface for mma_array_r10 block

Signal Name	Dir	Default Val	Description
aw0_r10_pwr_pwr_log_iso_en	In	1'd0	pwr
aw0_r10_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw0_r10_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw0_r10_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw0_r10_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.103 Power interface for mma_array_r11 block

Signal Name	Dir	Default Val	Description
aw0_r11_pwr_pwr_log_iso_en	In	1'd0	pwr
aw0_r11_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw0_r11_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw0_r11_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw0_r11_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.104 aw1_aw_ccc

Signal Name	Dir	Default Val	Description
aw1_aw_ccc_ipcfg[24:0]	Out	25'd0	ip config pin

9.4.105 aw1_c7xv_wrap_tieoff_clusternum

Signal Name	Dir	Default Val	Description
aw1_c7xv_wrap_tieoff_clusternum_ipcfg[7:0]	In	8'd0	ip config pin

9.4.106 aw1_c7xv_wrap_tieoff_corepacnum

Signal Name	Dir	Default Val	Description
aw1_c7xv_wrap_tieoff_corepacnum_ipcfg[7:0]	In	8'd0	ip config pin

9.4.107 aw1_c7xv_wrap_tieoff_sys_mma_present

Signal Name	Dir	Default Val	Description
aw1_c7xv_wrap_tieoff_sys_mma_present_ipcfg	In	1'd0	ip config pin

9.4.108 spare ports for PD

Signal Name	Dir	Default Val	Description
aw1_spare_spare_out[15:0]	Out	16'd0	spare output ports
aw1_spare_spare_in[15:0]	In	16'd0	spare input ports

9.4.109 aw1_pbist_clkstop

Signal Name	Dir	Default Val	Description
aw1_pbist_clkstop_clkstop_req	In	1'd0	Clock stop request
aw1_pbist_clkstop_clkstop_ack	Out	pbist_clkstop_clkstop_req	Clock stop acknowledge

9.4.110 Power interface

Signal Name	Dir	Default Val	Description
aw1_c7xv_pwr_pwr_log_iso_en	In	1'd0	pwr
aw1_c7xv_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw1_c7xv_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw1_c7xv_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw1_c7xv_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr
aw1_c7xv_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
aw1_c7xv_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
aw1_c7xv_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.111 dft_compression

Signal Name	Dir	Default Val	Description
aw1_dft_compression_core_dft_block_en	In	1'd1	dft_block_en
aw1_dft_compression_core_dft_partition_en	In	1'd1	dft_partition_en
aw1_dft_compression_core_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw1_dft_compression_core_dft_compression_mode	In	1'd0	dft_compression_mode
aw1_dft_compression_core_dft_compression_spread	In	1'd0	dft_compression_spread
aw1_dft_compression_core_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw1_dft_compression_core_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw1_dft_compression_core_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw1_dft_compression_core_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw1_dft_compression_core_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw1_dft_compression_core_dft_struct_intest	In	1'd0	dft_struct_intest
aw1_dft_compression_core_dft_struct_extest	In	1'd0	dft_struct_extest
aw1_dft_compression_core_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw1_dft_compression_core_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw1_dft_compression_core_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.112 Power interface

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_pwr_pwr_log_iso_en	In	1'd0	pwr
aw1_c7xv_core_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw1_c7xv_core_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw1_c7xv_core_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw1_c7xv_core_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr
aw1_c7xv_core_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
aw1_c7xv_core_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
aw1_c7xv_core_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.113 MEM_TESTRWM busdef

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_testrwm_dft_mem_testrwm	In	1'd0	memory testrwm

9.4.114 MEM_TEST1 busdef

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_test1_dft_mem_test1	In	1'd0	memory test1

9.4.115 Debug Timestamp Interface

Signal Name	Dir	Default Val	Description
aw1_glbdbg_timestamp_shft[1:0]	In	2'd0	binary interface
aw1_glbdbg_timestamp_gray[63:0]	In	64'd0	gray encoded interface

9.4.116 aw1_dbg_rst

Signal Name	Dir	Default Val	Description
aw1_dbg_rst_mod_dbg_l_rst_req_n	Out	1'd1	Module level debug local reset req

9.4.117 aw1_dbg_c7xv_state

Signal Name	Dir	Default Val	Description
aw1_dbg_c7xv_state_attention	Out	1'd0	debug event status

9.4.118 aw1_dbg_c7xv_susp

Signal Name	Dir	Default Val	Description
aw1_dbg_c7xv_susp_dbgsusp	Out	1'd0	Debug suspend mode request non real time

9.4.119 ARM Trust Zone Interface

Signal Name	Dir	Default Val	Description
aw1_dbg_sec_dbgen	In	1'd0	Invasive debug enable
aw1_dbg_sec_niden	In	1'd0	Non-invasive debug enable
aw1_dbg_sec_spiden	In	1'd0	Secure privileged invasive debug enable
aw1_dbg_sec_spniden	In	1'd0	Secure privileged non-invasive debug enable

9.4.120 aw1_debug

Signal Name	Dir	Default Val	Description
aw1_debug_debugenable	In	1'd0	enable debug

9.4.121 aw1_dbg_chm

Signal Name	Dir	Default Val	Description
aw1_dbg_chm_cti_chnl[3:0]	Out	4'd0	arm channel interface for cross triggering
aw1_dbg_chm_cti_chnl_ack[3:0]	In	4'd0	arm channel async handshake interface for cross triggering

9.4.122 aw1_dbg_chs

Signal Name	Dir	Default Val	Description
aw1_dbg_chs_cti_chnl[3:0]	In	4'd0	arm channel interface for cross triggering
aw1_dbg_chs_cti_chnl_ack[3:0]	Out	4'd0	arm channel async handshake interface for cross triggering

9.4.123 DFT mem interface

Signal Name	Dir	Default Val	Description
aw1_mem_atpg_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.124 MEM_MARGIN busdef

Signal Name	Dir	Default Val	Description
aw1_dftmem_s2cr1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cr1p compiler
aw1_dftmem_s2dc1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2dc1p compiler
aw1_dftmem_s2sr1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2sr1p compiler
aw1_dftmem_s2ss1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2ss1p compiler
aw1_dftmem_s2du1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2du1p compiler
aw1_dftmem_s2cr2p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cr2p compiler
aw1_dftmem_s2cu2p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cu2p compiler
aw1_dftmem_s2cr1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cr1p compiler
aw1_dftmem_s2dc1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2dc1p compiler
aw1_dftmem_s2sr1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2sr1p compiler
aw1_dftmem_s2ss1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2ss1p compiler
aw1_dftmem_s2du1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2du1p compiler
aw1_dftmem_s2cr2p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cr2p compiler
aw1_dftmem_s2cu2p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cu2p compiler
aw1_dftmem_s2du1p_dft_mem_wa[2:0]	In	3'd0	dft_mem_write assist for s2cr2p compiler
aw1_dftmem_s2du1p_dft_mem_wpulse[2:0]	In	3'd0	dft_mem_write assist for s2cr2p compiler
aw1_dftmem_rom_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for rom compiler

9.4.125 MEM_TESTRWM busdef

Signal Name	Dir	Default Val	Description
aw1_dftmem_testrwm_dft_mem_testrwm	In	1'd0	memory testrwm

9.4.126 MEM_DFTCLKEN busdef

Signal Name	Dir	Default Val	Description
aw1_dftmem_dftclken_dft_mem_dftclken	In	1'd0	memory dftclken

9.4.127 MEM_TEST1 busdef

Signal Name	Dir	Default Val	Description
aw1_dftmem_test1_dft_mem_test1	In	1'd0	memory test1

9.4.128 MEM_TEST_RNM busdef

Signal Name	Dir	Default Val	Description
aw1_dftmem_testnm_dft_mem_test_rnm	In	1'd0	memory test_rnm

9.4.129 PWR BIAS busdef

Signal Name	Dir	Default Val	Description
aw1_pwr_bias_pwr_mem_bc0	In	1'h0	pwr
aw1_pwr_bias_pwr_mem_bc1	In	1'h0	pwr
aw1_pwr_bias_pwr_mem_bc2	In	1'h0	pwr

9.4.130 efuse_ctl

Signal Name	Dir	Default Val	Description
aw1_efuse_ctl_fclrz	In	1'd0	efuse clear
aw1_efuse_ctl_fclk	In	1'd0	efuse clock

9.4.131 efuse_data

Signal Name	Dir	Default Val	Description
aw1_efuse_data_fdi	In	1'd0	efuse data in
aw1_efuse_data_fdo	Out	1'd0	efuse data out

9.4.132 efuse_ctl

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_efuse_ctl_fclrz	In	1'd0	efuse clear
aw1_c7xv_core_efuse_ctl_fclk	In	1'd0	efuse clock

9.4.133 efuse_data

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_efuse_data_fdi	In	1'd0	efuse data in
aw1_c7xv_core_efuse_data_fdo	Out	1'd0	efuse data out

9.4.134 divh_clk1_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw1_divh_clk1_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.135 divh_clk4_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw1_divh_clk4_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.136 divh_clk1_core_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw1_divh_clk1_core_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.137 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_clkstop_clkstop_req	In	1'd0	Clock stop request
aw1_c7xv_core_clkstop_clkstop_ack	Out	c7xv_core_clkstop_clkstop_req	Clock stop acknowledge

9.4.138 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw1_c7xv_wrap_clkstop_clkstop_req	In	1'd0	Clock stop request
aw1_c7xv_wrap_clkstop_clkstop_ack	Out	c7xv_wrap_clkstop_clkstop_req	Clock stop acknowledge

9.4.139 c7xv_clk1 clock

Signal Name	Dir	Default Val	Description
aw1_c7xv_clk_clk	In	1'd0	Single ended clock
aw1_c7xv_clk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_jpg_clk_l1

9.4.140 c7xv_clk1 clock

Signal Name	Dir	Default Val	Description
aw1_c7xv_divh_clk4_obsclk_out_clk	Out	1'd0	Single ended clock

9.4.141 pll ctrl clock

Signal Name	Dir	Default Val	Description
aw1_pll_ctrl_main_clk_clk	In	1'd0	Single ended clock

9.4.142 dft_ip

Signal Name	Dir	Default Val	Description
aw1_dft_clk_force_en	In	1'd0	dft_clk_force_en
aw1_dft_scan_en	In	1'd0	dft_scan_en
aw1_dft_async_rst_sel	In	1'd0	dft_async_rst_sel
aw1_dft_test_async_rst_n	In	1'd1	dft_test_async_rst_n
aw1_dft_tft_mcp_dis	In	1'd0	Assert High to block MCP path
aw1_dft_tft_conf_vector[67:0]	In	68'd0	clkdiv dft_tft_conf_vector
aw1_dft_tft_start_en	In	1'd0	clkdiv dft_tft_start_en
aw1_dft_delay_fault_en	In	1'd0	clkdiv dft_delay_fault_en
aw1_dft_tft_done	Out	1'd1	clkdiv dft_tft_done
aw1_dft_div_scan_en	In	1'd0	clkdiv dft_div_scan_en
aw1_dft_bypass_div_en	In	1'd0	clkdiv dft_bypass_div_en
aw1_dft_local_clk_en	In	1'd0	dft_local_clk_en
aw1_dft_pbist_clk_en	In	1'd0	dft_pbist_clk_en
aw1_dft_mode_atpg	In	1'd0	ATPG
aw1_dft_mode_iddq	In	1'd0	Iddq mode
aw1_dft_mode_burnin	In	1'd0	Burnin
aw1_dft_embed_mode_pbist	In	1'd0	Embedded pBIST Mode

9.4.143 dft_core ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
aw1_dft_core_clk_force_en	In	1'd0	dft_clk_force_en

9.4.144 dft_core ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
aw1_dft_mma_clk_force_en	In	1'd0	dft_clk_force_en

9.4.145 dft_compression

Signal Name	Dir	Default Val	Description
aw1_dft_compression_mma_dft_block_en	In	1'd1	dft_block_en
aw1_dft_compression_mma_dft_partition_en	In	1'd1	dft_partition_en
aw1_dft_compression_mma_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw1_dft_compression_mma_dft_compression_mode	In	1'd0	dft_compression_mode
aw1_dft_compression_mma_dft_compression_spread	In	1'd0	dft_compression_spread
aw1_dft_compression_mma_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw1_dft_compression_mma_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw1_dft_compression_mma_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw1_dft_compression_mma_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw1_dft_compression_mma_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw1_dft_compression_mma_dft_struct_intest	In	1'd0	dft_struct_intest
aw1_dft_compression_mma_dft_struct_extest	In	1'd0	dft_struct_extest
aw1_dft_compression_mma_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw1_dft_compression_mma_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw1_dft_compression_mma_dft_struct_so[35:0]	Out	36'd0	dft_struct_so



9.4.146 PLL control align signals

Signal Name	Dir	Default Val	Description
aw1_pll_ctrl_align_align_clk_div_rst_n	In	1'd0	Export align_clk_div
aw1_pll_ctrl_align_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.147 dft_compression

Signal Name	Dir	Default Val	Description
aw1_dft_compression_dft_block_en	In	1'd1	dft_block_en
aw1_dft_compression_dft_partition_en	In	1'd1	dft_partition_en
aw1_dft_compression_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw1_dft_compression_dft_compression_mode	In	1'd0	dft_compression_mode
aw1_dft_compression_dft_compression_spread	In	1'd0	dft_compression_spread
aw1_dft_compression_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw1_dft_compression_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw1_dft_compression_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw1_dft_compression_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw1_dft_compression_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw1_dft_compression_dft_struct_intest	In	1'd0	dft_struct_intest
aw1_dft_compression_dft_struct_extest	In	1'd0	dft_struct_extest
aw1_dft_compression_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw1_dft_compression_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw1_dft_compression_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.148 Controls for the TFT Clock control module

Signal Name	Dir	Default Val	Description
aw1_dft_pll_ctrl_dft_mode_pll_atpg	In	1'd0	dft_mode_pll_atpg
aw1_dft_pll_ctrl_dft_pll_capture_en	In	1'd0	dft_pll_capture_en
aw1_dft_pll_ctrl_dft_shift_clk_sel	In	1'd0	dft_shift_clk_sel
aw1_dft_pll_ctrl_cs_func_clk_en	In	1'd0	enable clocks to selected scan paths

9.4.149 DFT PLL CLK BYPASS control for the TFT Clock control module

Signal Name	Dir	Default Val	Description
aw1_dft_pll_clk_bypass_pll_clk_bypass	In	1'd1	DFT WIR input to kp_ti_ipg_loc_tft_clk_ctrl

9.4.150 c7xv clk1 clock

Signal Name	Dir	Default Val	Description
aw1_c7xv_clk_ctl_align_clk_div_rst_n	In	1'd0	Export align_clk_div
aw1_c7xv_clk_ctl_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.151 Clken interface

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw1_c7xv_core_clken_clk_en_ack	Out	c7xv_core_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.152 Clken interface

Signal Name	Dir	Default Val	Description
aw1_c7xv_wrap_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw1_c7xv_wrap_clken_clk_en_ack	Out	c7xv_wrap_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.153 DFT clock clken control for the Dividers

Signal Name	Dir	Default Val	Description
aw1_dft_local_div_clk_en_div_clk_en	In	1'd1	DFT local clk enable contribution to kp_ti_ipg_clk_divhdist5_tft_mux

9.4.154 aw1_c7xv_core_rst

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_rst_mod_g_rst_n	In	1'd1	Module level main reset
aw1_c7xv_core_rst_mod_por_rst_n	In	1'd1	Module level por reset
aw1_c7xv_core_rst_mod_l_rst_n	In	1'd1	Module level local reset

9.4.155 aw1_c7xv_wrap_rst

Signal Name	Dir	Default Val	Description
aw1_c7xv_wrap_rst_mod_g_rst_n	In	1'd1	Module level main reset
aw1_c7xv_wrap_rst_mod_por_rst_n	In	1'd1	Module level por reset
aw1_c7xv_wrap_rst_mod_l_rst_n	In	1'd1	Module level local reset

9.4.156 Power interface

Signal Name	Dir	Default Val	Description
aw1_mma_pwr_pwr_log_iso_en	In	1'd0	pwr
aw1_mma_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw1_mma_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw1_mma_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw1_mma_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.157 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw1_clk_bypass_wir	In	1'd0	dft wir signal

9.4.158 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw1_dft_clk_en_wir	In	1'd0	dft wir signal

9.4.159 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw1_divfactor_sel_wir	In	1'd0	dft wir signal

9.4.160 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw1_core_clk4_divfactor_sel_wir	In	1'd0	dft wir signal

9.4.161 Clk4 clock gate disable control for dft mode

Signal Name	Dir	Default Val	Description
dft_aw1_core_clk4_disable_wir	In	1'd0	dft wir signal

9.4.162 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw1_scrub_off_wir	In	1'd0	dft wir signal

9.4.163 Target .vbusp_pbist_cfg power disable

Signal Name	Dir	Default Val	Description
aw1_c7xv_pbist_disable_scr_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw1_c7xv_pbist_disable_scr_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw1_c7xv_pbist_disable_scr_disable_done	In	1'd0	Ack to SCR to stop clocks
aw1_c7xv_pbist_disable_scr_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw1_c7xv_pbist_disable_scr_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw1_c7xv_pbist_disable_scr_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.164 aw1_dft_embed_pbist

Signal Name	Dir	Default Val	Description
aw1_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
aw1_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
aw1_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
aw1_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
aw1_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
aw1_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
aw1_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
aw1_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
aw1_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
aw1_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
aw1_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
aw1_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
aw1_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
aw1_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
aw1_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
aw1_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
aw1_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
aw1_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
aw1_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.165 rom_ase

Signal Name	Dir	Default Val	Description
aw1_rom_ase_dft_mem_rom_ase	In	1'd0	dft_mem_rom_ase

9.4.166 aw1_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
aw1_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.167 aw1_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
aw1_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.168 aw1_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
aw1_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
aw1_dft_pbist_pwr_clk_en_ack	Out	dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.169 aw1_pbist_tclk

Signal Name	Dir	Default Val	Description
aw1_pbist_tclk_clk	In	1'd0	Single ended clock
aw1_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.170 aw1_pbist_rst

Signal Name	Dir	Default Val	Description
aw1_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.171 EFUSE interface

Signal Name	Dir	Default Val	Description
aw1_dft_rom_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.172 aw1_core_pbist_clkstop

Signal Name	Dir	Default Val	Description
aw1_core_pbist_clkstop_clkstop_req	In	1'd0	Clock stop request
aw1_core_pbist_clkstop_clkstop_ack	Out	core_pbist_clkstop_clkstop_req	Clock stop acknowledge

9.4.173 dft_lbist

Signal Name	Dir	Default Val	Description
aw1_dft_lbist_dft_lbist_atpg_mode	In	1'd0	ATPG control from SoC WIR
aw1_dft_lbist_dft_lbist_runbist_mode[3:0]	In	4'd0	Runbist mode enable
aw1_dft_lbist_dft_lbist_bist_reset	In	1'd0	Resets LBIST macro
aw1_dft_lbist_dft_lbist_bist_run[3:0]	In	4'd0	Starts LBIST if all bits are 1. Additional bits added to ensure that a single bit flip will not cause unintended LBIST execution.
aw1_dft_lbist_dft_lbist_scan_pc_def[3:0]	In	4'd0	Number of chain test patterns to run
aw1_dft_lbist_dft_lbist_reset_pc_def[3:0]	In	4'd0	Number of reset patterns to run.
aw1_dft_lbist_dft_lbist_set_pc_def[3:0]	In	4'd0	Number of set patterns to run
aw1_dft_lbist_dft_lbist_static_pc_def[13:0]	In	14'd0	Number of stuck-at patterns up to 16k to run
aw1_dft_lbist_dft_lbist_dc_def[1:0]	In	2'd0	Clock delay after scan_enable switching
aw1_dft_lbist_dft_lbist_prpg_def[52:0]	In	53'd0	Initial seed for PRPG
aw1_dft_lbist_dft_lbist_misr_mux_ctl[9:0]	In	10'd0	Selects block of 32 MISR bits to return to the MMR block.
aw1_dft_lbist_dft_lbist_out_mux_sel[1:0]	In	2'd0	Select to return PID, CTRL ID, or MISR result.
aw1_dft_lbist_dft_lbist_bist_done	Out	1'd0	LBIST done signal
aw1_dft_lbist_dft_lbist_bist_running	Out	1'd0	LBIST is running
aw1_dft_lbist_dft_lbist_misr_results[31:0]	Out	32'd0	32-bit section of MISR value. Selected using misr_mux_ctl input

9.4.174 dft_lbist_ctrl_id

Signal Name	Dir	Default Val	Description
aw1_dft_lbist_ctrl_id_dft_lbist_ctrl_id[5:0]	In	6'd0	Tieoff ID for this instance of LBIST

9.4.175 dft_iso_en

Signal Name	Dir	Default Val	Description
aw1_dft_iso_en_dft_iso_en_lbist	In	1'd0	DFT iso_en for pins to LBISTCTL
aw1_dft_iso_en_dft_iso_en_pbist	In	1'd0	DFT iso_en for pins to PBIST
aw1_dft_iso_en_dft_iso_en_other	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST
aw1_dft_iso_en_dft_iso_en_lbist_delayed	In	1'd0	DFT iso_en for pins to LBISTCTL
aw1_dft_iso_en_dft_iso_en_pbist_delayed	In	1'd0	DFT iso_en for pins to PBIST
aw1_dft_iso_en_dft_iso_en_other_delayed	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST

9.4.176 dft_lbist

Signal Name	Dir	Default Val	Description
aw1_core_dft_lbist_dft_lbist_atpg_mode	In	1'd0	ATPG control from SoC WIR
aw1_core_dft_lbist_dft_lbist_runbist_mode[3:0]	In	4'd0	Runbist mode enable
aw1_core_dft_lbist_dft_lbist_bist_reset	In	1'd0	Resets LBIST macro
aw1_core_dft_lbist_dft_lbist_bist_run[3:0]	In	4'd0	Starts LBIST if all bits are 1. Additional bits added to ensure that a single bit flip will not cause unintended LBIST execution.
aw1_core_dft_lbist_dft_lbist_scan_pc_def[3:0]	In	4'd0	Number of chain test patterns to run
aw1_core_dft_lbist_dft_lbist_reset_pc_def[3:0]	In	4'd0	Number of reset patterns to run.
aw1_core_dft_lbist_dft_lbist_set_pc_def[3:0]	In	4'd0	Number of set patterns to run
aw1_core_dft_lbist_dft_lbist_static_pc_def[13:0]	In	14'd0	Number of stuck-at patterns up to 16k to run
aw1_core_dft_lbist_dft_lbist_dc_def[1:0]	In	2'd0	Clock delay after scan_enable switching
aw1_core_dft_lbist_dft_lbist_prpg_def[52:0]	In	53'd0	Initial seed for PRPG
aw1_core_dft_lbist_dft_lbist_misr_mux_ctl[9:0]	In	10'd0	Selects block of 32 MISR bits to return to the MMR block.
aw1_core_dft_lbist_dft_lbist_out_mux_sel[1:0]	In	2'd0	Select to return PID, CTRL ID, or MISR result.
aw1_core_dft_lbist_dft_lbist_bist_done	Out	1'd0	LBIST done signal
aw1_core_dft_lbist_dft_lbist_bist_running	Out	1'd0	LBIST is running
aw1_core_dft_lbist_dft_lbist_misr_results[31:0]	Out	32'd0	32-bit section of MISR value. Selected using misr_mux_ctl input
aw1_core_dft_lbist_dft_lbist_divide_ratio[4:0]	In	5'd0	clkdiv divide value during lbist
aw1_core_dft_lbist_dft_lbist_load_div	In	1'd0	clkdiv divide load enable

9.4.177 dft_lbist_ctrl_id

Signal Name	Dir	Default Val	Description
aw1_core_dft_lbist_ctrl_id_dft_lbist_ctrl_id[5:0]	In	6'd0	Tieoff ID for this instance of LBIST

9.4.178 EFUSE interface

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_dft_rom_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.179 dft_iso_en

Signal Name	Dir	Default Val	Description
aw1_core_dft_iso_en_dft_iso_en_lbist	In	1'd0	DFT iso_en for pins to LBISTCTL
aw1_core_dft_iso_en_dft_iso_en_pbist	In	1'd0	DFT iso_en for pins to PBIST
aw1_core_dft_iso_en_dft_iso_en_other	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST
aw1_core_dft_iso_en_dft_iso_en_lbist_delayed	In	1'd0	DFT iso_en for pins to LBISTCTL
aw1_core_dft_iso_en_dft_iso_en_pbist_delayed	In	1'd0	DFT iso_en for pins to PBIST
aw1_core_dft_iso_en_dft_iso_en_other_delayed	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST

9.4.180 bisor

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
aw1_c7xv_core_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
aw1_c7xv_core_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
aw1_c7xv_core_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
aw1_c7xv_core_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
aw1_c7xv_core_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
aw1_c7xv_core_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
aw1_c7xv_core_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.181 Target .vbusp_core_pbist_cfg power disable

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_pbist_disable_scr_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw1_c7xv_core_pbist_disable_scr_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw1_c7xv_core_pbist_disable_scr_disable_done	In	1'd0	Ack to SCR to stop clocks
aw1_c7xv_core_pbist_disable_scr_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw1_c7xv_core_pbist_disable_scr_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw1_c7xv_core_pbist_disable_scr_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.182 bisor

Signal Name	Dir	Default Val	Description
aw1_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
aw1_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
aw1_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
aw1_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
aw1_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
aw1_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
aw1_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
aw1_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.183 aw1_c7xv_core_dft_embed_pbist

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
aw1_c7xv_core_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
aw1_c7xv_core_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
aw1_c7xv_core_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
aw1_c7xv_core_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
aw1_c7xv_core_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
aw1_c7xv_core_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
aw1_c7xv_core_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
aw1_c7xv_core_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
aw1_c7xv_core_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
aw1_c7xv_core_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
aw1_c7xv_core_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
aw1_c7xv_core_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
aw1_c7xv_core_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
aw1_c7xv_core_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
aw1_c7xv_core_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
aw1_c7xv_core_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
aw1_c7xv_core_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
aw1_c7xv_core_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.184 aw1_c7xv_core_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.185 aw1_c7xv_core_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.186 aw1_c7xv_core_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
aw1_c7xv_core_dft_pbist_pwr_clk_en_ack	Out	c7xv_core_dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.187 aw1_c7xv_core_pbist_tclk

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_pbist_tclk_clk	In	1'd0	Single ended clock
aw1_c7xv_core_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.188 aw1_c7xv_core_pbist_rst

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.189 Power SCR Disable Interface

Signal Name	Dir	Default Val	Description
aw1_c7xv_core_pwr_dis_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw1_c7xv_core_pwr_dis_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw1_c7xv_core_pwr_dis_disable_done	In	1'd0	Ack to SCR to stop clocks
aw1_c7xv_core_pwr_dis_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw1_c7xv_core_pwr_dis_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw1_c7xv_core_pwr_dis_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.190 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw1_c7xv_mma_clkstop_clkstop_req	In	1'd0	Clock stop request
aw1_c7xv_mma_clkstop_clkstop_ack	Out	c7xv_mma_clkstop_clkstop_req	Clock stop acknowledge

9.4.191 Clken interface

Signal Name	Dir	Default Val	Description
aw1_c7xv_mma_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw1_c7xv_mma_clken_clk_en_ack	Out	c7xv_mma_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.192 aw1_c7xv_mma_rst

Signal Name	Dir	Default Val	Description
aw1_c7xv_mma_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.193 dft_compression for mma_array_r00 block

Signal Name	Dir	Default Val	Description
aw1_mma_r00_dft_block_en	In	1'd1	dft_block_en
aw1_mma_r00_dft_partition_en	In	1'd1	dft_partition_en
aw1_mma_r00_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw1_mma_r00_dft_compression_mode	In	1'd0	dft_compression_mode
aw1_mma_r00_dft_compression_spread	In	1'd0	dft_compression_spread
aw1_mma_r00_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw1_mma_r00_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw1_mma_r00_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw1_mma_r00_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw1_mma_r00_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw1_mma_r00_dft_struct_intest	In	1'd0	dft_struct_intest
aw1_mma_r00_dft_struct_extest	In	1'd0	dft_struct_extest
aw1_mma_r00_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw1_mma_r00_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw1_mma_r00_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.194 dft_compression for mma_array_r01 block

Signal Name	Dir	Default Val	Description
aw1_mma_r01_dft_block_en	In	1'd1	dft_block_en
aw1_mma_r01_dft_partition_en	In	1'd1	dft_partition_en
aw1_mma_r01_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw1_mma_r01_dft_compression_mode	In	1'd0	dft_compression_mode
aw1_mma_r01_dft_compression_spread	In	1'd0	dft_compression_spread
aw1_mma_r01_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw1_mma_r01_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw1_mma_r01_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw1_mma_r01_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw1_mma_r01_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw1_mma_r01_dft_struct_intest	In	1'd0	dft_struct_intest
aw1_mma_r01_dft_struct_extest	In	1'd0	dft_struct_extest
aw1_mma_r01_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw1_mma_r01_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw1_mma_r01_dft_struct_so[35:0]	Out	36'd0	dft_struct_so



9.4.195 dft_compression for mma_array_r10 block

Signal Name	Dir	Default Val	Description
aw1_mma_r10_dft_block_en	In	1'd1	dft_block_en
aw1_mma_r10_dft_partition_en	In	1'd1	dft_partition_en
aw1_mma_r10_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw1_mma_r10_dft_compression_mode	In	1'd0	dft_compression_mode
aw1_mma_r10_dft_compression_spread	In	1'd0	dft_compression_spread
aw1_mma_r10_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw1_mma_r10_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw1_mma_r10_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw1_mma_r10_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw1_mma_r10_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw1_mma_r10_dft_struct_intest	In	1'd0	dft_struct_intest
aw1_mma_r10_dft_struct_extest	In	1'd0	dft_struct_extest
aw1_mma_r10_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw1_mma_r10_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw1_mma_r10_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.196 dft_compression for mma_array_r11 block

Signal Name	Dir	Default Val	Description
aw1_mma_r11_dft_block_en	In	1'd1	dft_block_en
aw1_mma_r11_dft_partition_en	In	1'd1	dft_partition_en
aw1_mma_r11_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw1_mma_r11_dft_compression_mode	In	1'd0	dft_compression_mode
aw1_mma_r11_dft_compression_spread	In	1'd0	dft_compression_spread
aw1_mma_r11_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw1_mma_r11_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw1_mma_r11_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw1_mma_r11_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw1_mma_r11_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw1_mma_r11_dft_struct_intest	In	1'd0	dft_struct_intest
aw1_mma_r11_dft_struct_extest	In	1'd0	dft_struct_extest
aw1_mma_r11_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw1_mma_r11_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw1_mma_r11_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.197 Power interface for mma_array_r00 block

Signal Name	Dir	Default Val	Description
aw1_r00_pwr_pwr_log_iso_en	In	1'd0	pwr
aw1_r00_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw1_r00_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw1_r00_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw1_r00_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.198 Power interface for mma_array_r01 block

Signal Name	Dir	Default Val	Description
aw1_r01_pwr_pwr_log_iso_en	In	1'd0	pwr
aw1_r01_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw1_r01_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw1_r01_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw1_r01_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.199 Power interface for mma_array_r10 block

Signal Name	Dir	Default Val	Description
aw1_r10_pwr_pwr_log_iso_en	In	1'd0	pwr
aw1_r10_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw1_r10_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw1_r10_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw1_r10_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.200 Power interface for mma_array_r11 block

Signal Name	Dir	Default Val	Description
aw1_r11_pwr_pwr_log_iso_en	In	1'd0	pwr
aw1_r11_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw1_r11_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw1_r11_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw1_r11_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.201 aw2_aw_ccc

Signal Name	Dir	Default Val	Description
aw2_aw_ccc_ipcfg[24:0]	Out	25'd0	ip config pin

9.4.202 aw2_c7xv_wrap_tieoff_clusternum

Signal Name	Dir	Default Val	Description
aw2_c7xv_wrap_tieoff_clusternum_ipcfg[7:0]	In	8'd0	ip config pin

9.4.203 aw2_c7xv_wrap_tieoff_corepacnum

Signal Name	Dir	Default Val	Description
aw2_c7xv_wrap_tieoff_corepacnum_ipcfg[7:0]	In	8'd0	ip config pin

9.4.204 aw2_c7xv_wrap_tieoff_sys_mma_present

Signal Name	Dir	Default Val	Description
aw2_c7xv_wrap_tieoff_sys_mma_present_ipcfg	In	1'd0	ip config pin

9.4.205 spare ports for PD

Signal Name	Dir	Default Val	Description
aw2_spare_spare_out[15:0]	Out	16'd0	spare output ports
aw2_spare_spare_in[15:0]	In	16'd0	spare input ports

9.4.206 aw2_pbist_clkstop

Signal Name	Dir	Default Val	Description
aw2_pbist_clkstop_clkstop_req	In	1'd0	Clock stop request
aw2_pbist_clkstop_clkstop_ack	Out	pbist_clkstop_clkstop_req	Clock stop acknowledge

9.4.207 Power interface

Signal Name	Dir	Default Val	Description
aw2_c7xv_pwr_pwr_log_iso_en	In	1'd0	pwr
aw2_c7xv_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw2_c7xv_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw2_c7xv_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw2_c7xv_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr
aw2_c7xv_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
aw2_c7xv_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
aw2_c7xv_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.208 dft_compression

Signal Name	Dir	Default Val	Description
aw2_dft_compression_core_dft_block_en	In	1'd1	dft_block_en
aw2_dft_compression_core_dft_partition_en	In	1'd1	dft_partition_en
aw2_dft_compression_core_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw2_dft_compression_core_dft_compression_mode	In	1'd0	dft_compression_mode
aw2_dft_compression_core_dft_compression_spread	In	1'd0	dft_compression_spread
aw2_dft_compression_core_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw2_dft_compression_core_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw2_dft_compression_core_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw2_dft_compression_core_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw2_dft_compression_core_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw2_dft_compression_core_dft_struct_intest	In	1'd0	dft_struct_intest
aw2_dft_compression_core_dft_struct_extest	In	1'd0	dft_struct_extest
aw2_dft_compression_core_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw2_dft_compression_core_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw2_dft_compression_core_dft_struct_so[35:0]	Out	36'd0	dft_struct_so



9.4.209 Power interface

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_pwr_pwr_log_iso_en	In	1'd0	pwr
aw2_c7xv_core_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw2_c7xv_core_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw2_c7xv_core_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw2_c7xv_core_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr
aw2_c7xv_core_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
aw2_c7xv_core_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
aw2_c7xv_core_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.210 MEM_TESTRWM busdef

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_testrwm_dft_mem_testrwm	In	1'd0	memory testrwm

9.4.211 MEM_TEST1 busdef

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_test1_dft_mem_test1	In	1'd0	memory test1

9.4.212 Debug Timestamp Interface

Signal Name	Dir	Default Val	Description
aw2_glbdbg_timestamp_shft[1:0]	In	2'd0	binary interface
aw2_glbdbg_timestamp_gray[63:0]	In	64'd0	gray encoded interface

9.4.213 aw2_dbg_rst

Signal Name	Dir	Default Val	Description
aw2_dbg_rst_mod_dbg_l_rst_req_n	Out	1'd1	Module level debug local reset req

9.4.214 aw2_dbg_c7xv_state

Signal Name	Dir	Default Val	Description
aw2_dbg_c7xv_state_attention	Out	1'd0	debug event status

9.4.215 aw2_dbg_c7xv_susp

Signal Name	Dir	Default Val	Description
aw2_dbg_c7xv_susp_dbgsusp	Out	1'd0	Debug suspend mode request non real time

9.4.216 ARM Trust Zone Interface

Signal Name	Dir	Default Val	Description
aw2_dbg_sec_dbgen	In	1'd0	Invasive debug enable
aw2_dbg_sec_niden	In	1'd0	Non-invasive debug enable
aw2_dbg_sec_spiden	In	1'd0	Secure privileged invasive debug enable
aw2_dbg_sec_spniden	In	1'd0	Secure privileged non-invasive debug enable

9.4.217 aw2_debug

Signal Name	Dir	Default Val	Description
aw2_debug_debugenable	In	1'd0	enable debug

9.4.218 aw2_dbg_chm

Signal Name	Dir	Default Val	Description
aw2_dbg_chm_cti_chnl[3:0]	Out	4'd0	arm channel interface for cross triggering
aw2_dbg_chm_cti_chnl_ack[3:0]	In	4'd0	arm channel async handshake interface for cross triggering

9.4.219 aw2_dbg_chs

Signal Name	Dir	Default Val	Description
aw2_dbg_chs_cti_chnl[3:0]	In	4'd0	arm channel interface for cross triggering
aw2_dbg_chs_cti_chnl_ack[3:0]	Out	4'd0	arm channel async handshake interface for cross triggering

9.4.220 DFT mem interface

Signal Name	Dir	Default Val	Description
aw2_mem_atpg_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.221 MEM_MARGIN busdef

Signal Name	Dir	Default Val	Description
aw2_dftmem_s2cr1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cr1p compiler
aw2_dftmem_s2dc1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2dc1p compiler
aw2_dftmem_s2sr1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2sr1p compiler
aw2_dftmem_s2ss1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2ss1p compiler
aw2_dftmem_s2du1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2du1p compiler
aw2_dftmem_s2cr2p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cr2p compiler
aw2_dftmem_s2cu2p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cu2p compiler
aw2_dftmem_s2cr1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cr1p compiler
aw2_dftmem_s2dc1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2dc1p compiler
aw2_dftmem_s2sr1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2sr1p compiler
aw2_dftmem_s2ss1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2ss1p compiler
aw2_dftmem_s2du1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2du1p compiler
aw2_dftmem_s2cr2p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cr2p compiler
aw2_dftmem_s2cu2p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cu2p compiler
aw2_dftmem_s2du1p_dft_mem_wa[2:0]	In	3'd0	dft_mem_write assist for s2cr2p compiler
aw2_dftmem_s2du1p_dft_mem_wpulse[2:0]	In	3'd0	dft_mem_write assist for s2cr2p compiler
aw2_dftmem_rom_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for rom compiler

9.4.222 MEM_TESTRWM busdef

Signal Name	Dir	Default Val	Description
aw2_dftmem_testrwm_dft_mem_testrwm	In	1'd0	memory testrwm

9.4.223 MEM_DFTCLKEN busdef

Signal Name	Dir	Default Val	Description
aw2_dftmem_dftclken_dft_mem_dftclken	In	1'd0	memory dftclken

9.4.224 MEM_TEST1 busdef

Signal Name	Dir	Default Val	Description
aw2_dftmem_test1_dft_mem_test1	In	1'd0	memory test1

9.4.225 MEM_TEST_RNM busdef

Signal Name	Dir	Default Val	Description
aw2_dftmem_testnm_dft_mem_test_rnm	In	1'd0	memory test_rnm

9.4.226 PWR BIAS busdef

Signal Name	Dir	Default Val	Description
aw2_pwr_bias_pwr_mem_bc0	In	1'h0	pwr
aw2_pwr_bias_pwr_mem_bc1	In	1'h0	pwr
aw2_pwr_bias_pwr_mem_bc2	In	1'h0	pwr

9.4.227 efuse_ctl

Signal Name	Dir	Default Val	Description
aw2_efuse_ctl_fclrz	In	1'd0	efuse clear
aw2_efuse_ctl_fclk	In	1'd0	efuse clock

9.4.228 efuse_data

Signal Name	Dir	Default Val	Description
aw2_efuse_data_fdi	In	1'd0	efuse data in
aw2_efuse_data_fdo	Out	1'd0	efuse data out

9.4.229 efuse_ctl

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_efuse_ctl_fclrz	In	1'd0	efuse clear
aw2_c7xv_core_efuse_ctl_fclk	In	1'd0	efuse clock

9.4.230 efuse_data

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_efuse_data_fdi	In	1'd0	efuse data in
aw2_c7xv_core_efuse_data_fdo	Out	1'd0	efuse data out

9.4.231 divh_clk1_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw2_divh_clk1_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.232 divh_clk4_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw2_divh_clk4_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.233 divh_clk1_core_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw2_divh_clk1_core_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.234 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_clkstop_clkstop_req	In	1'd0	Clock stop request
aw2_c7xv_core_clkstop_clkstop_ack	Out	c7xv_core_clkstop_clkstop_req	Clock stop acknowledge

9.4.235 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw2_c7xv_wrap_clkstop_clkstop_req	In	1'd0	Clock stop request
aw2_c7xv_wrap_clkstop_clkstop_ack	Out	c7xv_wrap_clkstop_clkstop_req	Clock stop acknowledge

9.4.236 c7xv clk1 clock

Signal Name	Dir	Default Val	Description
aw2_c7xv_clk_clk	In	1'd0	Single ended clock
aw2_c7xv_clk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_jpg_clk_l1

9.4.237 c7xv clk1 clock

Signal Name	Dir	Default Val	Description
aw2_c7xv_divh_clk4_obsclk_out_clk	Out	1'd0	Single ended clock

9.4.238 pll ctrl clock

Signal Name	Dir	Default Val	Description
aw2_pll_ctrl_main_clk_clk	In	1'd0	Single ended clock

9.4.239 dft_ip

Signal Name	Dir	Default Val	Description
aw2_dft_clk_force_en	In	1'd0	dft_clk_force_en
aw2_dft_scan_en	In	1'd0	dft_scan_en
aw2_dft_async_rst_sel	In	1'd0	dft_async_rst_sel
aw2_dft_test_async_rst_n	In	1'd1	dft_test_async_rst_n
aw2_dft_tft_mcp_dis	In	1'd0	Assert High to block MCP path
aw2_dft_tft_conf_vector[67:0]	In	68'd0	clkdiv dft_tft_conf_vector
aw2_dft_tft_start_en	In	1'd0	clkdiv dft_tft_start_en
aw2_dft_delay_fault_en	In	1'd0	clkdiv dft_delay_fault_en
aw2_dft_tft_done	Out	1'd1	clkdiv dft_tft_done
aw2_dft_div_scan_en	In	1'd0	clkdiv dft_div_scan_en
aw2_dft_bypass_div_en	In	1'd0	clkdiv dft_bypass_div_en
aw2_dft_local_clk_en	In	1'd0	dft_local_clk_en
aw2_dft_pbist_clk_en	In	1'd0	dft_pbist_clk_en
aw2_dft_mode_atpg	In	1'd0	ATPG
aw2_dft_mode_iddq	In	1'd0	Iddq mode
aw2_dft_mode_burnin	In	1'd0	Burnin
aw2_dft_embed_mode_pbist	In	1'd0	Embedded pBIST Mode

9.4.240 dft_core ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
aw2_dft_core_clk_force_en	In	1'd0	dft_clk_force_en

9.4.241 dft_core ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
aw2_dft_mma_clk_force_en	In	1'd0	dft_clk_force_en

9.4.242 dft_compression

Signal Name	Dir	Default Val	Description
aw2_dft_compression_mma_dft_block_en	In	1'd1	dft_block_en
aw2_dft_compression_mma_dft_partition_en	In	1'd1	dft_partition_en
aw2_dft_compression_mma_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw2_dft_compression_mma_dft_compression_mode	In	1'd0	dft_compression_mode
aw2_dft_compression_mma_dft_compression_spread	In	1'd0	dft_compression_spread
aw2_dft_compression_mma_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw2_dft_compression_mma_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw2_dft_compression_mma_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw2_dft_compression_mma_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw2_dft_compression_mma_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw2_dft_compression_mma_dft_struct_intest	In	1'd0	dft_struct_intest
aw2_dft_compression_mma_dft_struct_extest	In	1'd0	dft_struct_extest
aw2_dft_compression_mma_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw2_dft_compression_mma_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw2_dft_compression_mma_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.243 PLL control align signals

Signal Name	Dir	Default Val	Description
aw2_pll_ctrl_align_align_clk_div_rst_n	In	1'd0	Export align_clk_div
aw2_pll_ctrl_align_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.244 dft_compression

Signal Name	Dir	Default Val	Description
aw2_dft_compression_dft_block_en	In	1'd1	dft_block_en
aw2_dft_compression_dft_partition_en	In	1'd1	dft_partition_en
aw2_dft_compression_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw2_dft_compression_dft_compression_mode	In	1'd0	dft_compression_mode
aw2_dft_compression_dft_compression_spread	In	1'd0	dft_compression_spread
aw2_dft_compression_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw2_dft_compression_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw2_dft_compression_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw2_dft_compression_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw2_dft_compression_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw2_dft_compression_dft_struct_intest	In	1'd0	dft_struct_intest
aw2_dft_compression_dft_struct_extest	In	1'd0	dft_struct_extest
aw2_dft_compression_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw2_dft_compression_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw2_dft_compression_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.245 Controls for the TFT Clock control module

Signal Name	Dir	Default Val	Description
aw2_dft_pll_ctrl_dft_mode_pll_atpg	In	1'd0	dft_mode_pll_atpg
aw2_dft_pll_ctrl_dft_pll_capture_en	In	1'd0	dft_pll_capture_en
aw2_dft_pll_ctrl_dft_shift_clk_sel	In	1'd0	dft_shift_clk_sel
aw2_dft_pll_ctrl_cs_func_clk_en	In	1'd0	enable clocks to selected scan paths

9.4.246 DFT PLL CLK BYPASS control for the TFT Clock control module

Signal Name	Dir	Default Val	Description
aw2_dft_pll_clk_bypass_pll_clk_bypass	In	1'd1	DFT WIR input to kp_ti_ipg_loc_tft_clk_ctrl

9.4.247 c7xv clk1 clock

Signal Name	Dir	Default Val	Description
aw2_c7xv_clk_ctl_align_clk_div_rst_n	In	1'd0	Export align_clk_div
aw2_c7xv_clk_ctl_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.248 Clken interface

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw2_c7xv_core_clken_clk_en_ack	Out	c7xv_core_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.249 Clken interface

Signal Name	Dir	Default Val	Description
aw2_c7xv_wrap_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw2_c7xv_wrap_clken_clk_en_ack	Out	c7xv_wrap_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.250 DFT clock clken control for the Dividers

Signal Name	Dir	Default Val	Description
aw2_dft_local_div_clk_en_div_clk_en	In	1'd1	DFT local clk enable contribution to kp_ti_ipg_clk_divhdist5_tft_mux

9.4.251 aw2_c7xv_core_rst

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_rst_mod_g_rst_n	In	1'd1	Module level main reset
aw2_c7xv_core_rst_mod_por_rst_n	In	1'd1	Module level por reset
aw2_c7xv_core_rst_mod_l_rst_n	In	1'd1	Module level local reset

9.4.252 aw2_c7xv_wrap_rst

Signal Name	Dir	Default Val	Description
aw2_c7xv_wrap_rst_mod_g_rst_n	In	1'd1	Module level main reset
aw2_c7xv_wrap_rst_mod_por_rst_n	In	1'd1	Module level por reset
aw2_c7xv_wrap_rst_mod_l_rst_n	In	1'd1	Module level local reset

9.4.253 Power interface

Signal Name	Dir	Default Val	Description
aw2_mma_pwr_pwr_log_iso_en	In	1'd0	pwr
aw2_mma_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw2_mma_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw2_mma_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw2_mma_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.254 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw2_clk_bypass_wir	In	1'd0	dft wir signal

9.4.255 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw2_dft_clk_en_wir	In	1'd0	dft wir signal

9.4.256 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw2_divfactor_sel_wir	In	1'd0	dft wir signal

9.4.257 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw2_core_clk4_divfactor_sel_wir	In	1'd0	dft wir signal

9.4.258 Clk4 clock gate disable control for dft mode

Signal Name	Dir	Default Val	Description
dft_aw2_core_clk4_disable_wir	In	1'd0	dft wir signal

9.4.259 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw2_scrub_off_wir	In	1'd0	dft wir signal

9.4.260 Target .vbusp_pbist_cfg power disable

Signal Name	Dir	Default Val	Description
aw2_c7xv_pbist_disable_scr_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw2_c7xv_pbist_disable_scr_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw2_c7xv_pbist_disable_scr_disable_done	In	1'd0	Ack to SCR to stop clocks
aw2_c7xv_pbist_disable_scr_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw2_c7xv_pbist_disable_scr_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw2_c7xv_pbist_disable_scr_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.261 aw2_dft_embed_pbist

Signal Name	Dir	Default Val	Description
aw2_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
aw2_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
aw2_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
aw2_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
aw2_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
aw2_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
aw2_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
aw2_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
aw2_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
aw2_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
aw2_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
aw2_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
aw2_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
aw2_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
aw2_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
aw2_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
aw2_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
aw2_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
aw2_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.262 rom_ase

Signal Name	Dir	Default Val	Description
aw2_rom_ase_dft_mem_rom_ase	In	1'd0	dft_mem_rom_ase

9.4.263 aw2_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
aw2_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.264 aw2_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
aw2_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.265 aw2_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
aw2_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
aw2_dft_pbist_pwr_clk_en_ack	Out	dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.266 aw2_pbist_tclk

Signal Name	Dir	Default Val	Description
aw2_pbist_tclk_clk	In	1'd0	Single ended clock
aw2_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.267 aw2_pbist_rst

Signal Name	Dir	Default Val	Description
aw2_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.268 EFUSE interface

Signal Name	Dir	Default Val	Description
aw2_dft_rom_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.269 aw2_core_pbist_clkstop

Signal Name	Dir	Default Val	Description
aw2_core_pbist_clkstop_clkstop_req	In	1'd0	Clock stop request
aw2_core_pbist_clkstop_clkstop_ack	Out	core_pbist_clkstop_clkstop_req	Clock stop acknowledge

9.4.270 dft_lbist

Signal Name	Dir	Default Val	Description
aw2_dft_lbist_dft_lbist_atpg_mode	In	1'd0	ATPG control from SoC WIR
aw2_dft_lbist_dft_lbist_runbist_mode[3:0]	In	4'd0	Runbist mode enable
aw2_dft_lbist_dft_lbist_bist_reset	In	1'd0	Resets LBIST macro
aw2_dft_lbist_dft_lbist_bist_run[3:0]	In	4'd0	Starts LBIST if all bits are 1. Additional bits added to ensure that a single bit flip will not cause unintended LBIST execution.
aw2_dft_lbist_dft_lbist_scan_pc_def[3:0]	In	4'd0	Number of chain test patterns to run
aw2_dft_lbist_dft_lbist_reset_pc_def[3:0]	In	4'd0	Number of reset patterns to run.
aw2_dft_lbist_dft_lbist_set_pc_def[3:0]	In	4'd0	Number of set patterns to run
aw2_dft_lbist_dft_lbist_static_pc_def[13:0]	In	14'd0	Number of stuck-at patterns up to 16k to run
aw2_dft_lbist_dft_lbist_dc_def[1:0]	In	2'd0	Clock delay after scan_enable switching
aw2_dft_lbist_dft_lbist_prpg_def[52:0]	In	53'd0	Initial seed for PRPG
aw2_dft_lbist_dft_lbist_misr_mux_ctl[9:0]	In	10'd0	Selects block of 32 MISR bits to return to the MMR block.
aw2_dft_lbist_dft_lbist_out_mux_sel[1:0]	In	2'd0	Select to return PID, CTRL ID, or MISR result.
aw2_dft_lbist_dft_lbist_bist_done	Out	1'd0	LBIST done signal
aw2_dft_lbist_dft_lbist_bist_running	Out	1'd0	LBIST is running
aw2_dft_lbist_dft_lbist_misr_results[31:0]	Out	32'd0	32-bit section of MISR value. Selected using misr_mux_ctl input

9.4.271 dft_lbist_ctrl_id

Signal Name	Dir	Default Val	Description
aw2_dft_lbist_ctrl_id_dft_lbist_ctrl_id[5:0]	In	6'd0	Tieoff ID for this instance of LBIST

9.4.272 dft_iso_en

Signal Name	Dir	Default Val	Description
aw2_dft_iso_en_dft_iso_en_lbist	In	1'd0	DFT iso_en for pins to LBISTCTL
aw2_dft_iso_en_dft_iso_en_pbist	In	1'd0	DFT iso_en for pins to PBIST
aw2_dft_iso_en_dft_iso_en_other	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST
aw2_dft_iso_en_dft_iso_en_lbist_delayed	In	1'd0	DFT iso_en for pins to LBISTCTL
aw2_dft_iso_en_dft_iso_en_pbist_delayed	In	1'd0	DFT iso_en for pins to PBIST
aw2_dft_iso_en_dft_iso_en_other_delayed	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST

9.4.273 dft_lbist

Signal Name	Dir	Default Val	Description
aw2_core_dft_lbist_dft_lbist_atpg_mode	In	1'd0	ATPG control from SoC WIR
aw2_core_dft_lbist_dft_lbist_runbist_mode[3:0]	In	4'd0	Runbist mode enable
aw2_core_dft_lbist_dft_lbist_bist_reset	In	1'd0	Resets LBIST macro
aw2_core_dft_lbist_dft_lbist_bist_run[3:0]	In	4'd0	Starts LBIST if all bits are 1. Additional bits added to ensure that a single bit flip will not cause unintended LBIST execution.
aw2_core_dft_lbist_dft_lbist_scan_pc_def[3:0]	In	4'd0	Number of chain test patterns to run
aw2_core_dft_lbist_dft_lbist_reset_pc_def[3:0]	In	4'd0	Number of reset patterns to run.
aw2_core_dft_lbist_dft_lbist_set_pc_def[3:0]	In	4'd0	Number of set patterns to run
aw2_core_dft_lbist_dft_lbist_static_pc_def[13:0]	In	14'd0	Number of stuck-at patterns up to 16k to run
aw2_core_dft_lbist_dft_lbist_dc_def[1:0]	In	2'd0	Clock delay after scan_enable switching
aw2_core_dft_lbist_dft_lbist_prpg_def[52:0]	In	53'd0	Initial seed for PRPG
aw2_core_dft_lbist_dft_lbist_misr_mux_ctl[9:0]	In	10'd0	Selects block of 32 MISR bits to return to the MMR block.
aw2_core_dft_lbist_dft_lbist_out_mux_sel[1:0]	In	2'd0	Select to return PID, CTRL ID, or MISR result.
aw2_core_dft_lbist_dft_lbist_bist_done	Out	1'd0	LBIST done signal
aw2_core_dft_lbist_dft_lbist_bist_running	Out	1'd0	LBIST is running
aw2_core_dft_lbist_dft_lbist_misr_results[31:0]	Out	32'd0	32-bit section of MISR value. Selected using misr_mux_ctl input
aw2_core_dft_lbist_dft_lbist_divide_ratio[4:0]	In	5'd0	clkdiv divide value during lbist
aw2_core_dft_lbist_dft_lbist_load_div	In	1'd0	clkdiv divide load enable

9.4.274 dft_lbist_ctrl_id

Signal Name	Dir	Default Val	Description
aw2_core_dft_lbist_ctrl_id_dft_lbist_ctrl_id[5:0]	In	6'd0	Tieoff ID for this instance of LBIST

9.4.275 EFUSE interface

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_dft_rom_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.276 dft_iso_en

Signal Name	Dir	Default Val	Description
aw2_core_dft_iso_en_dft_iso_en_lbist	In	1'd0	DFT iso_en for pins to LBISTCTL
aw2_core_dft_iso_en_dft_iso_en_pbist	In	1'd0	DFT iso_en for pins to PBIST
aw2_core_dft_iso_en_dft_iso_en_other	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST
aw2_core_dft_iso_en_dft_iso_en_lbist_delayed	In	1'd0	DFT iso_en for pins to LBISTCTL
aw2_core_dft_iso_en_dft_iso_en_pbist_delayed	In	1'd0	DFT iso_en for pins to PBIST
aw2_core_dft_iso_en_dft_iso_en_other_delayed	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST

9.4.277 bisor

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
aw2_c7xv_core_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
aw2_c7xv_core_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
aw2_c7xv_core_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
aw2_c7xv_core_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
aw2_c7xv_core_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
aw2_c7xv_core_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
aw2_c7xv_core_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.278 Target .vbusp_core_pbist_cfg power disable

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_pbist_disable_scr_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw2_c7xv_core_pbist_disable_scr_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw2_c7xv_core_pbist_disable_scr_disable_done	In	1'd0	Ack to SCR to stop clocks
aw2_c7xv_core_pbist_disable_scr_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw2_c7xv_core_pbist_disable_scr_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw2_c7xv_core_pbist_disable_scr_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.279 bisor

Signal Name	Dir	Default Val	Description
aw2_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
aw2_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
aw2_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
aw2_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
aw2_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
aw2_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
aw2_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
aw2_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.280 aw2_c7xv_core_dft_embed_pbist

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
aw2_c7xv_core_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
aw2_c7xv_core_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
aw2_c7xv_core_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
aw2_c7xv_core_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
aw2_c7xv_core_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
aw2_c7xv_core_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
aw2_c7xv_core_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
aw2_c7xv_core_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
aw2_c7xv_core_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
aw2_c7xv_core_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
aw2_c7xv_core_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
aw2_c7xv_core_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
aw2_c7xv_core_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
aw2_c7xv_core_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
aw2_c7xv_core_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
aw2_c7xv_core_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
aw2_c7xv_core_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
aw2_c7xv_core_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.281 aw2_c7xv_core_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.282 aw2_c7xv_core_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.283 aw2_c7xv_core_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
aw2_c7xv_core_dft_pbist_pwr_clk_en_ack	Out	c7xv_core_dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.284 aw2_c7xv_core_pbist_tclk

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_pbist_tclk_clk	In	1'd0	Single ended clock
aw2_c7xv_core_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.285 aw2_c7xv_core_pbist_rst

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.286 Power SCR Disable Interface

Signal Name	Dir	Default Val	Description
aw2_c7xv_core_pwr_dis_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw2_c7xv_core_pwr_dis_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw2_c7xv_core_pwr_dis_disable_done	In	1'd0	Ack to SCR to stop clocks
aw2_c7xv_core_pwr_dis_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw2_c7xv_core_pwr_dis_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw2_c7xv_core_pwr_dis_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.287 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw2_c7xv_mma_clkstop_clkstop_req	In	1'd0	Clock stop request
aw2_c7xv_mma_clkstop_clkstop_ack	Out	c7xv_mma_clkstop_clkstop_req	Clock stop acknowledge

9.4.288 Clken interface

Signal Name	Dir	Default Val	Description
aw2_c7xv_mma_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw2_c7xv_mma_clken_clk_en_ack	Out	c7xv_mma_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.289 aw2_c7xv_mma_rst

Signal Name	Dir	Default Val	Description
aw2_c7xv_mma_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.290 dft_compression for mma_array_r00 block

Signal Name	Dir	Default Val	Description
aw2_mma_r00_dft_block_en	In	1'd1	dft_block_en
aw2_mma_r00_dft_partition_en	In	1'd1	dft_partition_en
aw2_mma_r00_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw2_mma_r00_dft_compression_mode	In	1'd0	dft_compression_mode
aw2_mma_r00_dft_compression_spread	In	1'd0	dft_compression_spread
aw2_mma_r00_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw2_mma_r00_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw2_mma_r00_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw2_mma_r00_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw2_mma_r00_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw2_mma_r00_dft_struct_intest	In	1'd0	dft_struct_intest
aw2_mma_r00_dft_struct_extest	In	1'd0	dft_struct_extest
aw2_mma_r00_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw2_mma_r00_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw2_mma_r00_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.291 dft_compression for mma_array_r01 block

Signal Name	Dir	Default Val	Description
aw2_mma_r01_dft_block_en	In	1'd1	dft_block_en
aw2_mma_r01_dft_partition_en	In	1'd1	dft_partition_en
aw2_mma_r01_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw2_mma_r01_dft_compression_mode	In	1'd0	dft_compression_mode
aw2_mma_r01_dft_compression_spread	In	1'd0	dft_compression_spread
aw2_mma_r01_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw2_mma_r01_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw2_mma_r01_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw2_mma_r01_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw2_mma_r01_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw2_mma_r01_dft_struct_intest	In	1'd0	dft_struct_intest
aw2_mma_r01_dft_struct_extest	In	1'd0	dft_struct_extest
aw2_mma_r01_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw2_mma_r01_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw2_mma_r01_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.292 dft_compression for mma_array_r10 block

Signal Name	Dir	Default Val	Description
aw2_mma_r10_dft_block_en	In	1'd1	dft_block_en
aw2_mma_r10_dft_partition_en	In	1'd1	dft_partition_en
aw2_mma_r10_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw2_mma_r10_dft_compression_mode	In	1'd0	dft_compression_mode
aw2_mma_r10_dft_compression_spread	In	1'd0	dft_compression_spread
aw2_mma_r10_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw2_mma_r10_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw2_mma_r10_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw2_mma_r10_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw2_mma_r10_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw2_mma_r10_dft_struct_intest	In	1'd0	dft_struct_intest
aw2_mma_r10_dft_struct_extest	In	1'd0	dft_struct_extest
aw2_mma_r10_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw2_mma_r10_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw2_mma_r10_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.293 dft_compression for mma_array_r11 block

Signal Name	Dir	Default Val	Description
aw2_mma_r11_dft_block_en	In	1'd1	dft_block_en
aw2_mma_r11_dft_partition_en	In	1'd1	dft_partition_en
aw2_mma_r11_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw2_mma_r11_dft_compression_mode	In	1'd0	dft_compression_mode
aw2_mma_r11_dft_compression_spread	In	1'd0	dft_compression_spread
aw2_mma_r11_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw2_mma_r11_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw2_mma_r11_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw2_mma_r11_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw2_mma_r11_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw2_mma_r11_dft_struct_intest	In	1'd0	dft_struct_intest
aw2_mma_r11_dft_struct_extest	In	1'd0	dft_struct_extest
aw2_mma_r11_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw2_mma_r11_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw2_mma_r11_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.294 Power interface for mma_array_r00 block

Signal Name	Dir	Default Val	Description
aw2_r00_pwr_pwr_log_iso_en	In	1'd0	pwr
aw2_r00_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw2_r00_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw2_r00_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw2_r00_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.295 Power interface for mma_array_r01 block

Signal Name	Dir	Default Val	Description
aw2_r01_pwr_pwr_log_iso_en	In	1'd0	pwr
aw2_r01_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw2_r01_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw2_r01_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw2_r01_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.296 Power interface for mma_array_r10 block

Signal Name	Dir	Default Val	Description
aw2_r10_pwr_pwr_log_iso_en	In	1'd0	pwr
aw2_r10_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw2_r10_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw2_r10_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw2_r10_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.297 Power interface for mma_array_r11 block

Signal Name	Dir	Default Val	Description
aw2_r11_pwr_pwr_log_iso_en	In	1'd0	pwr
aw2_r11_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw2_r11_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw2_r11_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw2_r11_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.298 aw3_aw_ccc

Signal Name	Dir	Default Val	Description
aw3_aw_ccc_ipcfg[24:0]	Out	25'd0	ip config pin

9.4.299 aw3_c7xv_wrap_tieoff_clusternum

Signal Name	Dir	Default Val	Description
aw3_c7xv_wrap_tieoff_clusternum_ipcfg[7:0]	In	8'd0	ip config pin

9.4.300 aw3_c7xv_wrap_tieoff_corepacnum

Signal Name	Dir	Default Val	Description
aw3_c7xv_wrap_tieoff_corepacnum_ipcfg[7:0]	In	8'd0	ip config pin

9.4.301 aw3_c7xv_wrap_tieoff_sys_mma_present

Signal Name	Dir	Default Val	Description
aw3_c7xv_wrap_tieoff_sys_mma_present_ipcfg	In	1'd0	ip config pin

9.4.302 spare ports for PD

Signal Name	Dir	Default Val	Description
aw3_spare_spare_out[15:0]	Out	16'd0	spare output ports
aw3_spare_spare_in[15:0]	In	16'd0	spare input ports

9.4.303 aw3_pbist_clkstop

Signal Name	Dir	Default Val	Description
aw3_pbist_clkstop_clkstop_req	In	1'd0	Clock stop request
aw3_pbist_clkstop_clkstop_ack	Out	pbist_clkstop_clkstop_req	Clock stop acknowledge

9.4.304 Power interface

Signal Name	Dir	Default Val	Description
aw3_c7xv_pwr_pwr_log_iso_en	In	1'd0	pwr
aw3_c7xv_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw3_c7xv_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw3_c7xv_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw3_c7xv_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr
aw3_c7xv_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
aw3_c7xv_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
aw3_c7xv_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.305 dft_compression

Signal Name	Dir	Default Val	Description
aw3_dft_compression_core_dft_block_en	In	1'd1	dft_block_en
aw3_dft_compression_core_dft_partition_en	In	1'd1	dft_partition_en
aw3_dft_compression_core_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw3_dft_compression_core_dft_compression_mode	In	1'd0	dft_compression_mode
aw3_dft_compression_core_dft_compression_spread	In	1'd0	dft_compression_spread
aw3_dft_compression_core_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw3_dft_compression_core_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw3_dft_compression_core_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw3_dft_compression_core_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw3_dft_compression_core_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw3_dft_compression_core_dft_struct_intest	In	1'd0	dft_struct_intest
aw3_dft_compression_core_dft_struct_extest	In	1'd0	dft_struct_extest
aw3_dft_compression_core_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw3_dft_compression_core_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw3_dft_compression_core_dft_struct_so[35:0]	Out	36'd0	dft_struct_so



9.4.306 Power interface

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_pwr_pwr_log_iso_en	In	1'd0	pwr
aw3_c7xv_core_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw3_c7xv_core_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw3_c7xv_core_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw3_c7xv_core_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr
aw3_c7xv_core_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
aw3_c7xv_core_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
aw3_c7xv_core_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.307 MEM_TESTRWM busdef

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_testrwm_dft_mem_testrwm	In	1'd0	memory testrwm

9.4.308 MEM_TEST1 busdef

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_test1_dft_mem_test1	In	1'd0	memory test1

9.4.309 Debug Timestamp Interface

Signal Name	Dir	Default Val	Description
aw3_glbdbg_timestamp_shft[1:0]	In	2'd0	binary interface
aw3_glbdbg_timestamp_gray[63:0]	In	64'd0	gray encoded interface

9.4.310 aw3_dbg_rst

Signal Name	Dir	Default Val	Description
aw3_dbg_rst_mod_dbg_l_rst_req_n	Out	1'd1	Module level debug local reset req

9.4.311 aw3_dbg_c7xv_state

Signal Name	Dir	Default Val	Description
aw3_dbg_c7xv_state_attention	Out	1'd0	debug event status

9.4.312 aw3_dbg_c7xv_susp

Signal Name	Dir	Default Val	Description
aw3_dbg_c7xv_susp_dbgsusp	Out	1'd0	Debug suspend mode request non real time

9.4.313 ARM Trust Zone Interface

Signal Name	Dir	Default Val	Description
aw3_dbg_sec_dbgen	In	1'd0	Invasive debug enable
aw3_dbg_sec_niden	In	1'd0	Non-invasive debug enable
aw3_dbg_sec_spiden	In	1'd0	Secure privileged invasive debug enable
aw3_dbg_sec_spniden	In	1'd0	Secure privileged non-invasive debug enable

9.4.314 aw3_debug

Signal Name	Dir	Default Val	Description
aw3_debug_debugenable	In	1'd0	enable debug

9.4.315 aw3_dbg_chm

Signal Name	Dir	Default Val	Description
aw3_dbg_chm_cti_chnl[3:0]	Out	4'd0	arm channel interface for cross triggering
aw3_dbg_chm_cti_chnl_ack[3:0]	In	4'd0	arm channel async handshake interface for cross triggering

9.4.316 aw3_dbg_chs

Signal Name	Dir	Default Val	Description
aw3_dbg_chs_cti_chnl[3:0]	In	4'd0	arm channel interface for cross triggering
aw3_dbg_chs_cti_chnl_ack[3:0]	Out	4'd0	arm channel async handshake interface for cross triggering

9.4.317 DFT mem interface

Signal Name	Dir	Default Val	Description
aw3_mem_atpg_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.318 MEM_MARGIN busdef

Signal Name	Dir	Default Val	Description
aw3_dftmem_s2cr1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cr1p compiler
aw3_dftmem_s2dc1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2dc1p compiler
aw3_dftmem_s2sr1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2sr1p compiler
aw3_dftmem_s2ss1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2ss1p compiler
aw3_dftmem_s2du1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2du1p compiler
aw3_dftmem_s2cr2p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cr2p compiler
aw3_dftmem_s2cu2p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cu2p compiler
aw3_dftmem_s2cr1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cr1p compiler
aw3_dftmem_s2dc1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2dc1p compiler
aw3_dftmem_s2sr1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2sr1p compiler
aw3_dftmem_s2ss1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2ss1p compiler
aw3_dftmem_s2du1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2du1p compiler
aw3_dftmem_s2cr2p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cr2p compiler
aw3_dftmem_s2cu2p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cu2p compiler
aw3_dftmem_s2du1p_dft_mem_wa[2:0]	In	3'd0	dft_mem_write assist for s2cr2p compiler
aw3_dftmem_s2du1p_dft_mem_wpulse[2:0]	In	3'd0	dft_mem_write assist for s2cr2p compiler
aw3_dftmem_rom_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for rom compiler

9.4.319 MEM_TESTRWM busdef

Signal Name	Dir	Default Val	Description
aw3_dftmem_testrwm_dft_mem_testrwm	In	1'd0	memory testrwm

9.4.320 MEM_DFTCLKEN busdef

Signal Name	Dir	Default Val	Description
aw3_dftmem_dftclken_dft_mem_dftclken	In	1'd0	memory dftclken



9.4.321 MEM_TEST1 busdef

Signal Name	Dir	Default Val	Description
aw3_dftmem_test1_dft_mem_test1	In	1'd0	memory test1

9.4.322 MEM_TEST_RNM busdef

Signal Name	Dir	Default Val	Description
aw3_dftmem_testnm_dft_mem_test_rnm	In	1'd0	memory test_rnm

9.4.323 PWR BIAS busdef

Signal Name	Dir	Default Val	Description
aw3_pwr_bias_pwr_mem_bc0	In	1'h0	pwr
aw3_pwr_bias_pwr_mem_bc1	In	1'h0	pwr
aw3_pwr_bias_pwr_mem_bc2	In	1'h0	pwr

9.4.324 efuse_ctl

Signal Name	Dir	Default Val	Description
aw3_efuse_ctl_fclrz	In	1'd0	efuse clear
aw3_efuse_ctl_fclk	In	1'd0	efuse clock

9.4.325 efuse_data

Signal Name	Dir	Default Val	Description
aw3_efuse_data_fdi	In	1'd0	efuse data in
aw3_efuse_data_fdo	Out	1'd0	efuse data out

9.4.326 efuse_ctl

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_efuse_ctl_fclrz	In	1'd0	efuse clear
aw3_c7xv_core_efuse_ctl_fclk	In	1'd0	efuse clock

9.4.327 efuse_data

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_efuse_data_fdi	In	1'd0	efuse data in
aw3_c7xv_core_efuse_data_fdo	Out	1'd0	efuse data out

9.4.328 divh_clk1_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw3_divh_clk1_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.329 divh_clk4_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw3_divh_clk4_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.330 divh_clk1_core_tft_clk_en control

Signal Name	Dir	Default Val	Description
aw3_divh_clk1_core_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.331 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_clkstop_clkstop_req	In	1'd0	Clock stop request
aw3_c7xv_core_clkstop_clkstop_ack	Out	c7xv_core_clkstop_clkstop_req	Clock stop acknowledge

9.4.332 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw3_c7xv_wrap_clkstop_clkstop_req	In	1'd0	Clock stop request
aw3_c7xv_wrap_clkstop_clkstop_ack	Out	c7xv_wrap_clkstop_clkstop_req	Clock stop acknowledge

9.4.333 c7xv clk1 clock

Signal Name	Dir	Default Val	Description
aw3_c7xv_clk_clk	In	1'd0	Single ended clock
aw3_c7xv_clk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_jpg_clk_l1

9.4.334 c7xv clk1 clock

Signal Name	Dir	Default Val	Description
aw3_c7xv_divh_clk4_obsclk_out_clk	Out	1'd0	Single ended clock

9.4.335 pll ctrl clock

Signal Name	Dir	Default Val	Description
aw3_pll_ctrl_main_clk_clk	In	1'd0	Single ended clock

9.4.336 dft_ip

Signal Name	Dir	Default Val	Description
aw3_dft_clk_force_en	In	1'd0	dft_clk_force_en
aw3_dft_scan_en	In	1'd0	dft_scan_en
aw3_dft_async_rst_sel	In	1'd0	dft_async_rst_sel
aw3_dft_test_async_rst_n	In	1'd1	dft_test_async_rst_n
aw3_dft_tft_mcp_dis	In	1'd0	Assert High to block MCP path
aw3_dft_tft_conf_vector[67:0]	In	68'd0	clkdiv dft_tft_conf_vector
aw3_dft_tft_start_en	In	1'd0	clkdiv dft_tft_start_en
aw3_dft_delay_fault_en	In	1'd0	clkdiv dft_delay_fault_en
aw3_dft_tft_done	Out	1'd1	clkdiv dft_tft_done
aw3_dft_div_scan_en	In	1'd0	clkdiv dft_div_scan_en
aw3_dft_bypass_div_en	In	1'd0	clkdiv dft_bypass_div_en
aw3_dft_local_clk_en	In	1'd0	dft_local_clk_en
aw3_dft_pbist_clk_en	In	1'd0	dft_pbist_clk_en
aw3_dft_mode_atpg	In	1'd0	ATPG
aw3_dft_mode_iddq	In	1'd0	Iddq mode
aw3_dft_mode_burnin	In	1'd0	Burnin
aw3_dft_embed_mode_pbist	In	1'd0	Embedded pBIST Mode

9.4.337 dft_core ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
aw3_dft_core_clk_force_en	In	1'd0	dft_clk_force_en

9.4.338 dft_core ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
aw3_dft_mma_clk_force_en	In	1'd0	dft_clk_force_en

9.4.339 dft_compression

Signal Name	Dir	Default Val	Description
aw3_dft_compression_mma_dft_block_en	In	1'd1	dft_block_en
aw3_dft_compression_mma_dft_partition_en	In	1'd1	dft_partition_en
aw3_dft_compression_mma_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw3_dft_compression_mma_dft_compression_mode	In	1'd0	dft_compression_mode
aw3_dft_compression_mma_dft_compression_spread	In	1'd0	dft_compression_spread
aw3_dft_compression_mma_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw3_dft_compression_mma_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw3_dft_compression_mma_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw3_dft_compression_mma_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw3_dft_compression_mma_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw3_dft_compression_mma_dft_struct_intest	In	1'd0	dft_struct_intest
aw3_dft_compression_mma_dft_struct_extest	In	1'd0	dft_struct_extest
aw3_dft_compression_mma_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw3_dft_compression_mma_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw3_dft_compression_mma_dft_struct_so[35:0]	Out	36'd0	dft_struct_so



9.4.340 PLL control align signals

Signal Name	Dir	Default Val	Description
aw3_pll_ctrl_align_align_clk_div_rst_n	In	1'd0	Export align_clk_div
aw3_pll_ctrl_align_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.341 dft_compression

Signal Name	Dir	Default Val	Description
aw3_dft_compression_dft_block_en	In	1'd1	dft_block_en
aw3_dft_compression_dft_partition_en	In	1'd1	dft_partition_en
aw3_dft_compression_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw3_dft_compression_dft_compression_mode	In	1'd0	dft_compression_mode
aw3_dft_compression_dft_compression_spread	In	1'd0	dft_compression_spread
aw3_dft_compression_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw3_dft_compression_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw3_dft_compression_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw3_dft_compression_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw3_dft_compression_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw3_dft_compression_dft_struct_intest	In	1'd0	dft_struct_intest
aw3_dft_compression_dft_struct_extest	In	1'd0	dft_struct_extest
aw3_dft_compression_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw3_dft_compression_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw3_dft_compression_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.342 Controls for the TFT Clock control module

Signal Name	Dir	Default Val	Description
aw3_dft_pll_ctrl_dft_mode_pll_atpg	In	1'd0	dft_mode_pll_atpg
aw3_dft_pll_ctrl_dft_pll_capture_en	In	1'd0	dft_pll_capture_en
aw3_dft_pll_ctrl_dft_shift_clk_sel	In	1'd0	dft_shift_clk_sel
aw3_dft_pll_ctrl_cs_func_clk_en	In	1'd0	enable clocks to selected scan paths

9.4.343 DFT PLL CLK BYPASS control for the TFT Clock control module

Signal Name	Dir	Default Val	Description
aw3_dft_pll_clk_bypass_pll_clk_bypass	In	1'd1	DFT WIR input to kp_ti_ipg_loc_tft_clk_ctrl

9.4.344 c7xv clk1 clock

Signal Name	Dir	Default Val	Description
aw3_c7xv_clk_ctl_align_clk_div_rst_n	In	1'd0	Export align_clk_div
aw3_c7xv_clk_ctl_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.345 Clken interface

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw3_c7xv_core_clken_clk_en_ack	Out	c7xv_core_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.346 Clken interface

Signal Name	Dir	Default Val	Description
aw3_c7xv_wrap_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw3_c7xv_wrap_clken_clk_en_ack	Out	c7xv_wrap_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.347 DFT clock clken control for the Dividers

Signal Name	Dir	Default Val	Description
aw3_dft_local_div_clk_en_div_clk_en	In	1'd1	DFT local clk enable contribution to kp_ti_ipg_clk_divhdist5_tft_mux

9.4.348 aw3_c7xv_core_rst

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_rst_mod_g_rst_n	In	1'd1	Module level main reset
aw3_c7xv_core_rst_mod_por_rst_n	In	1'd1	Module level por reset
aw3_c7xv_core_rst_mod_l_rst_n	In	1'd1	Module level local reset

9.4.349 aw3_c7xv_wrap_rst

Signal Name	Dir	Default Val	Description
aw3_c7xv_wrap_rst_mod_g_rst_n	In	1'd1	Module level main reset
aw3_c7xv_wrap_rst_mod_por_rst_n	In	1'd1	Module level por reset
aw3_c7xv_wrap_rst_mod_l_rst_n	In	1'd1	Module level local reset

9.4.350 Power interface

Signal Name	Dir	Default Val	Description
aw3_mma_pwr_pwr_log_iso_en	In	1'd0	pwr
aw3_mma_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw3_mma_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw3_mma_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw3_mma_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.351 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw3_clk_bypass_wir	In	1'd0	dft wir signal

9.4.352 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw3_dft_clk_en_wir	In	1'd0	dft wir signal

9.4.353 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw3_divfactor_sel_wir	In	1'd0	dft wir signal

9.4.354 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw3_core_clk4_divfactor_sel_wir	In	1'd0	dft wir signal

9.4.355 Clk4 clock gate disable control for dft mode

Signal Name	Dir	Default Val	Description
dft_aw3_core_clk4_disable_wir	In	1'd0	dft wir signal

9.4.356 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_aw3_scrub_off_wir	In	1'd0	dft wir signal

9.4.357 Target .vbusp_pbist_cfg power disable

Signal Name	Dir	Default Val	Description
aw3_c7xv_pbist_disable_scr_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw3_c7xv_pbist_disable_scr_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw3_c7xv_pbist_disable_scr_disable_done	In	1'd0	Ack to SCR to stop clocks
aw3_c7xv_pbist_disable_scr_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw3_c7xv_pbist_disable_scr_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw3_c7xv_pbist_disable_scr_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.358 aw3_dft_embed_pbist

Signal Name	Dir	Default Val	Description
aw3_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
aw3_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
aw3_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
aw3_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
aw3_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
aw3_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
aw3_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
aw3_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
aw3_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
aw3_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
aw3_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
aw3_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
aw3_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
aw3_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
aw3_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
aw3_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
aw3_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
aw3_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
aw3_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.359 rom_ase

Signal Name	Dir	Default Val	Description
aw3_rom_ase_dft_mem_rom_ase	In	1'd0	dft_mem_rom_ase

9.4.360 aw3_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
aw3_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.361 aw3_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
aw3_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.362 aw3_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
aw3_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
aw3_dft_pbist_pwr_clk_en_ack	Out	dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.363 aw3_pbist_tclk

Signal Name	Dir	Default Val	Description
aw3_pbist_tclk_clk	In	1'd0	Single ended clock
aw3_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.364 aw3_pbist_rst

Signal Name	Dir	Default Val	Description
aw3_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.365 EFUSE interface

Signal Name	Dir	Default Val	Description
aw3_dft_rom_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.366 aw3_core_pbist_clkstop

Signal Name	Dir	Default Val	Description
aw3_core_pbist_clkstop_clkstop_req	In	1'd0	Clock stop request
aw3_core_pbist_clkstop_clkstop_ack	Out	core_pbist_clkstop_clkstop_req	Clock stop acknowledge

9.4.367 dft_lbist

Signal Name	Dir	Default Val	Description
aw3_dft_lbist_dft_lbist_atpg_mode	In	1'd0	ATPG control from SoC WIR
aw3_dft_lbist_dft_lbist_runbist_mode[3:0]	In	4'd0	Runbist mode enable
aw3_dft_lbist_dft_lbist_bist_reset	In	1'd0	Resets LBIST macro
aw3_dft_lbist_dft_lbist_bist_run[3:0]	In	4'd0	Starts LBIST if all bits are 1. Additional bits added to ensure that a single bit flip will not cause unintended LBIST execution.
aw3_dft_lbist_dft_lbist_scan_pc_def[3:0]	In	4'd0	Number of chain test patterns to run
aw3_dft_lbist_dft_lbist_reset_pc_def[3:0]	In	4'd0	Number of reset patterns to run.
aw3_dft_lbist_dft_lbist_set_pc_def[3:0]	In	4'd0	Number of set patterns to run
aw3_dft_lbist_dft_lbist_static_pc_def[13:0]	In	14'd0	Number of stuck-at patterns up to 16k to run
aw3_dft_lbist_dft_lbist_dc_def[1:0]	In	2'd0	Clock delay after scan_enable switching
aw3_dft_lbist_dft_lbist_prpg_def[52:0]	In	53'd0	Initial seed for PRPG
aw3_dft_lbist_dft_lbist_misr_mux_ctl[9:0]	In	10'd0	Selects block of 32 MISR bits to return to the MMR block.
aw3_dft_lbist_dft_lbist_out_mux_sel[1:0]	In	2'd0	Select to return PID, CTRL ID, or MISR result.
aw3_dft_lbist_dft_lbist_bist_done	Out	1'd0	LBIST done signal
aw3_dft_lbist_dft_lbist_bist_running	Out	1'd0	LBIST is running
aw3_dft_lbist_dft_lbist_misr_results[31:0]	Out	32'd0	32-bit section of MISR value. Selected using misr_mux_ctl input

9.4.368 dft_lbist_ctrl_id

Signal Name	Dir	Default Val	Description
aw3_dft_lbist_ctrl_id_dft_lbist_ctrl_id[5:0]	In	6'd0	Tieoff ID for this instance of LBIST

9.4.369 dft_iso_en

Signal Name	Dir	Default Val	Description
aw3_dft_iso_en_dft_iso_en_lbist	In	1'd0	DFT iso_en for pins to LBISTCTL
aw3_dft_iso_en_dft_iso_en_pbist	In	1'd0	DFT iso_en for pins to PBIST
aw3_dft_iso_en_dft_iso_en_other	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST
aw3_dft_iso_en_dft_iso_en_lbist_delayed	In	1'd0	DFT iso_en for pins to LBISTCTL
aw3_dft_iso_en_dft_iso_en_pbist_delayed	In	1'd0	DFT iso_en for pins to PBIST
aw3_dft_iso_en_dft_iso_en_other_delayed	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST

9.4.370 dft_lbist

Signal Name	Dir	Default Val	Description
aw3_core_dft_lbist_dft_lbist_atpg_mode	In	1'd0	ATPG control from SoC WIR
aw3_core_dft_lbist_dft_lbist_runbist_mode[3:0]	In	4'd0	Runbist mode enable
aw3_core_dft_lbist_dft_lbist_bist_reset	In	1'd0	Resets LBIST macro
aw3_core_dft_lbist_dft_lbist_bist_run[3:0]	In	4'd0	Starts LBIST if all bits are 1. Additional bits added to ensure that a single bit flip will not cause unintended LBIST execution.
aw3_core_dft_lbist_dft_lbist_scan_pc_def[3:0]	In	4'd0	Number of chain test patterns to run
aw3_core_dft_lbist_dft_lbist_reset_pc_def[3:0]	In	4'd0	Number of reset patterns to run.
aw3_core_dft_lbist_dft_lbist_set_pc_def[3:0]	In	4'd0	Number of set patterns to run
aw3_core_dft_lbist_dft_lbist_static_pc_def[13:0]	In	14'd0	Number of stuck-at patterns up to 16k to run
aw3_core_dft_lbist_dft_lbist_dc_def[1:0]	In	2'd0	Clock delay after scan_enable switching
aw3_core_dft_lbist_dft_lbist_prpg_def[52:0]	In	53'd0	Initial seed for PRPG
aw3_core_dft_lbist_dft_lbist_misr_mux_ctl[9:0]	In	10'd0	Selects block of 32 MISR bits to return to the MMR block.
aw3_core_dft_lbist_dft_lbist_out_mux_sel[1:0]	In	2'd0	Select to return PID, CTRL ID, or MISR result.
aw3_core_dft_lbist_dft_lbist_bist_done	Out	1'd0	LBIST done signal
aw3_core_dft_lbist_dft_lbist_bist_running	Out	1'd0	LBIST is running
aw3_core_dft_lbist_dft_lbist_misr_results[31:0]	Out	32'd0	32-bit section of MISR value. Selected using misr_mux_ctl input
aw3_core_dft_lbist_dft_lbist_divide_ratio[4:0]	In	5'd0	clkdiv divide value during lbist
aw3_core_dft_lbist_dft_lbist_load_div	In	1'd0	clkdiv divide load enable

9.4.371 dft_lbist_ctrl_id

Signal Name	Dir	Default Val	Description
aw3_core_dft_lbist_ctrl_id_dft_lbist_ctrl_id[5:0]	In	6'd0	Tieoff ID for this instance of LBIST

9.4.372 EFUSE interface

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_dft_rom_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.373 dft_iso_en

Signal Name	Dir	Default Val	Description
aw3_core_dft_iso_en_dft_iso_en_lbist	In	1'd0	DFT iso_en for pins to LBISTCTL
aw3_core_dft_iso_en_dft_iso_en_pbist	In	1'd0	DFT iso_en for pins to PBIST
aw3_core_dft_iso_en_dft_iso_en_other	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST
aw3_core_dft_iso_en_dft_iso_en_lbist_delayed	In	1'd0	DFT iso_en for pins to LBISTCTL
aw3_core_dft_iso_en_dft_iso_en_pbist_delayed	In	1'd0	DFT iso_en for pins to PBIST
aw3_core_dft_iso_en_dft_iso_en_other_delayed	In	1'd0	DFT iso_en for pins not related to LBISTCTL or PBIST

9.4.374 bisor

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
aw3_c7xv_core_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
aw3_c7xv_core_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
aw3_c7xv_core_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
aw3_c7xv_core_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
aw3_c7xv_core_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
aw3_c7xv_core_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
aw3_c7xv_core_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.375 Target .vbusp_core_pbist_cfg power disable

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_pbist_disable_scr_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw3_c7xv_core_pbist_disable_scr_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw3_c7xv_core_pbist_disable_scr_disable_done	In	1'd0	Ack to SCR to stop clocks
aw3_c7xv_core_pbist_disable_scr_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw3_c7xv_core_pbist_disable_scr_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw3_c7xv_core_pbist_disable_scr_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.376 bisor

Signal Name	Dir	Default Val	Description
aw3_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
aw3_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
aw3_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
aw3_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
aw3_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
aw3_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
aw3_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
aw3_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.377 aw3_c7xv_core_dft_embed_pbist

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
aw3_c7xv_core_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
aw3_c7xv_core_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
aw3_c7xv_core_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
aw3_c7xv_core_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
aw3_c7xv_core_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
aw3_c7xv_core_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
aw3_c7xv_core_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
aw3_c7xv_core_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
aw3_c7xv_core_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
aw3_c7xv_core_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
aw3_c7xv_core_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
aw3_c7xv_core_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
aw3_c7xv_core_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
aw3_c7xv_core_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
aw3_c7xv_core_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
aw3_c7xv_core_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
aw3_c7xv_core_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
aw3_c7xv_core_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.378 aw3_c7xv_core_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.379 aw3_c7xv_core_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.380 aw3_c7xv_core_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
aw3_c7xv_core_dft_pbist_pwr_clk_en_ack	Out	c7xv_core_dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.381 aw3_c7xv_core_pbist_tclk

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_pbist_tclk_clk	In	1'd0	Single ended clock
aw3_c7xv_core_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.382 aw3_c7xv_core_pbist_rst

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.383 Power SCR Disable Interface

Signal Name	Dir	Default Val	Description
aw3_c7xv_core_pwr_dis_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw3_c7xv_core_pwr_dis_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw3_c7xv_core_pwr_dis_disable_done	In	1'd0	Ack to SCR to stop clocks
aw3_c7xv_core_pwr_dis_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw3_c7xv_core_pwr_dis_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw3_c7xv_core_pwr_dis_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.384 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
aw3_c7xv_mma_clkstop_clkstop_req	In	1'd0	Clock stop request
aw3_c7xv_mma_clkstop_clkstop_ack	Out	c7xv_mma_clkstop_clkstop_req	Clock stop acknowledge

9.4.385 Clken interface

Signal Name	Dir	Default Val	Description
aw3_c7xv_mma_clken_clk_en	In	1'd1	Clock-gate enable to IP
aw3_c7xv_mma_clken_clk_en_ack	Out	c7xv_mma_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.386 aw3_c7xv_mma_rst

Signal Name	Dir	Default Val	Description
aw3_c7xv_mma_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.387 dft_compression for mma_array_r00 block

Signal Name	Dir	Default Val	Description
aw3_mma_r00_dft_block_en	In	1'd1	dft_block_en
aw3_mma_r00_dft_partition_en	In	1'd1	dft_partition_en
aw3_mma_r00_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw3_mma_r00_dft_compression_mode	In	1'd0	dft_compression_mode
aw3_mma_r00_dft_compression_spread	In	1'd0	dft_compression_spread
aw3_mma_r00_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw3_mma_r00_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw3_mma_r00_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw3_mma_r00_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw3_mma_r00_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw3_mma_r00_dft_struct_intest	In	1'd0	dft_struct_intest
aw3_mma_r00_dft_struct_extest	In	1'd0	dft_struct_extest
aw3_mma_r00_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw3_mma_r00_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw3_mma_r00_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.388 dft_compression for mma_array_r01 block

Signal Name	Dir	Default Val	Description
aw3_mma_r01_dft_block_en	In	1'd1	dft_block_en
aw3_mma_r01_dft_partition_en	In	1'd1	dft_partition_en
aw3_mma_r01_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw3_mma_r01_dft_compression_mode	In	1'd0	dft_compression_mode
aw3_mma_r01_dft_compression_spread	In	1'd0	dft_compression_spread
aw3_mma_r01_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw3_mma_r01_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw3_mma_r01_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw3_mma_r01_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw3_mma_r01_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw3_mma_r01_dft_struct_intest	In	1'd0	dft_struct_intest
aw3_mma_r01_dft_struct_extest	In	1'd0	dft_struct_extest
aw3_mma_r01_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw3_mma_r01_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw3_mma_r01_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.389 dft_compression for mma_array_r10 block

Signal Name	Dir	Default Val	Description
aw3_mma_r10_dft_block_en	In	1'd1	dft_block_en
aw3_mma_r10_dft_partition_en	In	1'd1	dft_partition_en
aw3_mma_r10_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw3_mma_r10_dft_compression_mode	In	1'd0	dft_compression_mode
aw3_mma_r10_dft_compression_spread	In	1'd0	dft_compression_spread
aw3_mma_r10_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw3_mma_r10_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw3_mma_r10_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw3_mma_r10_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw3_mma_r10_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw3_mma_r10_dft_struct_intest	In	1'd0	dft_struct_intest
aw3_mma_r10_dft_struct_extest	In	1'd0	dft_struct_extest
aw3_mma_r10_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw3_mma_r10_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw3_mma_r10_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.390 dft_compression for mma_array_r11 block

Signal Name	Dir	Default Val	Description
aw3_mma_r11_dft_block_en	In	1'd1	dft_block_en
aw3_mma_r11_dft_partition_en	In	1'd1	dft_partition_en
aw3_mma_r11_dft_compression_cmle	In	1'd0	dft_compression_cmle
aw3_mma_r11_dft_compression_mode	In	1'd0	dft_compression_mode
aw3_mma_r11_dft_compression_spread	In	1'd0	dft_compression_spread
aw3_mma_r11_dft_elastic_reset	In	1'd0	dft_elastic_reset
aw3_mma_r11_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
aw3_mma_r11_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
aw3_mma_r11_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
aw3_mma_r11_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
aw3_mma_r11_dft_struct_intest	In	1'd0	dft_struct_intest
aw3_mma_r11_dft_struct_extest	In	1'd0	dft_struct_extest
aw3_mma_r11_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
aw3_mma_r11_dft_struct_si[35:0]	In	36'd0	dft_struct_si
aw3_mma_r11_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.391 Power interface for mma_array_r00 block

Signal Name	Dir	Default Val	Description
aw3_r00_pwr_pwr_log_iso_en	In	1'd0	pwr
aw3_r00_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw3_r00_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw3_r00_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw3_r00_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.392 Power interface for mma_array_r01 block

Signal Name	Dir	Default Val	Description
aw3_r01_pwr_pwr_log_iso_en	In	1'd0	pwr
aw3_r01_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw3_r01_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw3_r01_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw3_r01_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.393 Power interface for mma_array_r10 block

Signal Name	Dir	Default Val	Description
aw3_r10_pwr_pwr_log_iso_en	In	1'd0	pwr
aw3_r10_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw3_r10_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw3_r10_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw3_r10_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.394 Power interface for mma_array_r11 block

Signal Name	Dir	Default Val	Description
aw3_r11_pwr_pwr_log_iso_en	In	1'd0	pwr
aw3_r11_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
aw3_r11_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
aw3_r11_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
aw3_r11_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.395 ccc_ac

Signal Name	Dir	Default Val	Description
ccc_ac_ipcfg[35:0]	In	36'd0	ip config pin

9.4.396 el2_rat_source_id_tieoff

Signal Name	Dir	Default Val	Description
el2_rat_source_id_tieoff_ipcfg[15:0]	In	16'd0	ip config pin

9.4.397 msmc0_el2 vbusmc patron interface

Signal Name	Dir	Default Val	Description
msmc0_el2_caddress[47:0]	In	48'd0	Transaction address
msmc0_el2_camode[1:0]	In	2'd0	Addressing mode
msmc0_el2_cbytecnt[9:0]	In	10'd0	Transaction bytecount
msmc0_el2_cchanid[11:0]	In	12'd0	Channel ID
msmc0_el2_ccinner[1:0]	In	2'd0	Inner cacheability
msmc0_el2_ccsize[2:0]	In	3'd0	Cache line size
msmc0_el2_ccouter[1:0]	In	2'd0	Outer cacheability
msmc0_el2_ccredittype[1:0]	In	2'd0	Credit type indicator
msmc0_el2_cdtype[1:0]	In	2'd0	Data type attribute
msmc0_el2_cemudbg	In	1'd0	Emulation level
msmc0_el2_cepriority[2:0]	In	3'd0	Elevated priority
msmc0_el2_cexcl	In	1'd0	Exclusive attribute
msmc0_el2_cflush	In	1'd0	Firewall flush
msmc0_el2_cid[11:0]	In	12'd0	Transaction ID
msmc0_el2_cinterest	In	1'd0	Debug marker
msmc0_el2_cmemtype[1:0]	In	2'd0	Memory region type
msmc0_el2_cnogap	In	1'd0	Contiguous byte-enable attribute
msmc0_el2_copcode[5:0]	In	6'd0	Transaction opcode
msmc0_el2_corderid[3:0]	In	4'd0	Order ID
msmc0_el2_cpable	In	1'd0	Prefetchable attribute
msmc0_el2_cpriority[2:0]	In	3'd0	Transaction priority
msmc0_el2_cpriv[1:0]	In	2'd0	Privilege level
msmc0_el2_cprivid[7:0]	In	8'd0	Privilege ID for security
msmc0_el2_cqos[2:0]	In	3'd0	Quality of Service level
msmc0_el2_creq	In	1'd0	Command request
msmc0_el2_crouteid[11:0]	In	12'd0	Unique ID for the Initiator
msmc0_el2_crsl[3:0]	In	4'd0	Region select driven by the infrastructure
msmc0_el2_csdomain[1:0]	In	2'd0	Sharability domain
msmc0_el2_csecure	In	1'd0	Security level
msmc0_el2_cstarve	In	1'd0	Credit starvation indicator
msmc0_el2_csubthread	In	1'd0	Sub-thread category
msmc0_el2_cthread[2:0]	In	3'd0	Thread category
msmc0_el2_cvirtid[11:0]	In	12'd0	Virtual ID
msmc0_el2_cgreturn	Out	1'd0	Global credit return
msmc0_el2_cs0return[2:0]	Out	3'd0	Sub-thread 0 credit return
msmc0_el2_cs1return[2:0]	Out	3'd0	Sub-thread 1 credit return
msmc0_el2_ctreturn[2:0]	Out	3'd0	Thread credit return
msmc0_el2_crreturn[2:0]	Out	3'd0	Response credit return
msmc0_el2_cgcredinit[5:0]	Out	6'd0	Global credit initialization value
msmc0_el2_ct0credinit[5:0]	Out	6'd0	Thread 0 credit initialization value
msmc0_el2_ct0s0credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 0 credit initialization value
msmc0_el2_ct0s1credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 1 credit initialization value

msmc0_el2_ct0credinit[5:0]	Out	6'd0	Thread 0 Response credit initialization value
msmc0_el2_ct1credinit[5:0]	Out	6'd0	Thread 1 credit initialization value
msmc0_el2_ct1s0credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 0 credit initialization value
msmc0_el2_ct1s1credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 1 credit initialization value
msmc0_el2_ct1rcredinit[5:0]	Out	6'd0	Thread 1 Response credit initialization value
msmc0_el2_ct2credinit[5:0]	Out	6'd0	Thread 2 credit initialization value
msmc0_el2_ct2s0credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 0 credit initialization value
msmc0_el2_ct2s1credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 1 credit initialization value
msmc0_el2_ct2rcredinit[5:0]	Out	6'd0	Thread 2 Response credit initialization value
msmc0_el2_dalign[9:0]	In	10'd0	Response alignment
msmc0_el2_dbfirst	In	1'd0	First data phase in burst
msmc0_el2_dblast	In	1'd0	Last data phase in burst
msmc0_el2_dbytecnt[9:0]	In	10'd0	Bytecount of entire data transfer
msmc0_el2_dbyten[63:0]	In	64'd0	Byten for write data
msmc0_el2_dcoh[1:0]	In	2'd0	Coherence state for response
msmc0_el2_dcreditttype[1:0]	In	2'd0	Credit type indicator
msmc0_el2_ddata[511:0]	In	512'd0	Data
msmc0_el2_ddatavalid	In	1'd0	Valid data indicator
msmc0_el2_ddirty	In	1'd0	Cache line dirty indicator
msmc0_el2_dedc[19:0]	In	20'd0	EDC protection for ddata
msmc0_el2_depriority[2:0]	In	3'd0	Elevated priority
msmc0_el2_dfirst	In	1'd0	First data phase in entire data transfer
msmc0_el2_did[11:0]	In	12'd0	Transaction ID
msmc0_el2_dinterest	In	1'd0	Debug marker
msmc0_el2_dlast	In	1'd0	Last data phase in entire data transfer
msmc0_el2_dopcode[5:0]	In	6'd0	Transaction opcode
msmc0_el2_dorderid[3:0]	In	4'd0	Order ID
msmc0_el2_dpriority[2:0]	In	3'd0	Transaction priority
msmc0_el2_dreq	In	1'd0	Data phase request
msmc0_el2_drouteid[11:0]	In	12'd0	Unique ID for the originating Initiator
msmc0_el2_dstatus[2:0]	In	3'd0	Response status
msmc0_el2_dsubthread	In	1'd0	Sub-thread category
msmc0_el2_dthread[2:0]	In	3'd0	Thread category
msmc0_el2_dtrace[3:0]	In	4'd0	Debug trace information
msmc0_el2_dxcnt[7:0]	In	8'd0	Number of bytes transferred in data phase
msmc0_el2_dgreturn	Out	1'd0	Global credit return
msmc0_el2_ds0return[2:0]	Out	3'd0	Sub-thread 0 credit return
msmc0_el2_ds1return[2:0]	Out	3'd0	Sub-thread 1 credit return

msmc0_el2_dtreturn[2:0]	Out	3'd0	Thread credit return
msmc0_el2_drreturn[2:0]	Out	3'd0	Response credit return
msmc0_el2_dgcredinit[5:0]	Out	6'd0	Global credit initialization value
msmc0_el2_dt0credinit[5:0]	Out	6'd0	Thread 0 credit initialization value
msmc0_el2_dt0s0credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 0 credit initialization value
msmc0_el2_dt0s1credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 1 credit initialization value
msmc0_el2_dt0rcredinit[5:0]	Out	6'd0	Thread 0 Response credit initialization value
msmc0_el2_dt1credinit[5:0]	Out	6'd0	Thread 1 credit initialization value
msmc0_el2_dt1s0credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 0 credit initialization value
msmc0_el2_dt1s1credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 1 credit initialization value
msmc0_el2_dt1rcredinit[5:0]	Out	6'd0	Thread 1 Response credit initialization value
msmc0_el2_dt2credinit[5:0]	Out	6'd0	Thread 2 credit initialization value
msmc0_el2_dt2s0credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 0 credit initialization value
msmc0_el2_dt2s1credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 1 credit initialization value
msmc0_el2_dt2rcredinit[5:0]	Out	6'd0	Thread 2 Response credit initialization value
msmc0_el2_sbytecnt[9:0]	In	10'd0	Bytecount for this write response
msmc0_el2_screditttype[1:0]	In	2'd0	Credit type indicator
msmc0_el2_sid[11:0]	In	12'd0	Transaction ID
msmc0_el2_sinterest	In	1'd0	Debug marker
msmc0_el2_sorderid[3:0]	In	4'd0	Order ID
msmc0_el2_sreq	In	1'd0	Data phase request
msmc0_el2_srouteid[11:0]	In	12'd0	Unique ID for the originating Initiator
msmc0_el2_sstatus[2:0]	In	3'd0	Response status
msmc0_el2_ssubthread	In	1'd0	Sub-thread category
msmc0_el2_sthread[2:0]	In	3'd0	Thread category
msmc0_el2_strace[3:0]	In	4'd0	Debug trace information
msmc0_el2_sgreturn	Out	1'd0	Global credit return
msmc0_el2_ss0return[2:0]	Out	3'd0	Sub-thread 0 credit return
msmc0_el2_ss1return[2:0]	Out	3'd0	Sub-thread 1 credit return
msmc0_el2_streturn[2:0]	Out	3'd0	Thread credit return
msmc0_el2_srreturn[2:0]	Out	3'd0	Response credit return
msmc0_el2_sgcredinit[5:0]	Out	6'd0	Global credit initialization value
msmc0_el2_st0credinit[5:0]	Out	6'd0	Thread 0 credit initialization value
msmc0_el2_st0s0credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 0 credit initialization value
msmc0_el2_st0s1credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 1 credit initialization value
msmc0_el2_st0rcredinit[5:0]	Out	6'd0	Thread 0 Response credit initialization value

msmc0_el2_st1credinit[5:0]	Out	6'd0	Thread 1 credit initialization value
msmc0_el2_st1s0credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 0 credit initialization value
msmc0_el2_st1s1credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 1 credit initialization value
msmc0_el2_st1rcredinit[5:0]	Out	6'd0	Thread 1 Response credit initialization value
msmc0_el2_st2credinit[5:0]	Out	6'd0	Thread 2 credit initialization value
msmc0_el2_st2s0credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 0 credit initialization value
msmc0_el2_st2s1credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 1 credit initialization value
msmc0_el2_st2rcredinit[5:0]	Out	6'd0	Thread 2 Response credit initialization value
msmc0_el2_caddress_safep[3:0]	In	4'b1111	Safe Transaction address
msmc0_el2_camode_safep	In	1'd1	Safe Addressing mode
msmc0_el2_cbytecnt_safep	In	1'd1	Safe Transaction bytecount
msmc0_el2_cchanid_safep	In	1'd1	Safe Channel ID
msmc0_el2_ccinner_safep	In	1'd1	Safe Inner cacheability
msmc0_el2_cclsize_safep	In	1'd1	Safe Cache line size
msmc0_el2_ccouter_safep	In	1'd1	Safe Outer cacheability
msmc0_el2_ccredittype_safep	In	1'd1	Safe Credit type indicator
msmc0_el2_cdtype_safep	In	1'd1	Safe Data type attribute
msmc0_el2_cemudbg_safep	In	1'd1	Safe Emulation level
msmc0_el2_cepriority_safep	In	1'd1	Safe Elevated priority
msmc0_el2_cexcl_safep	In	1'd1	Safe Exclusive attribute
msmc0_el2_cflush_safep	In	1'd1	Safe Firewall flush
msmc0_el2_cid_safep	In	1'd1	Safe Transaction ID
msmc0_el2_cinterest_safep	In	1'd1	Safe Debug marker
msmc0_el2_cmemtype_safep	In	1'd1	Safe Memory region type
msmc0_el2_cnogap_safep	In	1'd1	Safe Contiguous byte-enable attribute
msmc0_el2_copcode_safep	In	1'd1	Safe Transaction opcode
msmc0_el2_corderid_safep	In	1'd1	Safe Order ID
msmc0_el2_cpable_safep	In	1'd1	Safe Prefetchable attribute
msmc0_el2_cpriority_safep	In	1'd1	Safe Transaction priority
msmc0_el2_cpriv_safep	In	1'd1	Safe Privilege level
msmc0_el2_cprivid_safep	In	1'd1	Safe Privilege ID for security
msmc0_el2_cqos_safep	In	1'd1	Safe Quality of Service level
msmc0_el2_creq_safer	In	1'd1	Safe Command request
msmc0_el2_crouteid_safep	In	1'd1	Safe Unique ID for the Initiator
msmc0_el2_crssel_safep	In	1'd1	Safe Region select driven by the infrastructure
msmc0_el2_csdomain_safep	In	1'd1	Safe Sharability domain
msmc0_el2_csecure_safep	In	1'd1	Safe Security level
msmc0_el2_cstarve_safep	In	1'd1	Safe Credit starvation indicator
msmc0_el2_csubthread_safep	In	1'd1	Safe Sub-thread category
msmc0_el2_cthread_safep	In	1'd1	Safe Thread category



msmc0_el2_cvirtid_safer	In	1'd1	Safe Virtual ID
msmc0_el2_cgeturn_safer	Out	1'd1	Safe Global credit return
msmc0_el2_cs0return_safer[2:0]	Out	3'b111	Safe Sub-thread 0 credit return
msmc0_el2_cs1return_safer[2:0]	Out	3'b111	Safe Sub-thread 1 credit return
msmc0_el2_ctreturn_safer[2:0]	Out	3'b111	Safe Thread credit return
msmc0_el2_crreturn_safer[2:0]	Out	3'b111	Safe Response credit return
msmc0_el2_cgcredinit_safer	Out	1'd1	Global credit initialization value
msmc0_el2_ct0credinit_safer	Out	1'd1	Thread 0 credit initialization value
msmc0_el2_ct0s0credinit_safer	Out	1'd1	Thread 0 Sub-thread 0 credit initialization value
msmc0_el2_ct0s1credinit_safer	Out	1'd1	Thread 0 Sub-thread 1 credit initialization value
msmc0_el2_ct0rcredinit_safer	Out	1'd1	Thread 0 Response credit initialization value
msmc0_el2_ct1credinit_safer	Out	1'd1	Thread 1 credit initialization value
msmc0_el2_ct1s0credinit_safer	Out	1'd1	Thread 1 Sub-thread 0 credit initialization value
msmc0_el2_ct1s1credinit_safer	Out	1'd1	Thread 1 Sub-thread 1 credit initialization value
msmc0_el2_ct1rcredinit_safer	Out	1'd1	Thread 1 Response credit initialization value
msmc0_el2_ct2credinit_safer	Out	1'd1	Thread 2 credit initialization value
msmc0_el2_ct2s0credinit_safer	Out	1'd1	Thread 2 Sub-thread 0 credit initialization value
msmc0_el2_ct2s1credinit_safer	Out	1'd1	Thread 2 Sub-thread 1 credit initialization value
msmc0_el2_ct2rcredinit_safer	Out	1'd1	Thread 2 Response credit initialization value
msmc0_el2_dalign_safer	In	1'd1	Safe Response alignment
msmc0_el2_dbfirst_safer	In	1'd1	Safe First data phase in burst
msmc0_el2_dblast_safer	In	1'd1	Safe Last data phase in burst
msmc0_el2_dbytecnt_safer	In	1'd1	Safe Bytecount of entire data transfer
msmc0_el2_dbyten_safer[5:0]	In	6'b111111	Safe Byten for write data
msmc0_el2_dcoh_safer	In	1'd1	Safe Coherence state for response
msmc0_el2_dcreditype_safer	In	1'd1	Safe Credit type indicator
msmc0_el2_ddatavalid_safer	In	1'd1	Safe Valid data indicator
msmc0_el2_ddirty_safer	In	1'd1	Safe Cache line dirty indicator
msmc0_el2_dpriority_safer	In	1'd1	Safe Elevated priority
msmc0_el2_dfirst_safer	In	1'd1	Safe First data phase in entire data transfer
msmc0_el2_did_safer	In	1'd1	Safe Transaction ID
msmc0_el2_dinterest_safer	In	1'd1	Safe Debug marker
msmc0_el2_dlast_safer	In	1'd1	Safe Last data phase in entire data transfer
msmc0_el2_dopcode_safer	In	1'd1	Safe Transaction opcode
msmc0_el2_dorderid_safer	In	1'd1	Safe Order ID
msmc0_el2_dpriority_safer	In	1'd1	Safe Transaction priority
msmc0_el2_dreq_safer	In	1'd1	Safe Data phase request

msmc0_el2_drouteid_safep	In	1'd1	Safe Unique ID for the originating Initiator
msmc0_el2_dstatus_safep	In	1'd1	Safe Response status
msmc0_el2_dsubthread_safep	In	1'd1	Safe Sub-thread category
msmc0_el2_dthread_safep	In	1'd1	Safe Thread category
msmc0_el2_dtrace_safep	In	1'd1	Safe Debug trace information
msmc0_el2_dxcnt_safep	In	1'd1	Safe Number of bytes transferred in data phase
msmc0_el2_dgreturn_safer	Out	1'd1	Safe Global credit return
msmc0_el2_ds0return_safer[2:0]	Out	3'b111	Safe Sub-thread 0 credit return
msmc0_el2_ds1return_safer[2:0]	Out	3'b111	Safe Sub-thread 1 credit return
msmc0_el2_dtreturn_safer[2:0]	Out	3'b111	Safe Thread credit return
msmc0_el2_drreturn_safer[2:0]	Out	3'b111	Safe Response credit return
msmc0_el2_dgcredinit_safep	Out	1'd1	Global credit initialization value
msmc0_el2_dt0credinit_safep	Out	1'd1	Thread 0 credit initialization value
msmc0_el2_dt0s0credinit_safep	Out	1'd1	Thread 0 Sub-thread 0 credit initialization value
msmc0_el2_dt0s1credinit_safep	Out	1'd1	Thread 0 Sub-thread 1 credit initialization value
msmc0_el2_dt0credinit_safep	Out	1'd1	Thread 0 Response credit initialization value
msmc0_el2_dt1credinit_safep	Out	1'd1	Thread 1 credit initialization value
msmc0_el2_dt1s0credinit_safep	Out	1'd1	Thread 1 Sub-thread 0 credit initialization value
msmc0_el2_dt1s1credinit_safep	Out	1'd1	Thread 1 Sub-thread 1 credit initialization value
msmc0_el2_dt1credinit_safep	Out	1'd1	Thread 1 Response credit initialization value
msmc0_el2_dt2credinit_safep	Out	1'd1	Thread 2 credit initialization value
msmc0_el2_dt2s0credinit_safep	Out	1'd1	Thread 2 Sub-thread 0 credit initialization value
msmc0_el2_dt2s1credinit_safep	Out	1'd1	Thread 2 Sub-thread 1 credit initialization value
msmc0_el2_dt2credinit_safep	Out	1'd1	Thread 2 Response credit initialization value
msmc0_el2_sbytecnt_safep	In	1'd1	Safe Bytecount for this write response
msmc0_el2_scredittytype_safep	In	1'd1	Safe Credit type indicator
msmc0_el2_sid_safep	In	1'd1	Safe Transaction ID
msmc0_el2_sinterest_safep	In	1'd1	Safe Debug marker
msmc0_el2_sorderid_safep	In	1'd1	Safe Order ID
msmc0_el2_sreq_safer	In	1'd1	Safe Data phase request
msmc0_el2_srouteid_safep	In	1'd1	Safe Unique ID for the originating Initiator
msmc0_el2_sstatus_safep	In	1'd1	Safe Response status
msmc0_el2_ssubthread_safep	In	1'd1	Safe Sub-thread category
msmc0_el2_sthread_safep	In	1'd1	Safe Thread category
msmc0_el2_strace_safep	In	1'd1	Safe Debug trace information
msmc0_el2_sgreturn_safer	Out	1'd1	Safe Global credit return



msmc0_el2_ss0return_safer[2:0]	Out	3'b111	Safe Sub-thread 0 credit return
msmc0_el2_ss1return_safer[2:0]	Out	3'b111	Safe Sub-thread 1 credit return
msmc0_el2_streturn_safer[2:0]	Out	3'b111	Safe Thread credit return
msmc0_el2_srreturn_safer[2:0]	Out	3'b111	Safe Response credit return
msmc0_el2_sgcredinit_safep	Out	1'd1	Global credit initialization value
msmc0_el2_st0credinit_safep	Out	1'd1	Thread 0 credit initialization value
msmc0_el2_st0s0credinit_safep	Out	1'd1	Thread 0 Sub-thread 0 credit initialization value
msmc0_el2_st0s1credinit_safep	Out	1'd1	Thread 0 Sub-thread 1 credit initialization value
msmc0_el2_st0rcredinit_safep	Out	1'd1	Thread 0 Response credit initialization value
msmc0_el2_st1credinit_safep	Out	1'd1	Thread 1 credit initialization value
msmc0_el2_st1s0credinit_safep	Out	1'd1	Thread 1 Sub-thread 0 credit initialization value
msmc0_el2_st1s1credinit_safep	Out	1'd1	Thread 1 Sub-thread 1 credit initialization value
msmc0_el2_st1rcredinit_safep	Out	1'd1	Thread 1 Response credit initialization value
msmc0_el2_st2credinit_safep	Out	1'd1	Thread 2 credit initialization value
msmc0_el2_st2s0credinit_safep	Out	1'd1	Thread 2 Sub-thread 0 credit initialization value
msmc0_el2_st2s1credinit_safep	Out	1'd1	Thread 2 Sub-thread 1 credit initialization value
msmc0_el2_st2rcredinit_safep	Out	1'd1	Thread 2 Response credit initialization value

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Signal Name	Dir	Default Val	Description
el2_msmc0_caddress[47:0]	Out	48'd0	Transaction address
el2_msmc0_cbytecnt[9:0]	Out	10'd0	Transaction bytecount
el2_msmc0_cchanid[11:0]	Out	12'd0	Channel ID
el2_msmc0_ccinner[1:0]	Out	2'd0	Inner cacheability
el2_msmc0_ccouter[1:0]	Out	2'd0	Outer cacheability
el2_msmc0_ccredittype[1:0]	Out	2'd0	Credit type indicator
el2_msmc0_cdtype[1:0]	Out	2'd0	Data type attribute
el2_msmc0_cemudbg	Out	1'd0	Emulation level
el2_msmc0_cepriority[2:0]	Out	3'd0	Elevated priority
el2_msmc0_cflush	Out	1'd0	Firewall flush
el2_msmc0_cid[11:0]	Out	12'd0	Transaction ID
el2_msmc0_cinterest	Out	1'd0	Debug marker
el2_msmc0_cmemtype[1:0]	Out	2'd0	Memory region type
el2_msmc0_cnogap	Out	1'd0	Contiguous byte-enable attribute
el2_msmc0_copcode[5:0]	Out	6'd0	Transaction opcode
el2_msmc0_corderid[3:0]	Out	4'd0	Order ID
el2_msmc0_cpable	Out	1'd0	Prefetchable attribute
el2_msmc0_cpriority[2:0]	Out	3'd0	Transaction priority
el2_msmc0_cpriv[1:0]	Out	2'd0	Privilege level
el2_msmc0_cprivid[7:0]	Out	8'd0	Privilege ID for security
el2_msmc0_creg	Out	1'd0	Command request
el2_msmc0_crouteid[11:0]	Out	12'd0	Unique ID for the Initiator
el2_msmc0_csdomain[1:0]	Out	2'd0	Sharability domain
el2_msmc0_csecure	Out	1'd0	Security level
el2_msmc0_cstarve	Out	1'd0	Credit starvation indicator
el2_msmc0_csubthread	Out	1'd0	Sub-thread category
el2_msmc0_ctypead[2:0]	Out	3'd0	Thread category
el2_msmc0_cvirtid[11:0]	Out	12'd0	Virtual ID
el2_msmc0_cgreturn	In	1'd0	Global credit return
el2_msmc0_cs0return[2:0]	In	3'd0	Sub-thread 0 credit return
el2_msmc0_cs1return[2:0]	In	3'd0	Sub-thread 1 credit return
el2_msmc0_ctreturn[2:0]	In	3'd0	Thread credit return
el2_msmc0_crreturn[2:0]	In	3'd0	Response credit return
el2_msmc0_cgcredinit[5:0]	In	6'd0	Global credit initialization value
el2_msmc0_ct0credinit[5:0]	In	6'd0	Thread 0 credit initialization value
el2_msmc0_ct0s0credinit[5:0]	In	6'd0	Thread 0 Sub-thread 0 credit initialization value
el2_msmc0_ct0s1credinit[5:0]	In	6'd0	Thread 0 Sub-thread 1 credit initialization value
el2_msmc0_ct0rcredinit[5:0]	In	6'd0	Thread 0 Response credit initialization value
el2_msmc0_ct1credinit[5:0]	In	6'd0	Thread 1 credit initialization value
el2_msmc0_ct1s0credinit[5:0]	In	6'd0	Thread 1 Sub-thread 0 credit initialization value
el2_msmc0_ct1s1credinit[5:0]	In	6'd0	Thread 1 Sub-thread 1 credit initialization value

el2_msmc0_ct1rcredinit[5:0]	In	6'd0	Thread 1 Response credit initialization value
el2_msmc0_ct2credinit[5:0]	In	6'd0	Thread 2 credit initialization value
el2_msmc0_ct2s0credinit[5:0]	In	6'd0	Thread 2 Sub-thread 0 credit initialization value
el2_msmc0_ct2s1credinit[5:0]	In	6'd0	Thread 2 Sub-thread 1 credit initialization value
el2_msmc0_ct2rcredinit[5:0]	In	6'd0	Thread 2 Response credit initialization value
el2_msmc0_dalign[9:0]	Out	10'd0	Response alignment
el2_msmc0_dbfirst	Out	1'd0	First data phase in burst
el2_msmc0_dblast	Out	1'd0	Last data phase in burst
el2_msmc0_dbytecnt[9:0]	Out	10'd0	Bytecount of entire data transfer
el2_msmc0_dbyten[63:0]	Out	64'd0	Byten for write data
el2_msmc0_dcoh[1:0]	Out	2'd0	Coherence state for response
el2_msmc0_dcredittype[1:0]	Out	2'd0	Credit type indicator
el2_msmc0_ddata[511:0]	Out	512'd0	Data
el2_msmc0_ddatavalid	Out	1'd0	Valid data indicator
el2_msmc0_ddirty	Out	1'd0	Cache line dirty indicator
el2_msmc0_dedc[19:0]	Out	20'd0	EDC protection for ddata
el2_msmc0_depriority[2:0]	Out	3'd0	Elevated priority
el2_msmc0_dfirst	Out	1'd0	First data phase in entire data transfer
el2_msmc0_did[11:0]	Out	12'd0	Transaction ID
el2_msmc0_dinterest	Out	1'd0	Debug marker
el2_msmc0_dlast	Out	1'd0	Last data phase in entire data transfer
el2_msmc0_dopcode[5:0]	Out	6'd0	Transaction opcode
el2_msmc0_dorderid[3:0]	Out	4'd0	Order ID
el2_msmc0_dpriority[2:0]	Out	3'd0	Transaction priority
el2_msmc0_dreq	Out	1'd0	Data phase request
el2_msmc0_drouteid[11:0]	Out	12'd0	Unique ID for the originating Initiator
el2_msmc0_dstatus[2:0]	Out	3'd0	Response status
el2_msmc0_dsubthread	Out	1'd0	Sub-thread category
el2_msmc0_dthread[2:0]	Out	3'd0	Thread category
el2_msmc0_dtrace[3:0]	Out	4'd0	Debug trace information
el2_msmc0_dxcnt[7:0]	Out	8'd0	Number of bytes transferred in data phase
el2_msmc0_dgreturn	In	1'd0	Global credit return
el2_msmc0_ds0return[2:0]	In	3'd0	Sub-thread 0 credit return
el2_msmc0_ds1return[2:0]	In	3'd0	Sub-thread 1 credit return
el2_msmc0_dreturn[2:0]	In	3'd0	Thread credit return
el2_msmc0_drreturn[2:0]	In	3'd0	Response credit return
el2_msmc0_dgcredinit[5:0]	In	6'd0	Global credit initialization value
el2_msmc0_dt0credinit[5:0]	In	6'd0	Thread 0 credit initialization value
el2_msmc0_dt0s0credinit[5:0]	In	6'd0	Thread 0 Sub-thread 0 credit initialization value

el2_msmc0_dt0s1credinit[5:0]	In	6'd0	Thread 0 Sub-thread 1 credit initialization value
el2_msmc0_dt0rcredinit[5:0]	In	6'd0	Thread 0 Response credit initialization value
el2_msmc0_dt1credinit[5:0]	In	6'd0	Thread 1 credit initialization value
el2_msmc0_dt1s0credinit[5:0]	In	6'd0	Thread 1 Sub-thread 0 credit initialization value
el2_msmc0_dt1s1credinit[5:0]	In	6'd0	Thread 1 Sub-thread 1 credit initialization value
el2_msmc0_dt1rcredinit[5:0]	In	6'd0	Thread 1 Response credit initialization value
el2_msmc0_dt2credinit[5:0]	In	6'd0	Thread 2 credit initialization value
el2_msmc0_dt2s0credinit[5:0]	In	6'd0	Thread 2 Sub-thread 0 credit initialization value
el2_msmc0_dt2s1credinit[5:0]	In	6'd0	Thread 2 Sub-thread 1 credit initialization value
el2_msmc0_dt2rcredinit[5:0]	In	6'd0	Thread 2 Response credit initialization value
el2_msmc0_sbytecnt[9:0]	Out	10'd0	Bytecount for this write response
el2_msmc0_screditttype[1:0]	Out	2'd0	Credit type indicator
el2_msmc0_sid[11:0]	Out	12'd0	Transaction ID
el2_msmc0_sinterest	Out	1'd0	Debug marker
el2_msmc0_sorderid[3:0]	Out	4'd0	Order ID
el2_msmc0_sreq	Out	1'd0	Data phase request
el2_msmc0_srouteid[11:0]	Out	12'd0	Unique ID for the originating Initiator
el2_msmc0_sstatus[2:0]	Out	3'd0	Response status
el2_msmc0_ssubthread	Out	1'd0	Sub-thread category
el2_msmc0_sthread[2:0]	Out	3'd0	Thread category
el2_msmc0_strace[3:0]	Out	4'd0	Debug trace information
el2_msmc0_sgrturn	In	1'd0	Global credit return
el2_msmc0_ss0return[2:0]	In	3'd0	Sub-thread 0 credit return
el2_msmc0_ss1return[2:0]	In	3'd0	Sub-thread 1 credit return
el2_msmc0_streturn[2:0]	In	3'd0	Thread credit return
el2_msmc0_srreturn[2:0]	In	3'd0	Response credit return
el2_msmc0_sgcredinit[5:0]	In	6'd0	Global credit initialization value
el2_msmc0_st0credinit[5:0]	In	6'd0	Thread 0 credit initialization value
el2_msmc0_st0s0credinit[5:0]	In	6'd0	Thread 0 Sub-thread 0 credit initialization value
el2_msmc0_st0s1credinit[5:0]	In	6'd0	Thread 0 Sub-thread 1 credit initialization value
el2_msmc0_st0rcredinit[5:0]	In	6'd0	Thread 0 Response credit initialization value
el2_msmc0_st1credinit[5:0]	In	6'd0	Thread 1 credit initialization value
el2_msmc0_st1s0credinit[5:0]	In	6'd0	Thread 1 Sub-thread 0 credit initialization value
el2_msmc0_st1s1credinit[5:0]	In	6'd0	Thread 1 Sub-thread 1 credit initialization value
el2_msmc0_st1rcredinit[5:0]	In	6'd0	Thread 1 Response credit initialization value

el2_msmc0_st2credinit[5:0]	In	6'd0	Thread 2 credit initialization value
el2_msmc0_st2s0credinit[5:0]	In	6'd0	Thread 2 Sub-thread 0 credit initialization value
el2_msmc0_st2s1credinit[5:0]	In	6'd0	Thread 2 Sub-thread 1 credit initialization value
el2_msmc0_st2rcredinit[5:0]	In	6'd0	Thread 2 Response credit initialization value
el2_msmc0_caddress_safep[3:0]	Out	4'b1111	Safe Transaction address
el2_msmc0_cbytecnt_safep	Out	1'd1	Safe Transaction bytecount
el2_msmc0_cchanid_safep	Out	1'd1	Safe Channel ID
el2_msmc0_ccinner_safep	Out	1'd1	Safe Inner cacheability
el2_msmc0_ccouter_safep	Out	1'd1	Safe Outer cacheability
el2_msmc0_ccredittytype_safep	Out	1'd1	Safe Credit type indicator
el2_msmc0_cdtype_safep	Out	1'd1	Safe Data type attribute
el2_msmc0_cemudbg_safep	Out	1'd1	Safe Emulation level
el2_msmc0_cpriority_safep	Out	1'd1	Safe Elevated priority
el2_msmc0_cflush_safep	Out	1'd1	Safe Firewall flush
el2_msmc0_cid_safep	Out	1'd1	Safe Transaction ID
el2_msmc0_cinterest_safep	Out	1'd1	Safe Debug marker
el2_msmc0_cmemtype_safep	Out	1'd1	Safe Memory region type
el2_msmc0_cnogap_safep	Out	1'd1	Safe Contiguous byte-enable attribute
el2_msmc0_copcode_safep	Out	1'd1	Safe Transaction opcode
el2_msmc0_corderid_safep	Out	1'd1	Safe Order ID
el2_msmc0_cpable_safep	Out	1'd1	Safe Prefetchable attribute
el2_msmc0_cpriority_safep	Out	1'd1	Safe Transaction priority
el2_msmc0_cpriv_safep	Out	1'd1	Safe Privilege level
el2_msmc0_cprivid_safep	Out	1'd1	Safe Privilege ID for security
el2_msmc0_creq_safer	Out	1'd1	Safe Command request
el2_msmc0_crouteid_safep	Out	1'd1	Safe Unique ID for the Initiator
el2_msmc0_csdomain_safep	Out	1'd1	Safe Sharability domain
el2_msmc0_csecure_safep	Out	1'd1	Safe Security level
el2_msmc0_cstarve_safep	Out	1'd1	Safe Credit starvation indicator
el2_msmc0_csubthread_safep	Out	1'd1	Safe Sub-thread category
el2_msmc0_cthread_safep	Out	1'd1	Safe Thread category
el2_msmc0_cvirtid_safep	Out	1'd1	Safe Virtual ID
el2_msmc0_cgreturn_safer	In	1'd1	Safe Global credit return
el2_msmc0_cs0return_safer[2:0]	In	3'b111	Safe Sub-thread 0 credit return
el2_msmc0_cs1return_safer[2:0]	In	3'b111	Safe Sub-thread 1 credit return
el2_msmc0_ctreturn_safer[2:0]	In	3'b111	Safe Thread credit return
el2_msmc0_crreturn_safer[2:0]	In	3'b111	Safe Response credit return
el2_msmc0_cgcredinit_safep	In	1'd1	Global credit initialization value
el2_msmc0_ct0credinit_safep	In	1'd1	Thread 0 credit initialization value
el2_msmc0_ct0s0credinit_safep	In	1'd1	Thread 0 Sub-thread 0 credit initialization value
el2_msmc0_ct0s1credinit_safep	In	1'd1	Thread 0 Sub-thread 1 credit initialization value

el2_msmc0_ct0credinit_safep	In	1'd1	Thread 0 Response credit initialization value
el2_msmc0_ct1credinit_safep	In	1'd1	Thread 1 credit initialization value
el2_msmc0_ct1s0credinit_safep	In	1'd1	Thread 1 Sub-thread 0 credit initialization value
el2_msmc0_ct1s1credinit_safep	In	1'd1	Thread 1 Sub-thread 1 credit initialization value
el2_msmc0_ct1rcredinit_safep	In	1'd1	Thread 1 Response credit initialization value
el2_msmc0_ct2credinit_safep	In	1'd1	Thread 2 credit initialization value
el2_msmc0_ct2s0credinit_safep	In	1'd1	Thread 2 Sub-thread 0 credit initialization value
el2_msmc0_ct2s1credinit_safep	In	1'd1	Thread 2 Sub-thread 1 credit initialization value
el2_msmc0_ct2rcredinit_safep	In	1'd1	Thread 2 Response credit initialization value
el2_msmc0_dalign_safep	Out	1'd1	Safe Response alignment
el2_msmc0_dbfirst_safep	Out	1'd1	Safe First data phase in burst
el2_msmc0_dblast_safep	Out	1'd1	Safe Last data phase in burst
el2_msmc0_dbytecnt_safep	Out	1'd1	Safe Bytecount of entire data transfer
el2_msmc0_dbyten_safep[5:0]	Out	6'b111111	Safe Byten for write data
el2_msmc0_dcoh_safep	Out	1'd1	Safe Coherence state for response
el2_msmc0_dcreditttype_safep	Out	1'd1	Safe Credit type indicator
el2_msmc0_ddatavalid_safep	Out	1'd1	Safe Valid data indicator
el2_msmc0_ddirty_safep	Out	1'd1	Safe Cache line dirty indicator
el2_msmc0_depriority_safep	Out	1'd1	Safe Elevated priority
el2_msmc0_dfirst_safep	Out	1'd1	Safe First data phase in entire data transfer
el2_msmc0_did_safep	Out	1'd1	Safe Transaction ID
el2_msmc0_dinterest_safep	Out	1'd1	Safe Debug marker
el2_msmc0_dlast_safep	Out	1'd1	Safe Last data phase in entire data transfer
el2_msmc0_dopcode_safep	Out	1'd1	Safe Transaction opcode
el2_msmc0_dorderid_safep	Out	1'd1	Safe Order ID
el2_msmc0_dpriority_safep	Out	1'd1	Safe Transaction priority
el2_msmc0_dreq_safer	Out	1'd1	Safe Data phase request
el2_msmc0_drouteid_safep	Out	1'd1	Safe Unique ID for the originating Initiator
el2_msmc0_dstatus_safep	Out	1'd1	Safe Response status
el2_msmc0_dsubthread_safep	Out	1'd1	Safe Sub-thread category
el2_msmc0_dthread_safep	Out	1'd1	Safe Thread category
el2_msmc0_dtrace_safep	Out	1'd1	Safe Debug trace information
el2_msmc0_dxcnt_safep	Out	1'd1	Safe Number of bytes transferred in data phase
el2_msmc0_dgreturn_safer	In	1'd1	Safe Global credit return
el2_msmc0_ds0return_safer[2:0]	In	3'b111	Safe Sub-thread 0 credit return
el2_msmc0_ds1return_safer[2:0]	In	3'b111	Safe Sub-thread 1 credit return
el2_msmc0_dtreturn_safer[2:0]	In	3'b111	Safe Thread credit return

el2_msmc0_drreturn_safer[2:0]	In	3'b111	Safe Response credit return
el2_msmc0_dgcredinit_safer	In	1'd1	Global credit initialization value
el2_msmc0_dt0credinit_safer	In	1'd1	Thread 0 credit initialization value
el2_msmc0_dt0s0credinit_safer	In	1'd1	Thread 0 Sub-thread 0 credit initialization value
el2_msmc0_dt0s1credinit_safer	In	1'd1	Thread 0 Sub-thread 1 credit initialization value
el2_msmc0_dt0rcredinit_safer	In	1'd1	Thread 0 Response credit initialization value
el2_msmc0_dt1credinit_safer	In	1'd1	Thread 1 credit initialization value
el2_msmc0_dt1s0credinit_safer	In	1'd1	Thread 1 Sub-thread 0 credit initialization value
el2_msmc0_dt1s1credinit_safer	In	1'd1	Thread 1 Sub-thread 1 credit initialization value
el2_msmc0_dt1rcredinit_safer	In	1'd1	Thread 1 Response credit initialization value
el2_msmc0_dt2credinit_safer	In	1'd1	Thread 2 credit initialization value
el2_msmc0_dt2s0credinit_safer	In	1'd1	Thread 2 Sub-thread 0 credit initialization value
el2_msmc0_dt2s1credinit_safer	In	1'd1	Thread 2 Sub-thread 1 credit initialization value
el2_msmc0_dt2rcredinit_safer	In	1'd1	Thread 2 Response credit initialization value
el2_msmc0_sbytecnt_safer	Out	1'd1	Safe Bytecount for this write response
el2_msmc0_scredittyp SAfer	Out	1'd1	Safe Credit type indicator
el2_msmc0_sid_safer	Out	1'd1	Safe Transaction ID
el2_msmc0_sinterest_safer	Out	1'd1	Safe Debug marker
el2_msmc0_sorderid_safer	Out	1'd1	Safe Order ID
el2_msmc0_sreq_safer	Out	1'd1	Safe Data phase request
el2_msmc0_srouteid_safer	Out	1'd1	Safe Unique ID for the originating Initiator
el2_msmc0_sstatus_safer	Out	1'd1	Safe Response status
el2_msmc0_ssubthread_safer	Out	1'd1	Safe Sub-thread category
el2_msmc0_sthread_safer	Out	1'd1	Safe Thread category
el2_msmc0_strace_safer	Out	1'd1	Safe Debug trace information
el2_msmc0_sgreturn_safer	In	1'd1	Safe Global credit return
el2_msmc0_ss0return_safer[2:0]	In	3'b111	Safe Sub-thread 0 credit return
el2_msmc0_ss1return_safer[2:0]	In	3'b111	Safe Sub-thread 1 credit return
el2_msmc0_streturn_safer[2:0]	In	3'b111	Safe Thread credit return
el2_msmc0_srreturn_safer[2:0]	In	3'b111	Safe Response credit return
el2_msmc0_sgcredinit_safer	In	1'd1	Global credit initialization value
el2_msmc0_st0credinit_safer	In	1'd1	Thread 0 credit initialization value
el2_msmc0_st0s0credinit_safer	In	1'd1	Thread 0 Sub-thread 0 credit initialization value
el2_msmc0_st0s1credinit_safer	In	1'd1	Thread 0 Sub-thread 1 credit initialization value
el2_msmc0_st0rcredinit_safer	In	1'd1	Thread 0 Response credit initialization value



el2_msmc0_st1credinit_safe	In	1'd1	Thread 1 credit initialization value
el2_msmc0_st1s0credinit_safe	In	1'd1	Thread 1 Sub-thread 0 credit initialization value
el2_msmc0_st1s1credinit_safe	In	1'd1	Thread 1 Sub-thread 1 credit initialization value
el2_msmc0_st1rcredinit_safe	In	1'd1	Thread 1 Response credit initialization value
el2_msmc0_st2credinit_safe	In	1'd1	Thread 2 credit initialization value
el2_msmc0_st2s0credinit_safe	In	1'd1	Thread 2 Sub-thread 0 credit initialization value
el2_msmc0_st2s1credinit_safe	In	1'd1	Thread 2 Sub-thread 1 credit initialization value
el2_msmc0_st2rcredinit_safe	In	1'd1	Thread 2 Response credit initialization value

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Signal Name	Dir	Default Val	Description
msmc1_el2_caddress[47:0]	In	48'd0	Transaction address
msmc1_el2_camode[1:0]	In	2'd0	Addressing mode
msmc1_el2_cbytecnt[9:0]	In	10'd0	Transaction bytecount
msmc1_el2_cchanid[11:0]	In	12'd0	Channel ID
msmc1_el2_ccinner[1:0]	In	2'd0	Inner cacheability
msmc1_el2_ccsize[2:0]	In	3'd0	Cache line size
msmc1_el2_ccouter[1:0]	In	2'd0	Outer cacheability
msmc1_el2_ccredittype[1:0]	In	2'd0	Credit type indicator
msmc1_el2_cdtype[1:0]	In	2'd0	Data type attribute
msmc1_el2_cemudbg	In	1'd0	Emulation level
msmc1_el2_cepriority[2:0]	In	3'd0	Elevated priority
msmc1_el2_cexcl	In	1'd0	Exclusive attribute
msmc1_el2_cflush	In	1'd0	Firewall flush
msmc1_el2_cid[11:0]	In	12'd0	Transaction ID
msmc1_el2_cinterest	In	1'd0	Debug marker
msmc1_el2_cmemtype[1:0]	In	2'd0	Memory region type
msmc1_el2_cnogap	In	1'd0	Contiguous byte-enable attribute
msmc1_el2_copcode[5:0]	In	6'd0	Transaction opcode
msmc1_el2_corderid[3:0]	In	4'd0	Order ID
msmc1_el2_cpable	In	1'd0	Prefetchable attribute
msmc1_el2_cpriority[2:0]	In	3'd0	Transaction priority
msmc1_el2_cpriv[1:0]	In	2'd0	Privilege level
msmc1_el2_cprivid[7:0]	In	8'd0	Privilege ID for security
msmc1_el2_cqos[2:0]	In	3'd0	Quality of Service level
msmc1_el2_creq	In	1'd0	Command request
msmc1_el2_crouteid[11:0]	In	12'd0	Unique ID for the Initiator
msmc1_el2_crsl[3:0]	In	4'd0	Region select driven by the infrastructure
msmc1_el2_csdomain[1:0]	In	2'd0	Sharability domain
msmc1_el2_csecure	In	1'd0	Security level
msmc1_el2_cstarve	In	1'd0	Credit starvation indicator
msmc1_el2_csubthread	In	1'd0	Sub-thread category
msmc1_el2_cthread[2:0]	In	3'd0	Thread category
msmc1_el2_cvirtid[11:0]	In	12'd0	Virtual ID
msmc1_el2_cgreturn	Out	1'd0	Global credit return
msmc1_el2_cs0return[2:0]	Out	3'd0	Sub-thread 0 credit return
msmc1_el2_cs1return[2:0]	Out	3'd0	Sub-thread 1 credit return
msmc1_el2_ctreturn[2:0]	Out	3'd0	Thread credit return
msmc1_el2_crreturn[2:0]	Out	3'd0	Response credit return
msmc1_el2_cgcredinit[5:0]	Out	6'd0	Global credit initialization value
msmc1_el2_ct0credinit[5:0]	Out	6'd0	Thread 0 credit initialization value
msmc1_el2_ct0s0credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 0 credit initialization value
msmc1_el2_ct0s1credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 1 credit initialization value



msmc1_el2_ct0credinit[5:0]	Out	6'd0	Thread 0 Response credit initialization value
msmc1_el2_ct1credinit[5:0]	Out	6'd0	Thread 1 credit initialization value
msmc1_el2_ct1s0credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 0 credit initialization value
msmc1_el2_ct1s1credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 1 credit initialization value
msmc1_el2_ct1rcredinit[5:0]	Out	6'd0	Thread 1 Response credit initialization value
msmc1_el2_ct2credinit[5:0]	Out	6'd0	Thread 2 credit initialization value
msmc1_el2_ct2s0credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 0 credit initialization value
msmc1_el2_ct2s1credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 1 credit initialization value
msmc1_el2_ct2rcredinit[5:0]	Out	6'd0	Thread 2 Response credit initialization value
msmc1_el2_dalign[9:0]	In	10'd0	Response alignment
msmc1_el2_dbfirst	In	1'd0	First data phase in burst
msmc1_el2_dblast	In	1'd0	Last data phase in burst
msmc1_el2_dbytecnt[9:0]	In	10'd0	Bytecount of entire data transfer
msmc1_el2_dbyten[63:0]	In	64'd0	Byten for write data
msmc1_el2_dcoh[1:0]	In	2'd0	Coherence state for response
msmc1_el2_dcreditttype[1:0]	In	2'd0	Credit type indicator
msmc1_el2_ddata[511:0]	In	512'd0	Data
msmc1_el2_ddatavalid	In	1'd0	Valid data indicator
msmc1_el2_ddirty	In	1'd0	Cache line dirty indicator
msmc1_el2_dedc[19:0]	In	20'd0	EDC protection for ddata
msmc1_el2_depriority[2:0]	In	3'd0	Elevated priority
msmc1_el2_dfirst	In	1'd0	First data phase in entire data transfer
msmc1_el2_did[11:0]	In	12'd0	Transaction ID
msmc1_el2_dinterest	In	1'd0	Debug marker
msmc1_el2_dlast	In	1'd0	Last data phase in entire data transfer
msmc1_el2_dopcode[5:0]	In	6'd0	Transaction opcode
msmc1_el2_dorderid[3:0]	In	4'd0	Order ID
msmc1_el2_dpriority[2:0]	In	3'd0	Transaction priority
msmc1_el2_dreq	In	1'd0	Data phase request
msmc1_el2_drouteid[11:0]	In	12'd0	Unique ID for the originating Initiator
msmc1_el2_dstatus[2:0]	In	3'd0	Response status
msmc1_el2_dsubthread	In	1'd0	Sub-thread category
msmc1_el2_dthread[2:0]	In	3'd0	Thread category
msmc1_el2_dtrace[3:0]	In	4'd0	Debug trace information
msmc1_el2_dxcnt[7:0]	In	8'd0	Number of bytes transferred in data phase
msmc1_el2_dgreturn	Out	1'd0	Global credit return
msmc1_el2_ds0return[2:0]	Out	3'd0	Sub-thread 0 credit return
msmc1_el2_ds1return[2:0]	Out	3'd0	Sub-thread 1 credit return

msmc1_el2_dtreturn[2:0]	Out	3'd0	Thread credit return
msmc1_el2_drreturn[2:0]	Out	3'd0	Response credit return
msmc1_el2_dgcredinit[5:0]	Out	6'd0	Global credit initialization value
msmc1_el2_dt0credinit[5:0]	Out	6'd0	Thread 0 credit initialization value
msmc1_el2_dt0s0credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 0 credit initialization value
msmc1_el2_dt0s1credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 1 credit initialization value
msmc1_el2_dt0rcredinit[5:0]	Out	6'd0	Thread 0 Response credit initialization value
msmc1_el2_dt1credinit[5:0]	Out	6'd0	Thread 1 credit initialization value
msmc1_el2_dt1s0credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 0 credit initialization value
msmc1_el2_dt1s1credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 1 credit initialization value
msmc1_el2_dt1rcredinit[5:0]	Out	6'd0	Thread 1 Response credit initialization value
msmc1_el2_dt2credinit[5:0]	Out	6'd0	Thread 2 credit initialization value
msmc1_el2_dt2s0credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 0 credit initialization value
msmc1_el2_dt2s1credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 1 credit initialization value
msmc1_el2_dt2rcredinit[5:0]	Out	6'd0	Thread 2 Response credit initialization value
msmc1_el2_sbytecnt[9:0]	In	10'd0	Bytecount for this write response
msmc1_el2_screditttype[1:0]	In	2'd0	Credit type indicator
msmc1_el2_sid[11:0]	In	12'd0	Transaction ID
msmc1_el2_sinterest	In	1'd0	Debug marker
msmc1_el2_sorderid[3:0]	In	4'd0	Order ID
msmc1_el2_sreq	In	1'd0	Data phase request
msmc1_el2_srouteid[11:0]	In	12'd0	Unique ID for the originating Initiator
msmc1_el2_sstatus[2:0]	In	3'd0	Response status
msmc1_el2_ssubthread	In	1'd0	Sub-thread category
msmc1_el2_sthread[2:0]	In	3'd0	Thread category
msmc1_el2_strace[3:0]	In	4'd0	Debug trace information
msmc1_el2_sgreturn	Out	1'd0	Global credit return
msmc1_el2_ss0return[2:0]	Out	3'd0	Sub-thread 0 credit return
msmc1_el2_ss1return[2:0]	Out	3'd0	Sub-thread 1 credit return
msmc1_el2_streturn[2:0]	Out	3'd0	Thread credit return
msmc1_el2_srreturn[2:0]	Out	3'd0	Response credit return
msmc1_el2_sgcredinit[5:0]	Out	6'd0	Global credit initialization value
msmc1_el2_st0credinit[5:0]	Out	6'd0	Thread 0 credit initialization value
msmc1_el2_st0s0credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 0 credit initialization value
msmc1_el2_st0s1credinit[5:0]	Out	6'd0	Thread 0 Sub-thread 1 credit initialization value
msmc1_el2_st0rcredinit[5:0]	Out	6'd0	Thread 0 Response credit initialization value

msmc1_el2_st1credinit[5:0]	Out	6'd0	Thread 1 credit initialization value
msmc1_el2_st1s0credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 0 credit initialization value
msmc1_el2_st1s1credinit[5:0]	Out	6'd0	Thread 1 Sub-thread 1 credit initialization value
msmc1_el2_st1rcredinit[5:0]	Out	6'd0	Thread 1 Response credit initialization value
msmc1_el2_st2credinit[5:0]	Out	6'd0	Thread 2 credit initialization value
msmc1_el2_st2s0credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 0 credit initialization value
msmc1_el2_st2s1credinit[5:0]	Out	6'd0	Thread 2 Sub-thread 1 credit initialization value
msmc1_el2_st2rcredinit[5:0]	Out	6'd0	Thread 2 Response credit initialization value
msmc1_el2_caddress_safep[3:0]	In	4'b1111	Safe Transaction address
msmc1_el2_camode_safep	In	1'd1	Safe Addressing mode
msmc1_el2_cbytecnt_safep	In	1'd1	Safe Transaction bytecount
msmc1_el2_cchanid_safep	In	1'd1	Safe Channel ID
msmc1_el2_ccinner_safep	In	1'd1	Safe Inner cacheability
msmc1_el2_cclsize_safep	In	1'd1	Safe Cache line size
msmc1_el2_ccouter_safep	In	1'd1	Safe Outer cacheability
msmc1_el2_ccredittype_safep	In	1'd1	Safe Credit type indicator
msmc1_el2_cdtype_safep	In	1'd1	Safe Data type attribute
msmc1_el2_cemudbg_safep	In	1'd1	Safe Emulation level
msmc1_el2_cepriority_safep	In	1'd1	Safe Elevated priority
msmc1_el2_cexcl_safep	In	1'd1	Safe Exclusive attribute
msmc1_el2_cflush_safep	In	1'd1	Safe Firewall flush
msmc1_el2_cid_safep	In	1'd1	Safe Transaction ID
msmc1_el2_cinterest_safep	In	1'd1	Safe Debug marker
msmc1_el2_cmemtype_safep	In	1'd1	Safe Memory region type
msmc1_el2_cnogap_safep	In	1'd1	Safe Contiguous byte-enable attribute
msmc1_el2_copcode_safep	In	1'd1	Safe Transaction opcode
msmc1_el2_corderid_safep	In	1'd1	Safe Order ID
msmc1_el2_cpable_safep	In	1'd1	Safe Prefetchable attribute
msmc1_el2_cpriority_safep	In	1'd1	Safe Transaction priority
msmc1_el2_cpriv_safep	In	1'd1	Safe Privilege level
msmc1_el2_cprivid_safep	In	1'd1	Safe Privilege ID for security
msmc1_el2_cqos_safep	In	1'd1	Safe Quality of Service level
msmc1_el2_creq_safer	In	1'd1	Safe Command request
msmc1_el2_crouteid_safep	In	1'd1	Safe Unique ID for the Initiator
msmc1_el2_crsl_safep	In	1'd1	Safe Region select driven by the infrastructure
msmc1_el2_csdomain_safep	In	1'd1	Safe Sharability domain
msmc1_el2_csecure_safep	In	1'd1	Safe Security level
msmc1_el2_cstarve_safep	In	1'd1	Safe Credit starvation indicator
msmc1_el2_csubthread_safep	In	1'd1	Safe Sub-thread category
msmc1_el2_cthread_safep	In	1'd1	Safe Thread category



msmc1_el2_cvirtid_safer	In	1'd1	Safe Virtual ID
msmc1_el2_cgreturn_safer	Out	1'd1	Safe Global credit return
msmc1_el2_cs0return_safer[2:0]	Out	3'b111	Safe Sub-thread 0 credit return
msmc1_el2_cs1return_safer[2:0]	Out	3'b111	Safe Sub-thread 1 credit return
msmc1_el2_ctreturn_safer[2:0]	Out	3'b111	Safe Thread credit return
msmc1_el2_crreturn_safer[2:0]	Out	3'b111	Safe Response credit return
msmc1_el2_cgcredinit_safer	Out	1'd1	Global credit initialization value
msmc1_el2_ct0credinit_safer	Out	1'd1	Thread 0 credit initialization value
msmc1_el2_ct0s0credinit_safer	Out	1'd1	Thread 0 Sub-thread 0 credit initialization value
msmc1_el2_ct0s1credinit_safer	Out	1'd1	Thread 0 Sub-thread 1 credit initialization value
msmc1_el2_ct0rcredinit_safer	Out	1'd1	Thread 0 Response credit initialization value
msmc1_el2_ct1credinit_safer	Out	1'd1	Thread 1 credit initialization value
msmc1_el2_ct1s0credinit_safer	Out	1'd1	Thread 1 Sub-thread 0 credit initialization value
msmc1_el2_ct1s1credinit_safer	Out	1'd1	Thread 1 Sub-thread 1 credit initialization value
msmc1_el2_ct1rcredinit_safer	Out	1'd1	Thread 1 Response credit initialization value
msmc1_el2_ct2credinit_safer	Out	1'd1	Thread 2 credit initialization value
msmc1_el2_ct2s0credinit_safer	Out	1'd1	Thread 2 Sub-thread 0 credit initialization value
msmc1_el2_ct2s1credinit_safer	Out	1'd1	Thread 2 Sub-thread 1 credit initialization value
msmc1_el2_ct2rcredinit_safer	Out	1'd1	Thread 2 Response credit initialization value
msmc1_el2_dalign_safer	In	1'd1	Safe Response alignment
msmc1_el2_dbfirst_safer	In	1'd1	Safe First data phase in burst
msmc1_el2_dblast_safer	In	1'd1	Safe Last data phase in burst
msmc1_el2_dbytecnt_safer	In	1'd1	Safe Bytecount of entire data transfer
msmc1_el2_dbyten_safer[5:0]	In	6'b111111	Safe Byten for write data
msmc1_el2_dcoh_safer	In	1'd1	Safe Coherence state for response
msmc1_el2_dcreditype_safer	In	1'd1	Safe Credit type indicator
msmc1_el2_ddatavalid_safer	In	1'd1	Safe Valid data indicator
msmc1_el2_ddirty_safer	In	1'd1	Safe Cache line dirty indicator
msmc1_el2_dpriority_safer	In	1'd1	Safe Elevated priority
msmc1_el2_dfirst_safer	In	1'd1	Safe First data phase in entire data transfer
msmc1_el2_did_safer	In	1'd1	Safe Transaction ID
msmc1_el2_dinterest_safer	In	1'd1	Safe Debug marker
msmc1_el2_dlast_safer	In	1'd1	Safe Last data phase in entire data transfer
msmc1_el2_dopcode_safer	In	1'd1	Safe Transaction opcode
msmc1_el2_dorderid_safer	In	1'd1	Safe Order ID
msmc1_el2_dpriority_safer	In	1'd1	Safe Transaction priority
msmc1_el2_dreq_safer	In	1'd1	Safe Data phase request

msmc1_el2_drouteid_safep	In	1'd1	Safe Unique ID for the originating Initiator
msmc1_el2_dstatus_safep	In	1'd1	Safe Response status
msmc1_el2_dsubthread_safep	In	1'd1	Safe Sub-thread category
msmc1_el2_dthread_safep	In	1'd1	Safe Thread category
msmc1_el2_dtrace_safep	In	1'd1	Safe Debug trace information
msmc1_el2_dxcnt_safep	In	1'd1	Safe Number of bytes transferred in data phase
msmc1_el2_dgreturn_safer	Out	1'd1	Safe Global credit return
msmc1_el2_ds0return_safer[2:0]	Out	3'b111	Safe Sub-thread 0 credit return
msmc1_el2_ds1return_safer[2:0]	Out	3'b111	Safe Sub-thread 1 credit return
msmc1_el2_dtreturn_safer[2:0]	Out	3'b111	Safe Thread credit return
msmc1_el2_drreturn_safer[2:0]	Out	3'b111	Safe Response credit return
msmc1_el2_dgcredinit_safep	Out	1'd1	Global credit initialization value
msmc1_el2_dt0credinit_safep	Out	1'd1	Thread 0 credit initialization value
msmc1_el2_dt0s0credinit_safep	Out	1'd1	Thread 0 Sub-thread 0 credit initialization value
msmc1_el2_dt0s1credinit_safep	Out	1'd1	Thread 0 Sub-thread 1 credit initialization value
msmc1_el2_dt0credinit_safep	Out	1'd1	Thread 0 Response credit initialization value
msmc1_el2_dt1credinit_safep	Out	1'd1	Thread 1 credit initialization value
msmc1_el2_dt1s0credinit_safep	Out	1'd1	Thread 1 Sub-thread 0 credit initialization value
msmc1_el2_dt1s1credinit_safep	Out	1'd1	Thread 1 Sub-thread 1 credit initialization value
msmc1_el2_dt1credinit_safep	Out	1'd1	Thread 1 Response credit initialization value
msmc1_el2_dt2credinit_safep	Out	1'd1	Thread 2 credit initialization value
msmc1_el2_dt2s0credinit_safep	Out	1'd1	Thread 2 Sub-thread 0 credit initialization value
msmc1_el2_dt2s1credinit_safep	Out	1'd1	Thread 2 Sub-thread 1 credit initialization value
msmc1_el2_dt2credinit_safep	Out	1'd1	Thread 2 Response credit initialization value
msmc1_el2_sbytecnt_safep	In	1'd1	Safe Bytecount for this write response
msmc1_el2_scredittytype_safep	In	1'd1	Safe Credit type indicator
msmc1_el2_sid_safep	In	1'd1	Safe Transaction ID
msmc1_el2_sinterest_safep	In	1'd1	Safe Debug marker
msmc1_el2_sorderid_safep	In	1'd1	Safe Order ID
msmc1_el2_sreq_safer	In	1'd1	Safe Data phase request
msmc1_el2_srouteid_safep	In	1'd1	Safe Unique ID for the originating Initiator
msmc1_el2_sstatus_safep	In	1'd1	Safe Response status
msmc1_el2_ssubthread_safep	In	1'd1	Safe Sub-thread category
msmc1_el2_sthread_safep	In	1'd1	Safe Thread category
msmc1_el2_strace_safep	In	1'd1	Safe Debug trace information
msmc1_el2_sgreturn_safer	Out	1'd1	Safe Global credit return



msmc1_el2_ss0return_safer[2:0]	Out	3'b111	Safe Sub-thread 0 credit return
msmc1_el2_ss1return_safer[2:0]	Out	3'b111	Safe Sub-thread 1 credit return
msmc1_el2_streturn_safer[2:0]	Out	3'b111	Safe Thread credit return
msmc1_el2_srreturn_safer[2:0]	Out	3'b111	Safe Response credit return
msmc1_el2_sgcredinit_safep	Out	1'd1	Global credit initialization value
msmc1_el2_st0credinit_safep	Out	1'd1	Thread 0 credit initialization value
msmc1_el2_st0s0credinit_safep	Out	1'd1	Thread 0 Sub-thread 0 credit initialization value
msmc1_el2_st0s1credinit_safep	Out	1'd1	Thread 0 Sub-thread 1 credit initialization value
msmc1_el2_st0credinit_safep	Out	1'd1	Thread 0 Response credit initialization value
msmc1_el2_st1credinit_safep	Out	1'd1	Thread 1 credit initialization value
msmc1_el2_st1s0credinit_safep	Out	1'd1	Thread 1 Sub-thread 0 credit initialization value
msmc1_el2_st1s1credinit_safep	Out	1'd1	Thread 1 Sub-thread 1 credit initialization value
msmc1_el2_st1rcredinit_safep	Out	1'd1	Thread 1 Response credit initialization value
msmc1_el2_st2credinit_safep	Out	1'd1	Thread 2 credit initialization value
msmc1_el2_st2s0credinit_safep	Out	1'd1	Thread 2 Sub-thread 0 credit initialization value
msmc1_el2_st2s1credinit_safep	Out	1'd1	Thread 2 Sub-thread 1 credit initialization value
msmc1_el2_st2rcredinit_safep	Out	1'd1	Thread 2 Response credit initialization value

9.4.400 1 MBA Initiator Interface

Signal Name	Dir	Default Val	Description
el2_msmc1_caddress[47:0]	Out	48'd0	Transaction address
el2_msmc1_cbytecnt[9:0]	Out	10'd0	Transaction bytecount
el2_msmc1_cchanid[11:0]	Out	12'd0	Channel ID
el2_msmc1_ccinner[1:0]	Out	2'd0	Inner cacheability
el2_msmc1_ccouter[1:0]	Out	2'd0	Outer cacheability
el2_msmc1_ccredittype[1:0]	Out	2'd0	Credit type indicator
el2_msmc1_cdtype[1:0]	Out	2'd0	Data type attribute
el2_msmc1_cemudbg	Out	1'd0	Emulation level
el2_msmc1_cepriority[2:0]	Out	3'd0	Elevated priority
el2_msmc1_cflush	Out	1'd0	Firewall flush
el2_msmc1_cid[11:0]	Out	12'd0	Transaction ID
el2_msmc1_cinterest	Out	1'd0	Debug marker
el2_msmc1_cmemtype[1:0]	Out	2'd0	Memory region type
el2_msmc1_cnogap	Out	1'd0	Contiguous byte-enable attribute
el2_msmc1_copcode[5:0]	Out	6'd0	Transaction opcode
el2_msmc1_corderid[3:0]	Out	4'd0	Order ID
el2_msmc1_cpable	Out	1'd0	Prefetchable attribute
el2_msmc1_cpriority[2:0]	Out	3'd0	Transaction priority
el2_msmc1_cpriv[1:0]	Out	2'd0	Privilege level
el2_msmc1_cprivid[7:0]	Out	8'd0	Privilege ID for security
el2_msmc1_creg	Out	1'd0	Command request
el2_msmc1_croutid[11:0]	Out	12'd0	Unique ID for the Initiator
el2_msmc1_csdomain[1:0]	Out	2'd0	Sharability domain
el2_msmc1_csecure	Out	1'd0	Security level
el2_msmc1_cstarve	Out	1'd0	Credit starvation indicator
el2_msmc1_csubthread	Out	1'd0	Sub-thread category
el2_msmc1_ctypead[2:0]	Out	3'd0	Thread category
el2_msmc1_cvirtid[11:0]	Out	12'd0	Virtual ID
el2_msmc1_cgreturn	In	1'd0	Global credit return
el2_msmc1_cs0return[2:0]	In	3'd0	Sub-thread 0 credit return
el2_msmc1_cs1return[2:0]	In	3'd0	Sub-thread 1 credit return
el2_msmc1_ctreturn[2:0]	In	3'd0	Thread credit return
el2_msmc1_crreturn[2:0]	In	3'd0	Response credit return
el2_msmc1_cgcredinit[5:0]	In	6'd0	Global credit initialization value
el2_msmc1_ct0credinit[5:0]	In	6'd0	Thread 0 credit initialization value
el2_msmc1_ct0s0credinit[5:0]	In	6'd0	Thread 0 Sub-thread 0 credit initialization value
el2_msmc1_ct0s1credinit[5:0]	In	6'd0	Thread 0 Sub-thread 1 credit initialization value
el2_msmc1_ct0rcredinit[5:0]	In	6'd0	Thread 0 Response credit initialization value
el2_msmc1_ct1credinit[5:0]	In	6'd0	Thread 1 credit initialization value
el2_msmc1_ct1s0credinit[5:0]	In	6'd0	Thread 1 Sub-thread 0 credit initialization value
el2_msmc1_ct1s1credinit[5:0]	In	6'd0	Thread 1 Sub-thread 1 credit initialization value



el2_msmc1_ct1rcredinit[5:0]	In	6'd0	Thread 1 Response credit initialization value
el2_msmc1_ct2credinit[5:0]	In	6'd0	Thread 2 credit initialization value
el2_msmc1_ct2s0credinit[5:0]	In	6'd0	Thread 2 Sub-thread 0 credit initialization value
el2_msmc1_ct2s1credinit[5:0]	In	6'd0	Thread 2 Sub-thread 1 credit initialization value
el2_msmc1_ct2rcredinit[5:0]	In	6'd0	Thread 2 Response credit initialization value
el2_msmc1_dalign[9:0]	Out	10'd0	Response alignment
el2_msmc1_dbfirst	Out	1'd0	First data phase in burst
el2_msmc1_dblast	Out	1'd0	Last data phase in burst
el2_msmc1_dbytecnt[9:0]	Out	10'd0	Bytecount of entire data transfer
el2_msmc1_dbyten[63:0]	Out	64'd0	Byten for write data
el2_msmc1_dcoh[1:0]	Out	2'd0	Coherence state for response
el2_msmc1_dcredittype[1:0]	Out	2'd0	Credit type indicator
el2_msmc1_ddata[511:0]	Out	512'd0	Data
el2_msmc1_ddatavalid	Out	1'd0	Valid data indicator
el2_msmc1_ddirty	Out	1'd0	Cache line dirty indicator
el2_msmc1_dedc[19:0]	Out	20'd0	EDC protection for ddata
el2_msmc1_depriority[2:0]	Out	3'd0	Elevated priority
el2_msmc1_dfirst	Out	1'd0	First data phase in entire data transfer
el2_msmc1_did[11:0]	Out	12'd0	Transaction ID
el2_msmc1_dinterest	Out	1'd0	Debug marker
el2_msmc1_dlast	Out	1'd0	Last data phase in entire data transfer
el2_msmc1_dopcode[5:0]	Out	6'd0	Transaction opcode
el2_msmc1_dorderid[3:0]	Out	4'd0	Order ID
el2_msmc1_dpriority[2:0]	Out	3'd0	Transaction priority
el2_msmc1_dreq	Out	1'd0	Data phase request
el2_msmc1_drouteid[11:0]	Out	12'd0	Unique ID for the originating Initiator
el2_msmc1_dstatus[2:0]	Out	3'd0	Response status
el2_msmc1_dsubthread	Out	1'd0	Sub-thread category
el2_msmc1_dthread[2:0]	Out	3'd0	Thread category
el2_msmc1_dtrace[3:0]	Out	4'd0	Debug trace information
el2_msmc1_dxcnt[7:0]	Out	8'd0	Number of bytes transferred in data phase
el2_msmc1_dgreturn	In	1'd0	Global credit return
el2_msmc1_ds0return[2:0]	In	3'd0	Sub-thread 0 credit return
el2_msmc1_ds1return[2:0]	In	3'd0	Sub-thread 1 credit return
el2_msmc1_dreturn[2:0]	In	3'd0	Thread credit return
el2_msmc1_drreturn[2:0]	In	3'd0	Response credit return
el2_msmc1_dgcredinit[5:0]	In	6'd0	Global credit initialization value
el2_msmc1_dt0credinit[5:0]	In	6'd0	Thread 0 credit initialization value
el2_msmc1_dt0s0credinit[5:0]	In	6'd0	Thread 0 Sub-thread 0 credit initialization value

el2_msmc1_dt0s1credinit[5:0]	In	6'd0	Thread 0 Sub-thread 1 credit initialization value
el2_msmc1_dt0rcredinit[5:0]	In	6'd0	Thread 0 Response credit initialization value
el2_msmc1_dt1credinit[5:0]	In	6'd0	Thread 1 credit initialization value
el2_msmc1_dt1s0credinit[5:0]	In	6'd0	Thread 1 Sub-thread 0 credit initialization value
el2_msmc1_dt1s1credinit[5:0]	In	6'd0	Thread 1 Sub-thread 1 credit initialization value
el2_msmc1_dt1rcredinit[5:0]	In	6'd0	Thread 1 Response credit initialization value
el2_msmc1_dt2credinit[5:0]	In	6'd0	Thread 2 credit initialization value
el2_msmc1_dt2s0credinit[5:0]	In	6'd0	Thread 2 Sub-thread 0 credit initialization value
el2_msmc1_dt2s1credinit[5:0]	In	6'd0	Thread 2 Sub-thread 1 credit initialization value
el2_msmc1_dt2rcredinit[5:0]	In	6'd0	Thread 2 Response credit initialization value
el2_msmc1_sbytecnt[9:0]	Out	10'd0	Bytecount for this write response
el2_msmc1_scredittytype[1:0]	Out	2'd0	Credit type indicator
el2_msmc1_sid[11:0]	Out	12'd0	Transaction ID
el2_msmc1_sinterest	Out	1'd0	Debug marker
el2_msmc1_sorderid[3:0]	Out	4'd0	Order ID
el2_msmc1_sreq	Out	1'd0	Data phase request
el2_msmc1_srouteid[11:0]	Out	12'd0	Unique ID for the originating Initiator
el2_msmc1_sstatus[2:0]	Out	3'd0	Response status
el2_msmc1_ssubthread	Out	1'd0	Sub-thread category
el2_msmc1_sthread[2:0]	Out	3'd0	Thread category
el2_msmc1_strace[3:0]	Out	4'd0	Debug trace information
el2_msmc1_sgreturn	In	1'd0	Global credit return
el2_msmc1_ss0return[2:0]	In	3'd0	Sub-thread 0 credit return
el2_msmc1_ss1return[2:0]	In	3'd0	Sub-thread 1 credit return
el2_msmc1_streturn[2:0]	In	3'd0	Thread credit return
el2_msmc1_srreturn[2:0]	In	3'd0	Response credit return
el2_msmc1_sgcredinit[5:0]	In	6'd0	Global credit initialization value
el2_msmc1_st0credinit[5:0]	In	6'd0	Thread 0 credit initialization value
el2_msmc1_st0s0credinit[5:0]	In	6'd0	Thread 0 Sub-thread 0 credit initialization value
el2_msmc1_st0s1credinit[5:0]	In	6'd0	Thread 0 Sub-thread 1 credit initialization value
el2_msmc1_st0rcredinit[5:0]	In	6'd0	Thread 0 Response credit initialization value
el2_msmc1_st1credinit[5:0]	In	6'd0	Thread 1 credit initialization value
el2_msmc1_st1s0credinit[5:0]	In	6'd0	Thread 1 Sub-thread 0 credit initialization value
el2_msmc1_st1s1credinit[5:0]	In	6'd0	Thread 1 Sub-thread 1 credit initialization value
el2_msmc1_st1rcredinit[5:0]	In	6'd0	Thread 1 Response credit initialization value



el2_msmc1_st2credinit[5:0]	In	6'd0	Thread 2 credit initialization value
el2_msmc1_st2s0credinit[5:0]	In	6'd0	Thread 2 Sub-thread 0 credit initialization value
el2_msmc1_st2s1credinit[5:0]	In	6'd0	Thread 2 Sub-thread 1 credit initialization value
el2_msmc1_st2rcredinit[5:0]	In	6'd0	Thread 2 Response credit initialization value
el2_msmc1_caddress_safep[3:0]	Out	4'b1111	Safe Transaction address
el2_msmc1_cbytecnt_safep	Out	1'd1	Safe Transaction bytecount
el2_msmc1_cchanid_safep	Out	1'd1	Safe Channel ID
el2_msmc1_ccinner_safep	Out	1'd1	Safe Inner cacheability
el2_msmc1_ccouter_safep	Out	1'd1	Safe Outer cacheability
el2_msmc1_ccredittytype_safep	Out	1'd1	Safe Credit type indicator
el2_msmc1_cdtype_safep	Out	1'd1	Safe Data type attribute
el2_msmc1_cemudbg_safep	Out	1'd1	Safe Emulation level
el2_msmc1_cepriority_safep	Out	1'd1	Safe Elevated priority
el2_msmc1_cflush_safep	Out	1'd1	Safe Firewall flush
el2_msmc1_cid_safep	Out	1'd1	Safe Transaction ID
el2_msmc1_cinterest_safep	Out	1'd1	Safe Debug marker
el2_msmc1_cmemtype_safep	Out	1'd1	Safe Memory region type
el2_msmc1_cnogap_safep	Out	1'd1	Safe Contiguous byte-enable attribute
el2_msmc1_copcode_safep	Out	1'd1	Safe Transaction opcode
el2_msmc1_corderid_safep	Out	1'd1	Safe Order ID
el2_msmc1_cpable_safep	Out	1'd1	Safe Prefetchable attribute
el2_msmc1_cpriority_safep	Out	1'd1	Safe Transaction priority
el2_msmc1_cpriv_safep	Out	1'd1	Safe Privilege level
el2_msmc1_cprivid_safep	Out	1'd1	Safe Privilege ID for security
el2_msmc1_creq_safer	Out	1'd1	Safe Command request
el2_msmc1_crouteid_safep	Out	1'd1	Safe Unique ID for the Initiator
el2_msmc1_csdomain_safep	Out	1'd1	Safe Sharability domain
el2_msmc1_csecure_safep	Out	1'd1	Safe Security level
el2_msmc1_cstarve_safep	Out	1'd1	Safe Credit starvation indicator
el2_msmc1_csubthread_safep	Out	1'd1	Safe Sub-thread category
el2_msmc1_cthread_safep	Out	1'd1	Safe Thread category
el2_msmc1_cvirtid_safep	Out	1'd1	Safe Virtual ID
el2_msmc1_cgreturn_safer	In	1'd1	Safe Global credit return
el2_msmc1_cs0return_safer[2:0]	In	3'b111	Safe Sub-thread 0 credit return
el2_msmc1_cs1return_safer[2:0]	In	3'b111	Safe Sub-thread 1 credit return
el2_msmc1_ctreturn_safer[2:0]	In	3'b111	Safe Thread credit return
el2_msmc1_crreturn_safer[2:0]	In	3'b111	Safe Response credit return
el2_msmc1_cgcredinit_safep	In	1'd1	Global credit initialization value
el2_msmc1_ct0credinit_safep	In	1'd1	Thread 0 credit initialization value
el2_msmc1_ct0s0credinit_safep	In	1'd1	Thread 0 Sub-thread 0 credit initialization value
el2_msmc1_ct0s1credinit_safep	In	1'd1	Thread 0 Sub-thread 1 credit initialization value

el2_msmc1_ct0credinit_safep	In	1'd1	Thread 0 Response credit initialization value
el2_msmc1_ct1credinit_safep	In	1'd1	Thread 1 credit initialization value
el2_msmc1_ct1s0credinit_safep	In	1'd1	Thread 1 Sub-thread 0 credit initialization value
el2_msmc1_ct1s1credinit_safep	In	1'd1	Thread 1 Sub-thread 1 credit initialization value
el2_msmc1_ct1rcredinit_safep	In	1'd1	Thread 1 Response credit initialization value
el2_msmc1_ct2credinit_safep	In	1'd1	Thread 2 credit initialization value
el2_msmc1_ct2s0credinit_safep	In	1'd1	Thread 2 Sub-thread 0 credit initialization value
el2_msmc1_ct2s1credinit_safep	In	1'd1	Thread 2 Sub-thread 1 credit initialization value
el2_msmc1_ct2rcredinit_safep	In	1'd1	Thread 2 Response credit initialization value
el2_msmc1_dalign_safep	Out	1'd1	Safe Response alignment
el2_msmc1_dbfirst_safep	Out	1'd1	Safe First data phase in burst
el2_msmc1_dblast_safep	Out	1'd1	Safe Last data phase in burst
el2_msmc1_dbytecnt_safep	Out	1'd1	Safe Bytecount of entire data transfer
el2_msmc1_dbyten_safep[5:0]	Out	6'b111111	Safe Byten for write data
el2_msmc1_dcoh_safep	Out	1'd1	Safe Coherence state for response
el2_msmc1_dcredittyp_safep	Out	1'd1	Safe Credit type indicator
el2_msmc1_ddatavalid_safep	Out	1'd1	Safe Valid data indicator
el2_msmc1_ddirty_safep	Out	1'd1	Safe Cache line dirty indicator
el2_msmc1_depriority_safep	Out	1'd1	Safe Elevated priority
el2_msmc1_dfirst_safep	Out	1'd1	Safe First data phase in entire data transfer
el2_msmc1_did_safep	Out	1'd1	Safe Transaction ID
el2_msmc1_dinterest_safep	Out	1'd1	Safe Debug marker
el2_msmc1_dlast_safep	Out	1'd1	Safe Last data phase in entire data transfer
el2_msmc1_dopcode_safep	Out	1'd1	Safe Transaction opcode
el2_msmc1_dorderid_safep	Out	1'd1	Safe Order ID
el2_msmc1_dpriority_safep	Out	1'd1	Safe Transaction priority
el2_msmc1_dreq_safer	Out	1'd1	Safe Data phase request
el2_msmc1_drouteid_safep	Out	1'd1	Safe Unique ID for the originating Initiator
el2_msmc1_dstatus_safep	Out	1'd1	Safe Response status
el2_msmc1_dsubthread_safep	Out	1'd1	Safe Sub-thread category
el2_msmc1_dthread_safep	Out	1'd1	Safe Thread category
el2_msmc1_dtrace_safep	Out	1'd1	Safe Debug trace information
el2_msmc1_dxcnt_safep	Out	1'd1	Safe Number of bytes transferred in data phase
el2_msmc1_dgreturn_safer	In	1'd1	Safe Global credit return
el2_msmc1_ds0return_safer[2:0]	In	3'b111	Safe Sub-thread 0 credit return
el2_msmc1_ds1return_safer[2:0]	In	3'b111	Safe Sub-thread 1 credit return
el2_msmc1_dtreturn_safer[2:0]	In	3'b111	Safe Thread credit return

el2_msmc1_drreturn_safer[2:0]	In	3'b111	Safe Response credit return
el2_msmc1_dgcredinit_safer	In	1'd1	Global credit initialization value
el2_msmc1_dt0credinit_safer	In	1'd1	Thread 0 credit initialization value
el2_msmc1_dt0s0credinit_safer	In	1'd1	Thread 0 Sub-thread 0 credit initialization value
el2_msmc1_dt0s1credinit_safer	In	1'd1	Thread 0 Sub-thread 1 credit initialization value
el2_msmc1_dt0rcredinit_safer	In	1'd1	Thread 0 Response credit initialization value
el2_msmc1_dt1credinit_safer	In	1'd1	Thread 1 credit initialization value
el2_msmc1_dt1s0credinit_safer	In	1'd1	Thread 1 Sub-thread 0 credit initialization value
el2_msmc1_dt1s1credinit_safer	In	1'd1	Thread 1 Sub-thread 1 credit initialization value
el2_msmc1_dt1rcredinit_safer	In	1'd1	Thread 1 Response credit initialization value
el2_msmc1_dt2credinit_safer	In	1'd1	Thread 2 credit initialization value
el2_msmc1_dt2s0credinit_safer	In	1'd1	Thread 2 Sub-thread 0 credit initialization value
el2_msmc1_dt2s1credinit_safer	In	1'd1	Thread 2 Sub-thread 1 credit initialization value
el2_msmc1_dt2rcredinit_safer	In	1'd1	Thread 2 Response credit initialization value
el2_msmc1_sbytecnt_safer	Out	1'd1	Safe Bytecount for this write response
el2_msmc1_scredittyp SAfer	Out	1'd1	Safe Credit type indicator
el2_msmc1_sid_safer	Out	1'd1	Safe Transaction ID
el2_msmc1_sinterest_safer	Out	1'd1	Safe Debug marker
el2_msmc1_sorderid_safer	Out	1'd1	Safe Order ID
el2_msmc1_sreq_safer	Out	1'd1	Safe Data phase request
el2_msmc1_srouteid_safer	Out	1'd1	Safe Unique ID for the originating Initiator
el2_msmc1_sstatus_safer	Out	1'd1	Safe Response status
el2_msmc1_ssubthread_safer	Out	1'd1	Safe Sub-thread category
el2_msmc1_sthread_safer	Out	1'd1	Safe Thread category
el2_msmc1_strace_safer	Out	1'd1	Safe Debug trace information
el2_msmc1_sgreturn_safer	In	1'd1	Safe Global credit return
el2_msmc1_ss0return_safer[2:0]	In	3'b111	Safe Sub-thread 0 credit return
el2_msmc1_ss1return_safer[2:0]	In	3'b111	Safe Sub-thread 1 credit return
el2_msmc1_streturn_safer[2:0]	In	3'b111	Safe Thread credit return
el2_msmc1_srreturn_safer[2:0]	In	3'b111	Safe Response credit return
el2_msmc1_sgcredinit_safer	In	1'd1	Global credit initialization value
el2_msmc1_st0credinit_safer	In	1'd1	Thread 0 credit initialization value
el2_msmc1_st0s0credinit_safer	In	1'd1	Thread 0 Sub-thread 0 credit initialization value
el2_msmc1_st0s1credinit_safer	In	1'd1	Thread 0 Sub-thread 1 credit initialization value
el2_msmc1_st0rcredinit_safer	In	1'd1	Thread 0 Response credit initialization value



el2_msmc1_st1credinit_safe	In	1'd1	Thread 1 credit initialization value
el2_msmc1_st1s0credinit_safe	In	1'd1	Thread 1 Sub-thread 0 credit initialization value
el2_msmc1_st1s1credinit_safe	In	1'd1	Thread 1 Sub-thread 1 credit initialization value
el2_msmc1_st1rcredinit_safe	In	1'd1	Thread 1 Response credit initialization value
el2_msmc1_st2credinit_safe	In	1'd1	Thread 2 credit initialization value
el2_msmc1_st2s0credinit_safe	In	1'd1	Thread 2 Sub-thread 0 credit initialization value
el2_msmc1_st2s1credinit_safe	In	1'd1	Thread 2 Sub-thread 1 credit initialization value
el2_msmc1_st2rcredinit_safe	In	1'd1	Thread 2 Response credit initialization value

**9.4.401 VBUSP_FW_CFG_EXCEPTION subordinate VBUSP
Interface**

Signal Name	Dir	Default Val	Description
vbusp_dmesc_dsp_orderid[3:0]	In	4'd0	Order ID
vbusp_dmesc_dsp_routeid[11:0]	In	12'd0	Route ID
vbusp_dmesc_dsp_xid[11:0]	In	12'd0	Transaction ID
vbusp_dmesc_dsp_req	In	1'b0	Request
vbusp_dmesc_dsp_rsel[3:0]	In	4'd0	Region Select
vbusp_dmesc_dsp_dir	In	1'b0	Direction
vbusp_dmesc_dsp_address[19:0]	In	20'd0	Address
vbusp_dmesc_dsp_xcnt[2:0]	In	3'd4	Good Byte Count
vbusp_dmesc_dsp_byten[3:0]	In	{4{1'b1}}	Byte Enables
vbusp_dmesc_dsp_wdata[31:0]	In	32'd0	Write Data
vbusp_dmesc_dsp_wedc[6:0]	In	7'd0	ECC on write data
vbusp_dmesc_dsp_wready	Out	1'b1	Write Ready
vbusp_dmesc_dsp_rdatap[31:0]	Out	32'd0	Read Data (Pipelined)
vbusp_dmesc_dsp_redc[6:0]	Out	7'd0	ECC on read data
vbusp_dmesc_dsp_rstatus[2:0]	Out	3'd0	Read Status
vbusp_dmesc_dsp_rready	Out	1'b1	Read Ready
vbusp_dmesc_dsp_emudbg	In	1'b0	Emulation Debug Attribute
vbusp_dmesc_dsp_interest	In	1'b0	Debug marker
vbusp_dmesc_dsp_ew	In	1'd0	Command Power Mgt EW
vbusp_dmesc_dsp_sorderid[3:0]	Out	4'd0	Status Order ID
vbusp_dmesc_dsp_srouteid[11:0]	Out	12'd0	Status Route ID
vbusp_dmesc_dsp_sid[11:0]	Out	12'd0	Status Transaction ID
vbusp_dmesc_dsp_sreq	Out	1'b0	Status Request
vbusp_dmesc_dsp_sbytecnt[9:0]	Out	10'd0	Status Done Indicator
vbusp_dmesc_dsp_sstatus[2:0]	Out	3'd0	Status Information
vbusp_dmesc_dsp_sinterest	Out	1'b0	Debug Marker
vbusp_dmesc_dsp_sready	In	1'b1	Status Ready
vbusp_dmesc_dsp_req_safer	In	1'b1	Req Safety Redundancy
vbusp_dmesc_dsp_rready_safer	Out	1'b0	Rready Safety Redundancy
vbusp_dmesc_dsp_wready_safer	Out	1'b0	Wready Safety Redundancy
vbusp_dmesc_dsp_sreq_safer	Out	1'b1	Sreq Safety Redundancy
vbusp_dmesc_dsp_sready_safer	In	1'b0	Sready Safety Redundancy
vbusp_dmesc_dsp_dir_safer	In	1'b1	Direction Safety Parity
vbusp_dmesc_dsp_orderid_safer	In	1'b1	Order ID Safety Parity
vbusp_dmesc_dsp_routeid_safer	In	1'b1	Route ID Safety Parity
vbusp_dmesc_dsp_xid_safer	In	1'b1	Transaction ID Safety Parity
vbusp_dmesc_dsp_rsel_safer	In	1'b1	Region Select Safety Parity
vbusp_dmesc_dsp_address_safer[1:0]	In	{2 {1'b1}}	Address Safety Parity
vbusp_dmesc_dsp_xcnt_safer	In	~^vbusp_dmesc_dsp_xcnt	Good Byte Count Safety Parity
vbusp_dmesc_dsp_byten_safer	In	{1 {1'b1}}	Byte Enables Safety Parity
vbusp_dmesc_dsp_rstatus_safer	Out	1'b1	Read Status Safety Parity
vbusp_dmesc_dsp_emudbg_safer	In	1'b1	Emulation Debug Attribute Safety Parity
vbusp_dmesc_dsp_interest_safer	In	1'b1	Debug marker Safety Parity
vbusp_dmesc_dsp_sorderid_safer	Out	1'b1	Status Order ID Safety Parity
vbusp_dmesc_dsp_srouteid_safer	Out	1'b1	Status Route ID Safety Parity
vbusp_dmesc_dsp_sid_safer	Out	1'b1	Status Transaction ID Safety Parity

vbusp_dmsc_dsp_sbytecnt_safep	Out	1'b1	Status Done Indicator Safety Parity
vbusp_dmsc_dsp_sstatus_safep	Out	1'b1	Status Information Safety Parity
vbusp_dmsc_dsp_sinterest_safep	Out	1'b1	Debug Marker Safety Parity

9.4.402 EL2 Memory scrubber disable control

Signal Name	Dir	Default Val	Description
dft_el2_scrub_off_wir	In	1'd0	dft wir signal

9.4.403 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_el2_dft_clk_en_wir	In	1'd0	dft wir signal

9.4.404 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_el2_clk_bypass_wir	In	1'd0	dft wir signal

9.4.405 EL2 Clk grid disable control

Signal Name	Dir	Default Val	Description
dft_el2_divfactor_sel_wir	In	1'd0	dft wir signal

9.4.406 divh_clk1_tft_clk_en control

Signal Name	Dir	Default Val	Description
el2_divh_clk1_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.407 divh_clk4_tft_clk_en control

Signal Name	Dir	Default Val	Description
el2_divh_clk4_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.408 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
el2_wrap_clkstop_clkstop_req	In	1'd0	Clock stop request
el2_wrap_clkstop_clkstop_ack	Out	el2_wrap_clkstop_clkstop_req	Clock stop acknowledge

9.4.409 el2 clk1 clock

Signal Name	Dir	Default Val	Description
el2_clk_clk	In	1'd0	Single ended clock
el2_clk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.410 el2 clk1 clock

Signal Name	Dir	Default Val	Description
el2_divh_clk4_obsclk_out_clk	Out	1'd0	Single ended clock

9.4.411 pll ctrl clock

Signal Name	Dir	Default Val	Description
el2_pll_ctrl_main_clk_clk	In	1'd0	Single ended clock

9.4.412 dft_ip

Signal Name	Dir	Default Val	Description
el2_dft_clk_force_en	In	1'd0	dft_clk_force_en
el2_dft_scan_en	In	1'd0	dft_scan_en
el2_dft_async_rst_sel	In	1'd0	dft_async_rst_sel
el2_dft_test_async_rst_n	In	1'd1	dft_test_async_rst_n
el2_dft_tft_mcp_dis	In	1'd0	Assert High to block MCP path
el2_dft_tft_conf_vector[67:0]	In	68'd0	clkdiv dft_tft_conf_vector
el2_dft_tft_start_en	In	1'd0	clkdiv dft_tft_start_en
el2_dft_delay_fault_en	In	1'd0	clkdiv dft_delay_fault_en
el2_dft_tft_done	Out	1'd1	clkdiv dft_tft_done
el2_dft_div_scan_en	In	1'd0	clkdiv dft_div_scan_en
el2_dft_bypass_div_en	In	1'd0	clkdiv dft_bypass_div_en
el2_dft_local_clk_en	In	1'd0	dft_local_clk_en
el2_dft_pbist_clk_en	In	1'd0	dft_pbist_clk_en
el2_dft_mode_atpg	In	1'd0	ATPG
el2_dft_mode_iddq	In	1'd0	Iddq mode
el2_dft_mode_burnin	In	1'd0	Burnin
el2_dft_embed_mode_pbist	In	1'd0	Embedded pBIST Mode

9.4.413 PLL control align signals

Signal Name	Dir	Default Val	Description
el2_pll_ctrl_align_align_clk_div_rst_n	In	1'd0	Export align_clk_div
el2_pll_ctrl_align_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.414 dft_compression

Signal Name	Dir	Default Val	Description
el2_dft_compression_dft_block_en	In	1'd1	dft_block_en
el2_dft_compression_dft_partition_en	In	1'd1	dft_partition_en
el2_dft_compression_dft_compression_cmle	In	1'd0	dft_compression_cmle
el2_dft_compression_dft_compression_mode	In	1'd0	dft_compression_mode
el2_dft_compression_dft_compression_spread	In	1'd0	dft_compression_spread
el2_dft_compression_dft_elastic_reset	In	1'd0	dft_elastic_reset
el2_dft_compression_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
el2_dft_compression_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
el2_dft_compression_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
el2_dft_compression_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
el2_dft_compression_dft_struct_intest	In	1'd0	dft_struct_intest
el2_dft_compression_dft_struct_extest	In	1'd0	dft_struct_extest
el2_dft_compression_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
el2_dft_compression_dft_struct_si[35:0]	In	36'd0	dft_struct_si
el2_dft_compression_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.415 Controls for the TFT Clock control module

Signal Name	Dir	Default Val	Description
el2_dft_pll_ctrl_dft_mode_pll_atpg	In	1'd0	dft_mode_pll_atpg
el2_dft_pll_ctrl_dft_pll_capture_en	In	1'd0	dft_pll_capture_en
el2_dft_pll_ctrl_dft_shift_clk_sel	In	1'd0	dft_shift_clk_sel
el2_dft_pll_ctrl_cs_func_clk_en	In	1'd0	enable clocks to selected scan paths

9.4.416 DFT PLL CLK BYPASS control for the TFT Clock control module

Signal Name	Dir	Default Val	Description
el2_dft_pll_clk_bypass_pll_clk_bypass	In	1'd1	DFT WIR input to kp_ti_ipg_loc_tft_clk_ctrl

9.4.417 el2 clk1 clock

Signal Name	Dir	Default Val	Description
el2_clk_ctl_align_clk_div_rst_n	In	1'd0	Export align_clk_div
el2_clk_ctl_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.418 Clken interface

Signal Name	Dir	Default Val	Description
el2_wrap_clken_clk_en	In	1'd1	Clock-gate enable to IP
el2_wrap_clken_clk_en_ack	Out	el2_wrap_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.419 DFT clock clken control for the Dividers

Signal Name	Dir	Default Val	Description
el2_dft_local_div_clk_en_div_clk_en	In	1'd1	DFT local clk enable contribution to kp_ti_ipg_clk_divhdist5_tft_mux

9.4.420 el2_wrap_rst

Signal Name	Dir	Default Val	Description
el2_wrap_rst_mod_g_rst_n	In	1'd1	Module level main reset
el2_wrap_rst_mod_por_rst_n	In	1'd1	Module level por reset
el2_wrap_rst_mod_l_rst_n	In	1'd1	Module level local reset

9.4.421 Power interface

Signal Name	Dir	Default Val	Description
el2_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
el2_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
el2_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.422 DFT mem interface

Signal Name	Dir	Default Val	Description
el2_mem_atpg_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.423 MEM_MARGIN busdef

Signal Name	Dir	Default Val	Description
el2_dftmem_s2cr1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cr1p compiler
el2_dftmem_s2dc1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2dc1p compiler
el2_dftmem_s2sr1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2sr1p compiler
el2_dftmem_s2ss1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2ss1p compiler
el2_dftmem_s2du1p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2du1p compiler
el2_dftmem_s2cr2p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cr2p compiler
el2_dftmem_s2cu2p_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for s2cu2p compiler
el2_dftmem_s2cr1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cr1p compiler
el2_dftmem_s2dc1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2dc1p compiler
el2_dftmem_s2sr1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2sr1p compiler
el2_dftmem_s2ss1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2ss1p compiler
el2_dftmem_s2du1p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2du1p compiler
el2_dftmem_s2cr2p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cr2p compiler
el2_dftmem_s2cu2p_dft_mem_ra[1:0]	In	2'd0	dft_mem_read assist for s2cu2p compiler
el2_dftmem_s2du1p_dft_mem_wa[2:0]	In	3'd0	dft_mem_write assist for s2cr2p compiler
el2_dftmem_s2du1p_dft_mem_wpulse[2:0]	In	3'd0	dft_mem_write assist for s2cr2p compiler
el2_dftmem_rom_dft_mem_margin[3:0]	In	4'd0	dft_mem_margin for rom compiler

9.4.424 MEM_TESTRWM busdef

Signal Name	Dir	Default Val	Description
el2_dftmem_testrwm_dft_mem_testrwm	In	1'd0	memory testrwm

9.4.425 MEM_DFTCLKEN busdef

Signal Name	Dir	Default Val	Description
el2_dftmem_dftclken_dft_mem_dftclken	In	1'd0	memory dftclken



9.4.426 MEM_TEST1 busdef

Signal Name	Dir	Default Val	Description
el2_dftmem_test1_dft_mem_test1	In	1'd0	memory test1

9.4.427 MEM_TEST_RNM busdef

Signal Name	Dir	Default Val	Description
el2_dftmem_testrm_dft_mem_test_rnm	In	1'd0	memory test_rnm

9.4.428 PWR BIAS busdef

Signal Name	Dir	Default Val	Description
el2_pwr_bias_pwr_mem_bc0	In	1'h0	pwr
el2_pwr_bias_pwr_mem_bc1	In	1'h0	pwr
el2_pwr_bias_pwr_mem_bc2	In	1'h0	pwr

9.4.429 SOC Events in to CLEC

Signal Name	Dir	Default Val	Description
el2_soc_events_in_intr[127:0]	In	128'd0	Interrupt signal

9.4.430 Events out of CLEC - Level

Signal Name	Dir	Default Val	Description
el2_soc_events_out_level_intr[127:0]	Out	128'd0	Interrupt signal

9.4.431 Events out of CLEC - Pulse

Signal Name	Dir	Default Val	Description
el2_esm_events_out_pulse_intr[63:0]	Out	64'd0	Interrupt signal

9.4.432 ESM Events out of CLEC - Level

Signal Name	Dir	Default Val	Description
el2_esm_events_out_level_intr[63:0]	Out	64'd0	Interrupt signal

9.4.433 ARM event to CLEC

Signal Name	Dir	Default Val	Description
el2_arm_0_evento_in_intr	In	1'd0	Interrupt signal

9.4.434 ARM eventi from CLEC

Signal Name	Dir	Default Val	Description
el2_arm_0_eventi_out_intr	Out	1'd0	Interrupt signal

9.4.435 Shared Peripherhal Interrupt (SPI) inputs

Shared Peripherhal Interrupt (SPI) inputs

Signal Name	Dir	Default Val	Description
el2_gic_spi_intr[991:32]	In	960'd0	Interrupt signal

9.4.436 Triggers into the DRU

Signal Name	Dir	Default Val	Description
ac_cc_i_req	In	1'd0	Event request signal
ac_cc_i_index[15:0]	In	16'd0	Event index signal
ac_cc_i_updn	In	1'd0	Event up or down signal. 0 for down or inactive. 1 for up or active.
ac_cc_i_ready	Out	1'd1	Event ready signal

9.4.437 Events from the DRU

Signal Name	Dir	Default Val	Description
ac_cc_o_req	Out	1'd0	Event request signal
ac_cc_o_index[15:0]	Out	16'd0	Event index signal
ac_cc_o_updn	Out	1'd0	Event up or down signal. 0 for down or inactive. 1 for up or active.
ac_cc_o_ready	In	1'd1	Event ready signal

9.4.438 atb64_ac_mst_async

Signal Name	Dir	Default Val	Description
atb64_ac_mst_async_slv_safe_state	Out	1'b0	AMBA trace sync request
atb64_ac_mst_async_zero_pointer	In	1'b0	AMBA trace valid
atb64_ac_mst_async_atb_fwd_data[454:0]	Out	455'h0	AMBA trace bytes
atb64_ac_mst_async_atb_rev_data[4:0]	In	5'd0	AMBA trace bytes
atb64_ac_mst_async_syncreqs_ack	Out	1'b0	AMBA trace sync request
atb64_ac_mst_async_syncreqs_req	In	1'b0	AMBA trace flush ready

9.4.439 ARM Trust Zone Interface

Signal Name	Dir	Default Val	Description
el2_dbg_sec_dbgen	In	1'd0	Invasive debug enable
el2_dbg_sec_niden	In	1'd0	Non-invasive debug enable
el2_dbg_sec_spiden	In	1'd0	Secure privileged invasive debug enable
el2_dbg_sec_spniden	In	1'd0	Secure privileged non-invasive debug enable

9.4.440 Debug Timestamp Interface

Signal Name	Dir	Default Val	Description
el2_glbdbg_timestamp_shft[1:0]	In	2'd0	binary interface
el2_glbdbg_timestamp_gray[63:0]	In	64'd0	gray encoded interface

9.4.441 el2_dbg_chm

Signal Name	Dir	Default Val	Description
el2_dbg_chm_cti_chnl[3:0]	Out	4'd0	arm channel interface for cross triggering
el2_dbg_chm_cti_chnl_ack[3:0]	In	4'd0	arm channel async handshake interface for cross triggering

9.4.442 el2_dbg_chs

Signal Name	Dir	Default Val	Description
el2_dbg_chs_cti_chnl[3:0]	In	4'd0	arm channel interface for cross triggering
el2_dbg_chs_cti_chnl_ack[3:0]	Out	4'd0	arm channel async handshake interface for cross triggering

9.4.443 Power interface

Signal Name	Dir	Default Val	Description
el2_ss_pwr_pwr_log_iso_en	In	1'd0	pwr
el2_ss_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
el2_ss_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
el2_ss_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
el2_ss_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr
el2_ss_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
el2_ss_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
el2_ss_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.444 Power interface

Signal Name	Dir	Default Val	Description
dru_c_pwr_pwr_log_iso_en	In	1'd0	pwr
dru_c_pwr_pwr_log_ponin[7:0]	In	8'hFF	pwr
dru_c_pwr_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
dru_c_pwr_pwr_log_ponout[7:0]	Out	8'hFF	pwr
dru_c_pwr_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr
dru_c_pwr_pwr_mem_ds[7:0]	In	8'h00	pwr
dru_c_pwr_pwr_mem_ls[7:0]	In	8'h00	pwr
dru_c_pwr_pwr_mem_sd[7:0]	In	8'h00	pwr

9.4.445 dft_compression

Signal Name	Dir	Default Val	Description
dru_c_compression_dft_block_en	In	1'd1	dft_block_en
dru_c_compression_dft_partition_en	In	1'd1	dft_partition_en
dru_c_compression_dft_compression_cmle	In	1'd0	dft_compression_cmle
dru_c_compression_dft_compression_mode	In	1'd0	dft_compression_mode
dru_c_compression_dft_compression_spread	In	1'd0	dft_compression_spread
dru_c_compression_dft_elastic_reset	In	1'd0	dft_elastic_reset
dru_c_compression_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
dru_c_compression_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
dru_c_compression_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
dru_c_compression_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
dru_c_compression_dft_struct_intest	In	1'd0	dft_struct_intest
dru_c_compression_dft_struct_extest	In	1'd0	dft_struct_extest
dru_c_compression_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
dru_c_compression_dft_struct_si[35:0]	In	36'd0	dft_struct_si
dru_c_compression_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.446 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
el2_pbist_0_clkstop_req	In	1'd0	Clock stop request
el2_pbist_0_clkstop_ack	Out	pbist_0_clkstop_req	Clock stop acknowledge

9.4.447 Target .el2_pbist_cfg_0 power disable

Signal Name	Dir	Default Val	Description
pwr_el2_pbist_cfg_0_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
pwr_el2_pbist_cfg_0_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
pwr_el2_pbist_cfg_0_disable_done	In	1'd0	Ack to SCR to stop clocks
pwr_el2_pbist_cfg_0_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
pwr_el2_pbist_cfg_0_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
pwr_el2_pbist_cfg_0_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.448 el2_0_dft_embed_pbist

Signal Name	Dir	Default Val	Description
el2_0_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
el2_0_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
el2_0_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
el2_0_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
el2_0_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
el2_0_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
el2_0_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
el2_0_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
el2_0_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
el2_0_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
el2_0_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
el2_0_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
el2_0_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
el2_0_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
el2_0_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
el2_0_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
el2_0_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
el2_0_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
el2_0_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.449 el2_0_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
el2_0_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.450 el2_0_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
el2_0_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.451 el2_0_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
el2_0_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
el2_0_dft_pbist_pwr_clk_en_ack	Out	el2_0_dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.452 el2_0_pbist_tclk

Signal Name	Dir	Default Val	Description
el2_0_pbist_tclk_clk	In	1'd0	Single ended clock
el2_0_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.453 el2_0_pbist_rst

Signal Name	Dir	Default Val	Description
el2_0_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.454 bisor

Signal Name	Dir	Default Val	Description
el2_0_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
el2_0_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
el2_0_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
el2_0_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
el2_0_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
el2_0_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
el2_0_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
el2_0_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.455 aw0 power disable interface

Signal Name	Dir	Default Val	Description
aw0_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw0_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw0_disable_done	In	1'd0	Ack to SCR to stop clocks
aw0_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw0_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw0_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.456 el2_mem0 power disable interface

Signal Name	Dir	Default Val	Description
pwr_el2_mem0_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
pwr_el2_mem0_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
pwr_el2_mem0_disable_done	In	1'd0	Ack to SCR to stop clocks
pwr_el2_mem0_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
pwr_el2_mem0_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
pwr_el2_mem0_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.457 dft_mem0 ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
el2_dft_mem0_clk_force_en	In	1'd0	dft_clk_force_en

9.4.458 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
el2_pbist_1_clkstop_req	In	1'd0	Clock stop request
el2_pbist_1_clkstop_ack	Out	pbist_1_clkstop_req	Clock stop acknowledge

9.4.459 Target .el2_pbist_cfg_1 power disable

Signal Name	Dir	Default Val	Description
pwr_el2_pbist_cfg_1_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
pwr_el2_pbist_cfg_1_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
pwr_el2_pbist_cfg_1_disable_done	In	1'd0	Ack to SCR to stop clocks
pwr_el2_pbist_cfg_1_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
pwr_el2_pbist_cfg_1_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
pwr_el2_pbist_cfg_1_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.460 el2_1_dft_embed_pbist

Signal Name	Dir	Default Val	Description
el2_1_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
el2_1_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
el2_1_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
el2_1_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
el2_1_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
el2_1_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
el2_1_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
el2_1_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
el2_1_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
el2_1_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
el2_1_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
el2_1_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
el2_1_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
el2_1_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
el2_1_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
el2_1_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
el2_1_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
el2_1_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
el2_1_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.461 el2_1_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
el2_1_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.462 el2_1_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
el2_1_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.463 el2_1_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
el2_1_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
el2_1_dft_pbist_pwr_clk_en_ack	Out	el2_1_dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.464 el2_1_pbist_tclk

Signal Name	Dir	Default Val	Description
el2_1_pbist_tclk_clk	In	1'd0	Single ended clock
el2_1_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.465 el2_1_pbist_rst

Signal Name	Dir	Default Val	Description
el2_1_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.466 bisor

Signal Name	Dir	Default Val	Description
el2_1_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
el2_1_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
el2_1_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
el2_1_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
el2_1_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
el2_1_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
el2_1_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
el2_1_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.467 aw1 power disable interface

Signal Name	Dir	Default Val	Description
aw1_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw1_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw1_disable_done	In	1'd0	Ack to SCR to stop clocks
aw1_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw1_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw1_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.468 el2_mem1 power disable interface

Signal Name	Dir	Default Val	Description
pwr_el2_mem1_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
pwr_el2_mem1_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
pwr_el2_mem1_disable_done	In	1'd0	Ack to SCR to stop clocks
pwr_el2_mem1_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
pwr_el2_mem1_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
pwr_el2_mem1_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.469 dft_mem1 ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
el2_dft_mem1_clk_force_en	In	1'd0	dft_clk_force_en

9.4.470 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
el2_pbist_2_clkstop_req	In	1'd0	Clock stop request
el2_pbist_2_clkstop_ack	Out	pbist_2_clkstop_req	Clock stop acknowledge

9.4.471 Target .el2_pbist_cfg_2 power disable

Signal Name	Dir	Default Val	Description
pwr_el2_pbist_cfg_2_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
pwr_el2_pbist_cfg_2_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
pwr_el2_pbist_cfg_2_disable_done	In	1'd0	Ack to SCR to stop clocks
pwr_el2_pbist_cfg_2_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
pwr_el2_pbist_cfg_2_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
pwr_el2_pbist_cfg_2_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.472 el2_2_dft_embed_pbist

Signal Name	Dir	Default Val	Description
el2_2_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
el2_2_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
el2_2_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
el2_2_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
el2_2_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
el2_2_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
el2_2_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
el2_2_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
el2_2_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
el2_2_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
el2_2_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
el2_2_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
el2_2_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
el2_2_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
el2_2_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
el2_2_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
el2_2_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
el2_2_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
el2_2_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.473 el2_2_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
el2_2_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.474 el2_2_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
el2_2_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.475 el2_2_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
el2_2_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
el2_2_dft_pbist_pwr_clk_en_ack	Out	el2_2_dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.476 el2_2_pbist_tclk

Signal Name	Dir	Default Val	Description
el2_2_pbist_tclk_clk	In	1'd0	Single ended clock
el2_2_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.477 el2_2_pbist_rst

Signal Name	Dir	Default Val	Description
el2_2_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.478 bisor

Signal Name	Dir	Default Val	Description
el2_2_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
el2_2_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
el2_2_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
el2_2_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
el2_2_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
el2_2_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
el2_2_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
el2_2_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.479 aw2 power disable interface

Signal Name	Dir	Default Val	Description
aw2_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw2_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw2_disable_done	In	1'd0	Ack to SCR to stop clocks
aw2_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw2_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw2_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.480 el2_mem2 power disable interface

Signal Name	Dir	Default Val	Description
pwr_el2_mem2_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
pwr_el2_mem2_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
pwr_el2_mem2_disable_done	In	1'd0	Ack to SCR to stop clocks
pwr_el2_mem2_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
pwr_el2_mem2_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
pwr_el2_mem2_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.481 dft_mem2 ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
el2_dft_mem2_clk_force_en	In	1'd0	dft_clk_force_en

9.4.482 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
el2_pbist_3_clkstop_req	In	1'd0	Clock stop request
el2_pbist_3_clkstop_ack	Out	pbist_3_clkstop_req	Clock stop acknowledge

9.4.483 Target .el2_pbist_cfg_3 power disable

Signal Name	Dir	Default Val	Description
pwr_el2_pbist_cfg_3_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
pwr_el2_pbist_cfg_3_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
pwr_el2_pbist_cfg_3_disable_done	In	1'd0	Ack to SCR to stop clocks
pwr_el2_pbist_cfg_3_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
pwr_el2_pbist_cfg_3_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
pwr_el2_pbist_cfg_3_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.484 el2_3_dft_embed_pbist

Signal Name	Dir	Default Val	Description
el2_3_dft_embed_pbist_dft_pbist_ctrl_rst_n	In	1'b1	ATE Reset
el2_3_dft_embed_pbist_dft_pbist_resume	In	1'd0	dft_pbist_resume
el2_3_dft_embed_pbist_dft_pbist_prog[15:0]	In	16'd0	dft_pbist_prog
el2_3_dft_embed_pbist_dft_pbist_vlct_mode	In	1'd0	dft_pbist_vlct_mode
el2_3_dft_embed_pbist_dft_pbist_gonogo_ovrrd_en	In	1'd0	dft_pbist_gonogo_ovrrd_en
el2_3_dft_embed_pbist_dft_pbist_clk_mux_sel_dft[3:0]	In	4'd0	dft_pbist_clk_mux_sel_dft
el2_3_dft_embed_pbist_dft_pbist_wr_enable_to_comb	Out	1'd0	dft_pbist_wr_enable_to_comb
el2_3_dft_embed_pbist_dft_pbist_retention_to_comb	Out	1'd0	dft_pbist_retention_to_comb
el2_3_dft_embed_pbist_dft_pbist_ext_err	Out	1'd0	dft_pbist_ext_err
el2_3_dft_embed_pbist_dft_pbist_fail_to_comb	Out	1'b1	dft_pbist_fail_to_comb
el2_3_dft_embed_pbist_dft_pbist_done_to_comb	Out	1'd0	dft_pbist_done_to_comb
el2_3_dft_embed_pbist_dft_pbist_run_to_comb	Out	1'd0	dft_pbist_run_to_comb
el2_3_dft_embed_pbist_dft_pbist_logout_to_comb	Out	1'd0	dft_pbist_logout_to_comb
el2_3_dft_embed_pbist_dft_pbist_diag_clk_from_comb	In	1'd0	dft_pbist_diag_clk_from_comb
el2_3_dft_embed_pbist_dft_pbist_rom_mode_from_comb	In	1'd0	dft_pbist_rom_mode_from_comb
el2_3_dft_embed_pbist_dft_pbist_rdata_to_comb[15:0]	Out	16'd0	dft_pbist_rdata_to_comb
el2_3_dft_embed_pbist_dft_pbist_id[6:0]	In	7'd0	dft_pbist_id
el2_3_dft_embed_pbist_dft_pbist_ram_rgs_override	In	1'd0	dft_pbist_ram_rgs_override
el2_3_dft_embed_pbist_dft_pbist_ram_freeze	In	1'd0	dft_pbist_ram_freeze

9.4.485 el2_3_dft_pbist_cpu

Signal Name	Dir	Default Val	Description
el2_3_dft_pbist_cpu_intr	Out	1'd0	Interrupt signal

9.4.486 el2_3_dft_pbist_safety_error

Signal Name	Dir	Default Val	Description
el2_3_dft_pbist_safety_error_intr	Out	1'd0	Interrupt signal

9.4.487 el2_3_dft_pbist_pwr

Signal Name	Dir	Default Val	Description
el2_3_dft_pbist_pwr_clk_en	In	1'd1	Clock-gate enable to IP
el2_3_dft_pbist_pwr_clk_en_ack	Out	el2_3_dft_pbist_pwr_clk_en	Clock-gate enable acknowledge from IP

9.4.488 el2_3_pbist_tclk

Signal Name	Dir	Default Val	Description
el2_3_pbist_tclk_clk	In	1'd0	Single ended clock
el2_3_pbist_tclk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.489 el2_3_pbist_rst

Signal Name	Dir	Default Val	Description
el2_3_pbist_rst_mod_g_rst_n	In	1'd1	Module level main reset

9.4.490 bisor

Signal Name	Dir	Default Val	Description
el2_3_bisor_dft_self_repair_mode	In	1'd0	dft_self_repair_mode
el2_3_bisor_dft_shadow_fdi	In	1'd0	dft_shadow_fdi
el2_3_bisor_dft_shadow_fdo	Out	1'd0	dft_shadow_fdo
el2_3_bisor_dft_too_many_fails	Out	1'd0	dft_too_many_fails
el2_3_bisor_dft_shadow_chain_enable	In	1'd0	dft_shadow_chain_enable
el2_3_bisor_dft_diag_clk_enable	In	1'd0	dft_diag_clk_enable
el2_3_bisor_dft_shadow_shift_mode	In	1'd0	dft_shadow_shift_mode
el2_3_bisor_dft_pbist_bisor_reset_n	In	1'd0	dft_pbist_bisor_reset_n

9.4.491 aw3 power disable interface

Signal Name	Dir	Default Val	Description
aw3_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
aw3_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
aw3_disable_done	In	1'd0	Ack to SCR to stop clocks
aw3_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
aw3_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
aw3_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.492 el2_mem3 power disable interface

Signal Name	Dir	Default Val	Description
pwr_el2_mem3_disable_req	In	1'd0	LPSC requesting SCR to start clkstop sequence
pwr_el2_mem3_disable_ack	Out	1'd0	SCR has finished transactions, endpoints can now begin clock-stop sequence
pwr_el2_mem3_disable_done	In	1'd0	Ack to SCR to stop clocks
pwr_el2_mem3_disable_force	In	1'd0	LPSC in FORCE mode, ignore clkstop handshake
pwr_el2_mem3_disable_vbusreq	Out	1'd0	vbus_req from scr used for wakeup
pwr_el2_mem3_xfrtrck_idle	Out	1'd1	xfrtrck_idle

9.4.493 dft_mem3 ip interface for unique clk force en control

Signal Name	Dir	Default Val	Description
el2_dft_mem3_clk_force_en	In	1'd0	dft_clk_force_en

9.4.494 divh_clk2_tft_clk_en control

Signal Name	Dir	Default Val	Description
el2_divh_clk2_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.495 divh_clk4_soc_tft_clk_en control

Signal Name	Dir	Default Val	Description
el2_divh_clk4_soc_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.496 divp_clk1_msmc_tft_clk_en control

Signal Name	Dir	Default Val	Description
el2_divp_clk1_msmc_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.497 divh_clk4_msmc_tft_clk_en control

Signal Name	Dir	Default Val	Description
el2_divh_clk4_msmc_dft_tft_clk_en	In	1'd1	Clock-gate enable to IP

9.4.498 el2 clk1 clock

Signal Name	Dir	Default Val	Description
msmc_clk_clk	In	1'd0	Single ended clock
msmc_clk_cs_func_clk_en	In	1'd0	catscan enable on kp_ti_ipg_clk_l1

9.4.499 msmc clk1 clock

Signal Name	Dir	Default Val	Description
msmc_clk_ctl_align_clk_div_rst_n	In	1'd0	Export align_clk_div
msmc_clk_ctl_align_clk_div_nomask_rst_n	In	1'd0	Export align_clk_div_nomask

9.4.500 rom_ase

Signal Name	Dir	Default Val	Description
rom_ase_dft_mem_rom_ase	In	1'd0	dft_mem_rom_ase

9.4.501 EFUSE interface

Signal Name	Dir	Default Val	Description
dft_rom_dft_mem_atpgm	In	1'd0	ATPGM control of memory

9.4.502 ac_fw_bypass

Signal Name	Dir	Default Val	Description
ac_fw_bypass_fw_bypass	In	1'b0	Firewall off bit

9.4.503 ac_fw_id_0

Signal Name	Dir	Default Val	Description
ac_fw_id_0_ipcfg[15:0]	In	16'd0	ip config pin

9.4.504 firewall security exception bit

Signal Name	Dir	Default Val	Description
ac_fw_exp_exception_pending	Out	1'b0	Exception pending signal

9.4.505 firewall safety exception bit

Signal Name	Dir	Default Val	Description
ac_fw_exp_sf_exception_pending	Out	1'b0	Exception pending signal

9.4.506 dft_compression busdef

Signal Name	Dir	Default Val	Description
el2_mem0_dft_block_en	In	1'd1	dft_block_en
el2_mem0_dft_partition_en	In	1'd1	dft_partition_en
el2_mem0_dft_compression_cmle	In	1'd0	dft_compression_cmle
el2_mem0_dft_compression_mode	In	1'd0	dft_compression_mode
el2_mem0_dft_compression_spread	In	1'd0	dft_compression_spread
el2_mem0_dft_elastic_reset	In	1'd0	dft_elastic_reset
el2_mem0_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
el2_mem0_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
el2_mem0_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
el2_mem0_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
el2_mem0_dft_struct_intest	In	1'd0	dft_struct_intest
el2_mem0_dft_struct_extest	In	1'd0	dft_struct_extest
el2_mem0_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
el2_mem0_dft_struct_si[35:0]	In	36'd0	dft_struct_si
el2_mem0_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.507 el2_mem0_rst

Signal Name	Dir	Default Val	Description
el2_mem0_mod_g_rst_n	In	1'd1	Module level main reset

9.4.508 Clken interface

Signal Name	Dir	Default Val	Description
el2_mem0_clk_en	In	1'd1	Clock-gate enable to IP
el2_mem0_clk_en_ack	Out	el2_mem0_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.509 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
el2_mem0_clkstop_req	In	1'd0	Clock stop request
el2_mem0_clkstop_ack	Out	el2_mem0_clkstop_clkstop_req	Clock stop acknowledge

9.4.510 efuse_ctl busdef

Signal Name	Dir	Default Val	Description
el2_mem0_fclrz	In	1'd0	efuse clear
el2_mem0_fclk	In	1'd0	efuse clock

9.4.511 efuse_data busdef

Signal Name	Dir	Default Val	Description
el2_mem0_fdi	In	1'd0	efuse data in
el2_mem0_fdo	Out	1'd0	efuse data out

9.4.512 el2_pwr busdef

Signal Name	Dir	Default Val	Description
el2_mem0_pwr_log_iso_en	In	1'd0	pwr
el2_mem0_pwr_log_ponin[7:0]	In	8'hFF	pwr
el2_mem0_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
el2_mem0_pwr_log_ponout[7:0]	Out	8'hFF	pwr
el2_mem0_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.513 dft_compression busdef

Signal Name	Dir	Default Val	Description
el2_mem1_dft_block_en	In	1'd1	dft_block_en
el2_mem1_dft_partition_en	In	1'd1	dft_partition_en
el2_mem1_dft_compression_cmle	In	1'd0	dft_compression_cmle
el2_mem1_dft_compression_mode	In	1'd0	dft_compression_mode
el2_mem1_dft_compression_spread	In	1'd0	dft_compression_spread
el2_mem1_dft_elastic_reset	In	1'd0	dft_elastic_reset
el2_mem1_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
el2_mem1_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
el2_mem1_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
el2_mem1_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
el2_mem1_dft_struct_intest	In	1'd0	dft_struct_intest
el2_mem1_dft_struct_extest	In	1'd0	dft_struct_extest
el2_mem1_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
el2_mem1_dft_struct_si[35:0]	In	36'd0	dft_struct_si
el2_mem1_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.514 el2_mem1_rst

Signal Name	Dir	Default Val	Description
el2_mem1_mod_g_rst_n	In	1'd1	Module level main reset

9.4.515 Clken interface

Signal Name	Dir	Default Val	Description
el2_mem1_clk_en	In	1'd1	Clock-gate enable to IP
el2_mem1_clk_en_ack	Out	el2_mem1_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.516 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
el2_mem1_clkstop_req	In	1'd0	Clock stop request
el2_mem1_clkstop_ack	Out	el2_mem1_clkstop_clkstop_req	Clock stop acknowledge

9.4.517 efuse_ctl busdef

Signal Name	Dir	Default Val	Description
el2_mem1_fclrz	In	1'd0	efuse clear
el2_mem1_fclk	In	1'd0	efuse clock

9.4.518 efuse_data busdef

Signal Name	Dir	Default Val	Description
el2_mem1_fdi	In	1'd0	efuse data in
el2_mem1_fdo	Out	1'd0	efuse data out

9.4.519 el2_pwr busdef

Signal Name	Dir	Default Val	Description
el2_mem1_pwr_log_iso_en	In	1'd0	pwr
el2_mem1_pwr_log_ponin[7:0]	In	8'hFF	pwr
el2_mem1_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
el2_mem1_pwr_log_ponout[7:0]	Out	8'hFF	pwr
el2_mem1_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.520 dft_compression busdef

Signal Name	Dir	Default Val	Description
el2_mem2_dft_block_en	In	1'd1	dft_block_en
el2_mem2_dft_partition_en	In	1'd1	dft_partition_en
el2_mem2_dft_compression_cmle	In	1'd0	dft_compression_cmle
el2_mem2_dft_compression_mode	In	1'd0	dft_compression_mode
el2_mem2_dft_compression_spread	In	1'd0	dft_compression_spread
el2_mem2_dft_elastic_reset	In	1'd0	dft_elastic_reset
el2_mem2_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
el2_mem2_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
el2_mem2_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
el2_mem2_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
el2_mem2_dft_struct_intest	In	1'd0	dft_struct_intest
el2_mem2_dft_struct_extest	In	1'd0	dft_struct_extest
el2_mem2_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
el2_mem2_dft_struct_si[35:0]	In	36'd0	dft_struct_si
el2_mem2_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.521 el2_mem2_rst

Signal Name	Dir	Default Val	Description
el2_mem2_mod_g_rst_n	In	1'd1	Module level main reset

9.4.522 Clken interface

Signal Name	Dir	Default Val	Description
el2_mem2_clk_en	In	1'd1	Clock-gate enable to IP
el2_mem2_clk_en_ack	Out	el2_mem2_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.523 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
el2_mem2_clkstop_req	In	1'd0	Clock stop request
el2_mem2_clkstop_ack	Out	el2_mem2_clkstop_clkstop_req	Clock stop acknowledge

9.4.524 efuse_ctl busdef

Signal Name	Dir	Default Val	Description
el2_mem2_fclrz	In	1'd0	efuse clear
el2_mem2_fclk	In	1'd0	efuse clock

9.4.525 efuse_data busdef

Signal Name	Dir	Default Val	Description
el2_mem2_fdi	In	1'd0	efuse data in
el2_mem2_fdo	Out	1'd0	efuse data out

9.4.526 el2_pwr busdef

Signal Name	Dir	Default Val	Description
el2_mem2_pwr_log_iso_en	In	1'd0	pwr
el2_mem2_pwr_log_ponin[7:0]	In	8'hFF	pwr
el2_mem2_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
el2_mem2_pwr_log_ponout[7:0]	Out	8'hFF	pwr
el2_mem2_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

9.4.527 dft_compression busdef

Signal Name	Dir	Default Val	Description
el2_mem3_dft_block_en	In	1'd1	dft_block_en
el2_mem3_dft_partition_en	In	1'd1	dft_partition_en
el2_mem3_dft_compression_cmle	In	1'd0	dft_compression_cmle
el2_mem3_dft_compression_mode	In	1'd0	dft_compression_mode
el2_mem3_dft_compression_spread	In	1'd0	dft_compression_spread
el2_mem3_dft_elastic_reset	In	1'd0	dft_elastic_reset
el2_mem3_dft_lp_gating_enable	In	1'd0	dft_lp_gating_enable
el2_mem3_dft_lp_gating_load	In	1'd0	dft_lp_gating_load
el2_mem3_dft_smartscan_enable	In	1'd0	dft_smartscan_enable
el2_mem3_dft_smartscan_broadcast	In	1'd0	dft_smartscan_broadcast
el2_mem3_dft_struct_intest	In	1'd0	dft_struct_intest
el2_mem3_dft_struct_extest	In	1'd0	dft_struct_extest
el2_mem3_dft_gating_iso_en	In	1'd0	dft_gating_iso_en
el2_mem3_dft_struct_si[35:0]	In	36'd0	dft_struct_si
el2_mem3_dft_struct_so[35:0]	Out	36'd0	dft_struct_so

9.4.528 el2_mem3_rst

Signal Name	Dir	Default Val	Description
el2_mem3_mod_g_rst_n	In	1'd1	Module level main reset

9.4.529 Clken interface

Signal Name	Dir	Default Val	Description
el2_mem3_clk_en	In	1'd1	Clock-gate enable to IP
el2_mem3_clk_en_ack	Out	el2_mem3_clken_clk_en	Clock-gate enable acknowledge from IP

9.4.530 PMU Clock stp req/ack interface

Signal Name	Dir	Default Val	Description
el2_mem3_clkstop_req	In	1'd0	Clock stop request
el2_mem3_clkstop_ack	Out	el2_mem3_clkstop_clkstop_req	Clock stop acknowledge

9.4.531 efuse_ctl busdef

Signal Name	Dir	Default Val	Description
el2_mem3_fclrz	In	1'd0	efuse clear
el2_mem3_fclk	In	1'd0	efuse clock

9.4.532 efuse_data busdef

Signal Name	Dir	Default Val	Description
el2_mem3_fdi	In	1'd0	efuse data in
el2_mem3_fdo	Out	1'd0	efuse data out

9.4.533 el2_pwr busdef

Signal Name	Dir	Default Val	Description
el2_mem3_pwr_log_iso_en	In	1'd0	pwr
el2_mem3_pwr_log_ponin[7:0]	In	8'hFF	pwr
el2_mem3_pwr_log_pgoodin[7:0]	In	8'hFF	pwr
el2_mem3_pwr_log_ponout[7:0]	Out	8'hFF	pwr
el2_mem3_pwr_log_pgoodout[7:0]	Out	8'hFF	pwr

#INTERNAL#

10. Physical Implementation

10.1 Clock Diagram

Refer to [AW spec](#) Clock diagram (section 9.1) for Analytics wrap clocking diagrams

Below is the clock diagram for tda54_el2_wrap subchip.

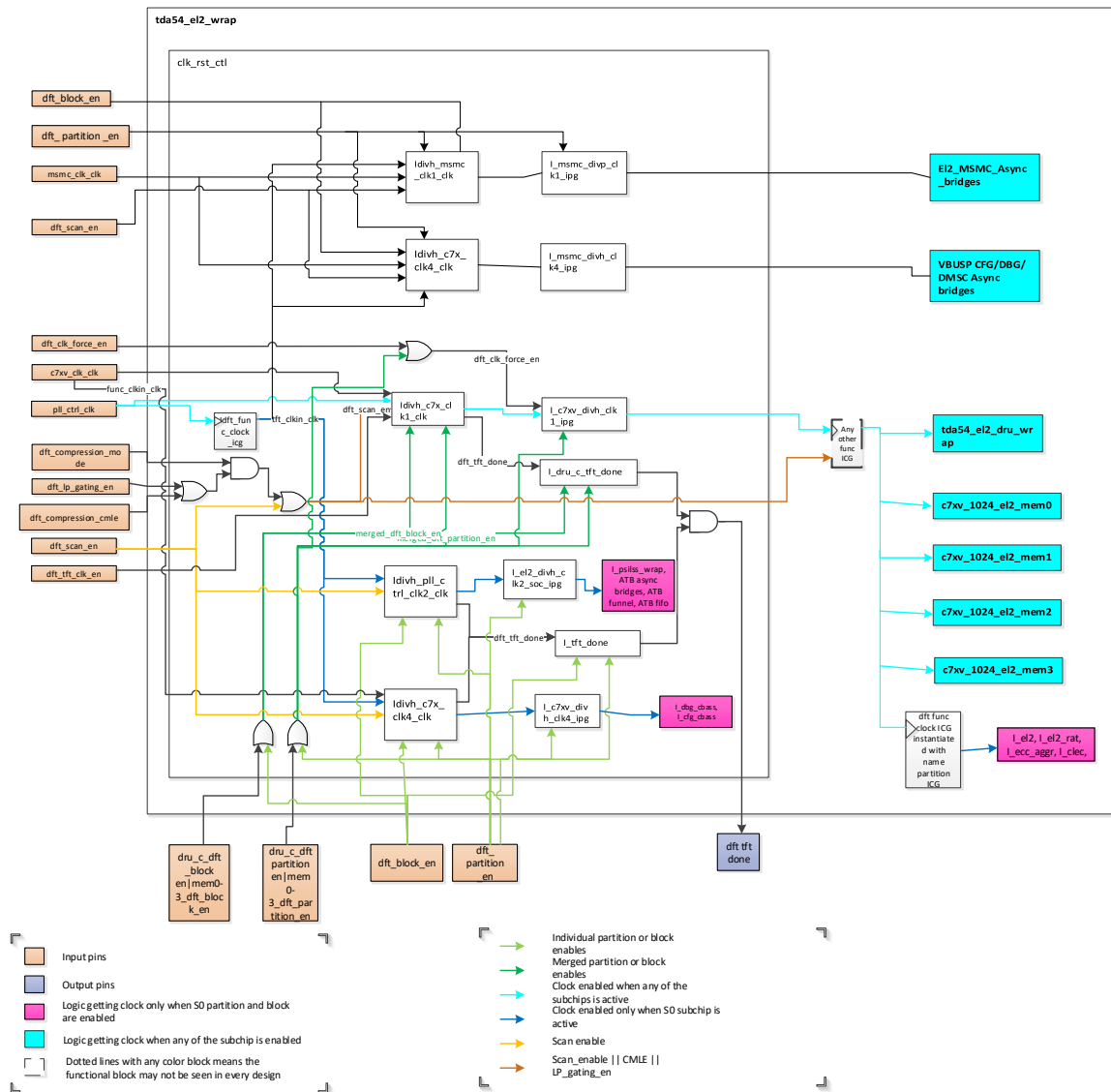


Figure 5 Clock diagram (tda54_el2_wrap)

10.2 Reset

Refer to [AW spec](#) Reset diagram (section 9.2) for Analytics wrap reset diagrams

Below is the reset diagram for tda54_el2_wrap subchip.

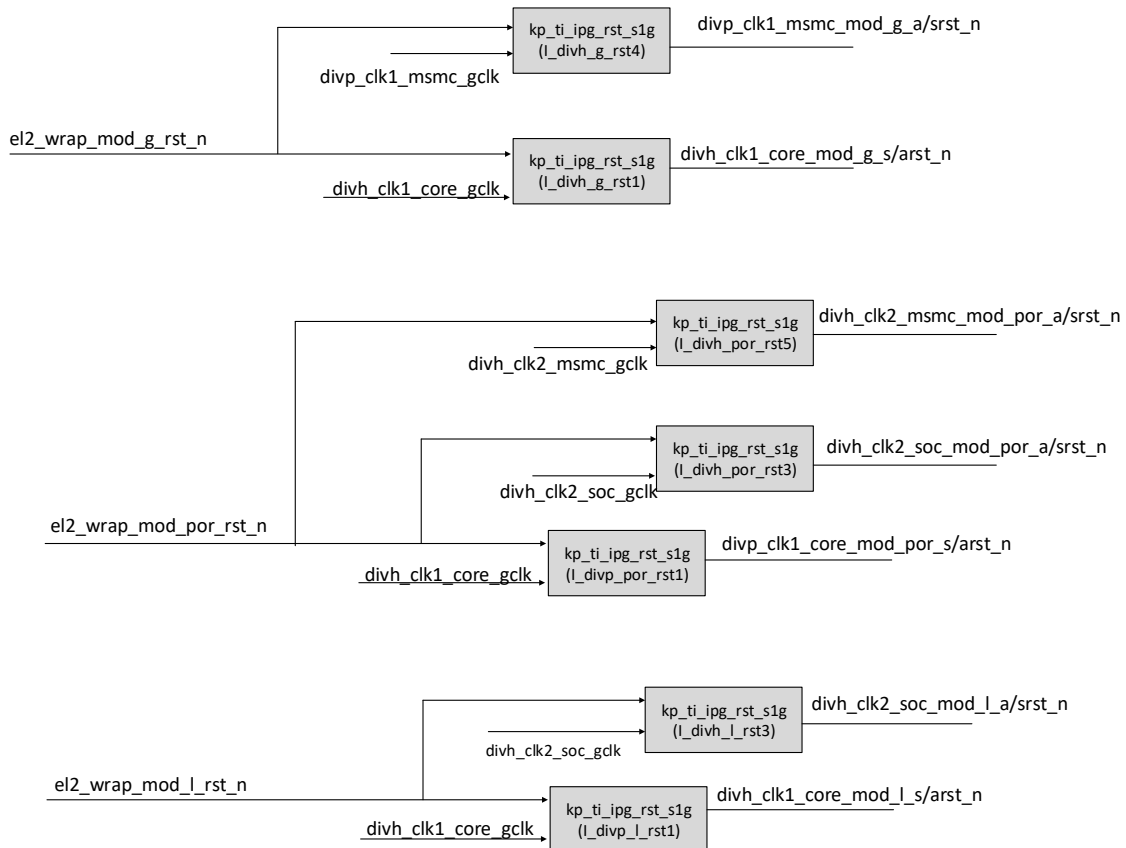


Figure 6 Reset diagram

10.3 Latches

There are no latches in the design.

10.4 Placement and Hierarchy Information

There are no special placement requirements for the design.

10.5 Hard Macro Information

Table below shows subchip partitioning within Analytics Cluster.

Soft wrapper	Top level Subchip name	Level 1 hierarchical subchips	Level 2 hierarchical subchips
tda54_analytics_cluster	tda54_analytics_wrap x 4	tda54_c7xv_core_wrap	
		tda54_mma_wrap	mma_array_wrap x 4
	tda54_el2_wrap	tda54_el2_dru_wrap	
		c7xv_1024_el2_mem0	
		c7xv_1024_el2_mem1	
		c7xv_1024_el2_mem2	
		c7xv_1024_el2_mem3	

10.6 MCP requirements

- EL2 SRAMs requires MCP=2 constraint for timing closure

10.7 Power/Voltage

- Design needs to be timing closed 1.5 GHz for 0.75v and high opp mode at 0.85v.

#/INTERNAL#

11. DFT

Note: This spec only covers details of tda54_el2_wrap subchip DFT. For Analytics wrap DFT clocking and other details, refer to [analytics wrap](#) spec.

11.1 Scan Implementation

This module adheres to the multiplexed scan methodology. There are no special scan considerations for this module.

11.2 IDDQ Modes

There are no special IDDQ considerations for this module.

11.3 LBIST requirements

Here is the list of subchips where LBIST is being inserted

- tda54_c7xv_core_wrap
- tda54_analytics_wrap

There is no LBIST inserted in tda54_mma_wrap subchip but LBIST isolation wrapper is present to enable isolation when we are running LBIST for tda54_analytics_wrap

11.4 PBIST for SRAMs

tda54_el2_wrap has 4 PBIST controllers to cover PBIST test for 24MB EL2, DRU and CLEC SRAM.

PBIST controller instance	Associated Memories
Pbist_0	EL2 MEM0 SRAM, DRU SRAM, CLEC SRAM
Pbist_1	EL2 MEM1 SRAM
Pbist_2	EL2 MEM2 SRAM
Pbist_3	EL2 MEM3 SRAM

#INTERNAL#

11.5 I/O DFT

This module does not contain any IODFT logic.

Subchip DFT clocking guidelines:

<https://confluence.itg.ti.com/pages/viewpage.action?spaceKey=SIDEVJ7&title=Subchip+DFT+Clocking+-+Architectural+intent+and+guidelines>

<https://confluence.itg.ti.com/display/SIDEVJ7/Subchip+DFT+Clocking+-+Hierarchical+DFT+subchips>

Dividers present in the tda54_el2_wrap:

<u>Divider module</u>	<u>Instance name</u>	<u>DFT connection case type (refer to above confluence page for details)</u>	<u>Div factor</u>
kp_ti_ipg_clk_divhdist5_tft_mux_v3	ldivh_c7x_clk1_clk	Case 3	func_divfactor = 0, tft_divfactor = 0. (both are 1 when dft_el2_divfactor_sel_wir is set)
kp_ti_ipg_clk_divhdist5_tft_mux_v3	ldivh_c7x_clk4_clk	Case 3	func_divfactor = 1, tft_divfactor = 1
kp_ti_ipg_clk_divhdist5_tft_mux_v3	ldivh_pll_ctrl_clk2_clk	Case 1	func_divfactor = 1, tft_divfactor = 1

dft tft done module instances:

<u>Instance name</u>	<u>Divider outputs combined in this instance</u>
kp_ti_ipg_dft_tft_done	divh_clk1_dft_tft_done, divh_clk4_dft_tft_done, divh_clk2_soc_dft_tft_done

#/INTERNAL#

12. Programmers Guide

12.1 Initialization sequence

CPU boot sequence

- Power up EL2_WRAP LPSC
- Power up EL2_MEMx LPSCs (All or few as per application needs)
- Power up Analytics Wrap LPSCs (All or few as per application needs)
- Set CPU configuration registers located at CCC DMSC boot config space
- Set CPU boot vectors
- Setup firewalls and RAT configurations as per application needs (this step can also be done anytime after Powering up EL2_WRAP LPSCs)
- Enable All or few MMA LPSCs
- Enable All or few C7xv core LPSCs
 - CPUs start booting from reset boot vector
 - Other initialization sequence is application specific, refer to CPU spec for more details on boot configuration

12.2 Real-time operating requirements

- NA

12.3 Power Management

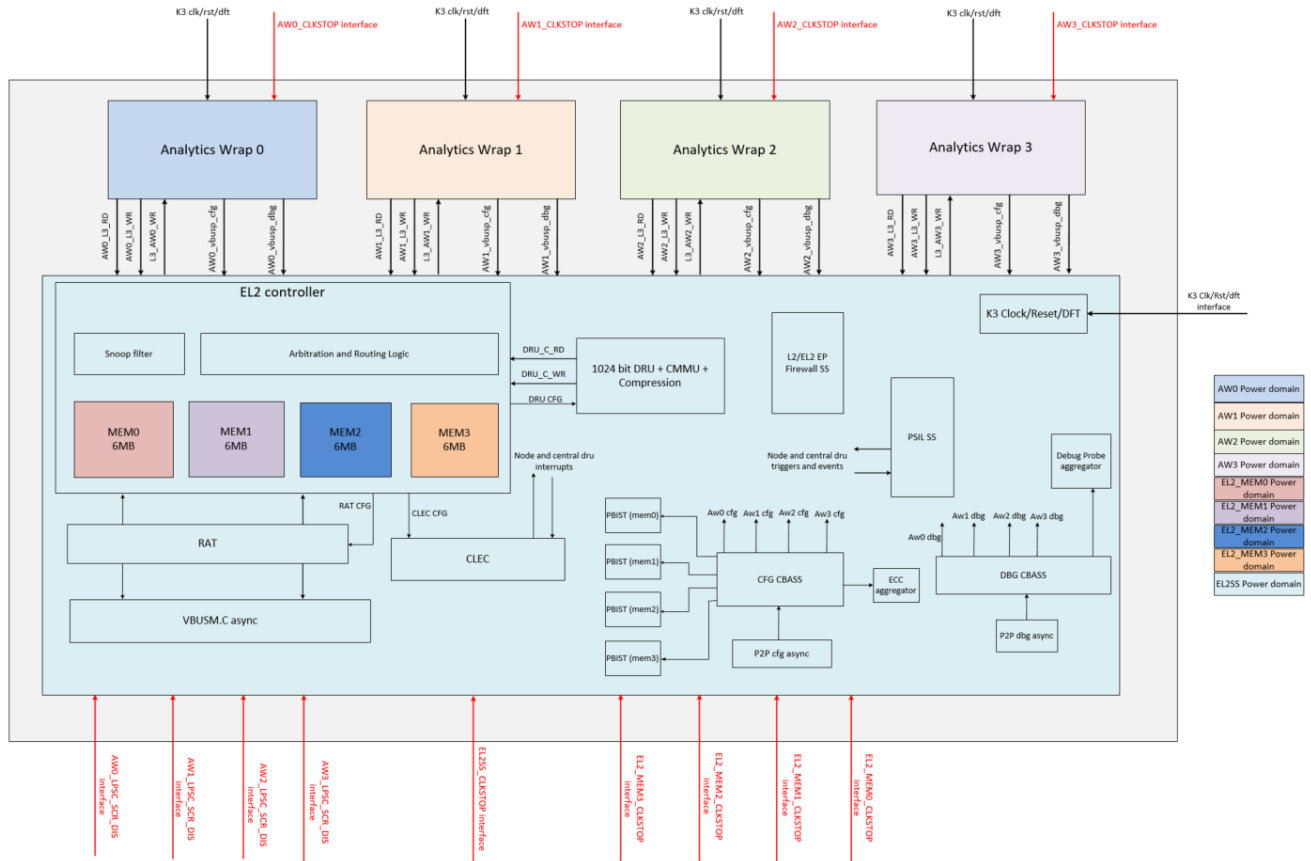


Figure 6 LPSC domains

12.3.1 Power domain to LPSC interface mapping

Power Domain	PWR interface	Subchip type (with regards to power domain)	CLK_EN/CLKSTOP interface
MMA Power domain	aw<0-3>_mma_pwr	Parent	aw<0-3>_c7xv_mma_clkstop aw<0-3>_c7xv_mma_clken
	aw<0-3>_r00_pwr	Child	
	aw<0-3>_r01_pwr	Child	
	aw<0-3>_r10_pwr	Child	
	aw<0-3>_r11_pwr	Child	
C7xv Core Power domain	aw<0-3>_c7xv_core_pwr	Parent	aw<0-3>_c7xv_core_clkstop aw<0-3>_c7xv_core_clken
Analytics Wrap Power domain	aw<0-3>_c7xv_pwr	Parent	aw<0-3>_c7xv_wrap_clkstop aw<0-3>_c7xv_wrap_clken
EL2 MEM<0-3>	el2_mem<0-3>_pwr	Parent	el2_mem<0-3>_clken el2_mem<0-3>_clkstop
EL2 WRAP	el2_ss_pwr	Parent	el2_wrap_clken el2_wrap_clkstop

12.3.2 EL2SS powerdown sequence

- 1 Powerdown each Analytics wrap following the steps mentioned in [Analytics wrap](#) spec section 11.3.3
- 2 Check EL2 IDLE
- 3 Initiate pwr_scr_disable handshake for EL2SS Wrap LPSC (This handshake will happen at MSMC level and once complete, MSMC should not send any accesses to Analytics Cluster)
- 4 Initiate clockstop_req/ack handshake for EL2 MEM0 to MEM3 blocks LPSCs
- 5 Initiate clockstop_req/ack handshake for EL2SS wrap LPSC

12.3.3 LPSC dependency Table

LPSC to Enable	State dependency on other LPSCs					
	AWx LPSC	AWx_core LPSC	EL2SS LPSC	EL2_MEMx LPSC	AWx/AWx core PBIST LPSC	EL2 PBIST LPSC
EL2SS LPSC	OFF	OFF	NA	OFF	OFF	OFF
EL2_MEMx LPSC*	OFF	OFF	ON	NA	OFF	OFF
AWx LPSC	NA	OFF	ON	Don't care	OFF	Don't care
AWx_core LPSC	ON	NA	ON	Don't care	OFF	Don't care
MMAx LPSC	ON	ON	ON	Don't care	Don't care	Don't care
AWx/AWx core PBIST LPSC	ON	ON	ON	ON	NA	Don't care
EL2 PBIST LPSC	Don't care	Don't care	ON	ON	Don't care	NA

*EL2 controller requires EL2SS and EL2_MEM0 power domains to be ON and OFF together and EL2_MEM2/3 to be ON and OFF together for proper functionality.

EL2_MEM0-3 LPSCs only support static powerdown during boot time, i.e. system has to decide at boot time whether MEM0, MEM0-1 or MEM0-3 are needed for the application.

12.3.4 Cluster Analytics Power Manager (CAPM)

Due to MMA's larger 128x128 panel, idle to active and active to idle transitions causes steep di/dt slopes which is an issue for PDN stability.

To resolve this CAPM (cluster APM) module is integrated inside EL2 wrap which tracks C7x/MMA activity for all instances inside Analytics cluster and can stall idle to active and active to idle transitions such that di/dt steps are within acceptable range.

To achieve this, 128x128 MMA is split into 4 64x64 slices with each slice havin git's own clock gate which can be controlled by FSM inside CAPM module.

Software can also disable un-used slices for the kernels which uses smaller geometries to save dynamic power.

More details about CAPM module can be found in CAPM IP [spec](#).

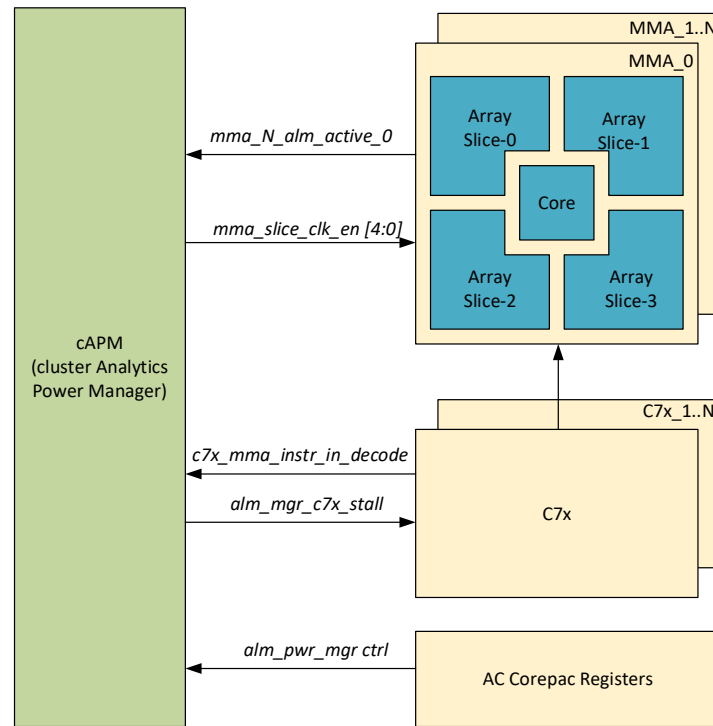


Figure 7 CAPM integration in cluster